GENERAL SCIENCE

TREND ANALYSIS (2016-2014)

PHYSICS

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	Mechanics	2	4	6	2
2	Heat	1	1	-	-
3	Properties of Matter	-	-	-	-
4	Waves	3	-	-	-
5	Optics	1	2	1	3
6	Electricity	1	1	3	2
7	Modern Physics	-	1	1	-
	Total	**8**	**9**	**11**	**7**

CHEMISTRY

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	General Chemistry	-	-	1	-
2	Atomic Structure	1	2	1	3
3	Radioactivity	1	-	-	1
4	Chemical Binding and Redox Reactions	-	-	-	-
5	Equilibrium , Gas Law and Catalyst	1	-	-	-
6	Solution	-	-	-	1
7	Acid, Base and Salt	1	-	-	1
8	Thermodynamics and Colloids	2	-	1	-
9	Electrochemistry	-	-	2	-
10	Inorganic Chemistry	3	3	3	1
11	Organic Chemistry	1	-	-	4
12	Man Made Materials	1	1	1	1
13	Environment and Its Pollution	-	3	2	1
	Total	**11**	**9**	**11**	**13**

BIOLOGY

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	Introduction	-	-	-	-
2	Cell and Inheritance	1	-	1	1
3	Origin of Life and Its Growth	-	2	1	1
4	Classifications of Plants and Animals	-	-	2	3
5	The Plant Life	1	1	1	1
6	Animal Physiology	8	2	3	3
7	Ecology and Environment	-	1	2	-
8	Applied Biology	1	4	-	1
	Total	**11**	**10**	**10**	**10**

01

PHYSICS

After analysing the previous year question papers, we have seen that 40-45 questions are asked from science section, out of these 20-22 questions are asked from Physics. From physics section, around 2 to 3 questions each are asked from topics like motion, sound, electric current and 1 to 2 questions each are asked from topics like optics, modern physics and heat.

MEASUREMENT, MOTION, WORK, ENERGY AND POWER

PHYSICAL QUANTITIES

All the quantities which can be measured directly or indirectly in terms of which law of physics are described and whose measurement is necessary, are called physical quantities. e.g. velocity, time, mass, etc.

Unit

The standard amount of a physical quantity chosen to measure the physical quantity of same kind is called a physical unit.

Fundamental and Derived Units

The unit of a defined set of physical quantities called fundamental quantities are known as fundamental units and the units for all other physical quantities, except fundamental quantities are known as derived units.

e.g. unit of speed can be derived from fundamental units as,

$$\text{Unit of speed} = \frac{\text{Unit of distance}}{\text{Unit of time}} = \frac{\text{m}}{\text{s}} = \text{ms}^{-1}$$

SI System

It is based on the following seven basic units and two supplementary units.

Name of Quantities	*Name of Units*	*Name of Quantities*	*Name of Units*
Basic Units			
Length	Metre	Thermodynamic temperature	Kelvin
Mass	Kilogram	Luminous intensity	Candela
Time	Second	Quantity of matter	Mole
Electric current	Ampere		
Supplementary Units			
Plane angle	Radian	Solid angle	Steradian

SCALAR AND VECTOR QUANTITIES

- Physical quantities which have magnitude only and no direction are called scalar quantities.
  e.g. mass, speed, volume, work, time, power, energy, etc.
- Physical quantities which have magnitude and direction both and which obey triangle law are called vector quantities.
  e.g. displacement, velocity, acceleration, force, momentum, torque, etc.

MECHANICS

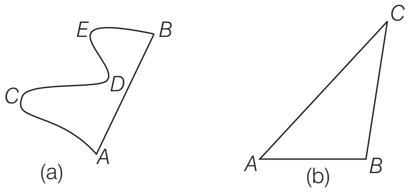
Mechanics deals with the study of motion of particles, rigid and deformable bodies and general system of particles.

Motion

A body is said to be in motion when its position changes continuously with respect to a stationary body taken as a reference point.

Distance and Displacement

- The **distance** covered by a body is the actual length of the path travelled by the body between the initial position and final position.
  - It is always positive.
  - It is a scalar quantity which has magnitude only. Its unit is metre.
- The **displacement** of a particle is the change in position of the particle in a particular direction and is given by a vector drawn from its initial position to its final position.
  - Displacement may be positive, negative or zero.
  - It is a vector quantity and its unit is also metre.
  - The magnitude of displacement may or may not be equal to the path length travelled by an object.
  - Displacement ≤ Distance



  - In Fig. (a) a particle starts from A and reach B through points C, D and E, then AB is the displacement and $(AC + CD + DE + EB)$ is the distance travelled.
  - Also, in Fig. (b), a particle starts from A and reach to C through point B, then AC is displacement and distance covered is $(AB + BC)$.

Speed

The path length or the distance covered by an object divided by the time taken by the object to cover that distance is called the speed of that object. Its unit is m/s or km/h.

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

- An object is said to be moving with a **uniform speed**, if it covers equal distances in equal intervals of time.
- An object is said to be moving with a **variable speed**, if it covers equal distances in unequal intervals of time or unequal distances in equal intervals of time.

Average Speed

- If a motion is the ratio of the total distance travelled by the object to the total time taken.
  i.e. average speed, $v_{\text{av}} = \dfrac{s_1 + s_2 + s_3 + \dots}{t_1 + t_2 + t_3 + \dots}$
- If the body covers equal distances with different speeds, then $v_{\text{av}} = \dfrac{2v_1v_2}{v_1 + v_2}$

  When the speed of an object is constantly changing, then the speed at a particular moment or instant of time is called the **instantaneous speed** of that object.

Velocity

The rate of change in position or displacement of a body with time is called the velocity of that body. It is a vector quantity.

$$\text{Velocity} = \frac{\text{Displacement in particular direction}}{\text{Time taken}}$$

- A body is said to be moving with **uniform velocity**, if equal displacements of the body take place in same direction in equal intervals of time.
- If a body moves in such a way that its speed or the direction or both changes with time, the body is said to have **variable velocity**.

Average Velocity

- If a body is defined as the net displacement divided by the time taken for displacement.

$$v_{av} = \frac{x_2 - x_1}{t_2 - t_1}$$

  As net displacement $= x_2 - x_1$ and time taken $= t_2 - t_1$
- Average velocity could be zero, positive or negative but average speed is always positive for moving body. When a particle returns to the starting point, its average velocity is zero but average speed is not zero.

Relative Velocity

- When two bodies are moving in the straight line, the speed (or velocity) of one with respect to another is known as its relative speed (or velocity).
- When two bodies are moving in the same direction, then relative velocity $= v_1 - v_2$. When the two bodies are moving in opposite directions, then relative velocity $= v_1 + v_2$.

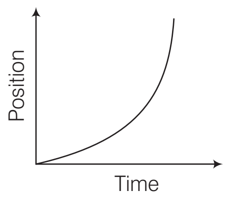
Acceleration

- It is the rate of change of velocity with respect to time. Its SI unit is m/s^2 and CGS is cm/s^2. It is a vector quantity.
- An object is said to be moving with **uniform acceleration**, if its velocity changes by equal amounts in equal intervals of time.
- An object is said to be moving with a **variable acceleration**, if its velocity changes by unequal amounts in equal intervals of time.
- If the velocity of an object increases without change in direction, it is said to be moving with positive acceleration.
- If the velocity of an object decreases without change in direction, the object is said to be moving with negative acceleration or deceleration or retardation.
- Acceleration of an object is zero, if it is at rest or moving with uniform velocity.

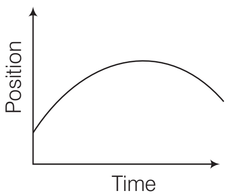
Graphical Representation of Motion

1. Position–Time Graphs

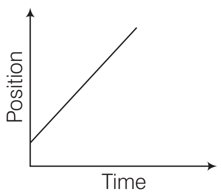
(*i*) When distance covered by a moving object goes on increasing with time, the object is said to have positive acceleration (i.e. velocity increases with time).



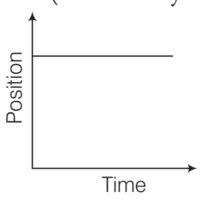
(*ii*) When distance covered by a moving object goes on decreasing with time, the object is said to have negative acceleration (i.e. velocity decreases with time).



(*iii*) When the moving object covers equal distance in equal time, the object is said to have zero acceleration i.e. uniform velocity.

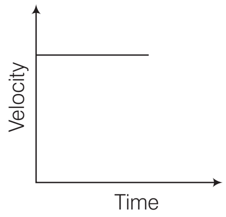


(*iv*) When the position of object does not change with time, the object is at rest (i.e. velocity is zero).

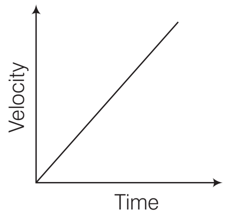


2. Velocity-Time Graphs

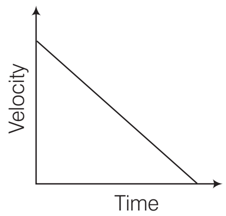
(*i*) In case of zero acceleration, the velocity of the object does not change with time.



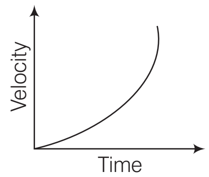
(*ii*) If the object is moving with uniform positive acceleration having zero initial velocity, then the velocity-time graph is a straight line starting from origin.



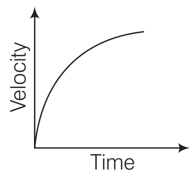
(*iii*) In case of negative and constant acceleration, the velocity of the object decreases linearly with time.



(*iv*) When the acceleration of the object is increasing, line bending towards velocity axis represent the increasing acceleration in the body.



(*v*) When the acceleration of the object is decreasing, line bending towards time axis represents the decreasing acceleration in the body.



Equations of Motion

For a moving object, the relations among its velocity, displacement, time and acceleration can be represented by equations. These equations are called equations of motion.

- For motion a straight line with constant acceleration a

  (i) $v = u + at$ (ii) $s = ut + \frac{1}{2}at^2$

  (iii) $v^2 = u^2 + 2as$

- Equation of motion under gravity, (downward direction)

  (i) $v = u + gt$ (ii) $h = ut + \frac{1}{2}gt^2$

  (iii) $v^2 = u^2 + 2gh$

Where, u = initial velocity

v = final velocity

g = gravitational acceleration

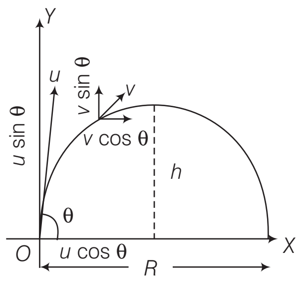
If an object is thrown vertically upward away from the earth, then we shall substitute $-g$ in place of g in the above equations. The value of g is 9.8 m/s^2.

➡ **Note** If a body starts from rest and moves with uniform acceleration, then distance covered by the body in t sec is proportional to t^2(i.e. $s \propto t^2$).

Projectile Motion

When a body moves under an acceleration whose direction is different from the direction of the initial velocity, then both the magnitude and direction of its velocity changes with time. Hence, the body moves on a curved path in a plane. This type of motion is called projectile motion.

- A bullet fired from a rifle and a body dropped from the window of a moving train shows the projectile motion.



- Flight time of projectile, $T = \frac{2u\sin\theta}{g}$
- Height of projectile, $h = \frac{u^2\sin^2\theta}{2g}$
- Range of projectile, $R = \frac{u^2\sin 2\theta}{g}$.
- We drop down a ball from a roof and at the same time throw another ball in a horizontal direction, then both the balls would strike the earth simultaneously at different places.
- The horizontal range is the same when the body is projected at θ and $(90° - \theta)$.
- The horizontal range of a projectile is maximum when angle of projection is 45°.
- When two balls of different masses are projected horizontally they will reach ground at same time.

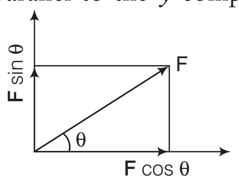
FORCE

Force is a push or pull which produces or tends to produce a change in uniform motion of body; stops or tends to stop a body which is in motion. Its unit is Newton.

- It is a vector quantity and the straight line along which a force is directed is called the line of action of the force.
- The rotational effect of a force on a body about an axis of rotation is described in terms of moment of force.
- CGS unit of force is dyne i.e. 1 dyne $= 10^{-5}$N.

Force is of two types

(i) **Component of Force** Forces acting at same angle from the coordinate axes can be resolved into mutually perpendicular forces called components. The component of a force parallel to the X-axis is called x-component, parallel to the y-component and so on.



Here,

- x-component of force **F** is $\mathbf{F}\cos\theta$.
- y-component of force **F** is $\mathbf{F}\sin\theta$.

(ii) **Contact and Non-contact Force** The forces which act on bodies when they are in physical contact are called **contact forces**. e.g. muscular force, frictional force, etc.

A force which can be exerted by an object on another object even from a distance (without physical contact with each other) is called a **non-contact force**. e.g. magnetic force, electrostatic force and gravitational force.

NEWTON'S LAWS OF MOTION

Newton has given three laws of motion as below:

First Law

It states that every body continues in its state of rest or in uniform motion in a straight line, unless it is compelled by an external force to change that state.

First law is also called **law of Galileo** or **law of inertia.**

- The property of bodies by virtue of which they oppose only change in their present state is called **inertia**.
- Mass is a measure of the inertia of a body.

Here some practical examples are as follows in which first law is used.

- Athlete runs some distance, before taking a long jump.
- A ball thrown upward in a train moving with uniform velocity returns to the hand of thrower.
- The mud from the wheels of a moving vehicle flies-off tangentially.
- When we shake the branch of a mango tree, the mangoes fall down.
- A man jumping from a moving train may fall down.
- If a cloth placed under a book is given a sudden pull, it comes out without disturbing the book.
- When a horse suddenly starts moving, the rider falls backward.
- When a running horse suddenly stops, the rider falls forwards.
- We hit a carpet with a stick to remove the dust.

Second Law

The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts. According to second law

$$F \propto \frac{dp}{dt} \text{ or } F = k\frac{dp}{dt}$$

where, k is constant of proportionality

$$F = k\frac{d}{dt}(mv) = kma$$

for simplicity let $k = 1$.

$$\Rightarrow \quad F = ma$$

$$1 \text{ newton } = 10^5 \text{ dyne}$$

- Newton's second law gives the magnitude of force.
- Newton's first law is contained in the second law.

Here some practical examples are as follows in which second law is used.

- China wires are wrapped in straw or paper before packing.
- A person falling on pucca floor (or frozen ice) is likely to receive more injuries than one falling on kuccha floor (loose earth).
- While catching a ball, a cricket player lowers his hands to save himself from getting hurt.
- Bogies of the trains are provided with buffers to avoid severe jerks during shunting of trains.

Third Law

According to this law, to every action there is an equal and opposite reaction. 'Action and reaction always act on the different bodies.' Here some practical examples are as follows in which third law is used

- In order to walk, we press the ground in backward direction with our feet. As a result, the ground pushes us in forward direction.
- It is difficult to drive a nail into a wooden block without holding the block.
- A jet plane moves on the principle of Newton's third law of motion. As exhaust gases come out from the nozzle at a greater speed, the reaction of the same moves the plane forwards.

Validity of Newton's Laws of Motion

Newton's laws are valid in classical physics only under certain conditions

- In general, the distances which one work must be much greater than the size of atoms and molecules. Else, quantum mechanics must be used in place of classical mechanics.
- In general, the speeds which one work with must be much less than the speed of light. Else, relativistic mechanics must be used in place of classical mechanics.
- Also, this is valid with respect to inertial frame of reference only.

Linear Momentum

- The linear momentum of a body is defined as, the product of its mass and its velocity i.e.
  Momentum = Mass × Velocity, $p = m \times v$
- It's direction is the same as the direction of velocity of the body. It is a vector quantity. It's SI unit is kg-m/s.
- Concept of momentum was introduced by Newton.
- A heavier body has a larger linear momentum than a lighter body moving with the same velocity. i.e. In the absence of external forces, the total momentum of the system is conserved.
  i.e. $m_1v_1 = m_2v_2$ [For single body]
  and $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ [For two bodies]

Applications of Conservation of Momentum

When a bullet is fired from a gun, the gun recoils or gives a sharp pull in backward direction. While firing a bullet, the gun must be held tight to the shoulder.

- When a man jump from a boat to the shore, the boat slightly moves away from the shore.
- Rocket works on the principle of conservation of momentum.
- If someone left on a frictionless floor desires to get out of it, he can do so by blowing air out of his mouth.

Impulse

- An impulse is a large force acting on a body for a short time to produce a finite change in momentum.
- Impulse = Force × Time interval i.e. $I = F \times \Delta t$
  It's SI unit is N-s or kg-m/s

Examples of momentum and impulse are as follows:

- In catching a ball, a player by drawing his hands backward increases the time of contact.
- An athlete is advised to come to stop slowly after finishing a fast race.
- The layer of bricks is broken by player.
- Vehicles like cars, buses, and scooters, are provided with shockers.
- Bogies of the trains are provided with buffers, because buffers increase the time duration of jerks during shunting. This reduces the force with which bogies push or pull each other and severe jerks are avoided.

FRICTION

The opposing force which comes into play when a body moves or tries to move the surface of another body is known as friction.

- The maximum value of the force of friction which comes into play before a body just begins to slide over the surface of another body is called **limiting friction**.
- The force of friction that comes into play between the surfaces of two bodies before the body actually starts moving is called **static friction**.
- The force of friction that opposes relative motion between two surfaces in contact is called **kinetic or sliding friction**.
- The frictional force developed, when a body rolls over a surface, is known as the **rolling friction**.
- Static friction is a self adjusting force and it adjusts itself, so that it became equal to the applied force.

Advantages of Friction

- Due to friction, we are able to move on the surface of the earth.
- The fibres of thread are held together due to force of friction.
- The brakes applied in automobiles work only due to friction.
- Sledges are used in Arctic region as friction is very low on the surface of ice.

Disadvantages of Friction

- A lot of energy is wasted in the form of heat due to friction that causes wear and tear of the moving parts.
- Due to friction, speed of automobiles cannot be increased beyond a certain limit.

Laws of Friction

Friction acts in a direction opposite to the direction of motion of the objects. It depends upon the nature of the two surfaces in contact.

- Friction is independent of the area of contact of the two surfaces. Rolling friction is less than sliding friction.
- Force of friction (F) is directly proportional to the normal reaction (R).

  i.e. $F \propto R$

  $F = \mu R$

  where, μ = coefficient of friction.

  Also, $\mu = \tan\alpha$, where α is the angle of friction.

  Angle of repose = Angle of friction i.e. $\theta = \alpha$

Methods for Reducing Friction

- By using lubricants e.g. grease, oil, etc.
- By using ball or roller bearings, which changes sliding into rolling.
- By using soap solution.
- By using powder.
- Friction due to air is considerably reduced by streamling the shape of the body moving at high speed in air. e.g. aircraft, jet-planes, fast cars are streamlined.
- The tyres are provided with treads which increases the friction between the tyres and the road.

➡ **Note Pulling is easier than pushing**
While pushing, there is one component of force that acting downward on the object adds to the weight of the body hence there is more friction opposing the effort. Whereas on pulling, the vertical component of force is against the weight of the body and hence there is less overall friction. So, it is easy to pull than push.

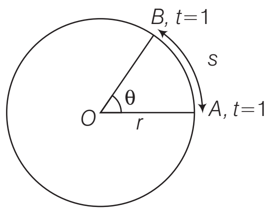
CIRCULAR MOTION

When an object moves along a circular path, then its motion is called circular motion as motion of top, etc. In circular motion force is always at right angles to the displacement therefore no work is done by the force on the particle.

- When an object moves along a circular path with uniform speed, its motion is called **uniform circular motion.**
- Circular motion is accelerated even, if the speed of the body is constant. The motion of a satellite is accelerated motion.

Angular Velocity

Angular velocity is the rate at which angle swept by the radius at the centre changes with time. Its unit is rad/s.



- Angular velocity, $\omega = \frac{\theta}{t} \Rightarrow s = v \times t$ and $\omega = \frac{\theta}{t}$,

$$\theta = \frac{s}{r}$$

So, $v = \omega \times r$

as v = linear speed,

ω = angular velocity,

r = radius of the circular path.

Thus, linear velocity = Angular velocity × Radius of circular path.

The angular velocity of revolution of a planet around the sun in an elliptical orbit increases. When the planet came closer to sun and *vice-versa*.

Angular Acceleration

- The rate of change of angular velocity is defined as angular acceleration.
- Angular acceleration, $\alpha = \frac{\omega}{t}$. Its unit is rad/$s^2$ or revolution/sec^2.
- Relation between angular acceleration and linear acceleration, $a = \alpha \times r$

Centripetal Acceleration

The acceleration is directed towards the centre and is given by $a = \frac{v^2}{r}$, where v is the speed and r is the radius. This is called centripetal acceleration.

Centripetal Force

A body performing circular motion is acted upon by a force which is always directed towards the centre of the circle. This is called **centripetal force**. Work done by centripetal force is always zero.

- If a body of mass m is moving on a circular path of radius R with uniform velocity v, then the required centripetal force, $F = \frac{mv^2}{R} = m\omega^2 R$
- Any of the forces found in nature such as frictional force, gravitational force, electrical force, magnetic force may act as a centripetal force.
- Cyclist bends his body towards the centre on a turn while turning to obtain the required centripetal force.
- Generally, in rain the scooter slips off the turning of a road because the friction between tyre and road is reduced. Due to this necessary centripetal force is not provided.
- Vehicles can move on circular path safely, if friction force ≥ required centripetal force.
- Roads are banked at turns to provide the required centripetal force for taking a turn.
- Work done by centripetal force is always zero.

Centrifugal Force

There are certain situations in which we feel that a body is acted upon by a force, but actually there is no force on the body. Such an apparent force is called a pseudo force.

- Centrifugal force is such a pseudo force. It is equal and opposite to centripetal force. When a person stand on a rotating platform, then he feels the centrifugal force.
- Cream separator and centrifugal drier work on the principle of centrifugal force.

➡ **Note** Centrifugal is a device to separate the particles of different masses present in liquid.

WORK

Work is said to be done, if a force acting on a body is able to actually move it through some distance in the direction of the force.

- Work = *Force* × *Distance*,
  i.e. $W = F \cdot s$
- If the force F is making an angle θ with the direction of displacement of the body, then the work done is $W = Fs\cos\theta$
- It's SI unit is joule (J).
- CGS unit of work is erg i.e. 1 J = 10^7 ergs

Types of Work

Work can be of following types

1. **Positive Work Done** Positive work means that force is parallel to displacement.
   Examples
   - When a body falls freely under gravitational pull.
   - When a horse pulls a cart on a level road.
2. **Negative Work Done** Negative work means that force is opposite to displacement.
   Examples
   - When a body is made to slide over a rough surface or when a positive charge is moved towards another positive charge.
3. **Zero Work Done** If the force is perpendicular to the displacement and if either the force or the displacement is zero, work done will be zero.
   Examples
   - When a coolie travels on a platform with a load on his head.
   - When a body is moved along a circular path. When a person does not move from his position but he may be holding any amount of heavy load.
   - If the force is perpendicular to the displacement and if either the force or the displacement is zero, work done is zero.

Conservative and Non-conservative Forces

- A force is said to be **conservative**, if the work done by the force in moving a body depends only upon the initial and final positions of the body and is independent of the path followed between the initial and final position, e.g. gravitational force, electrostatic force and magnetic force, etc.
- A force is said to be **non-conservative**, if the work done by the force or against the force, in moving a body from one position to another, depends upon the path followed between the two positions. e.g. friction force, etc.

ENERGY

The energy of an object is its capacity for doing work. It is a scalar quantity and its unit is joule (J).

- Energy can exist in different forms such as mechanical energy, heat energy, sound energy, light energy, chemical energy, etc.
- Mechanical energy is of two types i.e. kinetic energy and potential energy.

1. Kinetic Energy

The energy possessed by a body by virtue of its motion is called its kinetic energy.

- If a body of mass m is moving with velocity v, then kinetic energy $= \frac{1}{2}mv^2 = \frac{p^2}{2m}$
- When momentum is doubled kinetic energy becomes four times.
- If a body is moving in horizontal circle then its kinetic energy is same at all points, but if it is moving in vertical circle, then the kinetic energy is different at different points.

Applications of kinetic energy are given below

(*i*) Kinetic energy of air is used to run wind mills.
(*ii*) Kinetic energy of running water is used to run the water mills.
(*iii*) A bullet fired from a gun can pierce a target due to its kinetic energy.

2. Potential Energy

It is the energy possessed by a body by virtue of its position.

- Suppose a body is raised to a height h above the surface of the earth, then potential energy of body $= mgh$
- Also, here potential energy = work done $= mgh$
- When the body is projected upwards, then its kinetic energy goes on changing to potential energy and *vice-versa*.

Applications of potential energy are given below

(*i*) The potential energy of the wound spring of a clock is used to drive the hands of the clock.
(*ii*) Due to potential energy of the stretched bow, the arrow goes forward with a large velocity on releasing the bow.
(*iii*) The potential energy of water in dams is used to run turbines in order to produce electrical energy using the generators.

Principle of Conservation of Energy

According to the law of conservation of energy it can neither be created nor be destroyed but can only be transformed from one form to another.

- The total amount of energy in the universe remains constant. e.g. when an object falls, its potential energy decreases but at the same time kinetic energy increases by an equal amount.

Transformation of Energy

- In a heat engine, heat energy changes into mechanical energy. In the sun, mass changes into radiant energy.
- In an electrical heater, the electric energy is converted into heat energy.
- In an electric bulb, the electric energy is converted into light energy in burning coal, oil, etc. The chemical energy changes to heat energy.
- When we rub our hands, heat is generated. Here, mechanical energy of muscles is converted into heat energy.
- In an electromagnet the electric energy is converted into magnetic energy.

Einstein's Mass-Energy Equivalence

In 1905, Einstein proved the relation between mass and energy. $E = mc^2$, where c is the velocity of light.

$E = mc^2$ leads to verification of the two laws, law of conservation of mass and law of conservation of energy.

POWER

The time rate of doing work is called **power.**

If W is work done in time t then power, $P = \frac{W}{t}$.

For a constant force F, the

Power $(P) = F \cdot v \Rightarrow P = Fv\cos\theta$

where v = velocity of object

θ = angle between F and v

- Power is a scalar quantity with SI unit watt (W) or joule/s.
- Some other units of power are 1 watt hour = 3600 J

1 kilowatt hour $= 3.6 \times 10^6$ J

1 HP (Horse Power) = 746 W

ROTATIONAL MOTION AND GRAVITATION

ROTATIONAL MOTION

When a body rotates about a fixed axis, it is said to perform a rotational motion and the axis is called the **axis of rotation**. In rotational motion every particle of the body moves in a circle and the centres of all these circles lie at the axis of rotation.

- The rotating blades of an electric fan and the motion of a top are examples of rotational motion.
- Rigid body is the body in which there is no displacement among the particles of the body on applying an external force.

Centre of Mass [CM]

A point within the body or outside the body at which the whole mass of the body is assumed to be concentrated.

- The centre of mass of sphere, cylinder and ring is at its geometric centre.
- The centre of mass may lie outside the body where there is no material as in case of ring, hollow sphere, hollow cylinder, etc.

Moment of Inertia

Moment of inertia of a body with respect to axis of rotation is the product of the mass (m) and the square of distance from the axis of rotation (R).

i.e. moment of inertia $I = mR^2$

- The SI unit of moment of inertia is kg-m^2.
- Moment of inertia of a ring about an axis passing through its centre and perpendicular to its plane is
$$I = MR^2.$$
- Moment of inertia of a diameter about its diameter is
$$I = \frac{1}{2}MR^2.$$
- The moment of inertia depends not only on the mass and size of the body but also upon the manner in which the mass is distributed around the axis of rotation.

Practical applications of moment of inertia are given below

- The moment of inertia plays important role in our daily life.
- In cycle, rikshaw, bullockcart, etc., the moment of inertia of the wheels is increased by concentrating most of the mass at the rim of the wheel and connecting the rim to the axle of the wheel through spokes.
- It is due to the large moment of inertia of the wheels that when we stop cycling, the wheels of the cycle continue rotating for some time.

Radius of Gyration

Radius of gyration is defined as the distance from axis of rotation at which, if total mass of the body were supposed to be concentrated, the moment of inertia would be same.

$$I = MK^2$$

where, M = total mass of the body, K = radius of gyration

Torque or Moment of Force

The turning effect of a body is known as the moment of the force or torque.

- Moment of force or torque, $\tau = F \cdot d$, here d is the perpendicular distance from the axis of rotation.
- It is a vector quantity having SI unit N-m.
- A body of lesser weight can produce the same turning effect as the body of higher weight by adjusting their distance from the axis of rotation.
- Torque is equal to the product of moment of inertia and angular acceleration. i.e. $\tau = I \times \alpha$.

Practical applications of torque are given below

- The handle or the knob is fitted near the free edge of the door. So that, a heavy revolving door is opened or closed easily, when the force is applied at the edge of the door.
- A wrench with a long arm is used to unscrew a nut fitted tightly into a bolt and the handle of a screw driver is made wide.

ANGULAR MOMENTUM

- The angular momentum (J) of a body about an axis is equal to the product of the moment of inertia (I) of the body and its angular velocity (ω) about that axis. i.e. $J = I\omega$.
- If no external torque is acting upon a body rotating about an axis, then the angular momentum of the body remains constant. It is called conservation of angular momentum.

Examples related to conservation of angular momentum are given below

- When a diver jumps into water from a height, he does not keep his body straight but pulls in his arms and legs towards the centre of his body. On doing so, the moment of inertia of his body decreases. But, since the angular momentum remains constant, his angular velocity increases.
- A man with his arms outstretched and holding heavy dumb-bells in each hand, is standing at the centre of a rotating table. When the man pulls-in his arms, the speed of rotation of the table increases.
- The reason is that on pulling-in the arms, the distance of the dumb bells from the axis of rotation decreases and so the moment of inertia of the man decreases. Therefore by conservation of angular momentum, the angular velocity increases.
- The angular velocity of revolution of planet around the sun in elliptical orbit increases, when the planet comes closer to the sun.

- If a cat is dropped from a height, they almost always tend to land on their feet because they use the conservation of angular momentum to change their orientation.
- When a cat is in the air, no net external torque acts on it about its centre of mass, so the angular momentum about the cat's centre of mass cannot change.
- Stretching out its legs increases its rotational inertia and thus slows the cats angular speed. The conservation of angular momentum allows the cat to rotate its body and slow its rate of rotation enough, so that it lands on its feet easily.

ROLLING MOTION

The rolling motion can be regarded as the combination of pure rotation and pure translation.

- The kinetic energy of a body rolling without slipping is the sum of kinetic energies of translational and rotational motion.
  i.e. total KE of a rolling body
  $$= \text{Rotational KE} + \text{Translational KE}$$
  $$= \frac{1}{2} I \omega^2 + \frac{1}{2} m v_{cm}^2$$
- If a ring, a disc and a sphere, all of the same radius and mass roll down an inclined plane from the same height, the sphere will reach the bottom first and ring at last.

GRAVITATION

The phenomena of action of attractive forces between two bodies by virtue of their masses is known as gravitation.

Newton's Law of Gravitation

The force of gravitational attraction between two bodies is directly proportional to the product of their masses ($m_1 \cdot m_2$) and inversely proportional to the square of the distance (r) between them i.e. $F = G \frac{m_1 m_2}{r^2}$

where, m_1 is the mass of first body, m_2 is the mass of second body and r is the distance between them.

- The value of G is $6.67 \times 10^{-11} \text{N-m}^2/\text{kg}^2$.
- It is always attractive in nature, unlike electric and magnetic force which can be attractive or repulsive both.
- It acts along the line joining the centres of two interacting forces.
- The gravitational force between two masses is independent of the presence of any other object medium present between two masses.
- The predictions of solar and lunar eclipses on the basis of the Newton's law of gravitation come out in perfect agreement with actual observation.
- Tides are formed in oceans due to gravitational attraction between moon and ocean water.

Acceleration due to Gravity

It is the force by which the earth attracts a body towards its centre.

- The acceleration due to gravity is the rate of change of velocity of a body due to the gravitational force acting towards the earth. It is represented by $g = \frac{GM_e}{R_e^2}$
  where, M_e is the mass of the earth and R_e is the radius of the earth.
- The value of g at the surface of the earth is 9.8 m/s^2.
- Earth is surrounded by an atmosphere of gases due to gravity. The value of g on the moon is 1/6th of that of the earth.

Variation in the Value of (g)

- When we go above the surface of the earth, the acceleration due to gravity goes on decreasing.
- When we go below the surface of the earth, the acceleration due to gravity goes on decreasing and becomes zero at the centre of the earth.
- Decreasing the rotational motion of the earth, the value of g increases. When we go from the equator towards the poles, the value of g goes on increasing.
- If earth stops its rotation about its own axis, then at the equator, the value of g increases and consequently the weight of body lying there increases.
- The value of g is maximum on surface of the earth.

MASS AND WEIGHT

The mass of a body is the quantity of matter contained in it. It is a scalar quantity and its SI unit is kg. Mass does not change from place to place and remains constant.

The weight of the body is the force with which it is attracted towards the centre of the earth.

- Weight of the body $w = mg$
- Weight of a body is not constant, it changes from place to place. At the pole, the weight of a body will be maximum whereas at the equator it is minimum.
- Mass is measured by an ordinary equal arm balance, while weight is measured by a spring balance.

Weight of a Body at Moon

As mass and radius of moon is lesser than the earth, so the force of gravity at the moon is also less than the earth. Its value at the moon's surface is g/6.

Centre of Gravity

The centre of gravity of a body is that point, at which the whole weight of the body appears to act.

- It can be inside the material of the body or outside it.
- For regularly shaped body, the centre of gravity lies at its geometrical centre.

Weight of a Body in Lift

Some conditions are given below

- If lift is stationary or moving with uniform speed (either upward or downward), the apparent weight of a body is equal to its true weight.
- If lift is going up with acceleration, the apparent weight of a body is more than the true weight.
- If lift is going down with acceleration, the apparent weight of a body is less than the true weight.
- If the rod of the lift is broken, it falls freely. In this situation, the weight of a body in the lift becomes zero. This is the situation of weightlessness.
- While going down, if the acceleration of lift is more than acceleration due to gravity, then the body in the lift raise towards the ceiling of the lift.

SATELLITE

The heavenly body which revolves around the sun is called **planet** e.g. the earth, while the heavenly body which revolves round the planets is called **Satellite**. Moon is a natural satellite of the earth.

- Orbital speed of a satellite is independent of its mass and depends upon the radius of orbit.
  i.e. $v_0 = \sqrt{gR}$
- The orbital speed of a satellite revolving near the surface of the earth is 7.9 km/s.
- Time period of revolution of satellite revolving near the surface of the earth is 1 hr 24 min (84 min).
- Period of revolution of a satellite depends upon the height of satellite from the surface of the earth.
- Every body inside the satellite is in a state of weightlessness.
- Period of revolution of a satellite is independent of its mass.
- If a satellite revolves in equatorial plane in the direction of the earth's rotation i.e. from West to East with a period of revolution equal to time period of rotation of the earth on its own axis i.e. 24 h, then the satellite will appear stationary relative to the earth. Such a satellite is called **geo-stationary satellite**. Such a satellite revolves around the earth at a height of 3600 km.
- INSAT 2B and INSAT 2C are geo-stationary satellites of India. It is used to reflect TV signals and telecast TV programs from part of the work to another.
- **Polar satellite** revolves around the earth in polar orbit at a height of 800 km. Time period of the satellites is 84 min. These are used for weather forecasting and mapping, etc.

➡ **Note** The earth rotates on its axis from West to East. This rotation makes the sun and the stars appear to be moving across the sky from East to West.

Energy of a Satellite

A satellite revolving around the earth has potential energy as well as kinetic energy. It has potential energy because it remains in the gravitational field of earth and kinetic energy because it is in motion.

Total energy of the satellite,

$$E = \text{Kinetic energy} + \text{Potential energy}$$

$$E = \frac{GM_e m}{2R_e} + \left(-\frac{GM_e m}{R_e}\right)$$

$$E = -\frac{GM_e m}{2R_e}$$

where, m is the mass of satellite.

ESCAPE VELOCITY

Escape velocity is that minimum velocity with which a body should be projected from the surface of the earth, so as it goes out of gravitational field of earth and never returns to the earth. Escape velocity at the surface is 11.2 km/s.

Escape velocity is given by

$$v_e = \sqrt{\frac{2GM}{R_e}}$$

$$v_e = \sqrt{2gR_e} \qquad \left[\because\ g = \frac{GM}{R_e^2}\right]$$

- The value of the escape velocity of a body does not depend on its mass. Its value on the moon surface is 2.38 km/s. So, there is no atmosphere around the moon.
- Escape velocity is $\sqrt{2}$ times the orbital velocity.
- Satellites are launched with the escape velocity as needed.

KEPLER'S LAWS

Kepler's law of planetary motion are three scientific laws describing motion of planets around the sun.

- All planets move around the sun in elliptical orbits having the sun at one focus of the orbit.
- The areal speed of a planet around the sun is constant.
- The square of the period of revolution of any planet around the sun (T) is directly proportional to the cube of its mean distance from the sun (a) i.e.

$$T^2 \propto a^3.$$

Black Hole

A black hole is a region of spacetime exhibiting such strong gravitational effects that nothing not even particles and electromagnetic radiation such as light can escape from inside it. Event horizon is the boundary marking the limits of a black hole. Thus, nothing that enters a black hole can get out or can be observed from outside the event horizon.

PROPERTIES OF MATTER

ELASTICITY

Elasticity is the property of material of a body by virtue of which the body acquires its original shape and size after the removal of deforming force.

- Steel and ivory are more elastic than rubber and water is more elastic than air.
- The internal restoring force acting per unit area of cross-section of a deformed body is called **stress**. Its unit is N/m^2 or Pascal.
- The fractional change in configuration i.e. length, volume and shape is called **strain**.
- The ratio of stress to strain is a constant for the material and is called **modulus of elasticity.**

  i.e. $$E = \frac{Stress}{Strain}$$

  It is also called **Hooke's law**.
- The maximum deforming force up to which a body retains its property of elasticity is called the **limit of elasticity.**
- Putty, paraffin wax are nearly perfectly plastic bodies.
- Iron, copper, silver, aluminium, etc., are examples of ductile materials. Glass, dry clay, etc., are examples of brittle materials.
- Rubber is one example of elastomers.

PRESSURE

Pressure is defined as, force acting normally on unit area of the surface.

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

- Its unit is N/m^2 also called Pascal. It is a scalar quantity.
- The pressure exerted by liquid at depth h below the surface of liquid is given as,

  $p = hdg$, where d is density of liquid.
- It is measured by manometer.
- Pressure at a point in a static liquid has same value in all directions.
- Pressure at a point in a liquid is proportional to the density of the liquid and depth of the point from the free surface.
- Boiling point of all substances increases with the increase in pressure.

Atmospheric Pressure

The pressure exerted by the atmosphere is called atmospheric pressure.

- Atmospheric pressure $= 1.01$ bar $= 1.01 \times 10^5\ N/m^2 = 760$ torr
- Atmospheric pressure is measured by barometer.
- Atmospheric pressure decreases with altitude (height from the earth's surface). This is why
  - It is difficult to cook on the mountain.
  - The fountain pen of a passenger leaks in aeroplane at height.
- The slow rise in the barometric reading is the indication of clear weather.
- Sudden fall in barometric reading is the indication of storm.
- Slow fall in barometric reading is the indication of rain.

➡ **Note** Building and dams are laid on a larger area of ground so that, the height of the building or dam produces less pressure on ground.

PASCAL'S LAW

Pascal's law states that 'Pressure in a fluid in equilibrium is the same everywhere, if the effect of gravity can be neglected.'

- Hydraulic lift, hydraulic press, hydraulic brakes work on the basis of Pascal's law.
- Hydraulic lift is used to lift heavy loads.

ARCHIMEDES' PRINCIPLE

When a solid body is immersed wholly or partially in a liquid (in general, in a fluid), then there is some apparent loss in its weight. This loss in weight is equal to the weight of the liquid displaced by the body.

Applications of Archimedes' principle are given below

- Archimedes' principle is used in determining the relative density of a substance.
- Archimedes' principle is used in designing ships and submarines.
- The hydrometers used for determining the density of liquids are based on Archimedes' principle.
- The lactometers used for determining the purity of milk are based on Archimedes' principle.

DENSITY

Density is defined as mass per unit volume.

- Density, $\rho = \frac{Mass}{Volume} = \frac{M}{V}$. Its SI unit is kg/m^3.
- **Relative density** of substance is defined as the ratio of the density of that substance to the density of water at 4°C i.e.

  Relative density = $\frac{Density\ of\ material}{Density\ of\ water\ at\ 4°C}$
- Relative density has no unit. It is a scalar quantity.
- The density of water is maximum at temperature 4°C.
- Relative density of liquid or fluid is measured by **hydrometer.**
- The density of sea water is more than that of normal water. This explains why it is easier to swim in sea water.
- If ice floating in water in a vessel melts, the level of water in the vessel does not change.
- Ice floats on water surface as its density ($0.92\,g/cm^3$) is lesser than the density of water (1 g/cm^3).
- Iron will sink in mercury because density of mercury is more than the iron.

BUOYANT FORCE

When a body is immersed partially or wholly in a liquid, a force acts on the body by the liquid in the upward direction. This force is called buoyant force or force of buoyancy of upthrust.

- It is equal to the weight of liquid displaced by the body and acts at the centre of gravity of displaced liquid.
- The buoyant force exerted by a liquid depends on the volume of the solid object immersed in the liquid.
- The buoyant force exerted by a liquid depends on the density of the liquid in which the object is immersed.

Law of Floatation

When body floats in neutral equilibrium, the weight of the body is equal to the weight of displaced liquid.

- The centre of gravity of the body and centre of gravity of the displaced liquid should be in one vertical line.
- Three cases are possible when the body is immersed in the fluid.
  (*i*) If the weight of the body is greater than the upward force or upthrust by the fluid acting upwards, then the body sinks.
  (*ii*) If the weight of the body is equal to the upthrust or the weight of the body is just balanced by upthrust, then the body floats fully immersed.
  (*iii*) If the weight of the body is less than the upward force, then the body floats partially immersed. The centre of gravity of a body is that point at which the whole weight of the body appears to act.

Fluid Dynamics

(*i*) **Streamline Flow** or **Steady Flow** In this flow velocity at every point in the fluid will remains constant.
(*ii*) **Turbulent Flow** In this flow, the fluid does not maintain constancy of the velocity.
(*iii*) **Critical Velocity** The velocity above which flow will become turbulent.

PRINCIPLE OF CONTINUITY

It states that, when an incompressible and non-viscous liquid flows in streamlined motion through a tube of non-uniform cross-section, the product of the area-section and the velocity of flow is same at every point in the tube. For a tube of flow, between two points having area of cross-section A_1 and A_2 and velocities V_1 and V_2.

Then,

$$AV = \text{Constant}$$
$$A_1V_1 = A_2V_2$$

According to the equation of continuity, the speed of fluid flow becomes faster in the narrower pipe.

BERNOULLI'S THEOREM

When an incompressible and non-viscous liquid (or gas) flows in streamlined motion from one place to another, then at every point of its path the total energy per unit volume (Pressure energy + Kinetic energy + Potential energy) is constant.

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

where, ρ is density of fluid. This is the principal of conservation of energy for the non-viscous fluids i.e. there is no viscous force acting between the layers of the fluid.

Venturimeter, Pitot tube, Bunsen's burner, atomizor, filter pump and magnus effect are based upon the Bernoulli's theorem.

➡ **Note** Bernoulli's theorem is applicable only to streamline flow of a fluid. It is not valid for non-steady or turbulent flow.

SURFACE TENSION

Surface tension of a liquid is measured by the normal force acting per unit length on either side of an imaginary line drawn on the free surface of liquid and tangential to the free surface.

- If a force F acts on an imaginary line of length l, then surface tension $T = \frac{F}{l}$. Its SI unit is Newton/metre.
- Force of attraction applied between molecules of same substance is called **cohesive force**, while attractive force between molecules of different substances is called **adhesive force**.

- The surface tension of a liquid decreases with increase in temperature and becomes zero at the critical temperature.
- On mixing soap, the surface tension of water decreases.

Some Phenomena Based on Surface Tension

- Warm soup is tasty because at high temperature its surface tension is low and consequently the soup spreads on all parts of the tongue.
- Medicines used for washing wounds, as dettol have a surface tension lower than water. So, they reach the fine cavities formed in the wound.
- The surface tension of the tooth-paste scum is also less that it spreads quickly on the full areas of teeth and clean them.
- Insects and mosquitoes swim on the surface of water in ponds and lakes due to surface tension.
- Hair of a shaving brush cling together due to surface tension, when the brush is taken out from the water.
- The ends of a glass tube become rounded on heating. Small drops of mercury are spherical while large ones are flat. Formation of lead shots, liquid-drop model of nucleus. Floatation of needle on water. Dancing of camphor on water and helping. Detergent in cleaning the clothes all are due to surface tension.

ANGLE OF CONTACT

The angle inside the liquid between the tangent to the solid surface and the tangent to the liquid surface at the point of contact is called the angle of contact for that pair of solid and liquid.

- The angle of contact for pure water and clean glass is zero. For ordinary water and glass it is about 8°.
- The angle of contact for water and silver is 90°.
- The liquids which wet the solid have acute angle of contact. Meniscus of these liquids will be concave.
- The liquids which do not wet the solid have obtuse angle of contact as for mercury and glass the angle of contact is 135°. Meniscus of these liquids will be convex.

CAPILLARITY

If a capillary tube is dipped in a liquid, liquid ascends or descends in the capillary tube. This phenomenon is called capillarity.

- The oil in the wick of a lamp rises due to capillary action of threads in the wick.
- To prevent loss of water due to capillary action, the soil is loosened and split into pieces by the farmers.
- The root hairs of plants draws water from the soil through capillary action.
- Writing nib is split in the middle, so that a fine capillary is formed in it. When it is dipped in ink, the ink rises in the capillary.

➨ **Note** The liquids which wet glass for which the angle of contact is accute, rise up in capillary tube, while those which do not wet glass, for which the angle of contact is obtuse are depressed down in the capillary.

VISCOSITY

It is the property of the liquid by virtue of which, it opposes the relative motion between its adjacent layers.

- Viscosity is the property of liquids and gases both.
- With rise in temperature, viscosity of liquids decreases and that for gases increases.
- Viscosity of liquid increases with increase in pressure.
- Viscosity of a fluid is measured by its coefficient of viscosity. It's SI unit is ($N\text{-}s/m^{-2}$).

STOKES' LAW

When a small spherical body of radius r is moving slowly with a constant velocity v through a perfectly homogeneous medium (liquid or gas) of infinite extension and coefficient of viscosity η, then the retarding viscous force acting an the body is $F = 6\pi\eta r v$

- This law is used in Millikan's method for determining the electronic charge and for finding out the radii of small aid-drops by measuring their terminal velocity in air.

TERMINAL VELOCITY

When a body falls in viscous medium, its velocity first increases and finally becomes constant. This constant velocity is called terminal velocity. The terminal velocity of the sphere is directly proportional to the square of the radius of the sphere.

- The cloud particles fall down very slowly because of the viscosity of air and hence appear floating in the sky.
- Thin liquids like water, alcohol, etc are less viscous than thick liquids coal for blood, honey, glycerine, etc.
- Honey is less viscous than glycerine.

HEAT AND THERMODYNAMICS

TEMPERATURE

The measurement of hotness or coldness of a body is called its temperature. When two bodies are placed in contact, heat always flows from a body at higher temperature to the body at lower temperature.

THERMOMETER

An instrument used to measure the temperature of a body is called a thermometer. Mercury is generally used as, thermometric substance, because it is sensitive to slightest change in temperature which is notable when it is used in a thermometer. Mercury has a freezing point of $-38 \cdot 83°C$ and boiling point at $356 \cdot 7°C$, this gives a very big range to measure temperatures for many general purposes. Also, it does not stick to (wet) the capillary wall of the thermometers and it is easy to obtain in its elemental form.

Temperature Scales

Some temperature scales are as follows

1. **Celsius Scale** (C) The melting point of the ice at standard atmospheric pressure is regarded as 0°C and the boiling point of water as 100°C. This scale was designed by **Anders Celsius** in the year 1710.

   The degree celsius (°C) can refer to a specific temperature on the celsius scale as well as a unit to indicate a temperature interval.
2. **Fahrenheit Scale** (F) The melting point of ice is regarded as 32°F and the boiling point of water as 212°F. This scale was designed by **Gabriel Fahrenheit** in the year 1717.
3. **Reaumer Scale** (R) The melting point of ice is regarded as 0°R and the boiling point of water as 80°R. This scale was introduced by **RA Reaumer** in year 1730.

- Scales of temperature measurement,

$$\frac{C-0}{100-0} = \frac{F-32}{212-32} = \frac{R-0}{80-0} = \frac{K-273}{100-0}$$

$$\Rightarrow \quad \frac{C}{5} = \frac{F-32}{9} = \frac{R}{4} = \frac{K-273}{5}$$

- The normal temperature of a human body is 37°C or 98.4 F.
- $-40°$ is the temperature at which Celsius and Fahrenheit thermometers read same.
- The range of a laboratory thermometer is generally from $-10°C$ to $110°C$. A laboratory thermometer cannot be used to measure the human body temperature.

Total Radiation Pyrometer

Radiation pyrometer measures the temperature of a body by measuring the radiation emitted by the body. It cannot measure temperature below 800°C because at low temperature, emission of radiation is very small and cannot be detected. Sun's temperature is measured by it.

➠ **Note** Bolometer and thermopile are also used to measure the amount of heat radiated by a body.

HEAT

It is a form of energy which produces the sensation of warmth in our body.

- It is due to the kinetic energy of the molecules constituting the object or body. Its units are calorie, kilocalorie or joule.
- Calorie is the amount of heat required to raise the temperature of 1 g of water through 1°C (from 10°C to 11°C). It is represented by cal, 1 cal = 4.18 J.
- Air is a non-conductor of heat and silver is the best conductor of heat.
- Ebonite rubber, glass and air are bad conductors of heat.

Specific Heat

- Specific heat is the amount of heat required to raise the temperature of a unit mass of the substance by 1°C.
- If Q heat changes the temperature of m by ΔT, then specific heat

$$C = \frac{Q}{m\Delta T}$$

The SI unit of specific heat is J/kg-K.

- Specific heat of gold = 130 J/kg-K
- Specific heat of water = 4180 J/kg-K
- The specific heat of water is maximum.
- Mercury has low specific heat.

Thermal Capacity

It is the amount of heat required to raise the temperature of a body through 1K. Its units are J/K or cal/°C or kcal/°C.

Thermal capacity = mC

where, m is the mass of the body and C is the specific heat.

Water Equivalent

It is the mass of water, which absorbs or emits the same amount of heat as is done by the given body for the same rise or fall in temperature.

- The water equivalent of a body is denoted by W. Its SI unit is kg and CGS unit is g.
- Water equivalent, $W = mC$

  where, m is mass of the body and C is the specific heat.

Principle of Calorimetry

The heat lost by the hot body must be equal to the heat gained by the cold body. This is known as principle of calorimetry or principle of mixtures, i.e.

Heat gained = Heat lost

- Work and heat are two equivalent forms of energy.
- Calorimeter is a cylindrical vessel made of copper.

THERMAL EXPANSION

The phenomenon of change in dimensions of an object due to heat supplied is known as thermal expansion.

- A solid can undergo three types of expansions:
  - Linear expansion (expansion in length)
  - Superficial expansion (expansion in area)
  - Cubical expansion (expansion in volume)
- Relation between the coefficients of linear, superficial and cubical expansion : $\alpha : \beta : \gamma = 1 : 2 : 3$

Practical Applications of Thermal Expansion

- When hot tea is poured into a glass tumbler made of soft glass, it cracks because of thermal expansion of glass.
- Telegraph wires are given enough gap to allow the wires for contraction in winter.
- An ordinary pendulum clock runs faster in winter but slower in summer, because in summer the length of pendulum increases, while in winter it decreases.
- A metallic ball before heating can pass easily through the ring, but after heating the ball does not pass through the ring because of the thermal expansion of ball.
- A gap is provided between the iron rails of the railway track, so that rails can easily expand during summer and do not bend. In the construction of bridges, ends of steel gerders are not fixed but placed on rollers to allow free expansion and contraction in summer and winter respectively to avoid any damage to the bridge.
- A tightly fitted cork can be removed from a glass bottle without breaking it on heating its mouth carefully. The glass wall expand and the cork will come out easily.

Expansion of Liquids

- Liquids do not have their definite length and surface area. It always need a container. When they are heated, they expand and at the same time the container also expands.
- The expansion of a liquid, when the expansion of its container has not been taken into account is called **apparent-expansion**.
- The expansion of a liquid, when the expansion of its container is taken into account, is called **real expansion**.

Anamalous Expansion of Water

Almost every liquid expands with the increase in temperature. But when temperature of water is increased from 0°C to 4°C its volume decreases. If the temperature is increased above 4°C its volume starts increasing, clearly density of water is maximum at 4°C.

Interconversion of States of Matter

The temperature at which solid starts to liquify is known as melting point of that solid.

- The process in which a solid changes into a liquid on heating is called melting solid. Change of state takes place i.e. ice changes into water.
- The process in which a liquid changes into a solid is called **fusion.** The temperature at which liquid starts to freeze is known as **freezing point** of the liquid.
- The process in which a liquid changes into a vapour on heating is called **vaporisation**. The temperature at which the liquid starts to evaporate is called the **boiling point** where change of state takes place i.e. water changes into water vapour or steam.
- The process of change of state directly from solid to vapour (or gas) is known as **sublimation**.

LATENT HEAT

It is the amount of heat absorbed by a unit mass of the substance to change its state without change in temperature.

- SI unit of latent heat is joule per kilogram.
- Latent heat of fusion is 80 cal/g and vaporisation is 540 cal/g.
- Ice at 0°C is more effective in cooling a substance than water at 0°C.
- If Q is quantity of heat absorbed (or given out), then
  $$Q = mL$$
  where, m = mass of the substance
  L = latent heat of substance.
- Steam contains more heat, in the form of latent heat, then boiling water.
- The burns caused by steam are much more severe than those caused by boiling water though both of them are at the same temperature of 100°C.

EVAPORATION

It is the slow process of conversion of a liquid into its vapour even below its boiling temperature.

- Cooking is caused by evaporation due to expansion.
- Factors affecting the evaporation of a liquid are
  - the area of the liquid surface
  - the nature of the liquid
  - temperature of the liquid
  - dryness of air
  - the movement of the air
  - temperature of air or surrounding

RELATIVE HUMIDITY

The ratio of amount of water vapour (m) actually present in a certain volume of air at a given temperature to the amount of water vapour (M) required to saturate it, is called relative humidity, R_H.

$$R_H = \frac{m}{M} \times 100\%.$$

- The amount of water vapour in air is called as **humidity**.
- The amount of water vapour in the air varies as it depends on the rate of evaporation.
- The amount of water vapour present in 1 m^3 air is called its **absolute humidity.**
- Relative humidity is measured by **hygrometer**.
- Relative humidity increases with the increases of temperature.

THERMODYNAMICS

Thermodynamics is a branch of physics which deals with exchange of heat energy between bodies and conversion of the heat energy into mechanical energy and *vice-versa*.

First Law of Thermodynamics The amount of heat given to a system is used up in two ways, first to increase the internal energy and second to do the external work. i.e.

$$dQ = dU + dW.$$

Second Law of Thermodynamics The second law of thermodynamics is the outcome of human experience under which heat energy can be converted into mechanical energy. This law is based upon the two statements given below.

(*i*) It is impossible to construct a device which operates in a cycle that will take heat from a body and convert it completely into the work. (Kelvin-Planck's Statements)

(*ii*) It is impossible to construct a self acting device which operates in a cycle that will transfer heat from a cold body to a hot body without expenditure of work (Clausius Statement).

Nature of Internal Energy

- According to the kinetic modal of matter, every substance is composed of molecules which are in constant motion inside the substance. Hence, molecules posses kinetic energy.
- Molecules also exert force on one another which is called intermolecular force, hence they posses potential energy also. The kinetic energy and potential energy of molecules is the internal energy of the substance.
- The internal energy of an ideal gas is only the kinetic energy of its molecules which depends only upon the temperature of the gas.

Difference between Petrol Engine and Diesel Engine

Petrol engine	*Diesel engine*
It works with a spark plug.	It works with an oil plug.
Efficiency is smaller (47%).	Efficiency is larger (55%).
It is associated with the risk of explosion, because petrol vapour and air is compressed. So, low compression ratio is kept.	No risk of explosion, because only air is compressed. Hence, compression ratio is kept large.
Petrol vapour and air is created with spark plug.	Spray of diesel is obtained through the jet.

Transmission of Heat

Transfer of heat from one place to other place is called transmission of heat.

There are three processes by which transmission of heat takes place.

(a) Conduction (b) Convection (c) Radiation

- In solids, transmission of heat takes place by **conduction process**.
- In liquids and gases heat takes place by **convection process**.
- Heat from the sun reaches on the earth by **radiation**.
- Mercury though a liquid is heated by conduction and not by convection.
- In rooms, ventilators are provided to escape the hot air by the process of convection.
- Air is poor conductor of heat.
- On a cold night two thin blankets gives more hotness than a single thick blanket because the layer of air between the two blankets works as better insulator of heat.
- Cooking utensils are made up of aluminium, brass and steel, because these substances have low specific heat and high conductivity.

KIRCHHOFF'S LAW

Kirchhoff's law signifies that good absorbers are good emitters. If a shining metal ball with some black spot on its surface is heated to a high temperature and seen in dark, the shining ball becomes dull but the black spots shines brilliantly because black spot absorbs radiation during heating and emit in dark.

- White and light colours are bad absorbers and good reflectors of heat.
- In deserts, day temperature is very high and night temperature is extremely low because the specific heat of sand is very low. Therefore, it absorbs the heat readily and its temperature raises by a large degree during day. At night sand radiates the heat equally readily making the temperature loss.

Perfectly Black Body A perfectly black body is one which absorbs completely all the radiations falling on its surface, whatever be the wavelength.

Perfectly black body is a perfect absorber hence, according to Kirchhoff's law, it will also be a perfect radiator. It is not essential that a perfectly black body should appear black. Sun emits radiation of all wavelength, so it may be called a black body, even though it looks white.

Stefan's Law The radiant energy emitted by unit area of perfectly black body per unit time is directly proportional to the fourth power of its absolute temperature.

$$E \propto T^4 \Rightarrow E = \sigma T^4$$

where, σ is Stefan's constant.

➠ **Note** Wein's displacement law can be used to compute the temperature of the sun or of the stars.

Newton's Law of Cooling The rate of loss of heat by a body is directly proportional to the difference in temperature between the body and the surrounding.

If the door of refrigerator is kept open, it will not cool the room, it will increases the temperature of the room, because heat rejected by the refrigerator to the room, will be more than the heat taken by the refrigerator from the room.

OSCILLATIONS AND WAVES

PERIODIC MOTION

When a body repeats its motion continuously on a definite path in a definite interval of time, then its motion is called **periodic motion** and the interval of time is called **time period.** Motion of hands of a clock, motion of earth around the sun and motion of the needle of a sewing machine are examples of periodic motion.

Oscillatory Motion

If a body in periodic motion moves along the same path to and fro about a definite point (mean position or equilibrium position), then the motion of the body is a vibratory motion or oscillatory motion. Oscillatory motion is also called as harmonic motion.

- The motion of the pendulum of a wall clock, the motion of the bob of a simple pendulum, the motion of a loaded spring, the motion of liquid contained in a U-tube , the motion of the prongs of tuning fork and the motion of a bar magnet suspended in the earth's magnetic field are examples of oscillatory motion.
- All oscillatory motions are periodic motions, but all periodic motions are not oscillatory.

SIMPLE HARMONIC MOTION (SHM)

If a particle repeats its motion about a fixed point after a regular time interval in such a way that at any moment the acceleration of the particle is directly proportional to its displacement from the fixed point at that moment and is always directed towards the fixed point, then the motion of the particle is called simple harmonic motion.

- Vibrations of the prongs of a tuning fork, oscillations of a body. Suspended by a spring. Oscillations of a body partly immersed in a liquid and oscillations of a simple pendulum are examples of simple harmonic motion.
- When a particle executing SHM passes through the mean position, then
  - no force acts on the particle.
  - velocity is maximum.
  - acceleration is zero.
  - kinetic energy is maximum.
  - potential energy is zero.
- When a particle executing SHM is at the extreme end, then
  - velocity of particle is zero.
  - acceleration of the particle is maximum.
  - kinetic energy of particle is zero.
  - potential energy is maximum.
  - restoring force acting on particle is maximum.
- In case of spring block system, time period of oscillation is given by $T = 2\pi\sqrt{\frac{m}{k}}$, where m is the mass of the block.
- In case of spring block system, the restoring force $F = -kx$
  where, x is displacement of the block from mean position and k is spring constant.

RESTORING FORCE

When oscillating particle is displaced from its equilibrium position, then a periodic force acts upon it which is always directed towards the equilibrium position. This force is called **restoring force.**

Simple Pendulum

A simple pendulum consists of a small metal ball suspended by a long thread from a rigid support, such that the bob is free to swing back and forth.

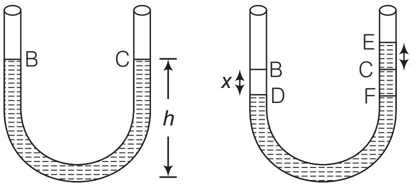
- The distance of the pendulum from its mean position is called its **displacement**.
- The maximum displacement of the pendulum on either side of its mean position is called the **amplitude** of oscillations.
- **Time period** is the time taken by the pendulum to complete one full oscillation.

- **Frequency** is the number of full oscillations completed by pendulum in one second.
- Second pendulum is one whose time period is 2s.
- Time period of simple pendulum $T = 2\pi\sqrt{\frac{L}{g}}$, where L is length of thread and g is the acceleration due to gravity.
- The time period of a simple pendulum of infinite length is 84.6 min. A pendulum clock goes slow in summer and fast in winter.
- If a simple pendulum is suspended in a lift descending down with acceleration, then time period of pendulum will increase. If lift is ascending, then time period of pendulum will decrease.
- If a lift falling freely under gravity, then the time period of the pendulum will be infinite.
- At moon, the time period of simple pendulum increases because acceleration due to gravity at moon is lesser.

➨ **Note** Every body inside a satellite remains in a state of weightlessness, that is the effective value of g remains zero. Hence, a pendulum inside a satellite will not oscillate. Therefore, the pendulum clock cannot work inside space satellites.

Oscillation of Liquid Column in a U-tube

If a liquid is taken in a U-tube, it maintains equal level in both the columns of the tube. When air is blown-in on side B, the liquid may go down upto point D. The displacement of the liquid is given by $BD = x$.



In the second column, the liquid goes upto E and the displacement is CE. As the applied force is withdrawn, the liquid level in the second column comes down because of gravity from E to C. But due to inertia, it overshoots the mark C and goes to F ($CF = x$), whereas the liquid in the other column also goes up by the same distance.

Thus, the liquid starts oscillating in both the columns. This motion of the liquid in the U-tube is SHM with time period, $T = 2\pi\sqrt{\frac{h}{g}}$ where, h is the height of liquid in each column.

WAVES

A wave is a disturbance which propagates energy from one place to the other without the transport of matter.

These are of two types

Mechanical Waves

The waves which require material medium (solid, liquid or gas) for their propagation are called mechanical waves or elastic waves. These are of two types

(*i*) **Longitudinal Waves** If the particles of the medium vibrate in the direction of propagation of wave, the wave is called longitudinal waves.

Waves on springs or sound waves in air are examples of longitudinal waves.

(*ii*) **Transverse Waves** If the particles of the medium vibrates perpendicular to the direction of propagation of wave, the wave is called transverse waves.

Waves on strings under tension, waves on the surface of water are examples of transverse waves.

Electromagnetic Waves

The waves which do not require any medium for their propagation i.e. which can propagate even through the vacuum are called electromagnetic waves. They propagate as transverse wave.

- Electromagnetic waves of wavelength range 10^{-3} m to 10^{-2} m are called **microwaves**.
- Microwaves are absorbed by water molecules and living tissue internal heating will damage or kill cells.
- Microwaves are used in communications, satellites, telephony, heating water and food.

➨ **Note** Cathode rays, canal rays, α-rays, β-rays are not electromagnetic waves. Light and heat are examples of electromagnetic waves.

Spectrum of Electromagnetic Waves

Electromagnetic Waves	*Discoverer*	*Wavelength range (in metre)*	*Frequency range*	*Uses*
γ-rays	Henry Becqueral	10^{-14} to 10^{-10}	10^{20} to 10^{18}	Medical tracers, killing cancerous cells, steralisation, imaging defects in metal.
X-rays	W Roenetgen	10^{-10} to 10^{-8}	10^{18} to 10^{16}	Imaging defects in bones, hidden devices.
Ultraviolet rays	Ritter	10^{-8} to 10^{-7}	10^{16} to 10^{14}	Making vitamin D, making ions.
Visible radiation	Newton	3.9×10^{-7} to 7.8×10^{-7}	10^{14} to 10^{12}	Photography
Infrared rays	Hershel	7.8×10^{-7} to 7.8×10^{-3}	10^{12} to 10^{10}	TV remote control
Short radio waves or Hertz hertzian waves	Heinrich	10^{-3} to 1	10^{10} to 10^{8}	Communication, radio, TV.
Long radio waves	Marcony	1 to 10^{4}	10^{8} to 10^{6}	Communication, radio, TV.

Important Terms Related to Waves

- **Amplitude** (A) Maximum displacement of a vibrating particle of medium from its mean position is called amplitude.
- **Wavelength** (λ) Wavelength is the distance between any two nearest particle of the medium, vibrating in the same phase.
- **Frequency** (f) Frequency of vibration of a particle is defined as the number of vibrations completed by particle in one second.

$$\text{Frequency} = \frac{1}{\text{Time period}}$$

- Velocity of wave (v) = Frequency (f) × Wavelength (λ)

Waves in Different Frequency Range

According to their frequency range, longitudinal mechanical waves are divided into the following categories:

- The longitudinal mechanical waves which lie in the range 20 Hz to 20000 Hz are called **audible** or **sound waves**. Human ears are sensitive to the waves in this frequency range.
- The longitudinal mechanical waves having frequencies less than 20 Hz are called **infrasonic**. These are produced by earthquakes, volcanic eruption, ocean waves and elephants and whales.
- The longitudinal mechanical waves having frequencies greater than 20000 Hz are called **ultrasonic waves**.
- Human ear cannot detect the ultrasonic waves. But certain creatures like dog, cat, bat, mosquito can detect these waves. Bat also produces ultrasonic waves.
- Ultrasonic waves are used for sending signals, measuring the depth of sea, cleaning clothes and machinery parts of clocks removing lamp shoot from chimney of factories and in ultrasonography.
- The use of ultrasonic waves to investigate the action of the heart is called **echocardiography.**

Sound Waves

Sound wave is the form of energy which make us hear and having frequency and high wavelength. These can not travel in vacuum. It is of two types i.e. longitudinal wave and mechanical wave.

- The rebouncing back of sound when it strikes a hard surface is called **reflection of sound**. The reflection of sound is utilised in working of devices such as megaphone, bulbhorn, stethoscope and sound board.
- The repetition of sound caused by the reflection of sound waves is called an **echo**. In addition to curtains, carpets and sofa-sets in our rooms also reduce the formation of echoes by absorbing sound waves.
- The persistence of sound in a big hall due to repeated reflections from walls, ceiling and floor of the hall is **reverberation**.
- To hear echo, the minimum distance between the observer and reflection should be 17. Persistance of ear is 1/10 s.
- At the moon, the echo is not heared due to absence of atmosphere.

SONAR

It stands for sound navigation and ranging. It is used to measure the depth of a sea and to locate the submarines and shipwrecks. The transmitter of a sonar produces pulses of ultrasonic sound waves of frequency of about 50000 Hz. The reflected sound waves are received by the receiver.

Speed of Sound

The distance travelled by sound wave in one second is known as speed of sound wave. i.e.

Speed of sound, $v = \text{Frequency} \times \text{Wavelength}$

- Speed of sound basically depends upon elasticity and density of medium. The speed of sound in a medium is given by $v = \sqrt{\frac{E}{d}}$, where E is modulus of elasticity of the medium and d = density.
- For gases, $v = \sqrt{\frac{\gamma p}{d}}$

  where, p is pressure and γ is ratio of specific heats.
- Speed of sound is maximum in solids and minimum in gases. Speed of sound in air is 332 m/s, 1483 m/s in water and 5130 m/s in iron.
- When sound enters from one medium to another medium, its speed and wavelength changes but frequency remains unchanged.
- Speed of sound remains unchanged by the increase or decrease of pressure.
- The speed of sound increases with the increase of temperature of the medium. The speed of sound in air increases by 0.61 m/s, when the temperature is increased by 1°C.
- The speed of sound is more in humid air than in dry air because the density of humid air is less than the density of dry air.
- The speed of sound in air is very slower as compared to the speed of light in air. Therefore, in rainy season, the flash of lightning is seen first and the sound of thunder is heard a little later.

➠ **Note** The astronauts who land on moon, talk to one another through wireless sets using radio waves.

Characteristics of Sound Waves

There are following characteristics of sound waves.

Intensity

Intensity of sound at any point in space is defined as amount of energy passing normally per unit area held around that point per unit time. SI unit of intensity is W/m^2.

- Intensity of sound at a point is inversely proportional to the square of the distance of point from the source and directly proportional to the square of amplitude of vibration, square of frequency and density of the medium.
- The loudness depends on intensity as well as on sensitivity of ear.
- The loudness of sound is measured in decibel (dB).
- Sound level, $\beta = 10 \log_{10}\left(\frac{I}{I_0}\right)$

where, I_0 = the minimum intensity that can be heard called threshold of hearing 10^{-12} W/m^2 at 1 kHz.

Standard value of noise pollution

Areas	*Day time (dB)*	*Night time (dB)*
Industrial area	75	70
Commercial area	65	55
Residential area	55	45
Silence	50	40

➨ **Note** Likewise intensity of sound wave, intensity of an earthquake defines the energy released by it which is indicated by the local effects and potential for damage produced on the earth's surface. It can be measured by a seismograph.

Pitch

It is that characteristic of sound, which distinguishes a sharp sound from a grave sound.

- Pitch depends upon frequency of sound waves.
- The pitch of female voice is higher than the pitch of male voice.
- The pitch of sound produced by roaring of lion is lower whereas the pitch of sound produced by mosquito whisper is high.

Quality (or Timbre)

It is that characteristic of sound which enables us to distinguish between sounds produced by two sources having the same intensity and pitch.

SHOCK WAVES

A body moving with supersonic speed in air leaves behind it a conical region of disturbance which spreads continuously. Such a disturbance is called **shock waves.**

- These waves carry huge energy and may even make cracks in window panes or even damage a building.
- The speed of supersonic wave is measured in **mach number**. One mach number is the ratio of velocity of source to the velocity of sound.
- Mach number = $\dfrac{\text{Velocity of source}}{\text{Velocity of sound}}$
  - If mach number > 1, body is called supersonic.
  - If mach number > 5, body is called hypersonic.
  - If mach number < 1, body is said to be moving with subsonic speed.

Resonance

- If however, the frequency of the external force is equal to the natural frequency of the body, then the amplitude of the forced oscillations of the body becomes quite large. This phenomenon is called **resonance**.
- A group of soldiers on a bridge are advised not to walk in steps because their movement causes the bridge to vibrate. If they walk in step, the frequency of vibration may match the natural frequency of the bridge structure and thus causing resonance. This resonance of frequency can cause the bridge to collapse.

Some Definitions Regarding Waves

Interference When two waves of same frequency, same wavelength, same velocity moves in the same direction. Their superimposition results in the interference. In interference, energy is neither created nor destroyed but is redistributed.

Beat When two sound waves of slightly different frequencies, travelling in a medium along the same direction, superimpose on each other, the intensity of the resultant sound at a particular position rises and falls regularly with time. This phenomenon of regular variation in intensity of sound with time at a particular position is called beats.

Progressive Waves The disturbance produced in the medium travels onward, it being handed over from one particle to the next. Each particle executes the same type of vibration as the preceding one, though not at the same time.

Stationary Waves When combination of two waves moving in opposite directions, each having the same amplitude and frequency.

DOPPLER'S EFFECT

If there is a relative motion between source of sound and observer, the apparent frequency of sound heard by the observer is different from the actual frequency of sound emitted by the source. This phenomenon is called Doppler's effect.

When the distance between the source and observer decreases, the apparent frequency increases and *vice-versa*.

Uses of Doppler's Effect

(*i*) By police to check over speeding of vehicles.
(*ii*) At airport to guide the aircraft.
(*iii*) To study heart beats and blood flow in different parts on the body.
(*iv*) It is used to determine the velocities of which stars and galaxies are moving toward or away from the earth.

OPTICS

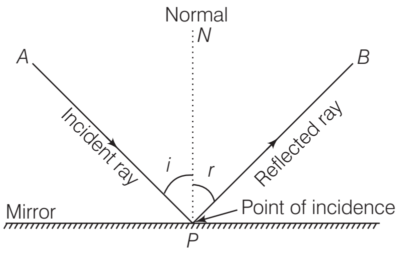
LIGHT

It is the radiation which makes our eyes able to see the object. Its speed is 3×10^8 m/s in vacuum.

- Light is the form of energy. It is a transverse wave and it takes 8 min 19 s to reach on the earth from the sun.
- The light reflected from the moon takes 1.28 s to reach the earth.
- It represents the phenomenon of reflection, refraction, interference, diffraction, scattering and polarisation.

Reflection of Light

Reflection is the phenomenon of change in the path of light without any change in its medium. In reflection, the frequency, speed and wavelength do not change, but a phase change may occur depending on the nature of reflecting surface.



Reflection of Light

Laws of Reflection

- The angle of reflection ($\angle r$) is always equal to the angle of incidence ($\angle i$) i.e. $\angle i = \angle r$.
- The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.

MIRROR

It is a polished surface (one side) like glass, which reflects almost all the light that is incident on it.

There are two types of mirror

Plane Mirror

If the reflecting surface of a mirror is plane, then the mirror is called a **plane mirror**.

- Size of image is always equal to the size of object.
- The minimum size of the mirror required to see the full image of an observer is half the height of the observer.
- If the plane mirror is rotated in the plane of incidence by an angle θ, then the reflected ray rotates by an angle 2θ.
- If the object is displaced by a distance x towards or away from the mirror, then its image will be displaced by a distance x towards or away from the mirror.
- Focal length of plane mirror is infinity i.e. power of the plane mirror is zero.
- Linear magnification produced by plane mirror is 1.
- When two plane mirrors are kept facing each other at an angle θ and an object is placed between them, then

  (*i*) Number of images, $n = \dfrac{360^\circ}{\theta} - 1$

  If $\dfrac{360^\circ}{\theta}$, is even or object lies symmetrically.

  (*ii*) Number of images, $n = \left(\dfrac{360^\circ}{\theta}\right)$. If $\left(\dfrac{360^\circ}{\theta}\right)$ is odd or the object lies symmetrically.

Uses of the Plane Mirror

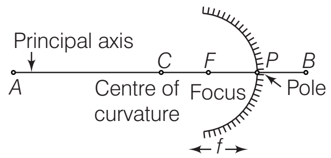
(*i*) Plane mirrors are used as looking glass.
(*ii*) Plane mirrors are used in constructing periscope which is used in submarines.
(*iii*) Plane mirror are used to make kaleidoscope.

Spherical Mirrors

Spherical mirrors can be regarded as a part of a hollow reflecting spheres of glass. These are of two types

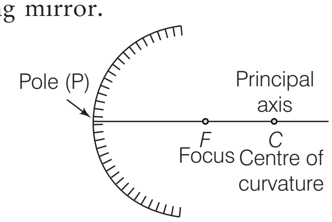
(*i*) Concave mirror (*ii*) Convex mirror

Concave mirror The spherical mirror whose reflecting surface is inwardly leaned, called concave mirror. It is also called converging mirror.



Concave mirror

Convex mirror The spherical mirror whose reflecting surface is outwardly leaned, called convex mirror. It is also called diverging mirror.



Convex mirror

Real Image

- Real images can be formed on a screen by the actual intersection of reflected (or refracted) light rays.
- The image formed on a cinema screen is a real image.
- A concave mirror forms real image, when an object is placed beyond its focus.

Virtual Image

- Virtual images cannot be formed on the screen by the apparent intersection of reflected (or refracted) light rays.
- The image of our face when we look into a plane mirror is a virtual image. A concave mirror forms a virtual image, when an object is placed between focus and pole of the mirror.

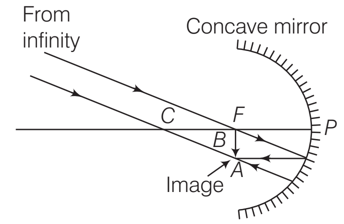
Position of object	*Position of image*	*Size of image*	*Nature of image*
At infinity	At *F*	Highly diminished	Real and inverted
Between infinity and *C*	Between *F* and *C*	Diminished	-do-
At *C*	At *C*	Same size	-do-
Between *F* and *C*	Between infinity and *C*	Enlarged	-do-
At *F*	At infinity	Highly enlarged	-do-
Between *F* and *P*	Behind the pole	Enlarged	Virtual and erect

Position of object	*Position of image*	*Size of image*	*Nature of image*
At infinity	At *F*	Highly diminished	Erect and virtual
Between infinity and pole	Between *F* and *P*	Diminished in size	Erect and virtual

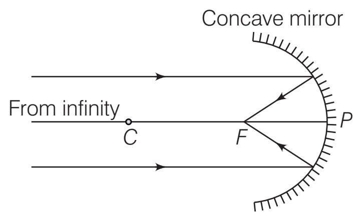
Image Formation by Concave Mirror

The following are the positions and nature of the images formed in a concave mirror for different positions of the object

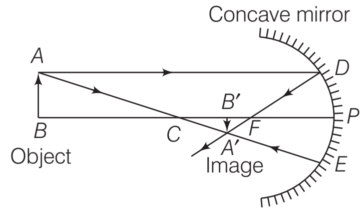
(*i*) **When the object is at infinity** Image is formed at *F*.



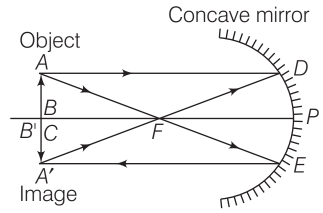
(*ii*) **When the rays are parallel to the principal axis** The image is formed at *F*.



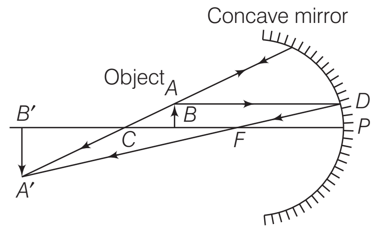
(*iii*) **When the object is between infinity and *C*** The image formed is real, inverted and smaller in size and is formed between *F* and *C*.



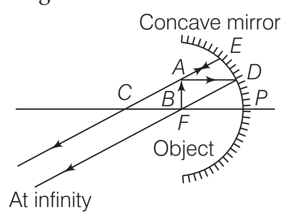
(*iv*) **When the object is at *C*** The image formed is inverted, real, equal in size and is formed at *C* itself.



(*v*) **When the object is between *F* and *C*** The image is formed beyond *C* and is inverted, real and larger than the object.



(*vi*) **When the object is at focus** The image is formed at infinity. The image is real, inverted and infinitely large in size.



(*vii*) **When the object is between *F* and *P*** The images formed behind the pole. The image is virtual, erect and large.

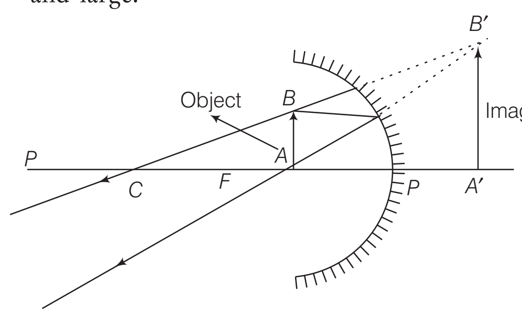
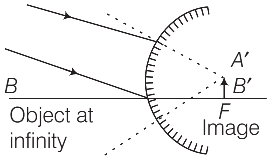
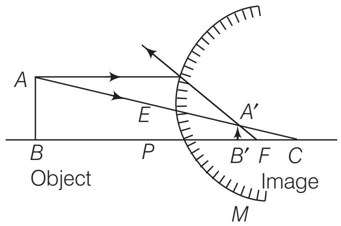


Image Formation by Convex Mirror

(*i*) **When the object is at infinity** The image is formed at focus.



(*ii*) **When the object is between infinity and pole** The image is formed between pole and focus.



MIRROR FORMULA

- If v is the image distance, u is the object distance and f is the focal length of a mirror, then $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}=\frac{2}{R}$
- The linear magnification (m) produced by a mirror is equal to the ratio of the image distance to the object distance with negative sign, $m=-\frac{v}{u}=\frac{f}{f-u}=\frac{f-v}{f}$
- If the magnification has positive sign, then the image is virtual and erect.
- If the magnification has negative sign, then the image is real and inverted.

Uses of Spherical Mirrors

Concave Mirrors

- These are used as reflectors in automobiles (car, buses, etc.), head light, search light, hand torches and table lamps.
- These are used as shaving mirrors.
- These are used by doctors for focussing intense light to examine inside of eye, ears, etc.
- Large concave mirrors are used in the application of solar energy, to focus the sun's rays for heating solar furnaces etc.

Convex Mirrors

- These are used as rear-view mirrors in automobiles as they give large view of the traffic.
- Big convex mirrors are used as shop security mirrors.
- These are used wherever we want to see erect images.

➨ **Note** We cannot use a concave mirror as a rear-view mirror in motor vehicles.

REFRACTION OF LIGHT

When a ray of light passes from one medium to other, it bends from its path. This phenomenon of bending of light is called as refraction of light.

- When light passes from rarer medium to a denser medium, it bends towards the normal at the point of incidence.
- When light passes from a denser medium to a rarer medium, it bends away from the normal.
- When a ray of light enters from one medium to other medium its frequency and phase do not change but wavelength and velocity change.

Laws of Refraction

- The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.
- The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for a given medium.

$$\frac{\sin i}{\sin r}=\mu \qquad \text{[Snell's law]}$$

where, μ is refractive index of the second medium with respect to first.

$$\because \quad \mu=\frac{\mu_2}{\mu_1}=\frac{c\ /v_2}{c\ /v_1}$$

$$\therefore \quad \mu=\frac{v_1}{v_2} \text{ or } {}_1\mu_2=\frac{v_1}{v_2}$$

where, v_1 is speed of light in first medium and v_2 is speed of light in second medium.

Some Illustrations of Refraction

- Twinkling of stars.
- Oval shape of sun in the morning and evening.
- Bending of a linear object when it is partially dipped in a liquid inclined to the surface of the liquid.
- A fish in a pond when viewed from air appears to be at a smaller depth than actual depth.
- A coin at the base of a vessel filled with water appears raised.

TOTAL INTERNAL REFLECTION

In case of propagation of light from denser to rarer medium through a plane boundary. **Critical angle** is the angle of incidence for which angle of refraction is 90°.

- If light is propagating from denser medium towards the rarer medium and angle of incidence is more than critical angle, then the light incident on the boundary is reflected back in the denser medium, obeying the law of reflection. This phenomenon is called total internal reflection.

Conditions of Total Internal Reflection

- Light must be propagating from denser to rarer medium.
- Angle of incidence must exceeds the critical angle.

• Sparkling of diamond, mirage and looming, shinning of air bubble in water and optical fibre are the examples of total internal reflection.

➨ **Note** Optical fibre is used for the transmitting and receiving the electrical signals after converting it into the light signal.

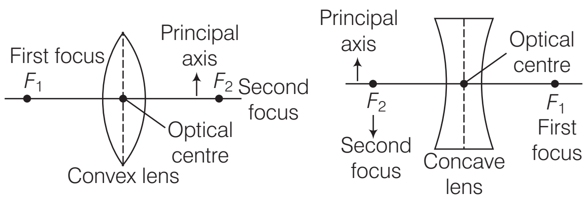
SPHERICAL LENSES

A lens is a piece of transparent glass bounded by two spherical surfaces.

There are two types of lenses

(*i*) **Convex lens (converging lens)** A lens which is thicker at the centre and thinner at its ends.

(*ii*) **Concave lens (diverging lens)** A lens which is thinner at the centre and thicker at its ends.



(a) (b)

• The centre point of a lens is known as its optical centre.

• The line joining the centres of the surfaces of the lens is called the principal axis of the lens.

• The distance of the first focus from the optical centre of the lens is called the first focal length of the lens.

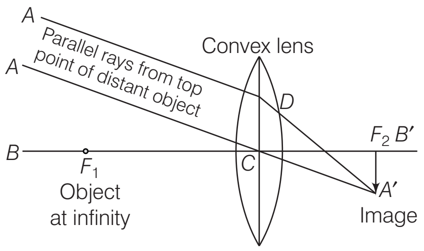
• The distance of the second focus from the optical centre of the lens is called the second focal length or the principal focal length.

Position of object	*Position of image*	*Size of image*	*Nature of image*
At infinity	At F_2	Highly diminished	Real and inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	-do-
At $2F_1$	At $2F_2$	Same size	-do-
Between $2F_1$ and F_1	Beyond $2F_2$	Enlarged	-do-
At F_1	At infinity	Highly enlarged	-do-
Between F_1 and lens	Behind the object, on the same side of the object	Enlarged	Virtual and erect

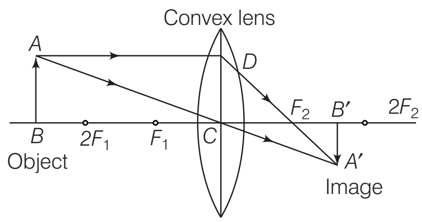
Position of object	*Position of image*	*Size of image*	*Nature of image*
At infinity	At F_2	Diminished	Erect and virtual
Between infinity and lens	Between F_2 and lens	Diminished	Erect and virtual

Image Formation by a Convex Lens

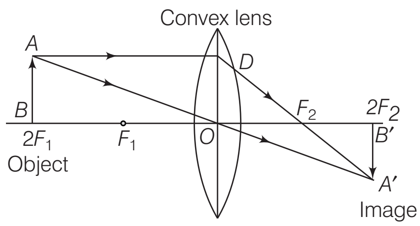
(*i*) **Object** at **infinity** The image formed is at second focus, real, inverted and highly diminished.



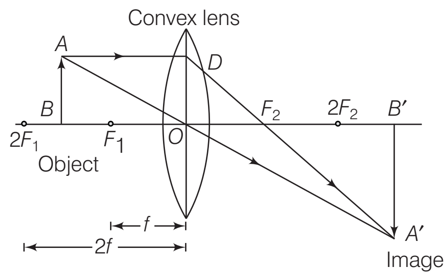
(*ii*) **Object beyond** $2F_1$ The image formed is between F_2 and $2F_2$ on the right side of the lens, real, inverted and diminished.



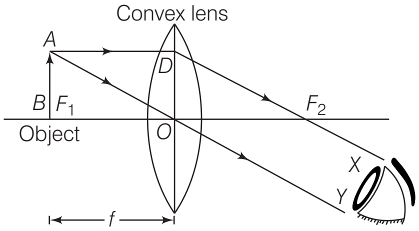
(*iii*) Object at $2F_1$ The image formed is at $2F_2$, real, inverted and of same size as that of the object.



(*iv*) Object between F_1 **and** $2F_1$ The image formed is beyond $2F_2$, real, inverted and magnified.



(*v*) **Object at F_1** The image formed is real, inverted, and highly enlarged at infinity.



(*vi*) **Object between F_1 and O** The image formed is behind the object, virtual, erect and enlarged.

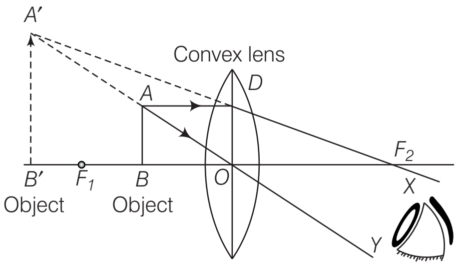
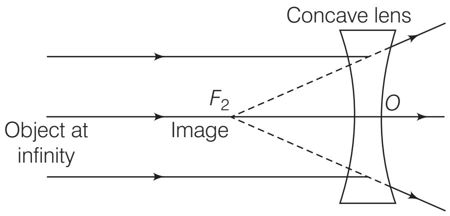
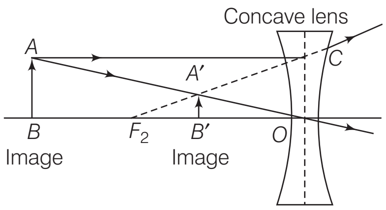


Image Formation by a Concave Lens

(*i*) **Object at infinity** The image formed is at F_2, erect, virtual and diminished.



(*ii*) **Object anywhere between infinity and lens** The image formed is between F_2 and lens, erect, virtual and diminished.



LENS FORMULA

- If v is the image distance, u is the object distance and f is the focal length of a lens, then

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Magnification, $$m = \frac{v}{u} = \frac{I}{O} = \frac{\textit{Size of image}}{\textit{Size of object}}$$

- If the magnification m has a positive value, the image is virtual and erect. And if the magnification m has a negative value the image will be real and inverted.
- If m is the total magnification of two lenses in contact having magnification m_1 and m_2, then

$$m = m_1 \times m_2$$

- Power (P) of a lens is the reciprocal of focal length of lens i.e. $$P = \frac{1}{f}$$

SI unit of power is dioptre (D).

- If two or more lenses are combined, then
  - there is an increase in the magnification of image.
  - the final image is erect.
  - the spherical aberration is reduced.
- If P is power of two lenses in contact having powers P_1 and P_2, then

$$P = P_1 + P_2 \text{ or } \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

- If two lenses of equal focal length but of opposite nature are in contact, then combination will behaves as a plane glass plate and $f = \infty$

➡ **Note** When lens is dipped in a liquid of higher refractive index the focal length increases and convex lens behaves as concave lens and *vice-versa*.
An air bubble trapped in water or glass appears as convex but behaves as concave lens.

Properties of Light

Dispersion of Light

Sir Isaac Newton observed that when a narrow beam of light is incident on a prism, the emergent beam is not only deviated but at the same time splits up into a coloured band of seven colours. This phenomenon is called dispersion of light.

- The seven colours of band are Violet, Indigo, Blue, Green, Yellow, Orange and Red (can be memorised as VIBGYOR).
- The colour of an object is determined by the colour of the light reflected by it.
- Violet colour deviates through maximum angle and red colour deviates through the minimum angle.
- Rainbow is formed due to the dispersion of light suffering refractions and total internal reflection in the droplets present in the atmosphere.

Primary Colours

Red, green and blue are called primary colours or basic colours.

Mixing of Colours

Red + Green + Blue = White
Red + Blue = Magenta
Blue + Green = Peacock blue (or Cyan)
Red + Green = Yellow

- Magenta, Cyan and Yellow, etc., are called secondary colours.
- A rose appears red, when day light falls on it because it absorbs all the constituent colour of white light except red which it reflects to us.
- If all the colours of white light are reflected back from the object, then it appears white.
- If all the colours of white light are absorbed by the object, then it appears black.
- The natural colours of transparent objects are due to their selective transmission of light.

Scattering of Light

- When sunlight passes through the earth's atmosphere, much of the light is absorbed by the fine dust particles and air molecules in the atmosphere which give out the absorbed light in some other direction. This is scattering of light.
- The scattering of light by particle in its path is called Tyndall effect.
- The blue coloured light present in white sunlight is scattered much more easily by air molecules causes the blue colour of the sky.
- The sun and the surrounding sky appear red at sunrise and at sunset because at that time most of blue colour present in sunlight has been scattered out and away from our line of sight, leaving behind mainly red colour in the direct sunlight beam that reaches our eyes.
- Dangers signals are of red colour because red colour of light scatter least and therefore signals can be see from far.
- If the earth had no atmosphere, there would have been no scattering of sunlight at all. In that case, no light from the sky would have entered our eyes and the sky would have looked dark and black to us.

Interference of Light

When two waves of exactly same frequency travels in a medium, in the same direction simultaneously, then due to their superposition at same points, intensity of light is maximum while at some other points intensity is minimum. This phenomenon is called interference of light.

- Thin layer of oil on water surface and soap bubbles show various colours in white light due to interference of waves reflected from the two surfaces of the film.
- Polarisation is the only phenomenon which proves that light is a transverse wave.

Diffraction of Light

If an opaque obstacle is kept between a source of light and a screen, a sufficiently distinct shadow is obtained on the screen. This shows that light which travels the obstacle is small. There is a departure from straight line propagation and so the light bends round the corners of obstacle and enters in the geometrical shadow. This bending of the light is called diffraction.

HUMAN EYE

Human eye is an optical instrument which forms real image of the objects.

- The ability of an eye to focus the distant objects as well as the nearby objects on the retina by changing the focal length of its lens is called accommodation.
- Least distance of distinct vision is 25 cm.
- Far point of normal eye is at infinity.

Defect of Human Eye

The ability to see is called vision. The defects of vision are known as defects of eye. There are following defects of human eye

- **Myopia** A person suffering from myopia can see the near objects clearly while far objects are not clear; concave lens is used to remove it. The defect of eye called myopia is caused
  (*i*) due to high converging power of eye lens or
  (*ii*) due to eye ball being too long.
- **Hypermetropia** A person suffering from hypermetropia can see the distant objects clearly but not the near objects. Convex lens is used to remove it. The defect of eye called hypermetropia is caused
  (*i*) due to low converging power of eye lens or
  (*ii*) due to eye ball being too short.
- **Presbyopia** In the old age, the power of accomodation of the eye is so much lost that a person can neither see the near objects distinctly nor the distant objects. To correct this defect bifocal lens is used.
- **Astigmatism** In this defect, a person cannot distantly see the horizontal and vertical lines, simultaneously at a normal distance. To correct this defect, cylindrical lens is used.
- **Cataract** In this defect, an opaque, white membrane is developed on cornea due to which a person losses power of vision partially or completely. This defect can be removed by removing this membrane through surgery.

➡ **Note** Colour blindness is a genetic disorder which occurs by inheritance. John Dalton was colour blind.

Optical Magnifying Instruments Microscope

A microscope is an instrument used to see objects that are too small for the naked eye.

- A simple microscope consists of a convex lens of small focal length.
- It is used to see the details of very small objects.
- The magnifying power of a simple microscope is given by $M = \left[1 + \frac{D}{f}\right]$, D is least distance of distinct vision.
- For compound microscope $m = m_o \times m_e$, where m_o is magnification of objective lens and m_e is magnification of eye-piece.

Telescope

- Telescope is an optical instrument which is used for observing distinct images of heavenly bodies like stars, planets, etc., when the final image is formed at infinity.
- There are two types of telescopes
  - Astronomical telescope (invented by Kepler in 1611)
  - Galilean telescope (invented by Galileo in 1609)
- In an astronomical telescope, the objective lens is a convex lens of large focal length but eye-piece is a convex lens of short focal length.
- In Galilean telescope, the objective lens is a convex lens of large focal length but the eye-piece is a concave lens of short focal length.
- Magnifying power of telescope, $m = \frac{f_o}{f_e}$ where f_o is focal length of objective of the telescope and where f_e is focal length of eye-piece of the telescope.
- Length of telescope tube in astronomical telescope is
  $$L = f_o + f_e$$
  In Galilean telescope the length of telescope,
  $$L = f_o + f_e.$$

➨ **Note** Periscope consists of two plane mirrors inclined at an angle of 45°. The principle of working of periscope is based upon reflection and refraction.

ELECTRIC CURRENT

ELECTRIC CHARGE

Charge is the basic property associated with matter due to which it produces and experience electrical and magnetic effects. The SI unit of electric charge is Coulomb.

- Positively charged particles called protons and negatively charged particles called electrons. A proton possesses a positive charge of 1.6×10^{-19}C, whereas, an electron possesses a negative charge of 1.6×10^{-19}C.
- Similar charges repel each other and opposite charges attract each other.

Conservation of Charges

When two or more charged bodies come in contact, then total charge on all the bodies are conserved.

- **Conductors** are those substances which allow passage of electrical charges to flow through them and have very low electrical resistance e.g. copper, aluminium, gold, silver, etc.
- Human body and earth act like a conductor. Silver is the best conductor.
- Charges are always distributed on the surface of the conductor.
- **Resistors** offer high resistance to the flow of current through them e.g. eureka, nichrome, etc.
- **Insulators** have infinite resistance and do not allow the passage of current. e.g. rubber, glass, ebonite, etc.

➨ **Note** The presence of free electrons in a substance makes it a conductor.

Coulomb's Law

- The force of attraction or the force of repulsion acting between the two point charges is proportional to the product of the magnitudes of the two charges and inversely proportional to the square of the distance between them i.e. $F = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2}$

Electric Field

The region around an electric charge in which its effect can be experienced is called the electric field.

$$\text{Electric field intensity } (E) = \frac{F}{q_0}$$

Electric Potential

The electric potential at a point in an electric field is the work done in bringing a unit positive charge from infinity to that point.

- Electric potential $= \frac{\text{Work done}}{\text{Charge}}$. Its SI unit is volt (V).
- Potential difference $(V_A - V_B)$ between two points A and B is the work done in bringing a unit charge from point B to point A.
- Potential difference is a scalar quantity and is measured by means of voltmeter (a high resistance device).
- The electric potential inside a spherical surface is same at each point and is equal to the potential on the surface.
- Electrical potential on earth is considered to be zero.
- A voltmeter connected in parallel with conductor to measure the potential difference across its ends.
- Voltage is the other name for potential difference.

Electric Current

- The rate of flow of electric charges through any cross-section of conductor is called electric current.
- The direction of positive charges is same as direction of conventional current.
  $$\text{Current} = \frac{\text{Charge}}{\text{Time}} \Rightarrow I = \frac{Q}{t}$$
  The SI unit of electric current is ampere.
- Current is measured by an instrument called ammeter.
- An ammeter is connected in series with the conductor to measure the current passing through it.

- There are two types of electric current (*i*) Alternating current (AC) and (*ii*) Direct current (DC).
- Alternating current is used in houses and factories and its frequency is 50 Hz.
- Ordinary DC ammeter and DC voltmeter cannot measure alternating current/voltages. They record zero reading, when used in AC circuits, because average value of alternating current/voltage over a full cycle is zero.

Galvanometer

It is a device used to detect and measure electric current in a circuit. It can measure current up to 10^{-6} A.
A galvanometer can be converted into a voltmeter by connecting a very high resistance in its series.

➠ **Note** Conventionally the direction of electric current is opposite to that of electrons. Since, electrons flow from negative terminal of a cell to its positive terminal, electric current flows from positive terminal to negative terminal.

Capacitor

A capacitor or condenser is a device over which a large amount of charge can be stored.

- Capacitance of capacitor (C) $= \dfrac{\text{Charge }(Q)}{\text{Potential }(V)}$
- Its unit is coulomb/volt or farad.
- A capacitor is used in several electrical devices having an electric motor and in several electronic circuits.

OHM'S LAW

If the physical circumstances of the conductor. (length, temperature, etc) remains constant then the current flowing through the conductor is directly proportional to the potential difference across its ends.

i.e. $I \propto V$

or $V = IR$, where R is resistance.

Resistance

The resistance of a conductor is directly proportional to its length and inversely proportional to its cross-sectional area. If l and A are respectively length and cross-sectional area of a conductor and R is its resistance, then

$$R \propto \frac{l}{A} \Rightarrow R = \rho\frac{l}{A}$$

Unit of resistance is ohm (Ω).

where, ρ is a constant of material of conductor is called specific resistance or resistivity. Its SI unit is ohm-metre.

- On increasing the temperature of the metal, its resistance increases.
- Those materials whose electrical conductivity lies in between that of insulators and conductors are called semiconductors. On increasing the temperature of semiconductor, its resistance decreases.
- The reciprocal of resistance of a conductor is called the electrical conductance of the conductor.

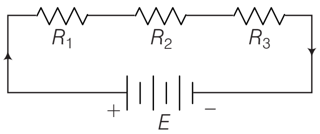
$$\text{Conductance} = \frac{1}{\text{Resistance}}$$

- Unit of conductance is mho or siemen.
- The specific resistance of the material depends only on the material of conductor and its temperature.
- Resistivity increases with temperature.
- Resistivity of a conductor change with impurity.
- Resistivity of an alloy is greater than the resistivity of its constituents and remains unchanged with temperature.
- If a wire is stretched or doubled on itself, its resistance will change, but its specific resistance will remains unaffected.
- The reciprocal of resistivity of a conductor is called its conductivity. Its SI unit is mho m^{-1} or siemen/metre (Sm^{-1}).

Combination of Resistances

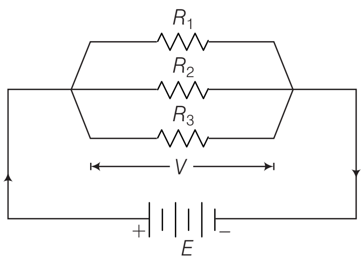
There are two combinations given below:

(*i*) **Series combination** $R = R_1 + R_2 + R_3$ and here current flows through each conductor is same.



The total resistance in the series combination is more than the greatest resistance in the circuit.

(*ii*) **Parallel combination** $\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3}$ and here potential across each conductor is same.



The total resistance in parallel combination is less than the least resistance of the circuit.

ELECTRIC POWER

Electrical power is the electrical work done per unit time.

$$P = \frac{W}{t} \quad \text{or} \quad P = V \times I$$

or $P = I^2R$ $\quad [\because V = IR]$

or $P = \dfrac{V^2}{R}$

Its SI unit is watt (W).

- 1 kilowatt hour =3600000 J = 3.6×10^6 J

  1 Horse power (HP) = 746 W

  1 Horse power (HP) = 550 foot-pound/second

  $$\text{Unit} = \frac{\text{volt} \times \text{ampere} \times \text{hour}}{1000}$$

  $$\text{Unit} = \frac{\text{watt} \times \text{hour}}{1000}$$

Heating Effect of Electric Current

Heat is produced, when electric current is passed through a conductor. This effect is called heating effect of electric current.

$$\text{Heat produced} = I^2Rt = VIt = \frac{V^2t}{R} \text{ joule}$$

➨ **Note** The filament of an electric bulb gets very hot and therefore easily oxidised. Thus, bulbs are sealed with inactive nitrogen and argon gases to prevent oxidation of the filament and increase its life.
In electric bulb, tungsten is used for the construction of filament because it has high melting point.

IMPORTANT POINTS BASED ON HEATING EFFECT

- In home appliances like electric iron, electric heater and heating rod, the heating element used is of a nichrome (an alloy of Ni and Ci) wire. Nichrome has high melting point and high resistivity. To avoid the risk of electric shock, the metal body of electrical appliances is earthed.
- An electric fuse is generally prepared from tin-lead alloy (63% tin + 37% lead). It should have high resistance and low melting point. It is connected in the series.
- Tubelight contains a long tube of glass which is linked internally with a fluorescent substance. It is filled with an inert gas like argon along with some mercury.

Magnetic Effect of Current

When electricity passes through insulated copper wire, it develops a magnetic field around it. Larger the amount of electricity, stronger the magnetic field. And when we place compass needle in the radius of the magnetic field it diverts from its North-South position.

MAGNETS

Magnet is a piece of iron or other materials that can attract iron containing objects and the property of attracting the magnetic substance by a magnet is called magnetism.

- The magnets which do not lose their magnetism with normal treatment are called permanent magnets.
- The materials which retain their magnetism for a long time are called hard magnetic materials.
- In bar, rod and horse-shoe magnets, North or South poles are either indicated by the letter N or S.
- When poles of two magnets are brought close together, they exert force on each other. This force is called intersection between the poles.
- If we cut a magnet in two parts, then each separate part will behave as a magnet.
- Permanent magnets are usually made of alloys such as carbon steel, chromium steel, cobalt steel, tungsten steel and alnico (alloy of aluminium, nickel, cobalt and iron).
- Such permanent magnets are used in microphones, loudspeakers, electric clocks, ammeters, voltmeters, speedometer and many other devices.

➨ **Note** The Magnetic Resonance Imaging (MRI) technique which is used to scan inner human body parts in hospitals also uses magnets for its working.

Magnetic Field

The space in the surrounding of a magnet or a current carrying conductor in which its magnetic effect can be experience is called magnetic field.

- Magnetic lines of force are imaginary lines in the magnetic field, which shows the direction of magnetic field continuously.
- The magnetic field lines originate from the North pole of a magnet and end at its South pole.
- The magnetic field lines do not intersect one another.

Magnetic Materials

According to behaviour of magnetic substances, they are classified into three classes

- Those substances when placed in an external magnetic field, acquire a very low magnetism in direction opposite to the field are called **diamagnetic substances**. e.g. copper, silver, bismuth, zinc, diamond, salt, water, mercury, etc.
- The permeability of diamagnetic substance is less than one.
- Those substances when placed in an external field, acquire a feeble magnetism in the direction of field are called **paramagnetic substances**. e.g. aluminium, sodium, potassium, oxygen, etc.
- The permeability of paramagnetic substance is slightly greater than one.
- Those substances, when placed in a magnetic field, acquire a strong magnetism in the direction of field are called **ferromagnetic substances**. These substances when brought near the end of a strong magnet get radially attracted towards it. e.g. iron, nickel, cobalt, magnetite, etc.
- The permeability of ferromagnetic substance is much greater than one.

Electromagnet

An electromagnet is a solenoid coil that attains magnetism due to flow of current. It works on the principle of magnetic effect of current. It is used in electric bells, electric motors, telephone, diaphragms, loudspeakers etc.

Magnetic Flux

The magnetic flux linked with a surface is equal to the total number of magnetic lines of force passing through that surface normally. Its unit is weber.

ELECTROMAGNETIC INDUCTION

When a change occurs in the magnetic flux linked with the coil, an emf is induced in the coil. The phenomenon is called electromagnetic induction.

- The phenomenon of production of induced emf in a circuit due to change in magnetic flux in its neighbouring circuit is called **mutual induction.**
- When the electric current flowing through a circuit changes, the magnetic field linked with circuit also change. As a result an induced emf is set up in the circuit. This phenomenon is called **self-induction**.
- Microphone converts sound energy into electrical energy and works on the principle of electromagnetic induction.

Electric Generator

It is a device which converts mechanical energy into electrical energy. An electric generator is based on the principle of electromagnetic induction. It is used to produce electric current by varying magnetic field through a coil.

- A generator can produced both alternating current and direct current.
- AC generator consists of slip rings, armature, magnet and brushes while DC generator consists of magnet commutator, armature and brushes.

➠ **Note** Electric motor is a rotating device which converts electrical energy into mechanical energy.

Transformer

Transformer is a device which converts low voltage AC into high voltage AC and high voltage AC into low voltage AC. It is based on mutual induction.

- Core of a transformer is made up of soft iron.
- Step-up transformer converts a low voltage of high current into a high voltage of low current.
- Step-down transformer converts a high voltage of low current into a low voltage of high current.
- The main energy losses in a transformer are given below:
  (a) Iron loss (b) Flux loss
  (c) Hysteresis loss (d) Humming loss
- Transformer is used in voltage regulators, refrigerators, computer, air conditioner.
- Step-down transformer is used for welding purpose.
- Audio transformer allowed telephone circuits to carry on a two-way conservation over a single pair of wires.
- Radio frequency transformers are used in radio communication.

➠ **Note** A mobile phone charger is one of the best example of step down transformer as it extracts the power from the home supply (AC 220V) and convert it to DC of required voltage.

MODERN PHYSICS

CATHODE RAYS

If the gas pressure in a discharge tube is 10^{-2} to 10^{-3} mm of Hg and a potential difference of 10^4 V is applied between the electrode, then a beam of electrons emerges from the cathode which is called cathode rays.

- Cathode rays are invisible and travel in a straight line.
- These rays are deflected by electric and magnetic fields.
- These rays can ionise gases.
- They can produce chemical change and thus affect a photographic plate.
- When they strike a target of heavy metal such as tungsten, they produce X-rays.

X-RAYS

When cathode rays strike on a heavy metal of high melting points, then a very small fraction of its energy converts into a new type of waves, called X-rays.

- X-rays are electromagnetic waves and these produce photoelectric effect.
- X-rays were discovered by scientist Roentgen.
- These produce illumination on falling the fluorescent substances and travel in a straight line with the speed of light.
- X-rays penetrate through different depth into different substances.
- Soft X-rays have greater wavelength and lower frequency and hard X-rays have lower wavelength and higher frequency.
- X-rays ionise gases through which they pass.
- X-rays are not deflected by electric and magnetic fields.
- They show all the important properties of light rays like, reflection, refraction, interference, diffraction and polarisation, etc.
- X-rays are used in surgery, radiotherapy, engineering department and searching.

➠ **Note** Barium and Iodine are some chemicals used as contrast mediums for *X*-ray based imaging method. i.e. they absorbs *X*-ray to highlight the targeted body part.

POSITIVE OR CANAL RAYS

These rays are positively charged particles and are called positive rays or canal rays. These rays were discovered by Goldstein.

- These rays can affect the photographic plates.
- These rays are capable of producing physical and chemical changes.
- These rays are deflected by electric and magnetic fields. These rays can produce fluorescence and phosphorescence.

Fluorescence

Some substances absorb the light of higher frequency or shorter wavelength and emits a light having lower frequency or higher wavelength in the presence of light source. This phenomenon of emission of light is called **fluorescence**. Such substances are called fluorescence substances.

- Fluoresphor, petrol, uranium oxide and barium platino cyanide are examples of fluorescence substance.
- Barium platino cyanide is used in search of X-rays.

Compact Fluorescent Lamp (CFL)

Compact fluorescent lamp converts electrical energy into radiant energy.

- It has two components
  - A glass tube filled with argon and mercury vapour and coated with layer of fluorescent materials.
  - An electronic ballast circuit.
- The ballast circuit (small circuit board with rectifiers, a filter capacitor and usually two switching transistors) takes a 220V input from external power source and sends a current into the fluorescent tube as output.
- When power supply is given to the CFL, filament attached with the cathode heats up and emits electrons in the tube. This ionises the argon and mercury vapour particles, causing it, ultraviolet light emission.

Phosphorescence

Some substances after being kept in sunlight for sometime emit light even after the light source is removed. This phenomenon is called **phosphorescence.** Zinc sulphide, calcium sulphide and barium sulphide are examples of phosphorescence.

PHOTOELECTRIC EFFECT

When the ultraviolet rays (or the visible light of low wavelength or any other kind of electromagnetic radiation) are made incident on a metal, the electrons are emitted from it. This phenomenon is called the photoelectric emission or photoelectric effect.

Photoelectric cell is based on photoelectric effect. It is a device by which light energy is converted into the electrical energy.

Applications of photoelectric effect are given below

(i) In reproduction of sound in cinema, television and photo telegraphy.
(ii) In automatic switches for street lights.
(iii) In photoelectric sorters.
(iv) To control the temperature in furnace and in chemical process.
(v) In industries for locating minor flows or holes in metallic sheets.

SEMICONDUCTOR

Semiconductors are those materials whose electrical conductivity at room temperature lies in between that of insulator and conductor. A semiconductor in an extremely pure form is known as intrinsic semiconductor.

- If a measured and small amount of chemical impurity is added to intrinsic semiconductor, it is called extrinsic semiconductor or doped semiconductor.
- External semiconductors are of two types
  (*i*) *n*-type semiconductor
  (*ii*) *p*-type semiconductor
- An extrinsic semiconductor in which electrons are majority charge carrier is called ***n*-type semiconductor**.
- An extrinsic semiconductor in which holes are the majority charge carrier is called ***p*-type semiconductor**.

Special Purpose Diodes

An arrangement consisting a *p*-type semiconductor brought into a close contact with *n*-type semiconductor is called a *p-n* junction or *p-n* junction diode.

- A device which converts alternating current (AC) or voltage into direct current (DC) or voltage is known as **rectifier**. DC can be converted into AC with the help of an inverter.
- LED's are specially designed diode made up of GaAsP, GaP and are used in electronic gadgets as indicator light. It is used as a rectifier which converts an alternating current into direct current.
- LED's light up very quickly. A typical red indicator LED will achieve full brightness in microseconds.
- LED or semiconductor laser is used to convert electrical energy into optical energy.
- Zener diode is a highly doped *p-n* junction diode which is not damaged by high reverse current. It can be used as voltage regulator.
- Photo diode is a special *p-n* junction diode fabricated with a transparent window to allow light to fall on the diode. It is used in reverse biasing.

➠ **Note** Diode valve acts as a rectifier and triode valve can be used as amplifier oscillator and detector.

Solar Cell

Solar cell is a *p-n* junction device which converts solar energy into electrical energy. It is consist of a silicon or gallium arsenide *p-n* junction.

Uses of Solar Cell

(i) Solar cells are used as solar cooker.
(ii) Solar cells are used in wrist watches, calculators and light meters.
(iii) Solar cells are also used in artificial satellite to provide power inside the satellite.
(iv) Solar cells are used for charging storage batteries.

Integrated Circuits

An IC can be defined generally as an arrangement of multifunction semiconductor devices. It consists of a single crystal chip of silicon, nearly 1.5 mm^2 in cross-section.

TV remote control is an electronic device containing an IC and few other components. If a key on a remote control is pressed, it emits infrared signals which are received by the electronic circuit of the TV receiver, etc.

LASER

The word LASER is the short form of Light Amplification by Stimulated Emission of Radiation. It is an optical device, which produces an intensive beam highly coherent, monochromatic light.

Applications of laser rays are described below

(i) Laser rays are used in treatment of detached retina, treatment of skin cancer and surgery of brain.
(ii) Laser beams are used in signals and radio communications.
(iii) The property of high intensity is used to cut hard metals.
(iv) The weather informations are collected with the help of radar, an instrument based on the properties of laser.
(v) The laser rays are parallel and do not spread. This property of laser rays is used in the measurement of long distance.

COMMUNICATION

The term communication refers to the transmitting receiving and processing of information by electronic means. A basic communication system consists of an information source, a transmitter, a link and a receiver.

- Electromagnetic waves which are used in radio, television and other communication system are radiowaves and micro waves.
- In **ground wave propagation**, the radio wave travel through atmosphere following the surface of the earth from transmitting antenna to receiving antenna. The ground wave propagation is useful for low range frequencies upto 1500 kHz. The ground wave propagation is generally used for local broadcasting.
- In **space wave propagation**, the radio wave from the transmitting antenna reach the receiving antenna either directly or after reflection from the ground or from troposphere. The space wave propagation utilises the electromagnetic waves of very high frequency band (between 30 MHz to 300 MHz), ultra high frequency band (between 300 MHz to 3000 MHz) and microwaves having frequencies range 100 GHz to 300 GHz.
- In **sky wave propagation** the radio waves from the transmitting antenna reach the receiving antenna after entering in ionosphere. The sky wave frequency lies between 2 MHz to 30 MHz.
- A communication satellite is a space craft, provided with microwave receiver and transmitter.
  - It is placed in an orbit around the earth.
  - The frequencies used in satellite communication lies in UHF/ microwave regions.
  - These waves can cross the ionosphere and reach the satellite.
- The satellite communication is used effectively in mobile communication, metrology and weather forecasting.
- Remote sensing is the technique to obtain information about an object by observing it from a distance and without coming to actual contact with it. It is used in development of weather forecasting system.
- Parallel wire line is used to connect an antenna with TV receiver. Parallel wire lines are never used for transmission of microwaves.
- Twisted pair wires are used to connect telephone system.
- Coaxial cables are used for long distance high frequency transmission. e.g. TV channel transmission.

➨ **Note** Maser is a device working on the same principle as a laser. Maser means 'Microwave Amplification by Stimulated Emission of Radiation, It is used to produce very high magnetic field and in radio astronomy and communications.

Optical Communication

It is a better communication system for large number of channels and higher bandwidth transmission. Now-a-days optical fibres of very small diameters ($\approx 10^{-5}$ m) are used in optical communication. Advantages of optical communication over the conventional two wires or coaxial cable systems are

(*i*) In optical fibres energy losses are less.
(*ii*) Secret information like banking, defence, etc., is more secure because optical signal is confined to the inside of fibre and cannot be tampered easily.
(*iii*) It can be used at high frequencies ($\approx 10^{14}$ Hz).

Phenomenon of total internal reflection is the principle behind optical fibres. Light can be transmitted along it with almost no loss because of total internal reflection. Important applications of optical fibres are in telecommunications and medicine.

PRACTICE EXERCISE

Measurement, Motion, Work, Energy and Power

1. A train travels a distance with a speed of 30 km/h and returns with a speed of 50 km/h. The average speed of the train is
(a) 36 km/h (b) 37.5 km/h
(c) 38 km/h (d) 40 km/h

2. Average velocity of an object is equal to the mean of its initial and final velocities, if the acceleration is
(a) uniform (b) variable
(c) Both a and b (d) None of these

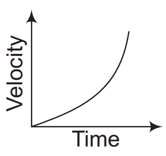
3. A water tank filled upto $\frac{2}{3}$ of its height is moving with a uniform speed. On sudden application of the brake, the water in the tank would
(a) move backward (b) move forward
(c) be unaffected (d) rise upwards

4. When the distance, an object travels is directly proportional to the length of time, it is said to travel with
(a) zero velocity (b) uniform velocity
(c) constant velocity
(d) constant acceleration

5. If the displacement of an object is proportional to square of time, then the object moves with
(a) uniform velocity
(b) uniform acceleration
(c) increasing acceleration
(d) decreasing acceleration

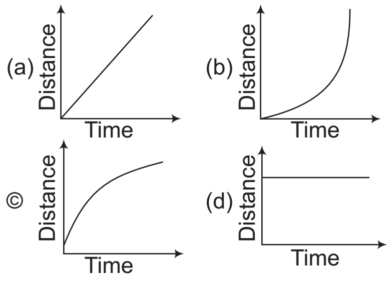
6. A passenger in a moving train tosses a five rupee coin. If the coin falls behind him, the train must be moving with a uniform
(a) acceleration
(b) deceleration
(c) speed
(d) velocity

7. The velocity-time graph for an object is shown below. Which one of the following statements helds true for this object?



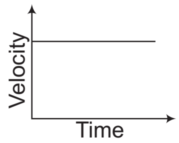
(a) The object is moving with uniform speed
(b) The object is in non-uniform accelerated
(c) The object is at rest
(d) The object is in non-uniform decelerated.

8. Which of the following figure represents uniform motion of a moving object correctly?



9. A body falls from rest. The velocity acquired by falling a distance $2h$ is
(a) $2\sqrt{gh}$ (b) $\sqrt{gh}$
(c) $\sqrt{2gh}$ (d) None of these

10. The velocity-time graph for an object is shown. Which one of the following statements holds true for this object?



(a) The object has uniform velocity
(b) The object has uniform acceleration
(c) The object has non-uniform acceleration
(d) None of the above

11. A boy throws four stones of same shape, size and weight with equal speed at different initial angles with the horizontal line. If the angles are 15°, 30°, 45° and 60°, at which angle the stone will cover the maximum distance horizontally?
(a) 15° (b) 30° (c) 45° (d) 60°

12. Which one of the following is a vector quantity?
(a) Force (b) Area
(c) Volume (d) Density

13. External forces acting on an object
(a) are always balanced
(b) are always unbalanced
(c) may or may not be balanced
(d) None of the above

14. When horse starts running all of a sudden, the rider on the horse falls back because
(a) he is afraid
(b) he is taken a back
(c) due to inertia of motion, the lower part of his body comes in motion
(d) due to inertia of rest, the upper part of his body remains at rest

15. A scooter driven over an oily road generally slips because on the oily road
(a) inertia of the scooter tyres increases
(b) inertia of the scooter tyres decreases
(c) friction between tyres and road increases
(d) friction between tyres and road decreases

16. Newton's second law of motion connects
(a) momentum and acceleration
(b) change of momentum and velocity
(c) rate of change of momentum and external force
(d) rate of change of force and momentum

17. When a ball drops onto the floor it bounces. Why does it bounce?
(a) Newton's third law implies that for every action (drop) there is a reaction (bounce)
(b) The floor exerts a force on the ball during the impact
(c) The floor is perfectly rigid
(d) The floor heats up on impact

18. If an object having mass of 1 kg is subjected to a force of 1 N, it moves with
(a) a speed of 1 m/s
(b) a speed of 1 km/s
(c) an acceleration of 10 m/s^2
(d) an acceleration of 1 m/s^2

19. Momentum has the same unit as that of
(a) couple
(b) torque
(c) impulse
(d) moment of momentum

20. Rocket works on the principle of conservation of
(a) mass (b) energy
(c) momentum (d) None of these

21. Two balls of different masses have the same kinetic energy. The ball having greater momentum will be
(a) lighter one
(b) heavier one
(c) both having equal masses
(d) cannot say

22. A goalkeeper in a game of football pulls his hands backwards after holding the ball shot at the goal. This enables the goalkeeper to
(a) exert larger force on the ball
(b) reduce the force exerted by the ball on hands
(c) increase the rate of change of momentum
(d) decrease the rate of change of momentum

23. A cricket player moves his hands along the direction of the moving ball while catching it in order to
(a) kill the googly of the ball
(b) conserve energy
(c) conserve mass
(d) reduce the force of impulse

24. Consider the following statements
1. The impulse given to a body is equal to the change in the momentum of the body.
2. The unit of the impulse is the same as that of the momentum.
3. The product of the force and velocity is called the impulse.
4. Impulse is a vector quantity.

Which of the following statements is/are correct?
(a) 1, 2 and 4 (b) Only 4
(c) Only 3 (d) 1, 2 and 3

25. It is easier to roll a barrel along the road than to pull it because
(a) rolling involves only one point of contact between the barrel and the earth
(b) rolling friction is considerably less than the sliding friction
(c) the full weight of the barrel does not become effective while rolling
(d) None of the above

26. It is difficult to walk on ice than on road because
(a) ice is harder than road
(b) road is harder than ice
(c) ice does not offer any reaction, when we push it, with our foot
(d) ice has a lesser friction than road

27. Frictional force
1. is self-adjusting force.
2. is a non-conservative force.
3. is a necessary evil.

Codes
(a) 1 and 2 (b) 1 and 3
(c) Only 1 (d) All of these

28. Consider the following statements
1. Friction force is conservative force.
2. Magnetic force and electrostatic force are non-conservative force.

Which of the following sentences is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

29. The chance of a vehicle to overturn while negotiating a circular path depends on
(a) speed of the vehicle only
(b) radius of the circular path only
(c) speed and height of the vehicle and radius of the circular path
(d) height of the vehicle only

30. Fat can be separated from milk in a cream separation because of
(a) cohesive force
(b) gravitational force
(c) centrifugal force
(d) centripetal force

31. Consider the following statements
1. If a body is rotating in a circular path, then after one rotation its distance will be zero.
2. distance and displacement are equal in straight line.
3. The speedometer of the car shows its instantaneous speed.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 2 and 3 (d) 1 and 3

32. Match the following columns.

Column I	Column II
A. We hit a carpet with a stick to remove the dust	1. Law of motion II
B. China wires are wrapped in straw or paper	2. Law of motion I
C. A jet plane moves	3. Law of motion III
D. Bogies of the train are provided with buffers	4. Impulse

Codes

	A	B	C	D		A	B	C	D
(a)	3	4	1	2	(b)	2	1	3	4
(c)	1	2	3	4	(d)	3	4	2	1

33. Match the following columns.

Column I (Situation)	Column II (Centripetal force)
A. Revolution of the earth around the sun	1. Tension in the string
B. Vehicle taking a turn on level road	2. Friction force exerted by the road on the tyres
C. A vechicle on a speed braker	3. Weight of the body
D. A particle tied to a string and whirled in a horizonal circle	4. Gravitation force exerted by the sun

Codes

	A	B	C	D		A	B	C	D
(a)	4	2	3	1	(b)	2	3	4	1
(c)	4	3	2	1	(d)	1	2	3	4

34. When work is done on the body
(a) it gains energy
(b) it loses energy
(c) its energy remains constant
(d) None of the above

35. **Statement I** When the force retards the motion of a body, the work done is zero.
Statement II Work done depends on angle between force and displacement.

Codes
(a) Both the statements are correct and statement II is the correct explanation of statement I
(b) Both the statements are correct, but statement II is not the correct explanation of statement I
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

36. An object of mass 2 kg is lifted vertically through a distance of 1.5 m. The work done in process is
(a) 29.4 J (b) 19.4 J
(c) 17.4 J (d) 20.4 J

37. The work done on an object does not depends upon the
(a) displacement (b) force applied
(c) final velocity (d) initial velocity

38. Consider the following statements
1. When a body falls freely under gravity the work done by gravity is positive.
2. If a force of 1 dyne produces a displacement of 1 cm in the direction of the force, the work done is called 1 erg.
3. Work is scalar quantity.
4. Work done is always positive.

Which of the following statements is/are correct?
(a) 1, 2 and 4 (b) 2 and 3
(c) Only 4 (d) 1, 2 and 3

39. If the velocity of a particle is reduced to half of its initial value, then the kinetic energy of the particle will
(a) get doubled
(b) become four times
(c) reduce to half of its original value
(d) reduce to one-fourth of its original value

40. A horse and a dog are running with the same speed. Then, the ratio of their kinetic energies, if the weight of the horse is ten times that of dog, is
(a) 10 : 1 (b) 100 : 1
(c) 1 : 9 (d) 9.8 : 98

41. If acceleration due to gravity is 10 m/s^2, then the potential energy of a body of mass 1 kg kept at a height of 5 m, is
(a) 50 J (b) 100 J (c) 500 J (d) 50 KJ

42. When a body falls freely
(a) its PE is converted into KE
(b) its KE is converted into PE
(c) its mechanical energy is converted into heat energy
(d) None of the above

43. Consider the following statements
1. When a body falls its potential energy decrease but at the same time kinetic energy increases by an equal amount.
2. The total amount of energy in the universe remains constant.

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) None of these

44. If acceleration due to gravity is 9.8 m/s^2. A bag of wheat weighs, 200 kg. The height to which it should be raised, so that its potential energy may be 9800 J, is
(a) 5 m (b) 4 m (c) 3 m (d) 10 m

45. Two bodies of equal mass move with the velocity of $2v$ and $3v$, respectively. Then, the ratio of their kinetic energies is
(a) 4 : 9 (b) 2 : 3 (c) 1 : 2 (d) 8 : 11

46. Power of the moving body is stored in the form of
(a) work and time
(b) force and distance
(c) force and velocity
(d) distance and acceleration

Previous Years' Questions

47. **Statement I** Pulling a lawn roller is easier than pushing it.
Statement II Pushing increases the apparent weight and hence the force of friction. **2012 II**
(a) Both the statements are correct and statement II is the correct explanation of statement I
(b) Both the statements are correct, but statement II is not the correct explanation of statement I
(c) Statement I is correct, but statement II is incorrect
(d) Statement I is incorrect, but statement II is correct

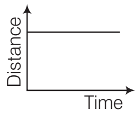
48. A bus travels at a speed of 50 km/h to go from its origin of its destination at a distance of 300 km and travels at a speed of 60 km/h to return to the origin. What is the average speed of the bus? **2012 II**
(a) 54.55 km/h (b) 55 km/h
(c) 55.55 km/h (d) 54 km/h

49. Magnetic, electrostatic and gravitational forces came under the category of **2013 II**
(a) non-contact forces
(b) contact forces
(c) frictional forces
(d) non-frictional forces

50. You are asked to jog in a circular track of radius 35 m. Right one complete round on the circular track, your displacement and distance covered by you respectively **2014 I**
(a) zero and 220 m (b) 220 m and zero
(c) zero and 110 m (d) 110 m and 220 m

51. In cricket match, while catching a fast moving ball, a fielder in the ground gradually pulls his hands backwards with the moving ball to reduce the velocity to zero. The act represents **2014 I**
(a) Newton's first law of motion
(b) Newton's second law of motion
(c) Newton's third law of motion
(d) Law of conservation of energy

52. The distance-time graph for an object is shown below. Which one of the following statements holds true for this object? **2014 II**



(a) The object is moving with uniform speed
(b) The object is at rest
(c) The object is having non-linear motion
(d) The object is moving with non-uniform speed

53. An electron and a proton are circulating with same speed in circular paths of equal radius. Which one among the following will happen, if the mass of a proton is about 2,000 times that of an electron? **2015 I**
(a) The centripetal force required by the electron is about 2,000 times more than that required by the proton
(b) The centripetal force required by the proton is about 2,000 times more than that required by the electron
(c) No centripetal force is required for any charged particle
(d) Equal centripetal force acts on both the particles as they rotate in the same circular path

54. An object is raised to a height of 3 m from the ground. It is then allowed to fall on to a table 1 m high from ground level. In this context, which one among the following statements is correct? **2015 I**
(a) Its potential energy decreases by two-thirds its original value of total energy
(b) Its potential energy decreases by one-third its original value of total energy
(c) Its kinetic energy increases by two-third, while potential energy increases by one-third
(d) Its kinetic energy increases by one-third, while potential energy decreases by one-third

55. **Statement I** When a gun is fired it recoils i.e. it pushes back, with much less velocity than the velocity of the bullet.
Statement II Velocity of the recoiling gun is less because the gun is much heavier than the bullet. **2015 I**
(a) Both the statements are correct and statement II is the correct explanation of statement I
(b) Both the statements are correct, but statement II is not the correct explanation of statement I
(c) Statement I is correct but statement II is incorrect
(d) Statement I is incorrect but statement II is correct

56. A body is falling freely under the action of gravity alone in vacuum. Which one of the following remains constant during the fall? **2015 II**
(a) Potential energy
(b) Kinetic energy
(c) Total linear momentum
(d) Total mechanical energy

57. Newton's laws of motion do not hold good for objects **2015 II**
(a) at rest
(b) moving slowly
(c) moving with high velocity
(d) moving with velocity comparable to velocity of light

58. A brick is thrown vertically from an aircraft flying 2 km above the Earth. The brick will fall with a **2015 II**
(a) constant speed
(b) constant velocity
(c) constant acceleration
(d) constant speed for sometime, then with constant acceleration as it nears the Earth

59. The rate of change of momentum of a body is equal to the resultant **2016 I**
(a) energy (b) power
(c) force (d) impulse

60. The SI unit of mechanical energy is **2016 I**
(a) Joule (b) Watt
(c) Newton-second (d) Joule-second

Rotational Motion and Gravitation

61. The centre of mass of a system
(a) is always at its geometric centre
(b) is always anywhere inside it
(c) is always outside it
(d) may be inside or outside it

62. **Assertion** (A) To unscrew a rusted nut, we need a wrench with longer arm.
Reason (R) Wrench with longer arm reduces the torque of the arm.
(a) Both A and R are correct and R is the correct explanation of the A
(b) Both A and R are correct but R is not the correct explanation of the A
(c) A is correct but R is incorrect
(d) A is incorrect but R is correct

63. Consider the following statements
1. Torque is equal to the product of moment of inertia and angular acceleration.
2. Torque is a vector quantity.
3. The rate of change of angular acceleration is equal to the torque.
4. The turning effect of a body is known as torque.

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 4
(c) 3 and 4 (d) 1, 2 and 4

64. Consider the following quantities.
I. Angular momentum
II. Kinetic energy

Which of the quantities given above will remain constant for a satellite in orbit?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

65. A boy suddenly comes and sits on a circular rotating table. What will remain conserved?
(a) Angular momentum
(b) Potential energy
(c) Linear momentum
(d) Kinetic energy

66. A man is sitting on a revolving stool with his arm outstretched. When he suddenly pulls his arms inside, then
(a) his angular velocity decreases
(b) his moment of inertia decreases
(c) his angular velocity remains constant
(d) his angular momentum increases

67. The cat can survive fall from a height much more than human or any other animal. It is because the cat
(a) can immediately adjust itself to land on all four paws and bend the legs to absorb the impact of falling
(b) has elastic bones
(c) has thick and elastic skin
(d) also gets injury equally with other animals, but has tremendous endurance, resistance and speedy recovery

68. An athlete diving off high springboard can perform a variety of exercise in tube air before entering the water body, which one of the following parameters will remain constant during the fall?
(a) The athlete's linear momentum
(b) The athlete's angular momentum
(c) The athlete's kinetic engine
(d) The athlete's moment of inertia

69. A sphere of mass 40 kg is attracted by a second sphere of mass 60 kg with a force equal to 4×10^{-5} N. If G is 6×10^{-11} N-m^2/k^2, what is the distance between them?
(a) 7 cm (b) 6 cm (c) 10 cm (d) 12 cm

70. If suddenly the gravitational force of attraction between the earth and a satellite revolving around it becomes zero, then the satellite will
(a) continue to move in this orbit with same velocity
(b) move tangentially to the original orbit in the same velocity
(c) became stationary in its orbit
(d) move towards the earth

71. The force of gravitation between two bodies does not depend on
(a) their separation
(b) the product of their masses
(c) the sum of their masses
(d) the gravitational constant

72. If the distance between two objects is doubled, the force of attraction between them will
(a) become $\frac{1}{4}$ times (b) become 2 times
(c) become $\frac{1}{8}$ times (d) remains same

73. Consider the following statements
1. Gravitational force is always attractive in nature.
2. Gravitational force is a central force.
3. Tides are formed in oceans due to gravitational attraction between moon and ocean water.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) All of these

74. The acceleration due to gravity
(a) has the same value everywhere in space
(b) has the same value everywhere on the earth
(c) varies with the latitude on the earth
(d) is greater on the moon due to its smaller diameter

75. Two objects of different masses falling freely near the surface of the moon would
(a) have same velocities at any instant
(b) have different accelerations
(c) experience forces of same magnitude
(d) undergo a change in their media

76. An apple falls from a tree because of gravitational attraction between the earth and apple. If F_1 is the magnitude of force exerted by the earth on the apple and F_2 is the magnitude of force exerted by apple on the earth, then
(a) F_1 is very much greater than F_2
(b) F_2 is very much greater than F_1
(c) F_1 is only a little greater than F_2
(d) F_1 and F_2 are equal

77. When an object is thrown up, the force of gravity
(a) is opposite to the direction of motion
(b) is in the same direction as the direction of motion
(c) decreases as it rises up
(d) increases as it rises up

78. **Assertion** (A) Moon's gravity is equivalent to one-eighth of the earth's gravity.
Reason (R) Due to less gravity, moon is unable to retain its atmosphere.
(a) Both A and R are correct and R is the correct explanation of A
(b) Both A and R are correct, but R is not the correct explanation of A
(c) A is correct, but R is incorrect
(d) A is incorrect, but R is correct

79. A body has a mass of 6 kg on the earth, when measured on the moon, its mass would be
(a) nearly 1 kg (b) less than 1 kg
(c) less than 6 kg (d) 6 kg

80. Match the following columns.

Column I	Column II
A. g is maximum	1. At the moon
B. g is minimum	2. At pole
C. g is zero	3. At the equator
D. g is 1/6 of original value	4. At centre of the earth

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	2	3	4	1
(c)	4	3	2	1	(d)	4	2	3	1

81. Consider the following statements
1. Mass of a body is not constant it changes from place to place.
2. Mass is measured by an ordinary equal arm balance.
3. At the equator, the mass of a body is maximum.

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 2
(c) 1 and 3 (d) All of these

82. A line drawn from the sun to a planet, moving around it, sweeps over a fixed area in a given interval of time. This is according to
(a) Kepler's law (b) Ohm's law
(c) Lenz's law (d) Newton's law

83. The earth revolves around the sun in an elliptical orbit, its speed
(a) is greatest when it is farthest from the sun
(b) is greatest when it is closest of the sun
(c) remains the same at all points on the orbit
(d) goes on decreasing continuously

84. The square of time period of revolution of a planet round the sun is
(a) directly proportional to its average distance from the sun
(b) directly proportional to the square of its average distance from the sun
(c) directly proportional to the cube of its average distance from the sun
(d) inversely proportional to its average distance from the sun

85. Consider the following statements
1. Time period of geo-stationary satellite is 24 h.
2. Geo- stationary satellite revolves in equatorial plane in the direction of the earth's rotation i.e. from West to East
3. Geo-stationary satellite revolves around the earth at 800 km.
4. INSAT 2B is a polar satellite.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 2, 3 and 4

86. Two satellites are revolving in the same circular orbit, their
1. masses are same
2. velocities are same

Codes
(a) 1 and 2
(b) Only 2
(c) Only 1
(d) None of the above

87. A communication relay satellite is able to telecast programmes continuously from one part of the world to another, because
(a) its period of revolution is greater than the period of rotation of the earth on its axis
(b) its period of revolution is equal to the period of rotation of the earth on its axis
(c) its period of revolution is less than the period of rotation of the earth on its axis
(d) its mass is less than the mass of the earth

88. A missile is launched with a velocity less than the escape velocity. The sum of its kinetic and potential energy is
(a) positive (b) negative
(c) zero
(d) may be positive or negative depending upon its initial velocity

89. The escape velocity of a projectile from the surface of the earth is approximately
(a) 11.3 km/s (b) 11.2 km/s
(c) 11.5 km/s (d) 11.4 km/s

90. The velocity with which a projectile must be fired so that it escapes the earth's gravitation does not depend on
(a) mass of the earth
(b) mass of the projectile
(c) radius of the projectile's orbit
(d) gravitational constant

91. The escape velocity of a body on the surface of the earth is 11.2 km/s. If the earth's mass increases to twice its present value and the radius of the earth become half the escape velocity would become
(a) 5.6 km/s
(b) 11.2 km/s (remain unchanged)
(c) 22.4 km/s (d) 44.8 km/s

92. The escape velocity at the moon
1. is 2.38 km/s
2. is less than the average velocity of the gas molecules at the temperature of the moon

Codes
(a) Only 1 (b) Only 2
(c) 1 and 2
(d) None of the above

93. Match the following columns.

Column I (Quantity)	Column II (Definition)
A. Mass	1. Attractive force only
B. Gravitational force	2. The attractive force exerted by the earth
C. Escape velocity	3. It is a quantity of matter
D. Weight	4. Body must be projected up

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	3	1	4	2
(c)	1	2	3	4	(d)	2	1	3	4

94. Satellite having the same orbital period as the period of rotation of the earth about its own axis is known as
(a) polar satellite
(b) stationary satellite
(c) geostationary satellite
(d) INSAT

95. A typical black hole is always specified by
(a) a (curvature) singularity
(b) a horizon
(c) either a (curvature) singularity or a horizon
(d) a charge

96. The motion of the wheel of a cycle is
(a) rectilinear
(b) translatory and rotatory
(c) rotatory
(d) translatory

Previous Years' Questions

97. In respect of the difference of the gravitational force from electric and magnetic forces, which one of the following statements is true? **2015 I**
(a) Gravitational force is stronger than the other two
(b) Gravitational force is attractive only, whereas the electric and the magnetic forces are attractive as well as repulsive
(c) Gravitational force has a very short range
(d) Gravitational force is a long range force, while the other two are short range forces

98. A brick is thrown vertically from an aircraft flying 2 km above the earth. The brick will fall with a **2015 II**
(a) constant speed
(b) constant velocity
(c) constant acceleration
(d) constant speed for sometime, then with constant acceleration as it nears the Earth

99. Which one of the following statements is not correct? **2015 II**
(a) Weight of a body is different on different planets
(b) Mass of a body on the earth, on the moon and in empty space is the same
(c) Weightlessness of a body occurs when the gravitational forces acting on it is counter-balanced
(d) Weight and mass of a body are equal at sea level on the surface of the earth

Properties of Matter

100. What is the value of stress in a wire of steel having radius of 2 mm if 10 KN of force is applied on it?
(a) 7.96×10^8 N/m^2
(b) 8.00×10^7 N/m^2
(c) 10×10^7 N/m^2
(d) 10×10^8 N/m^2

101. Why the spring is made up of steel in comparison of copper?
(a) Copper is more costly than steel
(b) Copper is more elastic than steel
(c) Steel is more elastic than copper
(d) None of the above

102. Match the following columns.

Column I (Property)	Column II (Example)
A. Elastic	1. Iron
B. Plastic	2. Steel
C. Ductile	3. Quartz
D. Perfectly elastic	4. Paraffin wax

Codes

	A	B	C	D		A	B	C	D
(a)	2	4	1	3	(b)	1	2	3	4
(c)	4	3	2	1	(d)	3	4	2	1

103. Consider the following statements
1. The internal restoring force acting per unit area of cross-section of the deformed body is called stress.
2. The ratio of stress to strain is a constant for a material and is called modulus of elasticity
3. Air is more elastic than water.

Which of the following statements is/are correct?
(a) 1 and 3 (b) 2 and 3
(c) 1 and 2 (d) Only 3

104. A 50 kg girl wearing high heal shoes balances on a single heal. The heal is circular with a diameter 1.0 cm. The pressure exerted by the heal on the horizontal floor is (g = 9.8 N) kg
(a) 6.2×10^6 N / m^2
(b) 2.6×10^5 N / m^2
(c) 6.2×10^8 N / m^2
(d) 2.6×10^6 N / m^2

105. The pressure exerted on the ground by a man is greatest
(a) when he lies down on the ground
(b) when he stands on the toes of one foot
(c) when he stands with both foot flat on the ground
(d) All of the above yield the same pressure

106. Consider the following statements
1. Pressure is measured by barometer.
2. Atmospheric pressure is measured by manometer.
3. Atmospheric pressure decrease with altitude.
4. Atmospheric pressure is 700 torr.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) Only 3 (d) 1 and 4

107. The relative density of silver is 10.8. If the density of water be 1.0×10^3 kg/m^3 the density of silver will be
(a) 10.8×10^3 kg/m^3 (b) 9×10^2 kg/m^3
(c) 8×10^3 kg/m^3 (d) 5×10^2 kg/m^3

108. The apparent weight of a steel sphere immersed in various liquids is measured using a spring balance. The greatest reading is obtained for the liquid
(a) having the smaller density
(b) having the largest density
(c) in which the sphere was submerged deepest
(d) having the greatest volume

109. Hydraulic brakes in automatic vehicles are based on
(a) Archimedes' principle
(b) Toricellis equation
(c) Pascal's law
(d) Bernoulli's equation

110. If a ship moves from fresh water into sea water, it will
(a) sink completely
(b) sink a little bit
(c) rise a little higher
(d) remain unaffected

111. Archimedes' principle is used
1. determining the relative density of a substance.
2. in designing ships and submarines
3. measuring a terminal velocity in air.

Which of the following statements is/are correct?
(a) 1 and 3 (b) 2 and 3
(c) 1 and 2 (d) Only 2

112. A needle made of steel floats in water, if it is placed horizontally because of
(a) capillary action
(b) interference
(c) viscosity
(d) surface tension

113. If the temperature of a liquid is raised, then its surface tension is
(a) decreased (b) increased
(c) irregular (d) equal to viscosity

114. More liquid rises in a thin tube because of
(a) larger value of radius
(b) larger value of surface tension
(c) small value of surface tension
(d) small value of radius

115. Coatings used on raincoat are waterproof because
(a) water is absorbed by the coating
(b) cohesive force becomes greater
(c) water is not scattered away by the coating
(d) angle of contact decrease

116. Small droplets of a liquid are spherical in shape because of
(a) viscosity (b) surface tension
(c) density (d) conductivity

117. A U shaped wire is dipped in a soap solution and removed. The thin soap film formed between the wire and a light slider supports a weight of 1.5×10^{-2}N. The length of the slider is 30 m. What is the surface tension of the film?
(a) 0.025 N/m (b) 0.052 N/m
(c) 25 N/m (d) None of these

118. Due to surface tension
1. Dancing of camphor on water.
2. Floatation of needle on water.
Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) Both are wrong

119. Cohesive force is experienced between
(a) magnetic substance
(b) molecules of different substances
(c) molecules of same substances
(d) None of the above

120. Which one of the following statements is correct?
(a) The angle of contact of water with glass is acute while that of mercury with glass is obtuse
(b) The angle of contact of water with glass is obtuse while that of mercury with glass is acute
(c) Both the angle of contact of water with glass and that of mercury with glass are acute
(d) None of the above

121. In an oil lamp, the oil rises up in the wick because
(a) oil is volatile (b) oil is very light
(c) of the capillary action
(d) of the surface tension

122. The rise of a liquid in a capillary tube is due to
(a) diffusion (b) osmosis
(c) surface tension (d) viscosity

123. The action of a nib spilt at the top is explained by
(a) gravity flow (b) diffusion of fluid
(c) capillary action (d) osmosis of liquid

124. Choose the wrong statement among the followings.
(a) Surface tension is a molecular phenomenon
(b) Tiny drops of a liquid are spherical due to spherical tension
(c) Oil rises in the wick due to capillarity
(d) Cold drink rises in the straw due to capillarity

125. Due to capillary action
1. the oil in the wick of the lamp rises.
2. refill pen writes
Which of the following statements is/are correct?
(a) Only 1
(b) Only 2
(c) 1 and 2
(d) None of the above

126. Machine parts get jammed in winter, because
(a) the viscosity of the lubricant used in machine parts increases with the decrease in temperature
(b) the viscosity of lubricant used in machine parts decreases with the decrease in temperature
(c) the inertia of machine parts increases with the decrease in temperature
(d) the inertia of machine parts increases with the decrease in temperature

127. After rising a short distance the smooth column of smoke from a cigarette breaks up into an irregular and random pattern. In a similar fashion, a stream of fluid flowing past an obstacle breaks up into eddies and vortices which give the flow irregular velocity components transverse to the flow direction. Other examples include the wakes left in water by moving ships the sound produced by whistling and by wind instruments. These examples are the results of
(a) laminar flow of air
(b) streamline flow of air
(c) turbulent flow of air
(d) viscous flow at low speed

128. Consider the following statements.
1. The property of the liquid by virtue of which it opposes the relative motion between its adjacent layers is known as viscosity.
2. Viscosity of liquid increases with increase in pressure.
3. Viscosity of a fluid is measurement by its coefficient of viscosity.
4. Viscosity is not property of gases.
Which of the following statements is/are correct?
(a) 1, 2 and 3 (b) 3 and 4
(c) 2 and 4 (d) Only 4

129. Hair of a shaving brush cling together when the brush is removed from water due to
(a) viscosity (b) surface tension
(c) friction (d) elasticity

130. Match the following columns.

Column I	Column II
A. Hook's law	1. Lactometers
B. Pascal Law	2. Elasticity
C. Archimedes' Principle	3. Hydraulic press
D. Stokes' law	4. Terminal velocity

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	1	4	(b)	4	3	2	1
(c)	3	2	1	4	(d)	1	2	3	4

Previous Years' Questions

131. A liquid is kept in a regular cylindrical vessel upto a certain height. If this vessel is replaced by another cylindrical vessel having half the area of cross-section of the bottom, the pressure on the bottom will **2012 I**
(a) remain unaffected
(b) be reduced to half the earlier pressure
(c) be increased to twice the earlier pressure
(d) be reduced to one-fourth the earlier pressure

132. It is difficult to cut things with a blunt knife because **2012 II**
(a) the pressure exerted by knife for a given force increases with increase in bluntness
(b) a sharp edge decreases the pressure exerted by knife for a given force

(c) a blunt knife decreases the pressure for a given force
(d) a blunt knife decreases the area of intersection

133. When deep sea fishes are brought to the surface of the sea, their bodies burst. This is because the blood in their bodies flows at very **2012 II**
(a) high speed
(b) high pressure
(c) low speed
(d) low pressure

134. Which type/types of pen uses/use capillary action in addition to gravity for flow of ink? **2013 I**
(a) Fountain pen
(b) Ballpoint pen
(c) Gel pen
(d) Both ballpoint and gel pens

135. After rising a short distance the smooth column of smoke from a cigarette breaks up into an irregular and random pattern. In a similar fashion, a stream of fluid flowing fast an obstacle breaks up into eddies and vortices which give the flow irregular velocity components transverse to the flow direction. Other examples include the wakes left in water by moving ships the sound produced by whistling and by wind instruments. These examples are the results of **2013 II**
(a) laminar flow of air
(b) streamline flow of air
(c) turbulent flow of air
(d) viscous flow at low speed

136. Dirty cloths containing grease and oil stains are cleaned by adding detergents to water. Stains are removed because detergent **2013 II**
(a) reduces drastically the surface tension between water and oil
(b) increases the surface tension between water and oil
(c) increases the viscosity of water and oil
(d) decreases the viscosity of detergent mixed water

Heat

137. The value of – 30°C on Fahrenheit scale is
(a) –54° F (b) 32° F
(c) 22° F (d) –22° F

138. Consider the following statements
1. Fahrenheit is the smallest unit measuring temperature.
2. Fahrenheit was the first temperature scale used for measuring temperature.

Which of the following statements is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) None of the above

139. The temperature for which the reading on Celsius and Fahrenheit scales are identical, is
(a) – 40° C (b) 40° C
(c) – 30° C (d) 30° C

140. Temperature of the body measures
(a) hotness of the body
(b) amount of heat contained in the body
(c) degree of hotness of the body
(d) the average energy of molecules of the object

141. The instrument used to mark the highest fixed point of a thermometer is
(a) pyrometer (b) hydrometer
(c) spherometer (d) calorimeter

142. P and Q are two objects. The temperature of P is greater than that of Q. This means that
(a) the heat content of P will always be greater than that of Q
(b) the average potential energy of P is greater than the average potential energy of Q
(c) the total energy of P is greater than the total energy of the molecules of Q
(d) the molecules of P are moving faster on the average than the molecules of Q

143. Which one of the following statements is true?
(a) Temperatures differing 25° on the Fahrenheit (F) scale must differ by 45° on the celsius (C) scale
(b) 0°F corresponds to – 32° C
(c) Temperatures which differ by 10° on the celsius scale must differ by 18° on the Fahrenheit scale
(d) Water at 90°C is warmer than water at 202°F.

144. Consider the following statements
1. Mercury is used in clinical thermometer for measuring body temperature.
2. Mercury shines and is easily observable.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

145. Specific heat of a substance depends upon
(a) mass of the substance
(b) volume of the substance
(c) shape of the body
(d) nature of the substance

146. Vapour pressure of liquid depends upon the
(a) quantity of heat
(b) temperature of liquid
(c) specific heat of liquid
(d) density of liquid

147. Two bodies A and B are of same mass and same amount of heat is given to both of them. If the temperature of A decreases more than that of B because of heat addition, then
(a) the specific heat capacity of A is more than that of B
(b) the specific heat capacity of A is less than that of B
(c) both A and B have the same specific heat capacity but A has greater thermal conductivity
(d) both A and B have the same specific heat capacity but B has greater thermal conductivity

148. Consider the following statements
1. Specific heat is amount of heat required to raise the temperature of a unit of the substance by 1°C.
2. The specific heat of water is maximum
3. Gases have no specific heat.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 2 and 1 (d) 1 and 3

149. Consider the following statements
1. Cooking utensils have low specific heat and high conductivity.
2. Specific heat is the amount of heat required to raise the temperature of unit mass of mass of the substance by 1°C

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) None of these

150. Real expansion of a liquid is always
(a) greater than apparent expansion
(b) less than apparent expansion
(c) equal to apparent expansion
(d) None of the above

151. Match the following columns.

Column I	Column II
A. Bimetallic strip	1. Radiation from a hot body
B. Steam engine	2. Energy conversion
C. Incondescent lamp	3. Melting
D. Electric fuse	4. Thermal expansion of solids

Codes

	A	B	C	D		A	B	C	D
(a)	4	2	1	3	(b)	3	2	1	4
(c)	4	3	2	1	(d)	2	1	3	4

152. Match the following columns.

Column I	Column II
A. Freezing point of mercury	1. 0°C
B. Freezing point of ice	2. – 115°C
C. Freezing point of alcohol	3. – 39°C

Codes

	A	B	C		A	B	C
(a)	1	2	3	(b)	2	3	1
(c)	3	1	2	(d)	1	3	2

153. Latent heat of fusion of any body
(a) is independent of pressure
(b) is dependent of volume
(c) is always same
(d) varies under different conditions

154. Water is used in hot water bottles in preference to any other liquid because
(a) water has greatest latent heat
(b) water is good conductor of heat
(c) water has greatest specific heat
(d) water has high boiling point

155. Consider the following statements
1. Steam is more harmful for human body than the boiling water in case of burn.
2. Boiling water contains more heat than steam.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

156. Factors affecting evaporation of a liquid are
1. the nature of the liquid
2. temperature of surrounding
3. humidity of surrounding air
4. temperature of liquid

Which of the following statements is/are correct?
(a) Only 2
(b) Only 3
(c) 1, 2 and 3
(d) All of the above

157. Consider the following statements
1. Water kept in an open vessel will quickly evaporate on the surface of the moon.
2. The temperature at the surface of the moon is much higher than the boiling point of the water.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

158. Heat energy of an object is
(a) the average energy of the molecules of the object
(b) the total energy of the molecules of the object
(c) the average velocity of the molecules of the object
(d) the average potential energy of the molecules of the object

159. A glowing electric bulb becomes hot after some time because the heat from filament is transmitted to the glass bulb by
(a) conduction (b) convection
(c) radiation (d) another process

160. Diesel engine
1. works with a oil plug.
2. is associated with the risk of explosion.
3. has larger efficiency (58%)

Which of the following statements is/are correct?
(a) Only 1
(b) 2 and 3
(c) Only 2
(d) All of the above

161. If there were no atmosphere, the average temperature on the surface of the earth would be
(a) higher (b) lower
(c) same as today (d) 0° C

162. Transfer of heat energy from the sun to the moon takes place by
(a) radiation only
(b) radiation and conduction
(c) radiation and convection
(d) radiation, conduction and convection

163. Material used for making cooking utensils should have
(a) low specific heat and high thermal conductivity
(b) high specific heat and high thermal conductivity
(c) low specific heat and low thermal conductivity
(d) high specific heat and low thermal conductivity

164. The wet clothes dry more quickly on a warm day because
(a) air carries less moisture
(b) air can absorb moisture rather quickly
(c) high temperature of the air helps fast evaporation
(d) All of the above

165. A glass chimney stops an oil lamp from smoking because
(a) it increases the supply of oxygen to the flame by connection
(b) the heat produced ensures complete combustion of carbon particles
(c) Both 'a' and 'b'
(d) None of the above

166. Consider the following statements
1. The water in a vessel placed on fire becomes hot by convection.
2. Heat is also transmitted in the form of electromagnetic waves from a hot body.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

167. A perfect black body has the unique characteristic feature as
(a) a good absorber only
(b) a good radiator only
(c) a good absorber and a good radiator
(d) Neither a radiator nor an absorber

168. Nights are cooler in the deserts than in the plains because
(a) sand radiates heat more quickly than the earth
(b) the sky remains clear most of the time
(c) sand absorbs heat more quickly than the earth
(d) None of the above

169. We are asked to wear white clothes in summer because
(a) they look more graceful
(b) it is a convention
(c) they are visible from long distance
(d) None of the above

170. Half portion of a rectangular piece of ice is wrapped with a white piece of cloth while the other half with a black one. In this context, which one among the following statements is correct?
(a) Ice melts more easily under black wrap
(b) Ice melts more easily under white wrap
(c) No ice melts at all under the black wrap
(d) No ice melts at all under the white wrap

171. Mr. X was advised by an architect to make outer walls of his house with bricks. The correct reason is that such walls
(a) make the building stronger
(b) help keeping inside cooler in summer and warmer in winter
(c) prevent seepage of mositure from outside
(d) protect the building from lightning

172. The blackboard seems black because it
(a) reflects every colour
(b) does not reflect any colour
(c) absorbs black colour
(d) reflects black colour

173. Consider the following statements
1. Rough surface is a better radiator than smooth surface.
2. Highly polished surface is a very good radiator.
3. Black surface is a better absorber of radiation than a white one.
4. Black surface is a better radiator than white one.

Which of the following statements is/are correct?
(a) 1, 2 and 3 (b) 1 and 2
(c) 3 and 4 (d) 1, 3 and 4

174. A polished plate with a rough black spot is heated to a high temperature and taken in a dark room, then,
(a) spot will appear brighter than the plate
(b) spot will appear darker than the plate
(c) both will appear equally bright
(d) Neither the spot, nor the plate will be visible

175. The temperature of the stars can be estimated by
(a) Wien's displacement law
(b) Rayleigh-Jeans law
(c) Faraday's law
(d) Maxwell-Boltzmann law

176. A solid copper sphere of radius R and hollow sphere of the same material and outer radius R are heated to the same temperature and allowed to cool in the same environment, the
(a) hollow sphere cools faster
(b) solid sphere cools faster
(c) both spheres cool at the same rate
(d) in the beginning solid sphere cools faster, at the end hollow sphere cools faster

177. If the door of a running refrigerator in a closed room is kept open, what will be the net effect on the room?
(a) It will cool the room
(b) It will heat the room
(c) It will make no difference on the average
(d) It will make thte temperature go up and down

178. The gas used in a refrigerator is
(a) cooled down on flowing
(b) heated up on flowing
(c) cooled down, when compressed
(d) cooled down, when expanded

Previous Years' Questions

179. The celsius temperature is a/an
(a) relative temperature **2013 I**
(b) absolute temperature
(c) specific temperature
(d) approximate temperature

180. The gas used in a refrigerator is
(a) cooled down on flowing **2013 II**
(b) heated up on flowing
(c) cooled down when compressed
(d) cooled down when expanded

181. A ray of white light strikes the surface of an object. If all the colours are reflected the surface would appear **2013 II**
(a) black (b) white (c) grey (d) opaque

182. Two layers of a cloth of equal thickness provide warmer covering than a single layer of cloth with double the thickness. Why? **2014 I**
(a) Because of the air encapsulated between two layers
(b) Since effective thickness of two layers is more
(c) Fabric of the cloth plays the role
(d) Weaving of the cloth plays the role

183. Which one of the following physical quantities is the same for molecules of all gases at a given temperature? **2015 II**
(a) Speed (b) Mass
(c) Kinetic energy (d) Momentum

184. Two systems are said to be in thermal equilibrium, if and only if **2016 I**
(a) there can be a heat flow between them even if they are at different temperatures
(b) there cannot be a heat flow between them even if they are at different temperatures
(c) there is no heat flow between them
(d) their temperatures are slightly different

Oscillations and Waves

185. The force acting on a particle executing simple harmonic motion is
(a) directly proportional to the displacement and is directed away from the mean position
(b) inversely proportional to the displacement and is directed towards the mean position
(c) directly proportional to the displacement and is directed towards the mean position
(d) inversely proportional to the displacement and is directed away from the mean position

186. Three pendulum clocks A, B and C are synchronised on the surface of the earth at the sea-level. The clock A is taken to the top of a mountain and the clock B is taken under a deep mine. The temperature at the three locations are assumed to be the same. On the basis of the above, which one of the following is correct?
(a) Both A and B run faster than C
(b) Both A and B run slower than C
(c) A runs slower than B but runs faster than C
(d) B runs slower than C but A runs faster than C

187. The length of a simple pendulum becomes four times the original length. If the initial time period of pendulum is T, then the new time period is
(a) T (b) $2T$ (c) $3T$ (d) $4T$

188. If the amplitude of a simple pendulum is made one-fourth, its time period will become
(a) twice (b) half
(c) one-fourth (d) will not change

189. Water waves are
(a) transverse (b) longitudinal
(c) partly transverse, partly longitudinal
(d) sometimes longitudinal, sometimes transverse

190. Energy is not transferred by
(a) transverse progressive waves
(b) longitudinal progressive waves
(c) stationary waves
(d) electromagnetic waves

191. Waves inside a gas are
(a) longitudinal (b) transverse
(c) partly longitudinal and party transverse
(d) Neither longitudinal nor transverse

192. What type of vibrations are produced in a sitar wire?
(a) Longitudinal vibration
(b) Transverse vibration
(c) Both (a) and (b)
(d) None of the above

193. Consider the following statements
1. Waves on springs or sound waves in air are longitudinal waves.
2. Waves on surface of water is mechanical waves.
3. Mechanical waves are used in communication, telephony
4. Mechanical waves can propagate in solid, liquid and gas.

Which of the following statements is/are correct.
(a) 1, 2 and 4 (b) 1 and 2
(c) 2 and 4 (d) Only 3

194. The most familiar form of radiant energy in sunlight that causes tanning and sunburning of human skin, is called
(a) ultraviolet radiation
(b) visible radiation
(c) infrared radiation
(d) microwaves radiation

195. Microwaves are
1. used in photography.
2. absorbed by water molecules and living tissue internal heating will damage or kill cells.

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) Neither 1 nor 2

196. Match the following columns.

Column I	Column II
A. Infrared rays	1. Medical tracers
B. γ-rays	2. TV control
C. Radio waves	3. Making ions
D. Ultraviolet rays	4. Communication

Codes

	A	B	C	D		A	B	C	D
(a)	3	2	1	4	(b)	2	1	4	3
(c)	3	1	2	4	(d)	3	4	2	1

197. Sound waves travel with a speed of about 330 m/s. The wavelength of sound whose frequency is 550 Hz is
(a) 0.8 m (b) 8.4 m (c) 0.6 m (d) 0.2 m

198. Which one of the following statements is correct?
(a) Sound waves in air are longitudinal while light waves are transverse
(b) Sound waves in air are transverse while light waves are longitudinal
(c) Both sound and light waves in air are transverse
(d) Both sound and light waves in air are longitudinal

199. Microphone is a device in which
(a) amplification is not required at all
(b) sound waves are directly transmitted
(c) sound waves are converted into electrical energy and then reconverted into sound after transmission
(d) electrical energy is converted into sound waves

200. Sound travels faster when the medium is of
(a) low density (b) medium density
(c) high density (d) vacuum

201. A source of sound has a noise level of 20 dB. If the intensity of sound produced by the source increases by 100 times, the noise level becomes
(a) 30 dB (b) 40 dB
(c) 400 dB (d) 20000 dB

202. Which among the following statement is correct regarding the audible sound waves?
1. Its frequency is greater than 20000 Hz.
2. It is used in measuring the depth of sea.
3. The liquid from which the audible sound waves pass gets hot.

The correct statement(s) is/are
(a) 1 and 3 (b) 1 and 2
(c) Only 1
(d) None of the above

203. Match the following columns.

Column I (Waves)	Column II (Sources)
A. Ultrasonic waves	1. Ocean waves
B. Sound waves	2. Energetic disturbance
C. Infrasonic waves	3. Generated by vibrating bodies
D. Shock waves	4. Wave produce by bat

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	1	2	(b)	3	2	1	4
(c)	4	3	2	1	(d)	1	2	3	4

204. Which of the following statement is incorrect?
(a) Sound travels in straight line
(b) Sound is a form of energy
(c) Sound travels in the form of waves
(d) Sound travels faster in vacuum than in air

205. Sound travels in rocks in the form of
(a) longitudinal elastic waves only
(b) transverse elastic waves only
(c) both longitudinal and transverse elastic waves
(d) non-elastic waves

206. Ultrasonic waves are those waves
(a) to which man can hear
(b) man cannot hear
(c) are of high velocity
(d) of high amplitude

207. Consider the following statements
1. The frequency of sound waves is 20 Hz to 20000 Hz.
2. Sound waves are transverse mechanical waves.
3. Sound waves travel in vacuum.
4. Sound waves have low frequency and high wavelength.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2, 3 and 4
(c) 1 and 4 (d) Only 4

208. Ultrasonic waves are used
1. to investigate the action of heart.
2. for sending signals.
3. measuring the depth of sea

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) All of these

209. Consider the following statements
1. Speed of sound basically depends upon elasticity and density of medium .
2. Speed of sound is changed by the increase or decrease of pressure
3. The speed of sound is more in humid air than in dry air.

Which of the following statements is/are correct?
(a) 1 and 3 (b) 1 and 2
(c) 2 and 3 (d) None of these

210. A woman's voice is shriller than a man's due to
(a) higher frequency
(b) higher amplitude
(c) lower frequency
(d) weak vocal chords

211. The phenomenon of beats can take place
(a) for longitudinal waves only
(b) for transverse waves only
(c) for both longitudinal and transverse waves
(d) for sound waves only

212. Doppler's effect is used
1. to determine the velocities of which stars and galaxies are moving toward or away from the earth
2. to increase the pressure.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

Previous Years' Questions

213. The time period of a simple pendulum having a spherical wooden bob is 2 s. If the bob is replaced by a metallic one twice as heavy, the time period will be **2012 I**
(a) more than 2s (b) 2s
(c) 1s (d) less than 1s

214. Two identical piano wires have same fundamental frequency when kept under the same tension. What will happen if tension of one of the wire is slightly increased and both the wires are made to vibrate simultaneously? **2012 I**
(a) Noise (b) Beats
(c) Resonance (d) Non-linear effects

215. In SONAR, we use **2012 II**
(a) ultrasonic waves
(b) infrasonic waves
(c) radio waves
(d) audible sound waves

216. Motion of an oscillating liquid column in a U-tube is **2013 II**
(a) periodic but not simple harmonic
(b) non-periodic
(c) simple harmonic and time period is independent of the density of the liquid
(d) simple harmonic and time period depends on the density of the liquid

217. An oscilloscope is an instrument which allows us to see waves produced by **2014 II**
(a) visible light (b) X-rays
(c) sound (d) γ-rays

218. Which one of the following statements is correct?
The velocity of sound **2016 I**
(a) does not depend upon the nature of media
(b) is maximum in gases and minimum in liquids
(c) is maximum in solids and minimum in liquids
(d) is maximum in solids and minimum in gases

219. Which one of the following statements is not correct? **2016 I**
(a) Sound waves in gases are longitudinal in nature
(b) Sound waves having frequency below 20 Hz are known as ultrasonic waves
(c) Sound waves having higher amplitudes are louder
(d) Sound waves with high audible frequencies are sharp

220. A person rings a metallic bell near a strong concrete wall. He hears the echo after 0.3 s. If the sound moves with a speed of 240 m/s, how far is the wall from him? **2016 I**
(a) 102 m (b) 11 m (c) 51 m (d) 30 m

Optics

221. The mirror used in vehicles as rear view mirror is
(a) convex mirror (b) concave mirror
(c) plane mirror
(d) plano-concave mirror

222. Viewfinders, used in automobiles to locate the position of the vehicles behind, are made of
(a) plane mirror (b) concave mirror
(c) convex mirror (d) parabolic mirror

223. Concave mirrors are used
1. as reflectors in automobiles and hand torches.
2. in the field of solar energy to focus sun's ray for heating solar furnaces.
3. as rear-view mirror in automobiles.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 3 and 1 (d) Only 1

224. A coin immersed in water pond appears to be raised, when viewed from the top. What is this due to?
(a) Total internal reflection of light
(b) Scattering of light
(c) Reflection of light
(d) Refraction of light

225. Sun appears reddish during the rising and setting time, because
(a) the atmosphere absorbs short wavelengths more than long wavelengths
(b) red light is emitted in huge amount by it
(c) the atmosphere absorbs long wavelength more than short wavelengths
(d) light of shorter wavelengths are scattered to a greater extent than the longer wavelengths by the atmosphere

226. The phenomenon of mirage is due to
(a) change in refractive index of air with change in temperature
(b) total internal reflection
(c) absorption of light by air at high temperature
(d) polarisation of light on reflection

227. Which one of the following phenomenon cannot be attributed to the refraction of light?
(a) Twinkling of stars
(b) Mirage
(c) Rainbow
(d) Red shift

228. Match the following columns.

Column I	Column II
A. Intensity of light	1. Properties of the medium
B. Colour of light	2. Refractive index of medium
C. Velocity of light	3. Amplitude of light
D. Propagation of light	4. Frequency of light

Codes

	A	B	C	D		A	B	C	D
(a)	3	4	2	1	(b)	2	4	1	3
(c)	3	1	2	4	(d)	4	2	3	1

229. Which of the following illustrations is of refraction?
1. Twinkling of stars
2. Oval shape of sun in the morning and evening
3. Sparking of diamond
4. Mirage and looming

Which of the following statements is/are correct?
(a) 1 and 3 (b) 1 and 2
(c) 2 and 3 (d) Only 4

230. Efficient working of optical fibre is dependent on which one of the following?
(a) Reflection
(b) Interference
(c) Diffraction
(d) Total internal reflection

231. Condition of total internal reflection
1. Light must be propagating from denser to rarer medium.
2. The critical angle must exceeds the angle of incidence.

Which of the following statements is/are correct?
(a) Only 1
(b) Only 2
(c) 1 and 2
(d) None of the above

232. Where should an object be placed, so that a real and inverted image of the same size is obtained using a convex lens?
(a) Between the lens and its axis
(b) At the focus
(c) At twice the focal length
(d) At infinity

233. A thin lens has a focal length of –25 cm. Then, the nature of lens is
(a) concave lens of power 4 D
(b) convex lens of power 4 D
(c) convex lens of power 25 D
(d) convex lens of power $\frac{1}{4}$ D

234. Parallel light rays entering a convex lens always converge at
(a) centre of curvature
(b) the focal plane
(c) the principal focus
(d) a point on the principal axis

235. We always get real image in a convex lens, if object is situated beyond
(a) optical centre
(b) focus
(c) centre of curvature
(d) None of the above

236. The power of a lens of focal length 80 cm is
(a) 80 D (b) 40 D (c) 12.5 D (d) 1.25 D

237. A diffraction pattern is obtained using a beam of red light. Which one among the following will be the outcome, if the red light is replaced by blue light?
(a) Bands disappear
(b) Diffraction pattern becomes broader and further apart
(c) Diffraction pattern becomes narrower and crowded together
(d) No change

238. Dispersion produced by a prism depends on
(a) its refracting angle
(b) size of the prism
(c) height of the prism
(d) one of the angle at the base of the prism

239. When dispersion of white light takes place through a prism, the maximum deviation is suffered by which one of the following colour?
(a) Red (b) Yellow
(c) Green (d) Violet

240. In the visible spectrum, the colour having the shortest wavelength is
(a) blue (b) yellow
(c) red (d) violet

241. Two beams of light, one red and the other green, fall on the same spot on a white screen. The colour on the screen will appear to be
(a) magenta (b) blue
(c) yellow (d) cyan

242. Consider the following statements
1. The colour of the green flower seen through red glass appears to be dark.
2. Red glass transmits only red light.

Which of the following statements is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) None of the above

243. The focal length of convex lens is
(a) the same for all colours
(b) shorter for blue light than for red
(c) shorter for red light than for blue
(d) maximum for yellow light

244. Consider the following statements
1. The colour of an object is determined by the colour of the light reflected by it.
2. If all the colours of white light are absorbed by the object then it appears white.
3. Dangers signals are of red colour because red colour of light scatter least and therefore signals can be seen from far.

Which of the following statements is/are correct?
(a) 1 and 3
(b) Only 3
(c) Only 1
(d) 2 and 3

245. Rainbow is formed due to
1. dispersion of light
2. suffering refraction
3. total internal reflection
4. All of the above

Which of the following statements is/are correct?
(a) 1 and 2
(b) 3 and 1
(c) Only 4
(d) Only 1

246. Which one of the following sets of colour combinations is added in colour vision in TV?
(a) Yellow, green and blue
(b) White, black, red and green
(c) Red, green and blue
(d) Orange, pink and blue

247. If the light moving in a straight line bends by a small but fixed angle, it may be a case of
(a) diffraction
(b) dispersion
(c) reflection
(d) refraction and dispersion

248. If a glass rod is immersed in a liquid of the same refractive index, then it will
(a) look longer (b) disappear
(c) look bent (d) None of these

249. Why is it difficult to see through fog?
(a) Rays of light suffer total internal reflection from the fog droplets
(b) Rays of light are scattered by the fog droplets
(c) Fog droplets absorb light
(d) The refractive index of fog is extremely high

250. The colour of the sky is blue because of
(a) combination of various lights producing blue colour
(b) scattering of light by dust particles
(c) diffraction of light
(d) reflection of light by water droplets

251. Rainbow is formed due to a combination of
(a) refraction and scattering
(b) refraction and absorption
(c) dispersion and focusing
(d) total internal reflection and dispersion

252. As the sunlight passes through the atmosphere, the rays are scattered by tiny particles of dust, pollen, soot and other minute particulate matters present there. However, when we look up, the sky appears blue during mid-day because
(a) blue light is scattered most
(b) blue light is absorbed most
(c) blue light is reflected most
(d) ultraviolet and yellow component of sunlight combine

253. Consider the following statements
1. Clear sky appears blue due to poor scattering of blue wavelength of visible light.
2. Red part of light shows more scattering than blue light in the atmosphere.
3. In the absence of atmosphere, there would be no scattering of light and sky will look black.

Which of the statements given above is/are correct?
(a) Only 1 (b) 1 and 2
(c) Only 3 (d) All of these

254. We cannot see the details of a distant object because
(a) there is a defect in our eyes
(b) the light rays are absorbed and scattered in this way
(c) there is a limit of resolution for our eyes
(d) the objects are too far away

255. Match the following columns.

Column I	Column II
A. Myopia	1. Cylindrical
B. Hypermetropia	2. Bifocal
C. Presbyopia	3. Convex
D. Astigmatism	4. Concave

Codes
A B C D A B C D
(a) 4 3 2 1 (b) 1 2 3 4
(c) 2 4 3 1 (d) 2 1 4 3

256. The principle of working of periscope is based on
(a) reflection only
(b) refraction only
(c) reflection and interference
(d) reflection and refraction

257. For a telescope, the focal lengths of two lenses are 30 cm and 10 cm. One acts as objective and other as eyepiece. The length of the telescope is
(a) 40 cm (b) 3 cm
(c) 200 cm (d) 30 cm

258. Consider the following statements
1. Periscope consists of two plane mirror inclined at an angle of 45°.
2. The principle of working of periscope is based upon reflection and refraction.

Which of the following statements is/are correct?
(a) Only 1 (b) 1 and 2
(c) Only 3 (d) None of these

Previous Years' Questions

259. Dispersion process forms spectrum due to white light falling on a prism. The light wave with shortest wavelength **2013 II**
(a) refracts the most
(b) does not change the path
(c) refracts the least
(d) is reflected by the side of the prism

260. A ray of white light strikes the surface of an object. If all the colours are reflected the surface would appear **2013 II**
(a) black (b) white (c) grey (d) opaque

261. Which one among the following colours has the highest wavelength? **2013 II**
(a) Violet (b) Green (c) Yellow (d) Red

262. No matter how far you stand from a mirror, your image appears erect. The mirror is likely to be **2014 I**
(a) either plane or convex
(b) plane only
(c) concave (d) convex only

263. The position, relative size and nature of the image formed by a concave lens for an object placed at infinity are respectively **2014 I**
(a) at focus, diminished and virtual
(b) at focus, diminished and real
(c) between focus and optical centre, diminished and virtual
(d) between focus and optical centre, magnified and real

264. In the phenomenon of dispersion of light, the light wave of shortest wavelength is **2014 II**
(a) accelerated and refracted the most
(b) slowed down and refracted the most
(c) accelerated and refracted the least
(d) slowed down and refracted the least

265. The upper and lower portions in common type of bi-focal lenses are respectively **2014 II**
(a) concave and convex
(b) convex and concave
(c) both concave of different focal lengths
(d) both convex of different focal lengths

266. A person standing 1m in front of a plane mirror approaches the mirror by 40 cm. The new distance between the person and his image in the plane mirror is **2015 I**
(a) 60 cm (b) 1.2 m (c) 1.4 m (d) 2.0 m

267. The outside rear view mirror of modern automobiles is marked with warning 'objects in mirror are closer than they appear'. Such mirrors are **2015 II**
(a) plane mirrors
(b) concave mirrors with very large focal lengths
(c) concave mirrors with very small focal lengths
(d) convex mirrors

268. The focal length of the lens of a normal human eye is about **2015 II**
(a) 25 cm (b) 1 m
(c) 2.5 mm (d) 2.5 cm

269. A myopic person has a power of – 1.25 D. What is the focal length and nature of his lens? **2016 I**
(a) 50 cm and convex lens
(b) 80 cm and convex lens
(c) 50 cm and concave lens
(d) –80 cm and concave lens

Electric Current

270. Which of the following is a good conductor of heat but a bad conductor of electricity?
(a) Mica (b) Aluminium
(c) Mercury (d) Platinum

271. Consider the following statements
1. The direction of positive charges is same as direction of conventional current.
2. The rate of flow of electric current charges through any cross-section of conductor is called as electric current.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

272. The work done in moving a unit positive charge across two points in an electric circuit is a measure of
(a) potential difference
(b) electrostatic force
(c) current (d) power

273. Potential difference between two points of a wire carrying 2A current is 0.1 V. The resistance of the wire is
(a) 0.05 Ω (b) 0.5 Ω
(c) 5 Ω (d) 0.25 Ω

274. Match the following columns.

Column I	Column II
A. Electrical force between two charged particles	1. Volt
B. Electric charge	2. Newton
C. Electric potential	3. Farad
D. Electrical capacity	4. Coulomb

Codes
A B C D
(a) 2 4 1 3
(b) 2 4 3 1
(c) 1 2 3 4
(d) 2 1 3 4

275. In electric supply lines in India, which parameter is kept constant?
(a) Voltage (b) Current
(c) Frequency (d) Power

276. Which one of the following is not correctly matched?
(a) Voltmeter — Potential difference
(b) Potentiometer — EMF
(c) Ammeter — Electric current
(d) Meter bridge — Electrical resistance

277. A DC voltmeter is capable of measuring a maximum of 300 V. If it is used to measure the voltage across a device operating at 220 V AC supply, the reading of the voltmeter will be
(a) 0 V (b) 110 V (c) 220 V (d) 300 V

278. Galvanometer
1. is a device used to detect and measure electric current in a circuit.
2. can be converted into a voltmeter by connecting a very high resistance in its parallel.

Which of the following statements is/are correct?
(a) Only 1 (b) 1 and 2
(c) Only 2 (d) None of these

279. When you pull out the plug connected to an electrical appliance, you often observe a spark. To which property of the appliance is this related?
(a) Resistance (b) Inductance
(c) Capacitance (d) Wattage

280. Match the columns.

Column I		Column II	
A.	Voltage	1.	$\frac{q}{t}$
B.	Current	2.	$\rho\frac{l}{A}$
C.	Capacity	3.	$\frac{c}{v}$
D.	Resistance	4.	IR

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	1	3	2
(c)	3	4	2	1	(d)	4	3	2	1

281. Two resistances when connected in parallel give the resultant value of 2 Ω, when connected in series the value becomes 9 ohm. The value of each resistance is
(a) 1 Ω and 8 Ω (b) 2 Ω and 7 Ω
(c) 3 Ω and 3 Ω (d) 3 Ω and 6 Ω

282. Consider the following statements
1. The property of a substance by virtue of which it opposes the flow of current through it, is known as the resistance.
2. Resistance of a conductor is directly proportional to it's length.
3. Resistance of a conductor is directly proportional to area of cross-section.
4. Resistance of a substance is always equal to resistivity.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 4
(c) 3 and 4 (d) 1, 2 and 3

283. Consider the following statements
1. Electric appliances with metallic body e.g. heater, presses, etc., have three pin connections whereas, an electric bulb has a two pin connection.
2. Three pin connections reduce heating of connecting cables.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

284. An electric bulb is marked as 240 V, 60 W. The resistance of its filament is
(a) 940 Ω (b) 245 Ω
(c) 950 Ω (d) 960 Ω

285. A tube light works on the principle of
(a) chemical effect of current
(b) magnetic effect of current
(c) heating effect of current
(d) discharge of electrons through gases

286. The wire used in the filaments of household bulbs has
(a) high resistance, high melting point
(b) high resistance, low melting point
(c) low resistance, high melting point
(d) low resistance, low melting point

287. Safety fuses are integral part of electric installations and instruments. This is so because safety fuses
(a) block the passage of current due to increase in their resistance and saves it
(b) switch off the service of electric supply through relay action
(c) provide alternative path to excess current as does a shunt
(d) switch off the supply, if current beyond a certain limit flows through the circuit

288. In a filament type high bulb, most of the electric power consumed appear as
(a) visible light
(b) infrared rays
(c) ultraviolet rays
(d) fluorescent light

289. Why are inner lining of hot water geysers made up of copper?
(a) Copper has low heat capacity
(b) Copper has high electrical conductivity
(c) Copper does not react with steam
(d) Copper is good conductor of both heat and electricity

290. Consider the following statements
1. Fuse wire is made up of tin lead alloy.
2. Fuse wire have low resistivity.
3. Fuse wire have low melting point.
4. Fuse wire is used in parallel as a safety device in an electric circuit.

Which of the following statements is/are correct?
(a) 1 and 3 (b) 2 and 4
(c) Only 4 (d) 1 and 2

291. A bird sitting on a high power line
(a) gets a fatal shock
(b) gets a mild shock
(c) gets killed instantly
(d) is not affected practically

292. The total resistance of 3 resistors, each of 3 Ω, connected in parallel, is
(a) 3 Ω (b) 2 Ω
(c) 1 Ω (d) 9 Ω

293. The circuit element where the impressed voltage is always in phase with the resulting current, is
(a) an ideal resistor
(b) an ideal coil
(c) an ideal capacitor
(d) an ideal transformer

294. The tape of a tape recorder is coated with
(a) zinc oxide
(b) mica
(c) copper sulphate
(d) ferromagnetic powder

295. Consider the following statements
1. Permanent magnets are usually made of alloys such as carbon steel, chromium steel, tungsten steel and alnico.
2. Alnico is alloy of gold, silver and zinc.

3. Permanent magnets are used in microphone etc.

Which of the following statements is/are correct?

(a) 1 and 2 (b) 2 and 3
(c) Only 2 (d) 1 and 3

296. An electric generator converts
(a) electric energy into sound energy
(b) electric energy into light energy
(c) electric energy into mechanical energy
(d) mechanical energy into electric energy

297. The essential difference between an AC generator and a DC generator has permanent magnet.
(a) AC generator has an electromagnet while a DC generator has permanent magnet
(b) DC generator will generate a higher voltage
(c) AC generator will generate a higher voltage
(d) AC generator has slip rings while DC generator has a commutator

298. The core of a transformer is laminated so that
(a) ratio of the voltages across the secondary and primary is doubled
(b) the weight of the transformer can be kept low
(c) the rusting of the core is prevented
(d) energy loss due to eddy currents is minimised

299. Which one of the following common devices works on the basis of the principle of mutual induction?
(a) Tubelight (b) Transformer
(c) Photodiode (d) LED

300. Consider the following statements
1. Core of a transformer is made up of soft iron.
2. Step-up transformer converts a low voltage of high current into a high voltage of low current.
3. Transformer is used for charging voltage only.

Which of the following statements is/are correct?

(a) 1 and 2 (b) 2 and 3
(c) Only 3 (d) None of these

301. Transformer is a kind of appliance that can
1. increase power
2. increase voltage
3. decrease voltage
4. measure current and voltage

Select the correct answer using the codes given below.

(a) 1 and 4 (b) Only 4
(c) 2 and 3 (d) 2, 3 and 4

Previous Years' Questions

302. In step-down transformer, the AC output gives the **2012 II**
(a) current more than the input current
(b) current less than the input current
(c) current equal to the input current
(d) voltage more than the input voltage

303. A rectifier is an electronic device used to convert **2012 II**
(a) AC voltage into DC voltage
(b) DC voltage into AC voltage
(c) sinusoidal pulse into square pulse
(d) None of the above

304. A device which is used in our TV set, computer, radio set for storing the electric charge, is **2012 II**
(a) resistor (b) inductor
(c) capacitor (d) conductor

305. If two conducting spheres are separately charged and then brought in contact **2013 I**
(a) the total energy of the two spheres is conserved
(b) the total charge on the two spheres is conserved
(c) both the total energy and the total charge are conserved
(d) the final potential is always the mean of the original potential of the two spheres

306. When an incandescent electric bulb glows **2014 I**
(a) the electric energy is completely converted into light
(b) the electric energy is partly converted into light energy and partly into heat energy
(c) the light energy is converted into electric energy
(d) The electric energy is converted into magnetic energy

307. A mobile phone charger is **2014 I**
(a) an inverter (b) an UPS
(c) a step-down transformer
(d) a step-up transformer

308. Inactive nitrogen and argon gases are usually used in electric bulbs in order to **2014 II**
(a) increase the intensity of light emitted
(b) increase the life of the filament
(c) make the emitted light coloured
(d) make the production of bulb economical

309. Tungsten is used for the construction of filament in electric bulb because of its **2014 II**
(a) high specific resistance
(b) low specific resistance
(c) high light emitting power
(d) high melting point

310. Electricity is produced through dry cell from **2015 I**
(a) chemical energy
(b) thermal energy
(c) mechanical energy
(d) nuclear energy

311. A wire-bound standard resistor uses manganin or constantan. It is because **2015 I**
(a) these alloys are cheap and easily available
(b) they have high resistivity
(c) they have low resistivity
(d) they have resistivity which almost remains unchanged with temperature

312. Two pieces of conductor of same material and of equal length are connected in series with a cell. One of the two pieces has cross-sectional area double that of the other. Which one of the following statements is correct in this regard? **2015 I**
(a) The thicker one will allow stronger current to pass through it
(b) The thinner one would allow stronger current to pass through it
(c) Same amount of electric current would pass through both the pieces producing more heat in the thicker one
(d) Same amount of electric current would pass through both the pieces producing more heat in the thinner one

313. Which one of the following statements about bar magnet is correct? **2016 I**
(a) The pole strength of the North pole of a bar magnet is larger than that of the South pole.
(b) When a piece of bar magnet is bisected perpendicular to its axis, the North and South poles get separated.
(c) When a piece of bar magnet is bisected perpendicular to its axis, two new bar magnets are formed.
(d) The poles of a bar magnet are unequal in magnitude and opposite in nature.

Modern Physics

314. Which is not true with respect to the cathode rays?
(a) A stream of electrons
(b) Charged particles
(c) Move with speed same as that of light
(d) Can be deflected by magnetic fields

315. Consider the following statements X-rays
1. can pass through aluminium.
2. can be deflected by magnetic field.
3. move with a velocity less than the velocity of ultraviolet ray in vacuum.

Which of the statements given above is/are correct?
(a) 2 and 3 (b) Only 1
(c) 1 and 3 (d) 1 and 2

316. Consider the following statements
1. Soft and hard X-rays differ in frequency as well as velocity.
2. The penetrating power of hard X-rays is more than the penetrating power of soft X-rays.

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) None of these

317. X-ray will travel minimum distance in
(a) air (b) iron (c) wood (d) water

318. X-ray beam can be deflected by
(a) magnetic field
(b) electric field
(c) Both 'a' and 'b'
(d) None of the above

319. X-rays were discovered by
(a) Johnson (b) Milikan
(c) Roentgen (d) Thomson

320. Positive rays were discovered by
(a) Thomson (b) Goldstein
(c) W Crookes (d) Rutherford

321. Consider the following statements
1. Positive rays are capable of producing physical and chemical changes.
2. Positive rays can produce fluorescence and phosphorescence.
3. Positive rays are not deflected by electric field and magnetic field.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) Only 3 (d) Only 2

322. Match the following columns.

Column I	Column II
A. Cathode rays	1. Electromagnetic waves
B. X-rays	2. Beam of electron
C. Positive-rays	3. Positively charged particles

Codes

	A	B	C		A	B	C
(a)	3	1	2	(b)	2	1	3
(c)	1	2	3	(d)	1	3	2

323. Consider the following statements
1. Diode laser are used as optical sources in optical communication.
2. Diode laser consume less energy.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

324. An ordinary tubelight used for lighting purpose contains
(a) fluorescent material and an inert gas
(b) one filament, reflective material and mercury vapour
(c) fluorescent material and mercury vapour
(d) two filaments, fluorescent material and mercury vapour

325. Fire fly gives us cold light by virtue of the phenomenon of
(a) fluorescence
(b) phosphorescence
(c) chemiluminescence
(d) effervescence

326. Photoelectric cell is a device of
(a) collecting the photons
(b) measuring the intensity of light
(c) changing the photon energy to mechanical energy
(d) substituting the accumulators by storing the electrical energy

327. What is the difference between a CFL and an LED lamp
1. To produced light, a CFL uses mercury vapour and phosphor while an LED lamp uses semiconductor materials.
2. The average life span of a CFL is much longer than that of an LED lamp.
3. A CFL is less energy efficient as compared to an LED lamp.

Which of the following statements is/are correct?
(a) Only 1 (b) 2 and 3
(c) 1 and 3 (d) All of these

328. Photoelectric cells are used
1. to control the temperature in furnace and in chemical process
2. in automatic switches for street light
3. in transformer
4. in photoelectron sorters

Which of the following statements is/are correct?
(a) 1, 2 and 4 (b) Only 1
(c) Only 2 (d) 1 and 3

329. The majority charge carriers in p-type semiconductors are
(a) electrons (b) protons
(c) holes (d) neutrons

330. If the temperature of a semiconductor is raised, its resistivity will
(a) increase (b) decrease
(c) remains same (d) reduce to zero

331. Consider the following statements
1. Solar cells are used as solar cooker.
2. Solar cells are used in wrist watches, calculators and light meters.
3. Solar cells are used for charging storage batteries.
4. Solar cells are used in artificial satellite

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 1, 3 and 4 (d) All of these

332. Laser is a device to produce
(a) a beam of white light
(b) coherent light
(c) microwaves (d) X-rays

333. Which of the following is the correct combination of the inventors and the inventions?
(a) Elisha G Otis — Windmill
(b) Galileo Galilei — Transistor
(c) Sir Frank Whittle — Laser
(d) J L Baird — Television

334. Match the following Columns.

Column I	Column II
A. Ground wave propagation	1. Radio waves received from troposphere
B. Space wave propagation	2. Radio waves travel through atmosphere
C. Sky wave propagation	3. Microwaves cross the ionosphere
D. Satellite communication	4. Radio waves received from ionosphere

Codes

	A	B	C	D
(a)	2	1	4	3
(b)	1	3	4	2
(c)	3	2	1	4
(d)	2	1	3	4

335. The television set receives
(a) magnetic waves
(b) radio waves
(c) microwaves
(d) None of the above

336. Optical fibres are mainly used for which of the following?
(a) Communication
(b) Eye surgery
(c) Weaving
(d) Food industry

Previous Years' Questions

337. Before X-ray examination (coloured X-ray) of the stomach, patients are given suitable salt of barium because **2012 I**
(a) barium salts are white in colour and this helps stomach to appear clearly
(b) barium is a good absorber of X-rays and helps stomach to appear clearly
(c) barium salts are easily available
(d) barium allows X-rays to pass through the stomach

338. Light Emitting Diode (LED) converts **2013 I**
(a) light energy into electrical energy
(b) electrical energy into light energy
(c) thermal energy into light energy
(d) mechanical energy into electrical energy

339. Dual energy X-ray Absorptiometry (DEXA) is used to measure **2013 II**
(a) Spread of solid tumour
(b) bone density
(c) ulcerous growth in stomach
(d) extent of brain haemorrhage

340. X-rays are **2015 II**
(a) deflected by an electric field, but not by a magnetic field
(b) deflected by a magnetic field, but not by an electric field
(c) deflected by both a magnetic field and an electric field
(d) not deflected by an electric field and a magnetic field

ANSWERS

1	b	2	a	3	b	4	b	5	b	6	a	7	b	8	a	9	a	10	a
11	c	12	a	13	c	14	d	15	d	16	c	17	a	18	d	19	c	20	c
21	b	22	d	23	d	24	a	25	b	26	d	27	d	28	d	29	c	30	c
31	c	32	b	33	c	34	a	35	b	36	a	37	d	38	d	39	d	40	a
41	a	42	a	43	a	44	a	45	a	46	c	47	a	48	a	49	a	50	a
51	b	52	b	53	b	54	a	55	a	56	d	57	d	58	c	59	c	60	a
61	d	62	c	63	d	64	a	65	a	66	b	67	a	68	b	69	b	70	b
71	c	72	a	73	d	74	c	75	a	76	d	77	a	78	d	79	d	80	b
81	b	82	a	83	c	84	c	85	a	86	b	87	b	88	a	89	b	90	b
91	b	92	a	93	b	94	c	95	b	96	b	97	b	98	c	99	d	100	a
101	c	102	a	103	c	104	a	105	b	106	c	107	a	108	c	109	c	110	c
111	c	112	d	113	a	114	d	115	b	116	b	117	a	118	a	119	c	120	a
121	c	122	c	123	c	124	d	125	a	126	a	127	c	128	a	129	b	130	a
131	c	132	c	133	a	134	a	135	c	136	a	137	d	138	a	139	a	140	c
141	a	142	d	143	c	144	a	145	d	146	b	147	b	148	a	149	a	150	a
151	a	152	c	153	c	154	c	155	a	156	d	157	a	158	b	159	c	160	a
161	b	162	a	163	a	164	d	165	b	166	c	167	c	168	a	169	d	170	a
171	b	172	b	173	d	174	a	175	a	176	a	177	b	178	d	179	c	180	d
181	b	182	a	183	c	184	c	185	c	186	b	187	b	188	d	189	a	190	c
191	a	192	b	193	a	194	a	195	c	196	b	197	c	198	a	199	c	200	c
201	b	202	d	203	a	204	d	205	c	206	b	207	c	208	d	209	a	210	a
211	c	212	a	213	b	214	a	215	a	216	c	217	a	218	d	219	b	220	c
221	a	222	c	223	a	224	d	225	d	226	b	227	c	228	c	229	b	230	b
231	a	232	b	233	a	234	c	235	b	236	d	237	a	238	a	239	d	240	d
241	c	242	c	243	b	244	a	245	c	246	c	247	d	248	d	249	b	250	b
251	d	252	a	253	c	254	b	255	a	256	d	257	a	258	b	259	c	260	b
261	d	262	a	263	a	264	c	265	a	266	b	267	d	268	c	269	d	270	c
271	c	272	a	273	a	274	a	275	c	276	a	277	a	278	a	279	c	280	b
281	d	282	a	283	a	284	d	285	d	286	a	287	d	288	b	289	d	290	a
291	d	292	c	293	a	294	d	295	d	296	d	297	d	298	d	299	b	300	a
301	c	302	a	303	a	304	c	305	b	306	b	307	c	308	b	309	d	310	a
311	d	312	a	313	c	314	c	315	b	316	c	317	b	318	d	319	c	320	b
321	a	322	b	323	c	324	d	325	c	326	b	327	c	328	a	329	c	330	b
331	d	332	b	333	d	334	a	335	c	336	a	337	b	338	b	339	b	340	d

Solutions

1. (*b*) Given,

$v_1 = 30$ km/h

and $v_2 = 50$ km/h and $v_{av} = ?$

$$v_{av} = \frac{2v_1 v_2}{v_1 + v_2} = \frac{2 \times 30 \times 50}{30 + 50}$$

$$= \frac{3000}{80} = 37.5 \text{ km/h}$$

5. (*b*) From second equation of motion,

$$s = ut + \frac{1}{2}at^2$$

$$\Rightarrow \quad s = 0 + \frac{1}{2}at^2$$

$$s \propto t^2 \qquad (\text{if } a = \text{constant})$$

So, object moves with constant or uniform acceleration.

11. (*c*) For maximum cover horizontal range,

$$R_{\max} = \frac{u^2 \sin 2\theta}{g}$$

$$\Rightarrow \quad R_{\max} = \frac{u^2}{g} \qquad (\theta = 45^\circ)$$

18. (*d*) Given, $m = 1$ kg, $F = 1$ N

$F = ma$

$\Rightarrow \quad 1 = 1 \times a$

$\Rightarrow \quad a = 1 \text{ m/s}^2$

36. (*a*) Given,

$m = 2$ kg, $h = 1.5$ m and work done = ?

$g = 9.8 \text{ m/s}^2$

Work done = PE = mgh

$= 2 \times 9.8 \times 1.5 = 29.4$ J

40. (*a*) Let the weight of dog be m kg, then the weight of horse be 10 m kg

$\because \quad \text{KE} = \frac{1}{2}mv^2$ or $K \propto m$

$$\therefore \quad \frac{K_1}{K_2} = \frac{m_1}{m_2} = \frac{m}{10\ m}$$

$K_2 : K_1 = 10 : 1$

41. (*a*) Given, $g = 10 \text{ m/s}^2$, $m = 1$ kg

and $h = 5$ m, PE = ?

$\therefore \quad \text{PE} = mgh = 1 \times 10 \times 5 = 50$ J

44. (*a*) Given,

$g = 9.8 \text{ m/s}^2$,

$m = 200$ kg

PE = 9800 J and $h = ?$

$\because \quad \text{PE} = mgh$

$9800 = 200 \times 9.8 \times h$

$$\Rightarrow \quad h = \frac{9800}{200 \times 9.8} = 5 \text{ m}$$

45. (*a*) Given,

$m_1 = m_2 = m$, $v_1 = 2v$, $v_2 = 3v$

$\because \quad K = \frac{1}{2}mv^2 \Rightarrow K \propto v^2$

$$\therefore \quad \frac{K_1}{K_2} = \left(\frac{v_1}{v_2}\right)^2 = \left(\frac{2v}{3v}\right)^2$$

$\Rightarrow \quad K_1 : K_2 = 4 : 9$

48. (*a*) Time taken by the bus to go from origin to destination with speed of 50 km/h, is

$$t = \frac{300}{50} = 6 \text{ h}$$

($\because$ distance = 300 km)

Time taken by the bus to go from destination to origin with speed of 60 km/h, is

$$t_2 = \frac{300}{60} = 5 \text{ h}$$

$$\therefore \quad \text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$= \frac{300 + 300}{6 + 5} = \frac{600}{11}$$

$= 54.55$ km/h

50. (*a*) In this case initial and final position are same. So, displacement will be zero.

Distance travelled by object = $2\pi r$

$$= 2 \times \frac{22}{7} \times 35$$

$$= 70 \times \frac{22}{7} = 220 \text{ m}$$

53. (*b*) The centripetal force, $F = \frac{mv^2}{r}$

Because an electron and a proton are circulating with same speed in circular paths of equal radius

So, $\quad F \propto m$

and $\quad \frac{F_e}{F_p} = \frac{m_e}{m_p} = \frac{m}{2000\ m}$

$F_p = 2000\ F_e$

Therefore, the centripetal force required by the proton is about, 2000 times more than that required by the electron.

55. (*a*) From law of conservation of linear momentum

The total momentum of the system is conserved.

i.e. $\quad m_G v + m_B v_G = 0$

$$\Rightarrow \quad v_G = -\frac{m_B}{m_G} v_G \quad \text{or } v_G \propto \frac{1}{m_G}$$

Therefore, gun is much heavier, so velocity of the recoiling gun is less.

100. (*a*) Stress $= \frac{\text{Force}}{\text{Area}} = \frac{1 \times 10^4}{\pi \times (2 \times 10^{-3})^2}$

$$= \frac{1 \times 10^4}{12.56 \times 10^{-6}}$$

$= 7.96 \times 10^8 \text{ N/m}^2$

104. (*a*) Radius of the heal, $r = \frac{1.0}{2}$

$= 0.5 \text{ cm} = 0.005 \text{ m}$

Pressure, $P = \frac{F}{A} = \frac{mg}{\pi r^2}$

$$= \frac{50 \times 9.8}{3.14 \times (0.005)^2}$$

$= 6.2 \times 10^6 \text{ N/m}^2$

107. (*a*) Relative density $= \frac{\text{Density of substance}}{\text{Density of water}}$

$$10.8 = \frac{\text{Density of silver}}{1.0 \times 10^3}$$

Density of silver $= 10.8 \times 10^3 \text{ kg/m}^3$

117. (*a*) There are two free surfaces of the film.

So, surface tension, $T = \frac{F}{2l}$

$\Rightarrow \quad T \times 2l = F$

$\Rightarrow \quad T \times 2l = Mg$

$$T = \frac{Mg}{2l} = \frac{1.5 \times 10^{-2}}{2 \times (30 \times 10^{-2})}$$

$= 0.025$ N/m

137. (*d*) Given, Celsius, $T_C = -30^\circ$ C, $T_F = ?$

$$\because \quad \frac{T_C}{5} = \frac{T_F - 32}{9}$$

$$\therefore \quad \frac{-30}{5} = \frac{T_F - 32}{9}$$

$-6 \times 9 = T_F - 32$

$-54 = T_F - 32$

$T_F = 32 - 54$

Fahrenheit $= -22^\circ$ F

139. (*a*) $\quad T_C = -40^\circ$ C

$$\because \quad \frac{T_C}{5} = \frac{T_F - 32}{9}$$

$$\therefore \quad \frac{-40}{5} = \frac{T_F - 32}{9}$$

$-72 = T_F - 32$

$T_F = 32 - 72$

$T_F = -40^\circ$ F

187. (*b*) Given $l_1 = l$, $l_2 = 4l$ and $T' = ?$

$\because \quad T \propto \sqrt{l}$

$$\frac{T'}{T} = \sqrt{\frac{l_2}{l_1}}$$

$\Rightarrow \quad T' = \sqrt{\frac{4l}{l}}\, T$

$\Rightarrow \quad T' = 2T$

197. (*c*) Given, $v = 330$ m/s, $f = 550$ Hz

and $\quad \lambda = ?$

$\because \quad v = f\lambda$

$\therefore \quad 330 = 550 \times \lambda$

$\Rightarrow \quad \lambda = \frac{3}{5} = 0.6$ m

201. (*b*) Given, $I = 100\, I_0$

$\because \quad \beta = 10 \log_{10}\left(\frac{I}{I_0}\right)$

$\therefore \quad \beta = 10 \log_{10}\left(\frac{100\, I_0}{I_0}\right)$

$= 2 \times 10 \log_{10} 10$

$= 20$ dB

Total noise level = (20 dB + 20 dB)

= 40 dB

213. (*b*) According to the formula, $T = 2\pi\sqrt{\frac{l}{g}}$

Since, the time period of a simple pendulum does not depend on mass.

Therefore, the time period will be remain same.

220. (*c*) Total distance travelled by the sound in going to the surface and back

= speed × time

So, the distance of the wall from the sound source

$= \frac{1}{2} \times$ speed × time

$= \frac{1}{2} \times 340 \times 0.3$

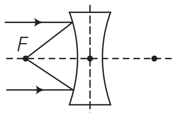
$= 51$ m

232. (*b*) We know that lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$\Rightarrow \quad \frac{1}{v} + \frac{1}{\infty} = \frac{1}{f}$

$\Rightarrow \quad v = f$



Clearly, image will be virtual diminished at focus.

233. (*a*) Given, $f = -25$cm (for concave lens)

Power of the lens,

$\because \quad P = \frac{1}{f(\text{m})} = \frac{100}{f\,(\text{cm})} = \frac{100}{25} = 4\text{D}$

So, the nature of lens is concave lens of power 4D.

236. (*d*) Given, the focal length of lens

$f = 80$ cm

The power of the lens,

$$P = \frac{100}{f\,(\text{cm})} = \frac{100}{80} = 1.25\text{ D}$$

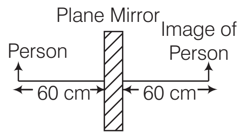
257. (*a*) Given, $f_o = 30$ cm, $f_e = 10$ cm

$\because \quad L = f_o + f_e$

$\therefore \quad L = 30 + 10 = 40$ cm

266. (*b*) The distance between the person and his image in the plane mirror

$= 60 + 60 = 120$ cm $= 1.2$ m



269. (*d*) Myopia is the defect of human eye by virtue of which one cannot see clearly far-off object.

Remedy : using concave lens.

Power of lens $(P) = \frac{1}{f}$

Here, $P = -1.25$ D

$\therefore \quad P = \frac{1}{f}$

$\Rightarrow \quad f = \frac{1}{P} = \frac{-1}{1.25} = \frac{-4}{5}$ m

$\Rightarrow \quad f = -\frac{4}{5} \times 100$ cm $= -80$ cm

or $\quad f = -80$ cm

Since, focal length of the lens is negative, so the lens is concave in nature.

273. (*a*) The resistance of wire

$R = \frac{V}{i}$

$\Rightarrow \quad R = \frac{0.1}{2} = 0.05\,\Omega$

281. (*d*) According to the question,

$R_1 + R_2 = 9 \qquad \ldots\text{(i)}$

and $\quad \frac{R_1 + R_2}{R_1 R_2} = \frac{1}{2}$

$\Rightarrow \quad \frac{R_1 R_2}{R_1 + R_2} = 2$

$\Rightarrow \quad R_1 R_2 = 18 \qquad \ldots\text{(ii)}$

We have,

$(R_1 - R_2)^2 = (R_1 + R_2)^2 - 4R_1R_2$

$= (9)^2 - 4 \times 18 = 81 - 72$

$(R_1 - R_2)^2 = 9$

$\Rightarrow \quad R_1 - R_2 = 3 \qquad \ldots\text{(iii)}$

On solving Eqs. (i) and (iii), we get

$R_1 = 6\ \Omega$ and $R_2 = 3\ \Omega$

284. (*d*) Power of electric bulb, $P = \frac{V^2}{R}$

$\therefore$ So, the resistance of bulb, $R = \frac{V^2}{P}$

$R = \frac{240 \times 240}{60}$

$\Rightarrow \quad R = 960\ \Omega$

292. (*c*) The effective resistance

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$\Rightarrow \quad \frac{1}{R} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$

$\frac{1}{R} = \frac{1+1+1}{3}$

$\Rightarrow \quad \frac{1}{R} = \frac{3}{3}$

$\Rightarrow \quad R = 1\,\Omega$

312. (*a*) In series, the current flows through each conductor is same.

Heat produced by resistance R, $H = i^2Rt$

Resistance of thinner wire, $R_1 = \rho\frac{l}{A}$

Resistance of thicker wire,

$R_2 = \rho\frac{l}{2A} \quad \Rightarrow \quad R_2 = \frac{R_1}{2}$

$\therefore$ Heat produced in the thinner wire,

$H_1 = i^2 R_1 t$

Heat produced in the thicker wire,

$H_2 = i^2 R_2 t$

$H_2 = \frac{i^2 R_1 t}{2}$

$= \frac{H_1}{2}$

So, same amount of electric current would pass through both the pieces producing more heat in the thinner one.

02

CHEMISTRY

After analysing the previous year question papers, we have seen that 40-50 questions are asked from science section, out of these 15-18 questions are asked from Chemistry. From chemistry section, around 2-3 questions are asked from matter, man-made materials, 1-2 questions from atomic structure, electrochemistry, radioactivity, acid base and 4-6 questions from inorganic, environment and its pollution and 1-2 question each from chemical bonding and redox reaction gas law and solution, surface chemistry, chemical thermodynamics, organic chemistry etc.

MATTER

As we look at our surroundings, we see a large variety of things with different shapes, sizes and textures. Everything in this universe is made up of material, which scientists have named “matter”.

MATTER

Anything that has mass, occupies space and can be felt by our one or more sense organs is called matter. Matter is found in five physical states-solid, liquid, gas, plasma and Bose-Einstein condensate. Out of which, three states, i.e. solid, liquid and gas are more common.

- **Solid** They have a fixed shape, volume and high density. The particles in solids are (closely) packed and held in rigid positions. They can be true or crystalline, (i.e. having ordered arrangement of constituent particles to a larger distance) like metals or amorphous or pseudo solids, (i.e. having short range arrangement of constituents particles) like non-metals, glass, most of the polymers, etc.
- **Liquid** They have a fixed volume but no fixed shape and have moderate to high densities. The particles in liquids are loosely packed and free to move. Their thermal conductivity decreases with rise in temperature with the exception of water.
- **Gas** They have neither a fixed shape nor a fixed volume and have very low density. The particles in gas are widely spaced apart and uniformly distributed in the container.

Two More States of Matter

- **Plasma** The state consists of super energetic and super excited particles which are in the form of ionised gases. Due to the presence of plasma, the sun and stars glow.
- **Bose-Einstein Condensate** (BEC) They are formed by cooling a gas of extremely low density to super low temperatures.

Properties of Matter

The properties of matter are as follow :

(i) **Density** It is the measure of relative heaviness of objects having constant volume and is defined as mass per unit volume. Mathematically, it can be written as :

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \text{g /mL}$$

At 4°C, density of water is maximum while its specific volume is minimum.

Gases possess very low density.

(ii) **Compression** Gases are more compressible because of the presence of large intermolecular spaces while solids are least compressible. Compressibility of liquids lies in between of the two other states.

(iii) **Thermal Expansion** It is minimum in case of solids but maximum in case of gases because intermolecular force of attraction is maximum in solids but minimum in gases.

Change of States of Matter

This is phenomenon of change of matter from one state to another and come back to original state by altering the original temperature and pressure.

(i) **Boiling Point** The process of conversion of liquid into vapours by heating is called boiling and the temperature at which the vapour pressure of a liquid becomes equal to the atmospheric pressure is called boiling point. Boiling point increases in the presence of impurity.

At high altitude, boiling point decreases because atmospheric pressure is low. Hence, food takes more time for cooking at higher altitudes. Similarly, in pressure cooker, food cooks early due to elevation in boiling point.

(ii) **Melting Point** The process of conversion of a solid into liquid by heating is called melting and the temperature at which a solid state to convert into its liquid state is called melting point.

(iii) **Sublimation** It is the process of heating a solid, so that it converts directly into gas. e.g. naphthalene, camphor.

(iv) **Freezing** It is the process of conversion of a liquid into solid at its freezing point and melting is the process of conversion of a solid into liquid at its melting point.

(v) **Evaporation** It is a process of conversion of a liquid into vapour at any temperature, while boiling is the conversion of a liquid into vapour at its boiling point. Evaporation causes cooling as the liquid takes energy for evaporation from surroundings.

(vi) **Latent Heat** When heat is supplied to a body either at its melting point or boiling point, the temperature of the body does not change. In this condition, heat supplied to the body is used up in changing its state and is called the latent heat.

$$\text{Latent heat } (L) = \frac{\text{Quantity of heat } (Q)}{\text{Mass } (m)}.$$

Its SI unit is Jkg^{-1}.

CLASSIFICATION OF MATTER

On the basis of chemical composition, matter is classified in the following manner:

Elements

The term element was first used by Robert Boyle in 1661. These are the simplest form of matter and therefore, cannot be split into simpler substances by any physical or chemical methods, e.g. sodium (Na), iron(Fe), mercury (Hg), etc. There are 118 elements known at present, out of which 98 occur in nature, while the remaining elements have been prepared artificially.

Compounds

These are made up of two or more elements combined in a fixed ratio. A compound cannot be separated into its components by physical methods.

- The properties of a compound are entirely different from those of its constituent elements.
- A compound has a fixed melting point, boiling point. A compound is a homogeneous substance. e.g. water, ammonia, sugar, etc.

Mixtures

A mixture is a substance which is made up of two or more elements or compounds, chemically combined together in any ratio (so no definite formula).

These can be separated into its constituents by the physical methods. A mixture does not have a fixed melting point, boiling point and shows the properties of its constituents.

There are two types of mixtures:

1. **Homogeneous Mixtures** In this type of mixture, the constituents are uniformly distributed through out. e.g. salt solution, sugar solution, air etc. These are also called true solutions.
2. **Heterogeneous Mixtures** In this type of mixture constituent does not have uniform composition. They may or may not settle on standing for a long time. They exhibit Tyndall effect. e.g. colloidal solutions, mixture of salt and sugar, iodised salt (mixture of potassium iodide and common salt) etc.

PHYSICAL AND CHEMICAL CHANGES

In physical changes, only the physical properties of matter like colour, hardness, density, etc., changes while the chemical properties and composition remain the same.

Crystallisation, boiling, dissolution of salt and sugar, vaporisation, melting of ice etc., are examples of physical change.

In chemical changes, the chemical composition as well as chemical properties of the matter changes and a new substance is formed. Burning of any substance, photosynthesis, ripening of fruits, etc., are examples of chemical change. Physical changes are reversible (i.e. can be reversed to obtain the original substance) while chemical changes are irreversible.

Laws of Chemical Combination

Combination of elements/atoms to form compounds is governed by following basic laws:

(i) **Law of Conservation of Mass** This law was put forth by **Antoine Lavoisier in 1789**. It states that, matter can neither be created nor destroyed but can be converted from one form to another.

In a chemical reaction, total mass of reactants is equal to total mass of products. In terms of energy, it is called law of conservation of energy.

(ii) **Law of Definite Proportions** A pure compound always contains same elements combined in same proportions by mass, whatever be its source. This is called law of definite proportions by weight and is given by **Joseph Proust**, e.g. water obtained from any source like tap water, well water, etc., always contain hydrogen and oxygen in 1:8 by mass.

(iii) **Law of Multiple Proportions** This law was proposed by **John Dalton in** 1803 . It states that when the fixed mass of an element combines with different masses of another element, then the masses of other elements (in two or more compounds) are in the ratio of small whole numbers..

e.g. H_2O 2 g : 16 g

H_2O_2 2 g : 32 g

or 1 g : 32 g.

Thus, ratio of O combining with same mass of H = 1 : 2.

(iv) **Gay Lussac's Law of Gaseous Volumes**The volume of reactants and products in large number of chemical reactions are related to each other by small integers, provided the volumes are measured at same temperature and pressure. These statements are considered as the law of definite proportions by volume given by Gay-Lussac.

MOLECULES

The term molecule was given by Avogadro. It is the smallest particle of a compound which can exist in free state. All the properties of a compound depend on its constituent molecules. Molecules of a compound contain two or more different types of atoms in a fixed ratio. e.g. ammonia (NH_3), water (H_2O), etc.

- **Molecular mass** is the mass of a molecule. It is an additive property and is calculated by adding the atomic masses of total atoms present in a molecule.

  e.g. $NH_3 = 14 + (1) \times 3 = 17$.

- **Equivalent weight** is obtained by dividing molecular or atomic mass by valency.

$$\text{Equivalent weight} = \frac{\text{Molecular mass or atomic mass}}{\text{Valency}}$$

(i.e. equivalent weight is affected by change in valency)

Mole Concept

The number of molecules present in 12 g of C-12 atom is called one mole. 1 mol $= 6.022 \times 10^{23}$ = Avogadro's number

- $$\text{Number of moles} = \frac{\text{Mass (in g)}}{\text{Atomic or molecular mass (g mol}^{-1})}$$

$$1 \text{ mol of atom} = 6.022 \times 10^{23} \text{ atoms}$$

$$1 \text{ mol of molecules} = 6.022 \times 10^{23} \text{ molecules}$$

- **Atomic mass** is the relative mass as compared with an atom of C-12 and is expressed in amu.

  (1 amu $= 1.6 \times 10^{-24}$ g).

ATOMIC STRUCTURE

Atoms are the basis of chemistry, even they are the basis for everything in the universe. **Maharishi Kanad** was one of the first person to propose that matter is made up of every small particles called **parmanu**. **John Dalton** called the particles by the name of atom.

DALTON'S ATOMIC THEORY

In 1808, **John Dalton** gave a theory, called atomic theory of matter which is based upon laws of chemical combination.

According to this theory,

- Atom is the smallest indivisible particle of an element, i.e. it can neither be created nor be destroyed.
- Atoms of different elements differ in mass, size and chemical properties.
- Atoms of same or different elements combine together to form compound or molecule. The 'number' and 'kinds' of atoms in a given molecule is fixed.
- Atoms of the same elements can combine in more than one ratio to form different compounds.

Atom and Molecule

Atom is the smallest particle of the element that can exist independently and retain all its capable of independent chemical reactions. Atoms are made up of subatomic particles like electrons, protons, neutrons, meson, positron, etc.

A molecule is the smallest particle of an element or a compound existence under ordinary conditions.

Properties of Atom

The properties of atom are as follow:

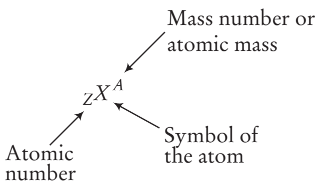
- **Atomic Number** For an element, it is the number of protons present inside the nucleus of its atom.

  Atomic number (Z) = number of protons
  = number of electrons
  (in case of neutral atom)

- **Mass Number** It is the total number of protons and neutrons (i.e nucleons) present in one atom of an element.

  Mass number (A)
  = number of protons + number of neutrons
  = atomic number + number of neutrons
  = number of electrons + number of neutrons

Representation of an Atom



Discovery of Subatomic Particles

The subatomic particles are discovered in many ways:

Cathode Rays

These rays travel in straight line and cast shadows of metallic object placed in their path. In the presence of electrical or magnetic field, the behaviour of cathode rays are similar to that of negatively charged particles, called **electrons.**

The first precise measurement of the charge on the electron was made by **Robert A. Millikan** in 1909.

Anode Rays

In 1886, **Engen Goldstein** discovered the canal rays. These rays travel in straight line and can also produce mechanical effects. These are positively charged, i.e. made up of **protons**.

Properties of Subatomic Particles

The properties of subatomic particles are as follow :

- **Electron** ($_{-1}e^0$) It is negatively charged particle. It was discovered by J. J. Thomson in 1897.

  Charge on electron $= -1.6 \times 10^{-19}$C
  Mass of electron $= 9.1 \times 10^{-31}$g

  The magnitude of negative charge on electron was determined by Millikan in his oil drop experiment.

- **Proton** ($_1H^1$) It was discovered by E. Goldstein. It is positively charged particle.

  Charge on proton $= 1.6 \times 10^{-19}$C
  Mass of proton ($_1H^1$) $= 1.6 \times 10^{-27}$kg

- **Neutron** ($_0n^1$) It was discovered by Chadwick. It has neutral particle, i.e having no charge.

  Mass of neutron $= 1.6 \times 10^{-27}$kg

  Hydrogen and protium is the only atom that does not possess neutrons.

Subatomic Particles

Particle	*Discoverer*	*Charge*	*Mass*
Electron	Thomson	–	1
Proton	Rutherford	+	1836
Neutron	Chadwick	0	1836
Meson	Yukawa	+, 0, –	273.8
Positron	Anderson	+	1
Neutrino	Fermi	0	20.04

NUCLEUS

Atom consists of a heavy and positively charged part at its centre, called the nucleus (dia $= 10^{-4}$m). Nucleus was discovered by Rutherford by α-particle scattering experiment. The negatively charged electrons revolve around the nucleus in well defined orbits. A nucleus consists of positively charged protons and electrically neutral neutrons.

Planck's Quantum Theory

(Particle Nature of Electromagnetic Radiation)

The radiant energy which is emited or absorbed discontinovsly in the form of small discrete packets of energy known as quantum and in care of light, the quantum of energy is called photon.

$$E = h\nu \qquad (c = \nu\lambda)$$

$$E = \frac{hc}{\lambda}$$

where, h = Planck's constant $= 6.63 \times 10^{-34}$ J-s

E = Energy of photon or quantum

If n is the number of quanta of a particular frequency and E_T be the total energy, then $E_T = nh\nu$. The energy possessed by one mole of quanta (or photon), i.e. Avogadro's number (N_0) of quanta, is called one Einstein of energy i.e., 1 Einstein of energy $(E) = N_0 h\nu = N_0 \frac{hc}{\lambda}$

Different Atomic Species

(*i*) **Isotopes** Atoms of the same element having the same atomic number (Z) but different mass number (A) are called isotopes,
e.g. ${}_1H^1$ (protium), ${}_1H^2$ (deuterium or heavy hydrogen) and ${}_1H^3$ (tritium) are isotopes of hydrogen.
${}_6C^{12}$, ${}_6C^{13}$, ${}_6C^{14}$ are isotopes of carbon.
${}_8O^{16}$, ${}_8O^{17}$, ${}_8O^{18}$ are isotopes of oxygen.
${}_{10}Ne^{20}$, ${}_{10}Ne^{21}$, ${}_{10}Ne^{22}$ are isotopes of neon.

(*ii*) **Isobars** Atoms of different elements having the same mass number (A) but different atomic number (Z) are called isobars, e.g. ${}_1H^3$ and ${}_2He^3$; ${}_{18}Ar^{40}$, ${}_{19}K^{40}$ and ${}_{20}Ca^{40}$; ${}_{52}Te^{130}$, ${}_{56}Ba^{130}$ and ${}_{54}Xe^{130}$, etc.

(*iii*) **Isoelectronic Species** Species having the same number of electrons but different nuclear charge are known as isoelectronic species. They also have same bond order. e.g. Mg^{2+} and Na^+, etc., are isoelectronic species as both have 10 electrons.

(*iv*) **Isotones** These have the same number of neutrons. e.g. ${}_6C^{14}$, ${}_7N^{15}$.

Bohr-Bury Scheme

It shows the distribution of electrons in different shells, i.e. orbitals or paths of different and definite energies in which the electrons revolve. According to this scheme,

- A shell can have a maximum of $2n^2$ electrons (where, n = number of shells). e.g. the maximum number of electrons in
  K (or first) shell $= 2n^2 = 2(1)^2 = 2$
  L (or second) shell $= 2n^2 = 2(2)^2 = 8$ and so on.
- The outermost shell cannot have more than 8 electrons.
- Electrons enter in the new shell only after filling the previous one completely.

Quantum Numbers

Quantum numbers are just like address of electrons. There are four types of quantum numbers which are given below

(*i*) Principal quantum number $(n) = 1, 2, 3, 4, \dots$

(*ii*) Azimuthal quantum number $(l) = 0$ to $n-1$ for a given value of n.

(*iii*) Magnetic quantum number (m or m_l) $= -1$ to $+1$ including '0' for a given value of m.

(*iv*) Spin quantum number (s or m_s) $= +\frac{1}{2}, -\frac{1}{2}$ for a given value of m.

Electronic Configuration

It is the arrangement of electrons in various shells, sub-shells and orbitals in an atom. It is written as 2,8,8,18,32.
e.g. the electronic configuration of

	K	L	M	N
Sodium (Na_{11})	2,	8,	1,	0
Calcium (Ca_{20})	2,	8,	8,	2

RADIOACTIVITY

All heavy elements and a few of lighter elements have naturally occuring isotopes, which possess the property of radioactivity.

These isotopes have unstable nuclei and attain stability through the phenomenon of radioactivity.

The unstability results in the emission of a complex type of powerful radiations known as alpha (α), beta (β) and gamma (γ) rays.

RADIOACTIVITY

Radioactivity was discovered by a French physicist **Henri Becquerel** in 1896. However, the term radioactivity was given by Marie Curie, the scientist who got Nobel Prize twice (for physics and chemistry).

The spontaneous emission of invisible radiations by disintegration of heavy elements into comparatively lighter elements is called **radioactivity**. The invisible rays emitted by radioactive elements consists of the following particles:

(*i*) **Alpha (α) particles**, i.e. ${}_2He^4$ (+ 2 unit charge and mass four units) They are deflected towards negative plate in the electric and magnetic field and have very high ionising power.

$${}_ZX^A \xrightarrow{\alpha\text{- particle}} {}_{Z-2}Y^{A-4}$$

(*ii*) **Beta (β) particles**, i.e. electrons (−1 charge and zero mass) They are deflected towards positive plate in the electric and magnetic field.

$${}_ZX^A \xrightarrow{\beta-\text{particle}} {}_{Z+1}Y^A$$

(*iii*) **Gamma** (γ) **rays** (no charge and no mass) They are not deflected from their path in the electric or magnetic field. These are electromagnetic radiations and have very high penetrating power. Emission of γ-rays is the secondary effect of radioactive charge.

Note

1. Stable nuclei are those for which number of neutrons and protons are equal.
2. The time taken by half of the atoms of a radioactive element to disintegrate is called its **half-life**. Its unit is time^{-1}.
3. X-rays were discovered by Roentgen in 1896. These are electromagnetic waves of very short wavelength and are used to detect cracks in fractured bones.
4. If the energies of α, β and γ-particles is same, then penerating power $\alpha < \beta < \gamma$.

Group Displacement Law

This law was put forward by Soddy, Fajan and Russel on the basis of following fact :

(*i*) When a radioactive element emits an α-particle, the atomic number of the resulting nuclide decreases by 4 units e.g.

$$^{238}_{92}\text{U} \longrightarrow \, ^{234}_{90}\text{Th} + \, ^{4}_{2}\text{He}$$

(*ii*) When a radioactive element emits a β-particle, the atomic number of the resulting nuclide increases by one unit but the atomic mass remains unchanged e.g.,

$$^{238}_{92}U \longrightarrow \, ^{238}_{93}Np + \, ^{0}_{-1}e$$

NUCLEAR REACTIONS

These are of following two types:

Nuclear Fission

The splitting of a heavy nucleus into two smaller of nearly comparable masses and release of about 200MeV of energy is called nuclear fission. Nuclear fission was discovered by 'Hahn' and 'Strassmann' in 1939.

- Atom bomb is the result of uncontrolled nuclear fission.
- The device in which controlled nuclear fission (chain reactions) is carried out is called **nuclear reactor**. The fission is controlled by absorbing neutrons by using cadmium or boron rods.
- Heavy water (D_2O, molecular weight 20) and graphite are used as moderator for slowing down the fast moving neutrons. U^{235} is used as a nuclear fuel.

Nuclear Fusion

The union of (two or more) lighter nuclei to form a heavier nucleus is called the nuclear fusion. It is also accompanied by release of energy because the total mass of products is lesser than the total mass of reactants.

- Nuclear fusion occurs only at extremely high temperature ($> 10^6$K), so it is also called **thermonuclear reactions**.
- **Hydrogen bomb** (mixture of deuterium oxide and tritium dioxide) is the result of nuclear fusion.
- Source of solar (sun) and stellar energy is nuclear fusion.
- The source of emission of large amount of energy during nuclear fission or fusion processes is conversion of mass into energy. It is given by the relation, $E = mc^2$ and calculated in MeV.

Applications of Radioisotopes

Isotopes of all the known elements with $Z > 83$ are radioactive and are called radioisotopes. These are used for various purposes, e.g. radiocarbon dating is used to determine the age of dead specimen with C^{14} content by comparing it with C-12 content.

$$N = N_0 \left(\frac{1}{2}\right)^n$$

where, n = total time$(T)/t_{1/2}$

N_0 = ratio of C^{14}/C^{12} in green plant

N = ratio of C^{14}/C^{12} in wood

Rock dating or uranium dating is used to determine the age of rocks or earth. It is based on $_{82}Pb^{206}$ and $_{92}U^{238}$ ratio.

Uses of other Radioisotopes

Radioisotope	*Uses*
I^{131} (Iodine-131)	(*i*) To study the structure and activity of thyroid gland (*ii*) For the treatment of thyroid disease
I^{123} (Iodine-123)	Brain imaging
Co^{60} (Cobalt-60)	Treatment of cancer
Na^{24} (Sodium-24)	To trace the flow of blood
P^{32} (Phosphorus-32)	For leukemia therapy (blood cancer)
C^{14} (Carbon-14)	To study the kinetics of photosynthesis

CHEMICAL BONDING AND REDOX REACTIONS

CHEMICAL BOND

The forces of attraction like interatomic, intermolecular or interionic forces which holds two or more constituents (atoms, molecules or ions) together are called chemical bonds and the process of their combination is called **chemical bonding**.

In general, it is believed that atoms combine in order to complete their octet, i.e. 8 electrons in their outermost shell (with the exception of hydrogen and helium in case of which outermost shell can have a maximum of two electrons).

Ions

An ion is an electrically charged species. A positively charged ion is called **cation**, e.g. Na^+, H^+, Mg^{2+} while a negatively charged ion is called **anion**, e.g. Cl^-, F^-, I^-.

- A cation contains less electrons than a normal atom while an anion contains more electrons than a normal atom.
- When an atom loses one electron, it carries one positive charge, i.e. converted into cation, and if it gains one electron, it carries one negative charge, i.e. converted into anion.
- There is no change in atomic number or number of protons when an atom forms ion.

Valence Electrons

An electron of an atom, located in the outermost shell (valence shell) of the atom, that can be transferred to or shared with another atom is called valence electrons.

- Elements having same number of valence electrons have similar chemical properties.
- Elements having 1, 2 or 3 valence electrons in their atoms are **metals** and form **cation** while those having 4, 5, 6 or 7 are non-metals and form **anion**.
- Elements with 8 valence electrons (except H and He which have 2 valence electrons) are inert, i.e. unreactive and exist as neutral atom.
- If number of valence electrons = 1, 2 or 3
  Valency = Number of valence electrons
  If number of valence electrons ≥ 4
  Valency = 8 - number of valence electrons

Types of Chemical Bonds

Chemical bonds can be of following types depending upon their mode of formation:

Electrovalent or Ionic Bond

The bond formed by the transfer of electrons from one atom (usually metal) to another (usually non-metal) is called electrovalent bond and the compound is called electrovalent or ionic compound. In other words, it is an intercoulombic forces of attraction between two oppositely charged species. The number of electrons transferred shows the electrovalency of the atom.

The properties of electrovalent compounds are as follow:

- They are usually crystalline solids, i.e. have definite shape and are hard and brittle.
- They have high melting and boiling points.
- They conduct electricity when dissolved in water or in molten state due to the presence of free or mobile ions.
- These are soluble in water and insoluble in organic solvents like alcohol, benzene, etc.
- If the electronegativity difference of two atoms is 1.7, the bond between them is 50% ionic.

Some Electrovalent Compounds

Name	*Formula*	*Ions present*
Aluminium oxide (alumina)	Al_2O_3	Al^{3+} and O^{2-}
Ammonium chloride	NH_4Cl	NH_4^+ and Cl^-
Copper sulphate	$CuSO_4$	Cu^{2+} and SO_4^{2-}
Magnesium chloride	$MgCl_2$	Mg^{2+} and Cl^-
Magnesium oxide	MgO	Mg^{2+} and O^{2-}
Sodium chloride	$NaCl$	Na^+ and Cl^-
Sodium hydroxide	$NaOH$	Na^+ and OH^-

Covalent Bond

The bond formed by the sharing of electrons between two atoms of same or different elements is called **covalent bond** and the compound is called **covalent compound**. When a non-metal combines with another non-metal, a covalent bond is formed. Covalent bond may be single, double or triple depending upon the number of sharing pairs of electrons. The number of electrons shared shows the covalency of the element.

The properties of covalent compounds are as follow:

- Normal covalent compounds are gases or liquid under ordinary conditions of temperature and pressure.
- These do not conduct electricity (due to the absence of free ions) and are insoluble in water but dissolve in organic solvents.
- Graphite although is covalent but conducts electricity because of the presence of free electrons (electrical conduction).
- These show stereoisomerism because covalent bond is directional in nature.

Some Covalent Compounds

Name	*Formula*	*Elements present*
Alcohol (ethanol)	$C_2H_5(OH)$	C, H and O
Ammonia	NH_3	N and H
Acetylene (ethyne)	C_2H_2	C and H
Carbon disulphide	CS_2	C and S
Carbon tetrachloride	CCl_4	C and Cl
Cane sugar	$C_{12}H_{22}O_{11}$	C, H and O
Ethylene	C_2H_4	C and H
Glucose	$C_6H_{12}O_6$	C, H and O

Coordinate or Dative Bond

This type of bond is formed by one sided sharing of one pair of electrons between two atoms. The atom having complete octet which provides the electron pair for sharing is known as donor atom. The other atom which accepts the electron pair is called the **acceptor atom**,

e.g. $NH_3 \longrightarrow BF_3, NH_4^+$

Hydrogen Bond

This bond is formed due to the electrostatic force of attraction between hydrogen atom highly electronegative atom.

i.e. N, O or F and any other electronegative atom which is present in the same or different molecules.

It is mainly of two types:

(i) **Intermolecular H-bonding** (e.g. H_2O, HF, NH_3 molecule). It occurs between different molecules of a compound.

(ii) **Intramolecular H-bonding** (e.g. *o*-nitrophenol)

It occurs with in different parts of a same molecule.

- Molecules having O—H, N—H or H—F bond show abnormal properties due to H—bond formation.

Redox Reactions

The reactions which involve both oxidation and reduction as its two half reactions are called **redox reactions**. When the same element is oxidised or reduced simultaneously, the reaction is called disproportionation reaction.

Oxidation

The process which involves gain of oxygen or any other electronegative element or loss of hydrogen or loss of one or more electrons (de-electronation) from an atom, ion or molecule is called oxidation,

e.g. $Mg \longrightarrow Mg^{2+} + 2e^-$.

The positive valency of an element increases by its oxidation.

RANCIDITY

The oils or fats containing materials get oxidised in the presence of oxygen or air and produce foul odour and taste. This process is called rancidity. It is the source of destructive free radicals. A sliced apple turns brown if kept open for sometimes due to oxidation of iron present in the apple. The chips packets are filled with inert gas nitrogen to protect them from being oxidised (or rancid).

Reduction

The process which involves the gain of hydrogen or electropositive element or one or more electrons (electronation) or loss of oxygen by an atom, ion or molecule is called reduction.

e.g. $S + 2e^- \longrightarrow S^{2-}$.

Reduction involves decrease in the positive valency of an element.

Oxidising Agent (Oxidant)

It is a substance which accepts electron in the chemical reaction, i.e. electron acceptors are oxidising agent. All the positively charged species behave like oxidising agents. Oxidising agents are Lewis acids.

Reducing Agent (Reductant)

The substance which donates electron in a chemical reaction is called reducing agent, i.e. electron donors are reducing agents. All the negatively charged species behave like reducing agents. Reducing agents are Lewis bases.

Some Oxidising and Reducing Agents

Oxidising agents	*Reducing agents*	*Both oxidising and reducing agents*
$KMnO_4$, $Cr_2O_7^{2-}$, H_2SO_4, HNO_3, O_2, CO_2 etc.	Na, Al, Fe, Zn, LiH, NaH, $(COOH)_2$ etc.	H_2O_2, SO_2, HNO_2, $NaNO_2$, O_3, Na_2SO_3 etc.

Oxidation State

It is the real or imaginary charge which an atom appears to have in its combined state. Oxidation state of an element may be positive, negative, zero or fractional.

Rules for Determining Oxidation State

- The oxidation state of an element in its free or uncombined state is zero. e.g. oxidation state of O in O_2 and O_3 is zero.
- Oxidation state of hydrogen in most of its compounds is +1 but in metallic hydrides it is –1.
- Oxidation state of oxygen in most of its compounds is – 2. Except peroxides (–1), superoxides (–1/2) and oxygen fluorides (+2 or +1).

- Oxidation state of elements of IA, IIA and IIIA subgroups in their compounds are + 1, + 2 and + 3, respectively.
- Oxidation state of any ion is equal to its charge present on it.
- The algebraic sum of oxidation states of all the elements in the neutral molecule is zero.
- The algebraic sum of the oxidation states of all elements present in polyatomic ion is equal to the charge on the ion.
- Oxidation state of fluorine (F) is always –1.

Note Compounds of all elements with oxygen are called oxides with the exception of fluorine in case of which it is called fluorides, e.g. oxygen fluoride (OF_2).

GAS LAWS AND SOLUTIONS

GAS LAWS

Gases exhibit dependency on temperature, pressure, volume and mass. These interrelations can be analysed by gas laws.

At STP/NTP, temperature, $(T) = 273.15$ K, pressure $(p) = 1$ atm $= 101.35$ kPa, volume, $(V) = 22.4$ L mol^{-1}

Some gas laws are given below

Boyle's Law (Robert Boyle-1662)

- At constant temperature, the volume of fixed mass of a gas is inversely proportional to its pressure.

$$V \propto \frac{1}{p} \quad \text{[at constant temperature and moles } (n)\text{]}$$

or $\quad p_1V_1 = p_2V_2$

Charles' Law (Jacques Charles-1787)

- At constant pressure, volume of fixed mass of a gas is directly proportional to the absolute temperature.

$$V \propto T \quad \text{(at constant pressure and moles)}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Gay Lussac's Law (J.Gay Lussac-1802)

- At constant volume, the pressure of fixed mass of a gas is directly proportional to its temperature.

$$p \propto T$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2} \quad \text{[at constant volume and moles]}$$

Avogadro's Law (Amedeo Avogadro-1811)

- At constant temperature and pressure, the volume of any gas is directly proportional to the number of moles of gas.

$$V \propto n \quad \text{[at constant } T \text{ and } P\text{]}$$

or $$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

GAS EQUATION

The three laws (Boyle's, Charle's and Avogadro's law) can be combined together in a single equation which is known as ideal gas equation.

$$V \propto 1/p \quad \text{(Boyle's law)}$$

$$V \propto T \quad \text{(Charles' law)}$$

$$V \propto n \quad \text{(Avogadro's law)}$$

On combining, $V \propto \dfrac{nT}{p}$

or $\quad V = R\dfrac{nT}{P}$

or $\quad pV = nRT \quad$ (where, R = gas constant)

For 1 mole, $pV = RT$

- Value of gas constant, R depends upon the units of measurements.

$$R = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$$
$$= 8.3143 \text{ J mol}^{-1} \text{ K}^{-1} = 5.189 \times 10^{19} \text{eV mol}^{-1} \text{ K}^{-1}$$
$$= 1.99 \text{ cal mol}^{-1} \text{ K}^{-1}$$

Gas constant, R for a single molecule is called Boltzmann constant (k) or $k = \dfrac{R}{N} = \dfrac{R}{6.022 \times 10^{23}}$

- Density of a gas is directly proportional to pressure and inversely related to temperature,

i.e. $\quad d = pM/RT \quad \left[\because n = \dfrac{Mass\ (m)}{Molecular\ weight\ (M)}\right]$

Graham's Law of Diffusion (or Effusion)

The rate of diffusion (r) of a gas at constant temperature and pressure is inversely proportional to the square root of its molecular mass, M or density, d.

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{d_2}{d_1}}$$

Maxwell's Distribution of Molecular Speeds (Velocities)

Maxwell and Boltzmann proposed that gas molecules are always in rapid random motion colliding with each other and with the walls of container because of which their velocity changes. On increasing temperature, the velocity or molecular motion increases because of which the rate of reaction increases.

SOLUTION

A solution is a homogeneous mixture of two or more substances on molecular level is called true solution or solution. e.g. lemonade, soda water. The substance which is dissolved in a liquid to make a solution is called 'solute' and the liquid in which the solute is dissolved is called 'solvent'.

A true solution does not scatter light and its particles cannot be seen even by microscope. e.g. salt solution, sea water, sugar solution, copper sulphate solution, vinegar, etc.

Depending upon the amount of solute present, the solutions can be classified as :

- A solution, in which more quantity of solute can be dissolved without raising its temperature is called an **unsaturated solution**.
- A solution, in which no more solute can be dissolved at that temperature is called a **saturated solution**.

Concentration of a Solution

The amount of a solute dissolved in unit weight or volume of solution is called strength (concentration) of a solution.

So, concentration of a solution

$$= \frac{\text{Amount of solute (in g)}}{\text{Weight of solution}} = \frac{\text{Amount of solute (in g)}}{\text{Volume of solution}}$$

Different Terms to Express Concentration

Parts Per Million It shows the parts of solute present per million parts of solution, i.e. ppm.

$$= \frac{\text{Mass of solute (in g)} \times 10^6}{\text{Volume of solution (in mL)}}$$

Molarity (M) The number of moles of solute dissolved in one litre of solution is called its molarity.

Thus, $$M = \frac{\text{Weight of solute (in g)} \times 1000}{\text{Molecular weight} \times \text{volume of solution (in mL)}}$$

Molality (m) The number of moles of solute dissolved in 1000 g of a solvent is called its molality.

Thus, $$m = \frac{\text{Weight of solute (in g)} \times 1000}{\text{Molecular weight} \times \text{weight of solvent (in g)}}$$

Normality (N) The number of gram equivalents of solute dissolved in one litre of solution is known as its normality.

$$N = \frac{\text{Weight of solute (in g)} \times 1000}{\text{Equivalent weight} \times \text{volume of solution (in mL)}}$$

➡ **Note** Normality and molarity are affected by temperature as these depend upon the volume whereas molality remains unaffected from temperature change.

Henry's Law

According to this law, 'at particular temperature, the solubility of a gas in a liquid is directly proportional to the pressure of the gas above the solvent.

Its main applications are as follow:

- Soft drinks and soda water bottles are sealed under high pressure in order to increase the solubility of CO_2 in them.
- To minimise the painful effects (bends) accompanying the decompression of deep sea divers, oxygen diluted with less soluble helium gas is used as breathing gas.
- Oxygen diluted with nitrogen cannot be used for this purpose due to high solubility of N_2 in blood.

Osmosis

It is the process of movement of solvent molecules from the solution of low concentration to high concentration through semipermeable membrane.

- If pressure greater than osmotic pressure (pressure require to stop osmosis) is applied on the solution of high concentration, **reverse osmosis** takes place. e.g. desalination of sea water.
- **Isotonic solutions** have same concentration and osmotic pressure. e.g. 0.91% solution of pure NaCl is isotonic with human red blood cells (RBCs).

Diffusion

- Diffusion is the process of spontaneous mixing of different gases and in this process, molecules of a substance move from higher concentration to lower concentration and goes on until a uniform mixture is formed. Its rate is highest for gases and lowest in solids. e.g. the smell of food being cooked, reaches us even from a considerable distance by the process of diffusion. Due to diffusion the scent spread all over the room if the lid is opened.

ACIDS, BASES AND SALTS

ACIDS

The substances which have sour taste and turn blue litmus to red are called **acids**.

- According to Arrhenius acids give hydrogen ion H^+ in an aqueous solution.

$$HCl\,(aq) \longrightarrow H^+(aq) + Cl^-\,(aq)$$

- According to Bronsted Lowry concept, acids donate proton (hydrogen ion).

$$\underset{\text{Acid}}{HCl} + NH_3 \longrightarrow \underset{\text{Conjugate base}}{Cl^-} + NH_4^+$$

(A conjugate acid-base pair differ by a proton)

- According to Lewis concept, acids have tendency to accept electrons, i.e. these behave as electrophiles. e.g. electron deficient species like BF_3 , $AlCl_3$, positive ions like Na^+, K^+ and molecules having multiple bond between dissimilar atoms (e.g. CO_2, SO_2, etc).

Sources of Some Acids

Acid	*Sources*
Citric acid	Lemon, orange, grapes (citrus fruits)
Maleic acid	Unripe apple
Tartaric acid	Tamarind , grapes
Lactic acid	Milk , curd
Acetic acid	Vinegar
Oxalic acid	Tomato , spinach
Hydrochloric acid	Stomach and other chemicals
Formic acid	Red ant sting

- Most of the acids contain hydrogen.
- HCl, H_2SO_4, HNO_3 are mineral acids and are much stronger than organic acids.
- Aqua-regia is a mixture of concentrated hydrochloric acid and concentrated nitric acid in the ratio 3 : 1. It can dissolve inert materials like platinum, gold, etc.

Properties of Acids

- Acids liberate hydrogen gas with more reactive metals.

e.g. $$\underset{\text{Metal}}{Zn} + 2HCl \longrightarrow \underset{\text{Salt}}{ZnCl_2} + \underset{\text{Hydrogen}}{H_2\uparrow}$$

- Acids evolve carbon dioxide (CO_2) gas with metal carbonates and bicarbonates. e.g.

$$\underset{\text{Metal carbonate}}{MCO_3} + \underset{\text{Acid}}{H_2SO_4} \longrightarrow M\,SO_4 + H_2O + CO_2\uparrow$$

- In aqueous solution, they are conductor of electricity due to the presence of free H^+ ions.

BASES

These substances which have bitter taste and turn red litmus to blue are called **bases**.

- According to Arrhenius, bases give hydroxyl ion (OH^-) in aqueous solution. e.g. NaOH, KOH, CsOH,$Mg(OH)_2$, etc.,

$$NaOH\,(aq) \longrightarrow Na^+(aq) + OH^-(aq)$$

- According to Bronsted Lowry concept, bases have a tendency to accept proton, e.g. NH_3, H_2O, etc.

$$\underset{\text{Base}}{NH_3} + H_2O \rightleftharpoons \underset{\text{Conjugate acid}}{NH_4^+} + OH^-$$

- According to Lewis concept, bases have a tendency to donate electron pair. e.g. simple anions like Cl^-, F^- OH^-, etc., molecules containing lone pairs like $\ddot{N}H_3$, $R-\ddot{\underset{\cdot\cdot}{O}}-H$, $R-\ddot{\underset{\cdot\cdot}{O}}-R$, etc.

Properties of Bases

- Base reacts with metal to form salt and liberates hydrogen gas. However, such reactions are not possible with all metals.
- Water soluble bases are called **alkali**. Thus, all alkalis are bases but all the bases are not alkali.
- In aqueous solution and molten state, they are good conductors of electricity.

Acid and Base in a Reaction

Acid-base pair

$$\underset{\substack{\text{Acid}\\\text{(as donates proton)}}}{CH_3COOH} + \underset{\substack{\text{Base}\\\text{(as accepts proton)}}}{H_2O} \longrightarrow \underset{\substack{\text{Conjugate}\\\text{acid}}}{H_3O} + \underset{\substack{\text{Conjugate}\\\text{base}}}{CH_3COO^-}$$

Acid-base pair

- Paraldehyde (a carbonyl compound) is used as a hypnotic.
- Clove oil can be used as indicator.

SALTS

- When acid and base react together, they form salt and water. This reaction is known as **neutralisation reaction,** e.g.

$$\underset{\text{Acid}}{HNO_3} + \underset{\text{Base}}{KOH} \longrightarrow \underset{\text{Salt}}{KNO_3} + \underset{\text{Water}}{H_2O}$$

- When acid and base both are strong, 13.7 kcal energy is released. However, if either the acid or the base is weak, energy released is less than 13.7 kcal because some part of energy is utilised to ionise weak acid or base.

- KNO_3, K_2SO_4, $NaNO_3$, KCl, NaCl, KNO_3 are salts of strong acids and strong bases so, they cannot be hydrolysed and their aqueous solution is neutral.
- Solutions of salts of Cl^-, SO_4^{2-}, NO_3^- with metals other than Na^+ and K^+ are acidic. e.g. NH_4Cl. In other words, salt of a strong acid and a weak base undergoes hydrolysis in water to give acidic solution.
- Solutions of salts of Na^+ and K^+ with anions (except SO_4^{2-}, NO_3^-, Cl^-) are basic, e.g. CH_3COONa.
  In other words, salt of a weak acid and a strong base undergoes hydrolysis in water to give basic solution.

pH SCALE

SPL Sorensen suggested a scale to express the acidity or basicity of an aqueous solution in terms of hydrogen ion concentration which is known as pH scale. This scale is based on the ionic product of water.

The pH of solution is defined as "the negative logarithm to the base 10 of the hydrogen ion concentration in moles per litre",

i.e. $\quad pH = -\log [H^+]$

or in other words, the logarithm to the base 10 of inverse of H^+ ion concentration in the solution is its pH. $\quad pH = \log \frac{1}{[H^+]}$

- The negative power to which 10 must be raised in order to express the H^+ ion concentration of a solution in moles per litre, is its pH.
  i.e. $\quad [H^+] = 10^{-pH}$
- In pure water or in neutral solution,
  $[H^+] = 10^{-7}$ so, pH = 7
- In an acidic solution, $[H^+] > 10^{-7}$ so, pH < 7
- In an alkaline solution, $[H^+] < 10^{-7}$ so, pH > 7

pH of Some Common Substances

Soft drinks	2.0-4.0
Tears	7.4
Lemon	2.2-2.4
Sea water	8.5
Human urine	4.8-8.4
Cow's milk,saliva	6.5
Rain water	6.0
Human blood	7.36-7.42

Importance of pH in Everyday Life

- Bee or ant sting (formic acid) leaves an acid into the skin which causes pain and irritation. Use of a mild base like baking soda ($NaHCO_3$) or calamine (zinc carbonate) is a remedy for it.
- Wasp sting introduced some base, so a mild acid is a remedy.
- Falling of pH of the mouth below 5.5 because of substances like chocolates and sweets lead to tooth decay which is naturally controlled by saliva (basic) to some extent but toothpaste being basic maintain this pH by neutralising excess acid.
- Excessive acid (hydrochloric acid) secreted in our stomach which causes acidity, is neutralised by mild bases called antacids like milk of magnesia (or magnesium hydroxide) $[Mg(OH)_2]$, aluminium hydroxide $[Al(OH)_3]$, etc.
- Every crop grows better in a particular pH range. e.g. slightly acidic soil is good for rice, neutral is good for sugarcane while citrus fruits generally grow in alkaline soil.

➨ **Note** Use of strong acid or base is not prescribed as these are corrosive in nature.
- Pickles are kept in glass jar because acid present in them reacts with the metal of metallic pot to give poisonous substances.
- Acidity of soft drinks is regulated by bicarbonate salts.

pOH Scale

The pOH value of an aqueous solution may be defined as the negative logarithm of the hydroxide ion concentration,

i.e. $\quad pOH = -\log [OH^-]$ and pH + pOH = 14.

➨ **Note**

1. When fresh ground water comes in contact of air its pH is slightly reduces. This is because of the presence of CO_2 in air which dissolves in water to give a weak acid, H_2CO_3 (carbonic acid). This acid provides H^+ ions and hence, increases the acidity of water due to which pH reduces.
2. pH of acidic solution increases when a base is added to it and pH of basic solution decreases when an acid is added to it.

Common Ion Effect

Addition of a strong electrolyte in the solution of weak acids or bases, decreases the ionisation of acid or base due to the presence of common ion. This is called common ion effect. e.g.

$$\underset{\text{Weak acid}}{H_2S} \rightleftharpoons 2H^+ + S^{2-} \qquad \ldots(i)$$

$$\underset{\text{Strong acid}}{HCl} \longrightarrow \underset{\substack{\text{Common}\\\text{ion (excess)}}}{H^+} + Cl^- \qquad \ldots(ii)$$

So, the reaction (i) goes in the reverse direction.

CHEMICAL THERMODYNAMICS AND SURFACE CHEMISTRY

THERMODYNAMICS

It deals with the study of heat and energy changes occurring during a physical and chemical processes.

An **open system** can exchange matter as well as energy with its surroundings, e.g. hot tea in a cup.

A **closed system** can exchange energy but not matter with its surroundings, hot water in a closed beaker.

An **isolated system** can exchange neither matter nor energy with its surroundings, e.g. thermos flask.

Thermodynamic Processes

Isothermal Process The process in which temperature of the system remains constant, i.e., $dT = 0$

Maximum work done during the isothermal expansion of 'n' moles of an ideal gas is given as

$$W_{\max} = -2.303\, nRT \log \frac{V_2}{V_1}$$

Adiabatic Process The process in which no heat enters or leaves the system during any step of process, i.e. $dq = 0$.

Isobaric Process The process which takes place at constant pressure, i.e. $dp = 0$.

Isochoric Process The process in which volume of system remains constant, i.e. $dV = 0$.

Cyclic Process In a cyclic process, a system in a given state goes through a series of different processes but finally returns to its initial state, i.e. dE or $dU = 0$.

Spontaneous Process A process, i.e physical or chemical change, which can proceed by itself is known as spontaneous process. There are also some processes which require some initiation before they proceeds spontaneously, i.e. physical or chemical change.

- It is also known as feasible and probable process.
- The process is spontaneous, if free energy change, ΔG is negative.

SURFACE CHEMISTRY

Surface chemistry deals with the phenomenon that occurs at the surface or interfaces.

Catalysis

The term catalyst is given by Berzelius. A catalyst is a substance which alters the rate of reaction without being consumed in it and the phenomenon is known as **catalysis.**

General Characteristics of Catalysts

- A catalyst remains unchanged in mass and chemical composition during a reaction.
- Only a very small amount of catalyst is sufficient to catalyse a reaction. A catalyst does not initiate a reaction.
- A catalyst does not change the equilibrium state of a reversible reaction as it increases the rate of forward as well as backward reaction to the same extent.
- When a reversible reaction is performed in a closed container, a state is reached when the rate of forward and backward reactions become equal. This state is called **chemical equilibrium** or equilibrium state.
- Free energy change ($\Delta G°$) at equilibrium is zero.

Types of Catalyst

1. **Positive and Negative Catalysts** A catalyst which increases the rate of a reaction (in forward direction) is called a **positive catalyst** while which decreases the rate of a reaction is called as **negative catalyst**.
2. **Induced Catalyst** The product of one reaction acts as catalyst for another reaction is called **induced catalyst**.
3. **Catalytic Promoters and Inhibitors** The substance which increases the activity of a catalyst are called **promoters** and which decreases the activity of a catalyst are called **inhibitors**.
4. **Catalytic Poisons** Catalytic poisons destroy the activity of a catalyst completely.

Enzyme Catalysis

The increase in the rate of reactions by the enzyme is known as enzyme catalysis. Since, they increase the rate of reaction occurring in living organisms, they are known by the name **biocatalysts**.

- All enzymes are protein in nature. Enzymes are highly specific in their nature. They are highly sensitive to temperature, i.e. the temperature at which the activity of enzymes is maximum, varies between 25-37°C.
- The rates of enzymatic reactions are very much affected by pH change.

Some Important Catalyst used in Different Processes

Processes	*Catalyst used*
Manufacture of ghee from vegetable oils	Nickel
Haber's process for the manufacture of ammonia	Fe (here Mo, acts as an activator)
Contact process for the manufacture of sulphuric acid	Pt powder
Conversion of proteins into peptide	Pepsin enzyme
Conversion of proteins into amino acids	Erepsin enzyme
Conversion of glucose into ethanol	Zymase enzyme
Conversion of starch into maltose	Diastase enzyme
Formation of vinegar from cane sugar	*Mycoderma aceti*
Conversion of sucrose into glucose and fructose	Invertase enzyme
Conversion of milk into curd	Lactase (lactobacilli)
Ostwald's process for the manufacture of nitric acid (HNO_3)	Platinum (Pt)

COLLOIDS

In a colloid, the size of solute particles is bigger than that of true solution but smaller than that of suspension, i.e. lies in between 1 to 100 nm. A colloidal solution is a heterogeneous system in which one substance is dispersed as very fine particles in another substance, called dispersion medium.

Properties of Colloidal Solutions

The important properties of the colloidal solutions are given below

Brownian Movement The rapid zig-zag movement executed by colloidal particle is termed as Brownian movement.

Tyndall Effect When a strong and converging beam of light is passed through a colloidal solution, its path becomes visible due to scattering of light by particles. It is called **Tyndall effect**. Blue appearance of sea water is an example of Tyndall effect.

Electrophoresis The phenomenon involving the migration of colloidal particles under the influence of electric field towards the oppositely charged electrode, is called electrophoresis.

➠ Colloidal solutions are purified by dialysis.

Emulsions

These are the colloidal solutions of two immiscible liquids in which the liquids act as the dispersed phase as well as the dispersion medium.

Types of Emulsions

There are two types of emulsions

- **Oil in water type** e.g. milk in which tiny droplets of liquid fat are dispersed in water.
- **Water in oil type** e.g. stiff greases, in which water being dispersed in lubricating oil.

Emulsifying Agents

During the preparation of emulsion, a small amount of some substances such as soap, gum, agar and protein etc are added to stabilise the emulsion. These substances are known as emulsifying agents.

- **Gel** is a colloidal system in which dispersed phase is liquid and dispersion medium is solid. e.g. butter, cheese. When a gel is allowed to stand for some time, it loses the dispersed phase (liquid), this phenomenon is called **syneresis** or **weeping of the gel**.
- Fog, clouds, mist are the colloidal systems of liquid in gas.
- Smoke is a colloidal systems of carbon (solid) in air (gas).
- Foams are the colloidal systems in which a gas is dispersed in liquid. e.g. alcohol, castor oil, shaving cream etc.
- Digestion of fats in intestine is aided by emulsification.
- **Coagulation** is the process of precipitation of colloid when it is shaked with some electrolyte. Its applications are
    1. Purification of muddy water by alum.
    2. Ferric chloride and alum are applied on wound to stop bleeding.

➠ **Note**

1. Blue colour of sky is due to the scattering of light by dust particles present in air.
2. The order of energy is 1 calorie > 1 joule > 1 erg.
3. Milk is a natural emulsion while paint is an artificial emulsion.

ELECTROCHEMISTRY

It is the branch of chemistry which deals with inter-conversion of chemical energies and electrical energies. It can be converted into electrical energy by means of an **electrochemical cell** (electrolysis).

Important Terms

- Chemical changes involving production or consumption of electricity are called **electrochemical changes.**
- The substance, which allow the electricity to pass through them are called **electrolytes**. e.g. common salt (NaCl), water (H_2O) etc.
- The substance which do not allow the electricity to pass through them are called **non-electrolytes**. e.g. sugar, wax, naphthalene etc.
- A number of metals such as Na, Mg, Ca and Al and a number of chemicals such as NaOH, Cl_2, Fe etc. are commercially produced by electrochemical methods.

Electrochemical Cell

It is a device, that produces an electric current from energy released by a spontaneous redox reaction (in short, which converts chemical energy into electrical energy).

This kind of cell includes the galvanic cell or voltaic cell. It has two conductive electrodes, i.e. anode or negative electrode (at which oxidation occurs) and cathode or positive electrode (at which reduction occurs). Electrolyte is filled in between the electrodes and contains freely moving ions.

BATTERY

An arrangement of one or more cells connected in a series, is called a battery. It is basically a galvanic cell.

Batteries are of two types :

1. **Primary Batteries** (non-rechargeable) These batteries act as galvanic cell and used only once. e.g. dry cell, mercury cell etc.

2. **Secondary Batteries** (rechargeable) These act as galvanic as well as electrolytic cell and can be reused. e.g. lead storage battery, nickel-cadmium battery.

Some Important Batteries

Battery	*Anode*	*Cathode*	*Electrolyte*	*Used in*
Leclanche cell (Dry cell)	Zinc	Carbon (Graphite) rod	Paste of ammonium chloride (NH_4Cl) and zinc chloride ($ZnCl_2$)	Transistors, clocks
Mercury cell	Zinc-mercury amalgam	Paste of HgO and C	Paste of KOH and ZnO	Hearing aids watches and camera
Lead storage cell	Lead	Lead packed with lead dioxide	38% solution of sulphuric acid	Automobiles, invertors
Nickel Cadmium cell	Cadmium	Nickel Dioxide	OH Solution	Torchlights, Electric Shavers

FUEL CELLS

These are galvanic cells, which use energy of combustion of fuels like hydrogen (H_2), methane (CH_4) , methanol (CH_3OH) etc., as the source to produce electrical energy. e.g. hydrogen-oxygen fuel cell. This cell was used for the first time in **Apollo space programme**. The byproduct of this cell, i.e. water vapours, were condensed and used by the astronauts for drinking.

Advantages associated with fuel cell are

- Fuel cells are more efficient than conventional power stations or batteries etc.
- With a fuel cell there are fewer stages in producing the useful energy, so there is less opportunity to lose potentially useful energy, e.g. waste heat, friction from moving parts etc.
- A hydrogen-oxygen fuel cell is a non-polluting clean fuel since, the only combustion product is water.

Disadvantages associated with fuel cell are

- Fuel cells cannot be used for large scale energy production, so conventional fossil fuels or nuclear power stations still have an important future.
- Hydrogen is a gas and requires a much larger storage volume as compared to fossil fuels like petrol.
- Safe storage is also an issue, especially as it would be stored under high pressure to decrease the storage space required. This immediately makes leaks and, accidents more likely to happen because hydrogen is a highly flammable explosive gas, too easily ignited.
- Efficient large scale technology is not yet developed to produce hydrogen on large scale, e.g. from electrolysis using solar power electricity, photovoltaic power system, wind turbines or hydroelectric power.
- Although water is cheap and plentiful, it requires expensive electrical energy to electrolyse water to split it into hydrogen and oxygen.

ELECTROLYSIS

The process, in which a non-spontaneous reaction is carried out by using electrical energy is called electrolysis and the instrument or container, in which the process of electrolysis is done, is called **electrolytic cell**.

Electrolysis of brine (the water saturated or nearly saturated with salt, usually sodium chloride) gives hydrogen and chlorine. The products are gaseous.

$$NaCl \longrightarrow Na^+ + Cl^-$$

$$H_2O \longrightarrow H^+ + OH^-$$

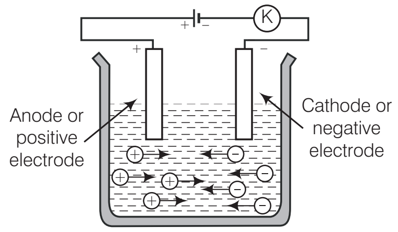
At anode $2Cl^- \xrightarrow{\text{Oxidation}} Cl_2 + 2e^-$

At cathode $2H^+ + 2e^- \xrightarrow{\text{Reduction}} H_2$

$$Na^+ + OH^- \longrightarrow NaOH$$

Net reaction

$$NaCl + 2H_2O \longrightarrow 2NaOH + H_2 + Cl_2$$



Electrolytic cell

➡ **Note** During electrolysis, electrolytes get decomposed at the electrodes.

Electrolysis is used in the following ways :

- In the production of oxygen for spacecraft and nuclear submarines.
- In layering metals to fortify them.
- In production of hydrogen for fuel.
- In electrolytic etching of metal surfaces like tools or knives with a permanent mark or logo.
- In electroplating, electrorefining, electroprinting, electro-metallurgy and industrial preparation.

FARADAY'S LAWS OF ELECTROLYSIS

First Law of Electrolysis

It states that "The amount of electricity (or charge) required for oxidation or reduction depends upon stoichiometry of the electrode reaction."

$$w \propto Q, \qquad w = ZQ = Zit$$

$\because$ Charge (Q) = Current (i) × time (t)

1 Faraday = 96500 C mol^{-1}

Second Law of Electrolysis

It states that the masses of the resulting separated elements are directly proportional to the atomic masses of the elements, when an appropriate integral divisor is applied.

$$W \propto E \qquad or \qquad \frac{W_1}{W_2} = \frac{E_1}{E_2}$$

Electrochemical Series

When metals are arranged in the increasing order of their standard reduction electrode potential, a series is obtained, which is called electrochemical series. It is also called **reactivity series** (as reactivity follows the reverse order). Lower is the value of reduction potential, greater would be its reducing power.

Depending upon the reactivity, the occurrence of metals in the reactivity series may be represented as shown in the figure.

	Element	
Metals above hydrogen	K, Ca, Na, Mg, Al	Highly reactive metals never found in free state
	Zn, Cr, Fe, Ni, Sn, Pb	Moderately reactive metals
	H	
Metals below hydrogen	Cu, Hg, Ag, Pd	Less reactive metals
	Pt, Au	Least reactive metals (found in the free state)

Thus, gold is the least reactive metal and hence, found in native state. The more reactive metals of the series can displace the less reactive metals from their salt solutions.

e.g. $Cu + FeSO_4 \longrightarrow$ No reaction

but $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$

Moreover, metal placed before H can evolve H_2 from dilute acids.

e.g. $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

The metals placed above are good reducing agent and reduces the metal oxides of the metal placed below them.

CORROSION

It is the process of oxidative deterioration of a metal surface by the action of environment to form unwanted corrosive products. e.g. conversion of iron into rust ($Fe_2O_3 \cdot xH_2O$). Corrosion of iron is called **rusting**. Tarnishing of silver (due to the formation of Ag_2S), development of green coating [of $Cu(OH)_2 \cdot CuCO_3$ (basic copper carbonate)] on copper and bronze. It is basically an electrochemical process.

It is accelerated by the presence of impurities, H^+, electrolytes such as NaCl and gases such as CO_2, SO_2, NO, NO_2 etc.

Prevention of Corrosion

- By coating with a suitable material (barrier protection *viz* paint, grease, metals (like Zn, Cu, Ni, Cr, Sn), adherent film of iron phosphate, adherent coating of magnetic oxide (Fe_3O_4).
- By alloying with suitable metals.
- By cathodic protection–the metal is made cathode by application of external circuit.
- By using artificial anode (sacrificial protection), more anodic metal (like Zn, Mg, Al etc.,) is connected to iron article by a wire. In such cases, current flows towards more reactive metal and prevents iron from corrosion.
- By galvanisation (covering of iron article with zinc).

INORGANIC CHEMISTRY

CLASSIFICATION OF ELEMENTS

- In 1896, Mendeleev gave a periodic law. According to which "the properties of elements are the periodic function of their atomic masses".
- There were seven periods (horizontal rows) and eight groups (vertical columns) in the periodic table of Mendeleev having 63 known elements at that time.
- Moseley modified Mendeleev's periodic law and proposed modern periodic law. According to **modern periodic law**, "the properties of elements are periodic function of their atomic numbers."
- There are eighteen vertical columns, known as **groups** and seven horizontal rows, known as **periods**, in the long form of periodic table.
- **Periodic Properties** The properties which are repeated at regular intervals are known as periodic properties. Atomic and ionic size, electron affinity, ionisation potential, electronegativity, electropositive character, acidic or basic character, metallic nature, etc., are some important periodic properties.

Characteristics of Periods

- The number of valence electrons in elements increases from 1 to 8 on moving from left to right in a period.
- The elements in a period have consecutive atomic numbers.
- The valency of the element increases from 1 to 4 and then decreases to 0 (zero) on moving from left to right in a period, with respect to hydrogen.
- Atomic size, electropositive character, metallic character, reducing nature of elements and basic nature of oxides all decrease from left to right in a period.

- Electronegative nature, non-metallic nature, acidic nature of oxides, ionisation potential increases from left to right in a period. In a period, electronegativity and electron affinity also increases from left to right.

Characteristics of Groups

- All the elements of a group of the periodic table have the same number of valence electrons and hence, have almost similar chemical properties.
- Atomic radius, electropositive nature, metallic nature, reducing nature of elements and the basic nature of oxides increase from top to bottom in a group.
- Electronegative nature, ionisation potential, electron affinity, electronegativity, non-metallic nature and acidic nature of oxides decrease down a group with increasing atomic number.

HYDROGEN

- Hydrogen is the first element in the periodic table with atomic number 1 and mass number 1.
- Hydrogen has three isotopes-protium or ordinary hydrogen $({}_1H^1)$, deuterium or heavy hydrogen (${}_1H^2$ or ${}_1D^2$), tritium (${}_1H^3$ or ${}_1T^3$). Tritium is the radioactive isotope of hydrogen.
- Hydrogen is a non-supporter of combustion and burns with 'pop' sound. It is an inflammable gas.

Compounds of Hydrogen

Water is one of the most important compound of hydrogen.

Water

- Water is the neutral oxide of hydrogen. Pure water freezes at 0° C and boils at 100° C. This abnormal high boiling point is due to the association of H_2O molecules through hydrogen bonding.
- Water is a polar compound (dipole moment 1.85 D) and possesses a high dielectric constant, i.e. 81. Its presence is tested by using anhydrous copper sulphate which turns blue in the presence of water.
- Water is a universal solvent because of its tendency to form H-bond and polar nature.
- Water has very high specific heat and thus, widely used as a coolant.
- Water which form lather with soap is called **soft** and which does not form lather is called **hard** water as the ions present in it form scum (precipitate) with the soap.

Hardness of Water

- Hardness of water may be due to the presence of bicarbonates of calcium or magnesium (**temporary hardness**) or due to the presence of chlorides or sulphates of Ca or Mg (**permanent hardness**).

Removal of Hardness of Water

- Temporary hardness is removed by boiling or by adding lime water (Clark's process).
- Permanent hardness is removed by adding sodium carbonate (washing soda,Na_2CO_3) or calgon (sodium hexa metaphosphate, $Na_6P_6O_{18}$ or zeolite which is also called permutit (hydrated sodium aluminium silicate, $Na_2Al_2Si_2O_8 \cdot xH_2O$).
- From sea water, pure water is obtained by a process, called **reverse osmosis**, i.e. by applying pressure higher than the osmotic pressure towards the solution side. This process is also called desalination of sea water.
- Boiling, chlorination and treatment with UV irradiation all kill the microorganisms present in water.

Heavy Water

- **Heavy water** (D_2O, molecular weight = 20) is the oxide of heavy hydrogen. It is used as a moderator in nuclear reactors.
- In hydrogen peroxide (H_2O_2) oxygen is in −1 oxidation state, which is the intermediate oxidation state of oxygen. H_2O_2 behaves as an oxidising, reducing and bleaching agent.
- Lead painting becomes blacken in air due to the formation of PbS by the action of H_2S present in air, H_2O_2 is used to wash it. H_2O_2 oxidises PbS to $PbSO_4$.

Aquifers

An aquifer is an underground layer of water bearing permeable rock, i.e. they have opening through which only liquids and gases can pass.

Clay layers although are poor aquifers. This is because clay minerals are dense impermeable materials. They act as aquiferal, i.e. a layer of material that is almost impenetrable to water.

Metallurgy

It is the process of extraction of metals from their ores is called **metallurgy**.

Terminology Related with Metallurgy

- ***Metals*** occur in the form of minerals in nature.
- ***Gangue*** or matrix are the impurities associated with ore.
- ***Roasting*** is the process of heating ore (mainly sulphide ore) in the excess of air whereas calcination is heating the ore (carbonate or hydroxide ore) in the absence or limited supply of air.
- ***Flux*** is added to gangue to convert it into slag.
- ***Ores*** are the minerals from which metal is extracted conveniently and beneficially.

Some Important Ores and their Uses

Metal	*Ores/minerals*	*Chemical composition*
Sodium	Rock salt Chile salt petre Borax, tincal or suhaga carnallite	$NaCl$ $NaNO_3$ $Na_2B_4O_7 \cdot 10H_2O$
Magnesium	Magnesite Asbestos Carnallite	$KCl, MgCl_2, 6\ H_2O, MgCO_3$ $CaSiO_3 \cdot 3MgSiO_3$ $KCl \cdot MgCl_2 \cdot 6H_2O$
Calcium	Lime stone Gypsum Fluorspar	$CaCO_3$ $CaSO_4 \cdot 2H_2O$ CaF_2
Aluminium	Bauxite Cryolite Feldspar Mica Granite	$Al_2O_3 \cdot 2H_2O$ Na_3AlF_6 $KAlSi_3O_8$ $KAlSi_2O_{10}(OH)_2$ SiO_2 and Al_2O_3 (73 : 14)
Iron	Haematite Magnetite Iron pyrites Siderite	Fe_2O_3 Fe_3O_4 FeS_2 $FeCO_3$
Copper	Copper glance Copper pyrites Malachite Azurite	Cu_2S $CuFeS_2$ $Cu(OH)_2 CuCO_3$ $2CuCO_3 \cdot Cu(OH)_2$
Silver	Silver glance Horn silver Ruby silver	Ag_2S $AgCl$ $Ag_2S \cdot Sb_2S_3$
Gold	Sylvanite	$AuAgTe_4$
Zinc	Zinc blende Calamine Zincite	ZnS $ZnCO_3$ ZnO
Mercury	Cinnabar	HgS
Tin	Cassiterite	SnO_2
Lead	Galena Cerrusite Anglesite	PbS $PbCO_3$ $PbSO_4$
Potassium	Nitre Sylvine	KNO_3 KCl

Alloy

- An alloy is a homogeneous mixture of two or more metals or a metal and a non-metal :

Some Important Alloys and their Uses

Alloys	*Composition*	*Important uses*
Solder	Tin and lead	Soldering
Bronze	Copper and tin	Making utensils, statues, coins, etc.
Type metal	Tin, lead and antimony	Used in printing
Bell metal	Copper, tin	Making bells
Gun metal	Copper, tin and zinc	Gears and bearing
Brass	Copper, zinc	Utensils, condenser, tubes, cartridge caps, etc.
Aluminium bronze	Copper and aluminium	Coins, picture, cheap jewellery, flames
German silver	Copper, zinc, nickel	Utensils, resistance wires
Constantan	Copper, nickel	Electrical apparatus
Dental alloy	Silver, mercury, tin, zinc	For filling teeth
Stainless steel	Iron, chromium. nickel	Utensils, bicycle parts, etc.
Magnalium	Magnesium and aluminium	Automobile and aeroplane parts
Nichrome	Nickel, iron, chromium, manganese	In making coils of heater
Misch metal	Cerium, lanthanum, neodymium, praseodymium and other lanthanoids	In making cigarette lighters

Elements of IA Group (Alkali Metals)

- Lithium, sodium, potassium, rubidium and caesium are alkali metals and are *s*-block elements.
- These metals are soft and can be cut with knife.
- Lithium is the lightest metal.
- Alkali metals are stored under kerosene or paraffins to protect them from the action of air.
- Lithium shows diagonal relationship with magnesium.
- In Castner's process, metallic sodium is prepared by electrolysis of molten NaOH.
- Sodium is used in yellow light lamps.
- Sodium chloride (NaCl) or common salt or table salt is used in our daily diet, as a preservative for pickles, meat and fish. It is also used in the manufacture of NaOH, Cl_2 gas and soap.
- Sodium hydroxide (NaOH) or caustic soda is used in the soap, dyes and artificial silk industries and in the refining of bauxite mineral.
- Sodium bicarbonate is used in the kitchen for making food tasty and crispy.
  - Sodium bicarbonate (Sodium hydrogen carbonate or baking soda) is used for making baking powder, which is a mixture of baking soda and a mild edible acid such as tartaric acid.
  - It is also an ingredient of antacids. Being alkaline, it neutralises excess acid in the stomach and provides relief.
  - It is also used in soda acid fire extinguishers.
- Sodium carbonate decahydrate ($Na_2CO_3 \cdot 10\ H_2O$) or washing soda is used in the manufacture of glass, soap, washing powder and for softening hard water.
- Sodium sulphate ($Na_2SO_4 \cdot 10H_2O$) is Glauber's salt. It is used as purgative.
- Sodium thiosulphate ($Na_2S_2O_3 \cdot 5H_2O$) is also known as hypo. It is used in photography as a fixing agent because it removes the undecomposed AgBr as soluble silver thiosulphate salt.

Elements of IB Group

- $_{29}Cu$, $_{47}Ag$, $_{79}Au$ are the elements of **I B group**. These elements are called coinage or currency metals. Their important properties are as follow:
  - They are *d*-block, transition elements.
  - These are very hard, malleable and ductile metals.
  - They have high density and high melting and boiling points.
- $CuSO_4$ is used as a mordant in dyeing. Copper sulphate pentahydrate is blue in colour and is called **blue vitriol**. $CuSO_4$ is used to test the presence of water.
- Copper is alloyed with gold and silver for making ornaments.
- $AgNO_3$ is called **lunar caustic**. It is stored in dark brown bottles because it is photosensitive.
- Au, Ag, Pt, etc., metals which remain unaffected by the action of other strong acids dissolve in aqua-regia (which is a mixture of HCl and HNO_3 in 3 : 1).

Elements of IIA Group
(Alkaline Earth Metals)

- II A subgroup elements are Be, Mg, Ca, Sr, Ba and Ra. They are called **alkaline earth metals** and are *s*-block elements.
- Calcium of this family is most abundant element in the earth's crust while magnesium is present in chlorophyll, the green pigment of the leaf.
- Beryllium shows diagonal relationship with aluminium.
- $Mg(OH)_2$ is called **milk of magnesia**. Being a mild base, it is used as an antacid to neutralise the excess acid present in the stomach.
- $MgSO_4$ is used as a mordant in dyeing and tanning industry. It is also used as a purgative.
- Calcium oxide (CaO), also called quicklime, gives hissing sound when dissolved in water. It is used in the manufacture of glass, calcium chloride, cement, mortar, bleaching powder, calcium carbide, slaked lime, etc. It is also used in the extraction of iron and as a drying agent for ammonia and alcohol.
- Calcium hydroxide (slaked lime) $[Ca(OH)_2]$, is used in the manufacturing of bleaching powder, caustic soda and soda lime and for softening of hard water.
- When $CaCO_3$ is dipped in water, the bubbles evolve due to the evolution of carbon dioxide.
- Calcium sulphate is gypsum ($CaSO_4 \cdot 2H_2O$). It loses a $\left(\frac{2}{3}\right)$rd part of its water of crystallisation when heated at 120°C to form $(CaSO_4)_2 \cdot H_2O$ which is known as Plaster of Paris.
- Gypsum is used in the preparation of building's plaster and anhydrous, $CaSO_4$.
- Anhydrous $CaSO_4$ is a dehydrating agent and is used for the manufacture of ammonium sulphate (Sindri fertiliser) and sulphuric acid.
- Plaster of Paris is a white powder, which sets into hard mass called gypsum on hydration with water, so it is used in making casting for statues, toys, for plastering fractured bones, etc.
- Cement has the following composition: Calcium oxide (CaO, 50-60%), alumina (Al_2O_3, 5-10%) and magnesium oxide (MgO, 2-3%). Raw materials used for its preparation are limestone and clay. When cement mixed with water, the setting of cement takes place to give a hard mass.
  This is due to the hydration of the molecules of the constituents and their rearrangement.
- Adding gypsum is only to slow down the process of setting time of the cement.
- Portland cement is a mixture of silicates and aluminates of calcium with small amount of gypsum. The chemical composition of portland are CaO, SiO_2, Al_2O_3, $SO_3 MgO, Fe_2O_3$, silica SiO_2 imparts strength to the cement due to the formation of dicalcium and tricalcium silicates.
- Alumina (Al_2O_3) imparts quick setting property to the cement. It acts as a flux and lowers the clinkering temperature.

Elements of IIB Group

- Zinc ($_{30}Zn$), cadmium ($_{48}Cd$) and mercury ($_{80}Hg$) are the three elements present in II B subgroup. These are *d*-block elements.
- Their properties are as follow:
  - Zinc is used in making alloys like brass, bronze, German silver, etc.
  - Zinc is deposited on the surface of iron articles by the process called galvanisation.
  - Mercury, Hg is a liquid at room temperature and is filled in CFL lamps.
  - Mercurous chloride or calomel(Hg_2Cl_2) is used for making calomel electrode which is a reference electrode.

Elements of III A Group (Boron Family)

- Elements are $_5B$, $_{13}Al$, $_{31}Ga$, $_{49}In$ and $_{81}TI$.
- Boron is a semi-metal. Rest of elements are metals.
- Aluminium is the most abundant metal (and third most adundant element) in the earth's crust.
- Boron halides are Lewis acids. The order of acidity is $BBr_3 > BCl_3 > BF_3$.
- Being lighter, all alloys are used in making aeroplanes.
- Aluminium sulphate is used as a mordant in dyeing.

Elements of IVA Group (Carbon Family)

- The IVA group consists of elements C, Si, Ge, Sn, Pb.
- All are *p*-block elements.
- The stability and volatility of tetrahalide of carbon decrease with increasing molecular weight of the tetrahalide e.g. $CF_4 > CCl_4 > CBr_4 > CI_4$.

Carbon

This is the element which forms highest number of compounds (except hydrogen) due to its catenation power (tendency of self linking) and tetravalency. It has two crystalline allotropes : Diamond and graphite.

Diamond

- It is the purest form of carbon and has rigid three dimensional network structure in which each carbon atom is bonded to four other carbon atoms in tetrahedral fashion.
- It is hardest substance and is a bad conductor of electricity.
- It is used in making jewellery.
- It is used as an abbrasive for sharpening hard tools, i.e. in making rock drilling machines , glass cutting etc.

Graphite

- It has layered two dimensional hexagonal structure.
- Large amounts of graphite are prepared artificially by Acheson process.
- Graphite is a good conductor of electricity.
- It is used as dry lubricant and in nuclear reactor as a moderator.

➨ **Note** Fullerenes form another class of carbon allotropes. The first one to be identified was C-60 which has carbon atoms arranged in the shape of football.

Coal and Charcoal

- Coal is the amorphous variety of carbon. It is used as fuel. Coals are of following types :
  (a) Peat (60% carbon).
  (b) Lignite or brown coal (70% carbon).
  (c) Bituminous (78-83% carbon). It is the most common variety.
  (d) Anthracite (90% carbon).
- Charcoal can be animal charcoal, wood charcoal, sugar charcoal and activated charcoal.
- It is used in the manufacture of fuel gases like-water gas, producer gas and semi-water gas.
- Water gas ($CO + H_2$), semiwater gas (water gas + producer gas), producer gas ($N_2 + CO$) are obtained from CO.
- Charcoal (animal charcoal) is used as a fuel and also as a deodorant in the purification of water, decolourising sugar solution and in gas masks.

Carbon Dioxide (CO_2)

- Air contains CO_2 to an extent of 0.03%-0.05% by the volume. CO_2 is an acidic oxide of carbon.
- Solid CO_2 is technically known as dry ice (dricold).
- It produces very low temperature (– 100°C) with ether, hence gives an excellent freezing mixture.
- It is a non-supporter of combustion so generally used as fire extinguisher.
- When carbon dioxide is passed through lime water, it turns milky due to the formation of calcium carbonate (which is milky white in colour). However, in excess of the gas, the precipitate dissolves because of the formation of soluble calcium bicarbonate.

$$CO_2 + \underset{\text{Lime water}}{Ca(OH)_2} \longrightarrow \underset{\text{White}}{CaCO_3} + H_2O$$

$$CaCO_3 + \underset{\text{Excess}}{CO_2} + H_2O \longrightarrow \underset{\text{Soluble}}{Ca(HCO_3)_2}$$

- **Carbon dioxide** is the product of calcination and respiration processes.

Silicon

- It is the second most abundant element in the earth's crust.
- Crystalline form of SiO_2 (silica) is quartz. SiO_2 is the major component of glass and is soluble in HF. That's why HF cannot be stored in glass bottles.

Glass

Glass is an amorphous and transparent solid, obtained by solidification of various silicates and borates.

- Various varieties of glass are : Coloured glass, hard glass, high refractive index glass, pyrex glass, Crook's glass.

Types of Glasses and their Uses

Types of Glasses	*Uses*
Soft glass	Window panes, bottles, test tubes, etc.
Hard glass	Combustion tubes, etc.
Crook's glass	Absorbs UV rays, used for making lenses
Pyrex glass	Cooking utensils
High refractive index glass	Lenses, cut glasses
Glass laminates	Bullet proof materials

- Coloured glass can be obtained by adding following compounds.

Compound added	*Colour imparted*
Cobalt oxide	Blue
Cuprous oxide	Red
Cadmium sulphide	Lemon yellow
Chromium oxide	Green
Auric chloride	Ruby
Manganese dioxide	Purple

Glass Wool

It is an insulating material, obtained from fibre glass arranged into a texture similar to wood. It is produced or rolled in slabs with different thermal and mechanical properties.

Elements of VA Group

- Elements present in this group are $_7N$, $_{15}P$, $_{33}As$, $_{51}Sb$ and $_{83}Bi$.
- Phosphorus is the most abundant element of this family in the earth's crust.
- Nitrogen is an essential component of explosives. It is generally filled in chips packets or electric bulbs to increase their life because of its inert nature.
- Ammonia NH_3 is an alkaline gas, extremely soluble in water and is dried over quicklime.
- Oxides of nitrogen when dissolve in water give oxyacids, e.g. NO_2 when dissolves in water, gives HNO_3 (nitric acid).
- **Nitric acid** is a strong acid and used in the synthesis of fertilisers and explosives.
- Red phosphorus is largely used in match industry and in the manufacture of phosphor bronze-an alloy of Cu, Sn and P.
- The ignition temperature (543K) is much higher than that of white phosphorus (303K). Hence, it does not catch fire easily.

Elements of VIA Group

(Oxygen Family-Chalcogens)

- Elements in this group are $_8O$, $_{16}S$, $_{34}Se$, $_{52}Te$ and $_{84}Po$.
- Most abundant element of the family is oxygen.
- Common oxidation state of oxygen is (-2). In gaseous form oxygen is a supporter of combustion.
- Oil of vitriol or king of chemicals is H_2SO_4.
- Sulphuric acid (H_2SO_4) is a dibasic acid and is a dehydrating agent. It is used in making explosives.
- Ozone (O_3) is an allotrope of oxygen. It is diamagnetic gas.
- SO_2 is a bleaching agent. Its bleaching action is due to reduction and is temporary.
- Oxides of sulphur causes acid rain in the atmosphere.

Elements of VIIth Group (Halogens)

- Elements present in this group are $_9F$, $_{17}Cl$, $_{35}Br$, $_{53}I$ and $_{85}At$. Out of these, bromine (Br_2) exists in liquid state and iodine (I_2) in solid state.
- Fluorine 'F' has maximum electronegativity.
- Chlorine is prepared by oxidation of HCl with MnO_2, $KMnO_4$, $K_2Cr_2O_7$ etc.
- Chlorine has maximum electron affinity in the halogens.
- Order of reactivity of halogens, $F_2 > Cl_2 > Br_2 > I_2$.
- Order of electronegativity, F > Cl > Br > I.
- Order of acidic strength, HI > HBr > HCl > HF.
- Iodine is a powerful antiseptic and used as a tincture of iodine (a solution of iodine in alcohol).

Elements of VIIIth Group

- Elements present in this group are $_{26}Fe$, $_{27}Co$, $_{28}Ni$ (Ferrous metals)
  $_{44}Ru$, $_{45}Rh$, $_{46}Pd$ (Light platinum metals)
  $_{76}Os$, $_{77}Ir$, $_{78}Pt$ (High platinum metals)
- All of these are *d*-block elements. Pig iron is obtained from blast furnace. It is most impure form of iron. It is very hard, but brittle. It cannot be welded. It contains maximum amount of carbon (2.5% to 4.3%).
- Wrought iron is the purest form of iron. It contains 0.1 to 0.25% carbon. Wrought iron is soft, ductile and malleable, so it can be welded.
- Steel is the most important form of iron. It contains 0.2 to 1.5% carbon.
- Iron is present in haemoglobin (blood) and its oxidation state in haemoglobin is +2.
- Fe_2O_3 is called jeweller's rough.
- Vitamin B_{12} is also called cobalamine and contains CO^{3+}ion. Its deficiency in diet causes pernicious anaemia.

Elements of Zero Group

- Elements present in this group are $_2He$, $_{10}Ne$, $_{18}Ar$, $_{36}Kr$, $_{54}Xe$ and $_{86}Rn$ which are called nobel or inert gases.
- Argon is the most abundant inert gas in air.
- Helium was discovered by Frankland and Lockyer. He-III is abundant on the lunar surface and holds the potential to put an end to the energy crisis of the earth.
- Argon was discovered by Rayleigh and Ramsay.
- Ne, Kr and Xe were discovered by Ramsay and Traverse.
- Radon was discovered by Dorn and is not present in air.
- All of these elements are colourless monoatomic gases.
- Their electronegativity and electron affinity are zero.

Gas	*Uses*
Helium	With oxygen for respiration in case of deep seadives, filling balloons and in low temperature applications.
Neon	In advertising signs.
Argon	Light bulbs
Krypton/Xenon	Photographic flash, TV tubes

Some Important Compounds

Common name	Chemical name	Formula	Uses
Aqua-fortis	Nitric acid	HNO_3	It is used especially in the production of fertilisers and explosives and rocket fuels.
Aqua-regia	Nitric acid + hydrochloric acid	Conc. HNO_3 + conc. HCl (in 1 : 3 ratio)	It is used for testing metals and dissolving platinum and gold.
Baking soda	Sodium bicarbonate	$NaHCO_3$	It is sodium bicarbonate that is added to baked goods to make them rise.
Brine	Sodium chloride solution	NaCl solution	The electrolysis of brine is a large-scale process used to manufacture chlorine from salt.
Blue vitriol	Copper sulphate	$CuSO_4 \cdot 5H_2O$	It is used as antiseptic agent, dyes, treatment of copper deficiency etc.
Bone ash	Calcium phosphate	$Ca_3(PO_4)_2$	Bone ash can be used alone as an organic fertiliser. It can be treated with sulphuric acid to form a ''single superphosphetic'' fertiliser which is more water soluble.
Bleaching powder	Calcium oxy-chloride	$CaOCl_2$	It is used as disinfectant and germicide in the sterilization of drinking water.
Caustic soda	Sodium hydroxide	NaOH	It is used as a cleansing agent and in the manufacturing of washing soda.
Chile salt petre	Sodium nitrate	$NaNO_3$	It is used in solid propellants, explosives, fertilizers etc.
Dry ice	Solid carbon dioxide	CO_2	It is the solid form of CO_2. It is used primarily as a cooling agent.
Epsom salt	Heptahydrate magnesium sulphate	$MgSO_4 \cdot 7H_2O$	This is used as an antacid or as a mineral supplement to maintain the body's magnesium balance and laxative.
Foul air	Nitrogen	N_2	It is used in food processing, in purging air conditioning and refrigeration.
Grain alcohol	Ethyl alcohol	C_2H_5OH	It is used as a solvent, in the synthesis of other organic chemicals. as an additive to automotive gasoline,
Grape sugar	Dextrose	$C_6H_{12}O_6$	Dextrose is a type of sugar better known today as glucose. It is used for treatment of hypoglycemia.
Gypsum	Calcium sulphate dihydrate	$CaSO_4 \cdot 2H_2O$	It is widely used as fertilizer and the main constituent in many forms of plaster, blackboard chalk and wallboard.
Green vitriol	Ferrous sulphate heptahydrate	$FeSO_4 \cdot 7H_2O$	It is also used in medicine to treat iron deficiency.
Gammexane	Benzene hexachloride (BHC)	$C_6H_6Cl_6$	It is also known as lindane. It is used as an agricultural inseticide and as pharmaceutical treatment for lice and scabies.
Heavy water	Deuterium oxide	D_2O	It is used in certain types of nuclear reactors. It acts as a neutron moderator to slow down neutrons.
Halite or table salt	Sodium chloride	NaCl	It is commonly used as a condiment and food preservative.
Lime stone (pearl)	Calcium carbonate	$CaCO_3$	It is a chemical or biolozical sedimentary rock that has many uses in agriculture and industry.
Laughing gas	Nitrous oxide	N_2O	This gas used as anaesthetic, as a propellant in whipped cream cans and as an oxidizing agent in racing cars.
Milk of magnesia	Magnesium hydroxide	$Mg(OH)_2$	It is used as a laxative to relieve occasional constipation and as an antacid to relieve indigestion.
Milk of lime (slaked lime)	Calcium hydroxide	$Ca(OH)_2$	It is used as lubricant carrier in wire drawing.
Oil of vitriol	Sulphuric acid	H_2SO_4	It uses in the production of fertilizers, manufacture of chemicals in petrolium refining etc.
Plaster of Paris	Calcium sulphate hemihydrate	$CaSO_4 \cdot \frac{1}{2} H_2O$	It is used for casts to hold broken limbs in place, modeling casts, sculptures and in plasterboard walls and ceilings etc.
Phosgene	Carbonyl chloride	$COCl_2$	It is a major industrial chemical used to make chemicals and pesticides.
Paris green	Double salt	$Cu(AsO_2)_2 \cdot Cu(C_2H_3O_2)_2$	It is used as a pigment and insecticide. It is used as preservative also.
Pearl white	Bismuthoxychloride	BiOCl	It is used as a yellow pigment for cosmetics and paints.
Quartz or sand	Silicon dioxide	SiO_2	Common uses of silica (quartz) is the manufacture of glass.
Quick silver	Mercury	Hg	It is the chemical element of mercury. It is used in batteries, fluorescent lights, thermometers etc.
Quicklime	Calcium oxide	CaO	It is used in the petroleum industry as an alkali in biodiesel production.
Red lead	Lead tetraoxide	Pb_3O_4	It is used in the manufacture of batteries, lead glass and rust-proof primer paints.
Stranger gas	Xenon	Xe	Xenon is used in photographic flash lamps, stroboscopic lamps, bactericidal lamps, high intensive arc lamps for motion picture protection etc.
Spirit of salt	Hydrochloric acid	HCl	An old fashioned name for hydrochloric acid. It is used in production of batteries. photoflash bulbs and fireworks etc.
Soda ash	Sodium carbonate	Na_2CO_3	It is used in the manufacturing of glass, detergents, chemicals and other industrial products.

Common name	*Chemical name*	*Formula*	*Uses*
Salt cake	Sodium sulphate	Na_2SO_4	It used in the manufacture of glass, ceramic glazes, soaps and sodium salts.
Tear gas	Chloropicrin	CCl_3NO_2 (or chloro compounds or NH_3)	It is used as a riot control agent.
Vinegar	Acetic acid	CH_3COOH	It is mainly used as cooking ingredient or in pickling.
Washing soda	Sodium carbonate decahydrate	$Na_2CO_3 \cdot 10\,H_2O$	(i) It is used in manufacturing of glass borax, paper etc. (ii) To remove parmanent hardness of water.
Water glass	Sodium silicate	Na_2SiO_3.	It is used as coagulant/deflocculant agent in wastewater treatment places.
White vitriol	Zinc sulphate	$ZnSO_4 \cdot 7\,H_2O$	It is used for preserving skins and wood, in the electro deposition of zinc etc.

ORGANIC CHEMISTRY

ORGANIC COMPOUNDS

- Chemistry of hydrocarbons and their derivatives are called organic compounds which deal with an organic chemistry.
- **Urea** *(NH_2CONH_2)* was the first organic compound prepared in laboratory. It was prepared by **Wohler**.
- **Acetic acid** *(CH_3COOH)* was the first organic compound synthesised from its elements by **Kolbe.**
- Main sources of organic compounds are plants, animals, coal and petroleum.

HYDROCARBONS

Hydrocarbons are the compounds made up of carbon and hydrogen only. These are of two types:

1. Saturated Hydrocarbons or Paraffins or Alkanes (Contain C—C Bond)

- **General formula** C_nH_{2n+2} where, $n = 1, 2, 3, \ldots$
- All the carbon atoms in alkanes are sp^3-hybridised. They show chain isomerism.
- Methane, CH_4 is the first and least reactive member of the series. Its source is wet and marshy land. It is also present in the air exhaled by animals whose food contains cellulose.
- Cavities in coal contains 90% methane. It is called **fire-damp**. It is responsible for the explosions occurring in coal mines.
- It is used as a gaseous fuel, for preparing carbon black and in the preparation of variety of organic compounds.

2. Unsaturated Hydrocarbons

Contains $>C{=}C<$ or $—C{\equiv}C—$Bond

These further are of two types :

(i) **Alkenes or Olefins** $\left(>C{=}C<\right)$

- **General formula** C_nH_{2n}
  Type formula $R—CH{=}CH_2$
- **Common name** Alkylene, **IUPAC name** Alkenes.
- First member is C_2H_4, ($CH_2{=}CH_2$) known as ethylene or ethene.
- Ethene is used in the manufacture of plastics, for artificial ripening of fruits (banana, apple and orange), as an anaesthesia etc.

(ii) **Alkynes or Acetylenes** ($—C{\equiv}C—$)

- **General formula** C_nH_{2n-2}
  Type formula $R—C{\equiv}C—H$
- **Common name** Acetylene, **IUPAC name** Alkynes.
- Acetylene (C_2H_2) is the first member of the series.
- It is used in oxyacetylene torch and for welding in acetylene lamps.
- Alkynes can be distinguished from alkenes by the reaction with Tollen's reagent (ammoniacal silver nitrate).

Aromatic Hydrocarbons

- **General formula** C_nH_{2n-6}
- **Common name** Arenes
- Most of these compounds contain benzene ring. e.g. benzene, toluene, naphthalene, etc.
- Benzene (C_6H_6) was first synthesised by Berthelot. It is used in the production of various organic compounds, as a solvent and fuel in motor vehicle when mixed with petrol, etc.

Hydrocarbon Derivatives

1. Haloalkanes

- **General formula** $C_nH_{2n+1} \cdot X$ (where, X = Cl, Br, I)
  Type formula These are of three types:
  Primary alkyl halide, $R—CH_2—X$

  Secondary alkyl halide, $R_1—\underset{\displaystyle |}{\overset{\displaystyle R_2}{}}\!\!CH—X$

Tertiary alkyl halide, $\begin{array}{c} R_2 \\ | \\ R_1\text{—}C\text{—}X \\ | \\ R_3 \end{array}$

- **Common name** Alkyl halide;
  IUPAC name Haloalkane.
- Trihalo methane are called **haloforms**, e.g. chloroform *($CHCl_3$)*, iodoform *(CHI_3)*, etc.
- Chloroform oxidises to $COCl_2$ (carbonyl chloride or phosgene gas) in the presence of air and light. So, it is kept in closed dark coloured bottles and mixed with ethyl alcohol which acts as negative catalyst and form non-poisonous, ethyl carbonate with $COCl_2$.
- Chloroform is used as a general anaesthetic.
- Chloropicrin *($CCl_3 \cdot NO_2$)* is used as an insecticide and also called **tear gas**. Carbon tetrachloride (CCl_4) is used for extinguishing fire under the name **pyrene**.
- Ethylene dibromide is added to petrol in place of tetraethyl lead to save the environment from lead pollution.
- Chlorofluorocarbons (freons or CFC) are used as refrigerants. They contain carbon, chlorine, hydrogen and fluorine and destroy the ozone layer.

2. Alcohols

- **General formula** $C_nH_{2n+1} \cdot OH$
- **Common name** Alkyl alcohol;
  IUPAC name Alkanol.
- Primary alcohol is R—CH_2OH, secondary alcohol is R_2CH —OH and tertiary alcohol is R_3C—OH.
- Lucas reagent (conc. HCl + anhydrous $ZnCl_2$) is used to distinguish between primary, *sec* and *tert*-alcohols.
- CH_3OH is produced by the destructive distillation of wood, hence also known as, wood spirit, methyl alcohol, methanol or carbinol. It is used as a solvent for paints and varnishes, in making dyes, drugs, perfumes and also used as motor fuel with petrol.
- The destructive distillation of wood yields wood gas (gaseous), tar (liquid), charcoal (solid residue), methyl alcohol (liquid) and acetic acid (vinegar).
- Various products obtained by the destructive distillation of coal are coal gas (H_2 + CH_4 + CO), ammoniacal liquor, coal tar and coke.
- CH_3CH_2OH is ethyl alcohol or ethanol or grain alcohol. It is made unfit for drinking purpose by adding methanol or pyridine. Such alcohol is called **methylated spirit** or **denatured alcohol**.
- Ethanol changes the colour of sodium or potassium dichromate from orange to green. It is also used in the manufacture of paints, varnishes, in making transparent soop in the preparation of chloroform, as a fuel in internal combustion engines. It is used in alcoholic drinks, whisky, wine, beer, etc.

3. Ethers

- **General formula** $C_nH_{2n+2} \cdot O$
- **Common name** Dialkyl ether
- **IUPAC name** Alkoxy alkane
- Ethers behave like Lewis bases.
- Diethyl ether ($C_2H_5OC_2H_5$) is used as an industrial solvent and also as an anaesthesia.

4. Aldehydes

- **General formula** $C_nH_{2n+1}CHO$ or $C_nH_{2n}O$;
- **IUPAC name** Alkanal
- 40% solution of formaldehyde is called **formalin** and used in the preservation of dead animals.
- Benzaldehyde is almond extract.
- Paraldehyde (a carbonyl compound) is used as a hypnotic.
- Eugenol is an active component of clove oil.

5. Ketones

- **General formula** $C_nH_{2n}O$ **or** $(C_nH_{2n+1})_2O$
- **IUPAC name** Alkanone**.**
- Ketones containing — $COCH_3$ (methyl ketonic group) gives yellow colour (iodoform) in iodoform test.
- Acetone (CH_3COCH_3) is the first member of this class.
- Acetone is used as nail polish remover. However, now-a-days, acetone free removers are in common use. Its main ingredient is ethyl acetate or butyl acetate.
- Methyl Ethyl Ketone (MEK) is used as paint stripper.
- Aldehydes and ketones having α-hydrogen atom give aldol condensation. Such carbonyl compounds which have α-hydrogen atom undergo condensation in the presence of dilute base to form aldol.

6. Carboxylic Acids

- **General formula** $C_nH_{2n+1} \cdot COOH$ or $C_nH_{2n}O_2$;
  IUPAC name Alkanoic acid.
- First member of the carboxylic acid series is formic acid, HCOOH, which is present in ant's or bee's sting.
- HCOOH is the strongest acid among the carboxylic acids.
- CH_3COOH is the ethanoic or acetic acid (vinegar).
- Vinegar contains 6-10 % acetic acid. Acetic acid is used in the manufacture of rubber from latex and casein from milk. It is used for coagulation.
- Lactic acid is α-hydroxy propanoic acid. It is present in milk and provides it sour taste.

- Citric acid is a hydroxy tricarboxylic acid. It is present in citrus fruits.
- Oxalic acid removes rust stains.
- Salicylic acid is used to prepare aspirin (*o*-acetyl salicylic acid).

7. Esters

- **General formula** $C_nH_{2n+1}COOR$ or $C_nH_{2n}O_2$
- **IUPAC name** Alkyl alkanoate.
- These are used in making artificial perfumes, flavours and essence used in cold drinks, ice-creams, etc.
- Sodium benzoate is used as a food preservative.
- These have characteristic sweet fruity smell.

Ester	*Flavour*	*Ester*	*Flavour*
Amyl acetate	Banana	Isoamyl valerate	Apple
Octyl acetate	Orange	Methyl butyrate	Pineapple

MAN-MADE MATERIALS

SOAPS

Soaps are the sodium or potassium salts of long chain carboxylic (fatty) acids (RCOONa) e.g. sodium palmitate ($C_{15}H_{31}COONa$), sodium stearate ($C_{17}H_{35}COONa$) and sodium oleate ($C_{17}H_{33}COONa$) etc. Animal fat or vegetable oil, sodium hydroxide, sodium chloride act as the raw materials for the manufacture of ordinary soap. The process of making soap is called **saponification**.

- Soaps do not form lather with hard water due to the formation of scum.
- When the medium is acidic, soaps form insoluble long chain fatty acids and hence, lost their cleansing action.

DETERGENTS

- These are also called "soap-less" soaps as they do not contain any soap.
- These are the long chain benzene sulphonic acid or the sodium salt of the long chain alkyl hydrogen sulphate.
- Long chain hydrocarbons, sulphuric acid and sodium hydroxide are the raw materials for the manufacturing of detergents.
- These are non-biodegradable and cause water pollution However, straight chain detergents are biodegradable.
- They form lather with soft as well as hard water.

Polymers

A polymer is a compound of high molecular weight formed by the union of a large number of molecules of one or two types of low molecular weight (known as monomers) and the process involving the formation of a polymer is called **polymerisation**.

Types of Polymers

On the basis of their origin, the polymers are classified in the following manner:

1. **Natural Polymers** These are found in nature, e.g. cellulose, starch, rubber, wool ,silk etc.
2. **Synthetic Polymers** These are prepared in the laboratory by synthetic means, e.g. polythene, nylon, orlon, dacron, etc.
3. **Semi-synthetic Polymers** These are synthesised by man from natural substances, e.g. rayon (a polymer of cellulose nitrate) is also called artificial silk due to its silk like appearance.

On the basis of intermolecular forces, these are classified as:

1. **Elastomers** In these polymers, the polymer chains are held together by weak van der Waals' forces, e.g. vulcanised rubber.
2. **Fibres** In this type, polymer chains are attached with one another through H-bonds, e.g. nylon-6,6.
3. **Thermoplastics** In these, the intermolecular forces are intermediate of elastomers and fibres. These are soften on heating and become hard on cooling, e.g. polystyrene, polythene, PVC, etc.
4. **Thermosetting** These are highly cross-linked, hard, non-fusible and insoluble polymers. These can only be moulded into desired shape on heating once. e.g. bakelite (phenol-formaldehyde resin), melamine, etc.

Some Important Synthetic Polymers

- **Polyethylene** (polythene) The monomer units are ethylene molecules. It is frequently used in making coats, milk cartons and electrical insulation.
- **Polystyrene** The monomer units are styrene molecules. It is a white thermoplastic material and is used for making toys, combs, lining material for refrigerators and TV cabinets.
- **Teflon** (Polytetrafluoroethylene) The monomer unit is tetrafluoroethylene molecule. It is very tough material. It is a bad conductor of electricity and is used in coating utensils, making seals, baskets, pipes, flooring, etc.
- **Polyvinyl Chloride** (PVC) The monomer units are vinyl chloride molecules. PVC is a hard, horny material. It is resistant to chemicals as well as heat. It is used for making raincoats, hand bags, electrical insulators and floor covering.
- **Nylon or Nylon-6, 6** It was first fully synthetic fibre and has good elasticity, low water absorption and wrinkle resistant. It has also high tensile strength. It is used in making fishing nets, tyre cord, parachute fabrics, ropes, etc.

- **Phenol-formaldehyde Resins** (Bakelite) These are made by the reaction of phenol and formaldehyde in the basic medium. Bakelite is a cross-linked thermosetting polymer. It is used for making combs, fountain pen barrels, electrical goods making electrical appliances, handles of crockery, etc. Sulphonated bakelites are used as ion-exchange resins for softening of hard water.
- **Rubber** It is a polymer of isoprene (2-methyl buta-1, 3-diene). It is insoluble in water, dilute acids and alkalis, absorbs a large amount of water, has low tensile strength and elasticity.
- **Neoprene** It is a synthetic rubber which resembles natural rubber in its properties. It is obtained by the polymerisation of chloroprene (2-chlorobuta-1, 3-diene). It is superior to natural rubber in its stability.
  Easy availability of raw materials from products of petroleum refining industries lead to growth of synthetic rubber. It is generally used for making hoses, shoe heels, stoppers, etc.
- **Buna-S** It is a copolymer of 1, 3-butadiene and styrene. It is also known as SBR (Styrene-Butadiene Rubber). It has slightly less tensile strength than natural rubber. It is used in the manufacturing of automobile tyres, rubber soles, etc.
- **Terylene** or **dacron** or **phthalate** It is a polymer of ethylene glycol and terephthalic acid, i.e. it is a polyester. It is used for making wash and wear fabrics, tyre cords, safety belts, tents, etc.
- **Vulcanisation of Rubber** It is the process of heating of rubber with sulphur. It makes the rubber hard, strong and more elastic.
- **Kevlar** It is a polymer of terephthalic acid and 1,4-diaminobenzene so, it is a polyamide. It is used for making bulletproof vests.
- **Lexan or polycarbonate** It is a polymer of diethyl carbonate and bi-phenol. It is used in making bulletproof windows and safety helmets.
- **Polyurethanes** It is a polymer of toluene diisocyanate and ethylene glycol. It is used for making washable and long lasting mattresses, cushions, pillows , etc.
- **Lucite** It is a polymer of methyl methacrylate. It is used for making contact lenses. Plexiglass, acrylite or perspex are other terms used for lucite.
- **Orlon** or **Acrilon Polyvinyl Cyanide** It is a polymer of vinyl cyanide or acrylonitrile. It is used as a substitute for wool in making synthetic blankets (acrylic fibres).

FERTILISERS

Repeated cultivation of crops makes the soil deficient in some elements, mainly nitrogen, phosphorus and potassium. The substances which are added to the soil to make up the loss of these elements are called **fertilisers**.

Some Important Fertilisers

- **Basic Calcium Nitrate** (Nitrate of lime) $[CaO \cdot Ca(NO_3)_2]$ It is a good nitrogeneous fertiliser, used in acidic soil.
- **Ammonium Sulphate** $[(NH_4)_2SO_4]$ It contains 21.2% nitrogen. Its repeated use makes the soil acidic and unfit for the germination of seeds. Hence, addition of lime to the soil becomes necessary.
- **Calcium Cynamide** $[CaCN_2]$ (Nitrolim) It is black in colour because of the presence of carbon. It contains 19% nitrogen.
- **Urea** $[NH_2CONH_2]$(Carbamide) It contains 46.6% nitrogen. It is the best nitrogeneous fertiliser as it leaves only carbon dioxide after ammonia has been assimilated by plants. Its repeated use does not change the pH of the soil.
- **Superphosphate of Lime** $[Ca(H_2PO_4)_2 + 2CaSO_4 \cdot H_2O]$ It is phosphatic fertiliser.
- **Vermicompost** It is the product or process of composting by using various species of worms, usually red wigglers, white worms and earthworms to create a heterogeneous mixture of decomposing vegetables or food, bedding material and vermicant containing water soluble nutrients. Vermicompost is an excellent, nutrient rich organic fertiliser and soil conditioner.
- Magnesium, manganese and sulphur are the macronutrients which are provided by the inorganic fertilisers.

FUELS

The substances which on combustion produces energy in the form of heat, is called a fuel, e.g. coal, wood, kerosene, petrol, diesel, cooking gas, etc. Fuel efficiency is expressed in terms of calorific value (kJg^{-1}).

- Sulphur compound (ethyl mercaptan or thioethanol) is added to odourless LPG (Liquefied Petroleum Gas) for imparting a detectable smell to the gas.
- LPG contains mainly butane alongwith some propane. CNG (Compressed Natural Gas) contains mainly methane and is used as a fuel in many vehicles because it produces less pollutants therefore , it is a better fuel than petrol or diesel.
- Petrol is used as a fuel to run cars and aeroplanes whereas diesel is used to run trucks, buses, trains and ships.
- Quality of petrol or gasoline is expressed in terms of **octane number** and of diesel in terms of **cetane number**.
- Synthetic rubber, liquid ammonia, liquid hydrogen are used as propellants.
- Acetylene is a fuel used in gas welding. It is used to weld and cut the metals that have a temperature of 300°C of a flame.

Renewable and Non-renewable Resources

- Natural resources such as coal, petroleum and natural gas take thousands of years to from naturally and cannot be replaced as fast as they are being consumed. It is a non-renewable resource.
- Natural resources such as solar energy, water and fisheries are renewable resources.

Petroleum

It is dark oily liquid also called rock oil, mineral oil, crude oil or black gold. On fractional distillation, it gives different substances at different temperatures.

Biogas or Gobar Gas

The gaseous mixture obtained by the degradation of animal and plant wastes by anaerobic microorganisms in the presence of water is called biogas. It is a convenient fuel for domestic use. It is used for street lighting.

- Constituents of biogas are methane (45-70%), carbon dioxide, hydrogen, hydrogen sulphide.
- Gobar gas is produced by the fermentation of cow dung. It contains CH_4 (main component), CO_2 and H_2.

➨ **Note** Bagasse is the fibrous matter remains after sugarcane stacks are crushed to extract their juice. It is used as a biofuel and as a substitute for wood in many tropical and subtropical countries for the production of pulp, paper, board, etc.

Flame

A flame is a region where combustion of gaseous substances take place. Innermost region of flame is black because of the presence of unburned carbon particles. Middle region is yellow luminous due to partial combustion. Outermost region is blue due to complete combustion of fuel which is the hottest part of the flame.

Paint

It is a mixture of four ingredients, *viz*, binder, solvent, pigment and additives (like driers, plasticisers, emulsifiers, corrosion inhibitors, etc).

- Binders are polymers (resins) forming a continuous film on the surface of the substrate. These are responsible for good adhesion. Resins, chlorinated rubber, latex (PVA) are the examples of binders.
- Solvent (water or organic solvents like toluene, ketone, alcohol, etc.,) is a medium where all other ingredients of paint are dispersed in molecular form (true solutions) or as colloidal dispersion (emulsions).
- Pigments are responsible for imparting colour to the paint. It may also protect the substrate from UV light.
- Ultramarine blue is the natural pigment made up of semiprecious mineral lapis lazuli. It is resistant towards fading.

Pigment added	*Colour of paint*
Chromium oxide	Green
Vermillion or cuprous oxide	Red
Cobalt oxide	Blue
Titanium dioxide	White
Benzidine yellow	Yellow to red
Iron (II) oxide	Black

ENVIRONMENT AND ITS POLLUTION

Air

- Composition of air is nitrogen (78.084%), oxygen (20.946%), argon (0.934%), carbon dioxide (0.033%) and small amount of other gases. Region of air present around the earth is called atmosphere.
- Main layers from the surface of earth towards upward
  (a) Troposphere (b) Stratosphere
  (c) Mesosphere (d) Thermosphere
- Most of the atmospheric air is present in troposphere.
- Ozone layer present in the stratospheric region (at a height of about 20 to 40 km) protects the living beings from the harmful UV radiations coming from the sun. Due to high concentration of ozone, this region is also known as ozonosphere.
- In 1775, French scientist Lavoisier performed experiments on composition of air.

Air Pollution

It is due to the presence of foreign undesirable substances in air. The substance which causes pollution is known as pollutant. Main air pollutants are SO_2, CO, nitrogen oxides, particulates, etc.

Carbon monoxide CO combines with the haemoglobin of the blood about 300 times more readily as compared to oxygen and forms a stable carboxyhaemoglobin complex due to which supply of oxygen is restricted which results in headache, nervousness, asphyxia and even death in some cases.

Major gaseous air pollutants:

(a) Oxides of carbon
(b) Oxides of nitrogen
(c) Oxides of sulphur
(d) Hydrocarbon (esp CH_4)

Methyl isocyanate (CH_3NC) is responsible for the Bhopal gas tragedy (in 1984). In which thousands of people were killed. It was a case of serious air pollution in which MIC (Methyl Isocyanate) gas released from a pesticide manufacturing plant of union carbide.

Smog

The word smog is derived from smoke and fog. It is of two types :

1. **Classical Smog** It is also called London smog. It is formed in cool humid climate and is reducing in nature.
2. **Photochemical Smog** It is also called Los Angeles smog. It is formed in warm, dry and sunny climate and is oxidising in nature.

Particulates

These are minute solid particles and liquid droplets dispersed in air such as mists, dusts, smoke , etc.

Diseases Caused by Particulates

Disease	*Cause*
Pneumoconiosis	Coal dust
Silicosis	Silica (from ceramics, glass and pottery industry)
Black lung disease	Coal mines
White lung disease	Textile industries
Asbestosis	Asbestos
Byssinosis	Cotton fibre dust
Melanosis	Hyperpigmentation due to increased concentration of melanin

Consequences of Air Pollution

- **Greenhouse effect** It is the heating up of earth and its objects due to trapping of infrared radiations (IR) by greenhouse gases present in atmosphere like CO_2, CH_4, NO, O_3, chlorofluorocarbon and water vapours.
- **Global Warming** It is due to increased concentration of greenhouse gases. It may lead to melting of ice caps and glaciers, spreading of several infectious diseases like malaria, sleeping sickness, etc.
- Since, the late 19th century, the temperature of atmosphere increases due to the increase in CO_2 concentration which is responsible for the expansion of water in the ocean.
- **Acid Rain** The pH of normal rain water is 5.6-6.5 due to the dissolution of carbon dioxide from the atmosphere. When the pH of rain water is below 5, it is called acid rain (by Robert Augus).
  - The main cause of acid rain is oxides of sulphur and nitrogen (H_2SO_4 and HNO_3).
  - Burning of sulphur alongwith coal results in the formation of oxides of sulphur as

$$S + O_2 \longrightarrow SO_2$$
$$SO_2 + \frac{1}{2}O_2 \longrightarrow SO_3$$

  when sulphur trioxide (SO_3) reacts with water of douds it forms acid, hence causes acid rain

$$SO_3 + H_2O \longrightarrow H_2SO_4$$

  - Acid rain damages the marble buildings (Taj Mahal) and monuments, corrodes metal pipes and results in several diseases.
  Acid rain reacts which buildings made from limestone and causes its decomposition

$$CaCO_3 + H_2SO_4 \longrightarrow CaSO_4 + CO_2 + H_2O$$

- **Ozone Layer Depletion** Thinning of ozone layer because of its reaction with chlorine free radicals usually generated by chlorofluorocarbons is called **ozone layer depletion.**
  - Hypochlorous acid and chlorine on photochemical decomposition also produce nascent chlorine ($Cl\bullet$ or chlorine free radical) which causes ozone depletion in the following manner:

$$HOCl(g) \xrightarrow{h\nu} \bullet OH(g) + Cl\bullet(g)$$
$$Cl_2(g) \xrightarrow{h\nu} 2Cl\bullet(g)$$
$$CF_2Cl_2 \longrightarrow \bullet CF_2 + Cl\bullet$$

  - The released chlorine free radical reacts with ozone breaking it into molecular oxygen.

$$Cl\bullet + O_3 \longrightarrow ClO\bullet + O_2$$
$$2ClO\bullet + O_2 \longrightarrow 2Cl\bullet + 2O_2$$

Water Pollution

Water pollution is due to the presence of foreign substances, i.e water pollutants in water.

- Cholera, dysentery, typhoid, etc., are water-borne diseases caused by bacteria.
- Mercury causes minimata disease, chromium and arsenic causes cancer and cadmium causes *itai-itai* disease.
- Presence of excessive fluoride leads to dental fluorosis, i.e. developmental disturbance of dental enamel.
- The usual effect of agricultural run off due to the excessive increase of nutrients such as nitrogen, phosphorus etc., in an ecosystem like ponds or results in excessive algal growth in affected water bodies. It is called **eutrophication**.
- For clean water, DO (Dissolved Oxygen) is 5-6 ppm and BOD (Biochemical Oxygen Demand) is less than 5 ppm.
- Recycling, sewage treatment, incineration are some strategy to control water pollution.

➠ **Presbycusis** is gradual hearing loss in both ears that commonly occurs in aged people but noise pollution also lead to it.

PRACTICE EXERCISE

Matter

1. During evaporation, particles of a liquid change into vapours only
(a) from the surface (b) from the bulk
(c) from both surface and bulk
(d) None of the above

2. Arrange the following in increasing order of density.
1. Air 2. Honey
3. Water 4. Iron
5. Chalk

Codes
(a) 1 < 2 < 3 < 4 < 5
(b) 1 < 2 < 3 < 5 < 4
(c) 1 < 3 < 2 < 5 < 4
(d) 1 < 3 < 2 < 4 < 5

3. Which one of the following mixtures is homogeneous?
(a) Starch and sugar
(b) Methanol and water
(c) Graphite and charcoal
(d) Calcium carbonate and calcium bicarbonate

4. Which one of the following properties changes with valency?
(a) Atomic weight (b) Equivalent weight
(c) Molecular weight (d) Density

5. 44 g of carbon dioxide at STP contains
(a) 6.02×10^{23} molecules
(b) 3.01×10^{23} molecules
(c) 6.02×10^{24} molecules
(d) 3.01×10^{24} molecules

6. Which of the following is a physical change?
(a) Decomposition (b) Oxidation
(c) Sublimation (d) Reduction

7. **Statement I** Ice floats over water.
Statement II Density of liquids is generally lower than that of the solids.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

8. Which of the following is a chemical change?
1. Burning of coal in air
2. Crystallisation of table salt from sea water.

Select the correct answer using the codes given below.
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

9. Which of the following is not a mixture?
1. Toothpaste 2. Baking soda
3. Vinegar

Choose the correct option
(a) Only 1 (b) Only 2
(c) 1 and 3 (d) 2 and 3

10. On addition of salt to water, its
1. boiling point increases
2. boiling point decreases
3. boiling point is not affected
4. freezing point decreases

The correct option(s) is/are
(a) 1 and 4 (b) 2 and 4
(c) Only 2 (d) Only 3

11. Match the following Columns.

Column I (Process)	Column II (Changes)
A. Evaporation	1. Liquid into gas
B. Sublimation	2. Gas into liquid
C. Freezing	3. Solid into gas
D. Melting	4. Solid into liquid
	5. Liquid into solid

Codes

	A	B	C	D
(a)	2	1	3	4
(b)	1	3	5	4
(c)	4	3	2	1
(d)	5	3	1	2

Previous Years' Questions

12. Iodised salt is a **2012 II**
(a) mixture of potassium iodide and common salt
(b) mixture of molecular iodine and common salt
(c) compound formed by the combination of potassium iodide and common salt
(d) compound formed by molecular iodine and common salt

13. **Statement I** All compounds contain more than one element.
Statement II All compounds are heterogeneous mixtures. **2012 II**
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

14. A liquid initially contracts when cooled down to 4°C but on further cooling down to 0°C, it expands. The liquid is **2013 II**
(a) alcohol (b) water
(c) molten iron (d) mercury

15. A metal screw top on a glass bottle which appears to be stuck could be opened by using the fact that **2014 I**
(a) the metal expands more than the glass when both are heated
(b) the metal and glass expand identically when heated
(c) the metal shrinks when heated
(d) both metal and glass shrink when cooled

16. Which one among the following is an example of chemical change? **2014 II**
(a) The melting of an ice cube
(b) The boiling of gasoline
(c) The frying of an egg
(d) Attraction of an iron nail to a magnet

17. Which one among the following compounds has same equivalent weight and molecular weight? **2015 I**
(a) H_2SO_4 (b) $CaCl_2$
(c) Na_2SO_4 (d) NaCl

18. Creation of something from nothing is against the law of **2015 I**
(a) constant proportions
(b) conservation of mass energy
(c) multiple proportions
(d) conservation of momentum

19. Match the following Columns. **2015 II**

Column I (Exponent)	Column II (Law)
A. John Dalton	1. Law of definite proportion by volume
B. Joseph Proust	2. Law of multiple proportion
C. Antoine Lavoisier	3. Law of definite proportion by weight
D. Joseph Louis Gay Lussac	4. Law of conservation of mass

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	4	1	(b)	2	4	3	1
(c)	1	4	3	2	(d)	1	3	4	2

20. Which one of the following is an example of chemical change? **2016 I**
(a) Burning of paper
(b) Magnetisation of soft iron
(c) Dissolution of cane sugar in water
(d) Preparation of ice cubes from water

21. A piece of ice, 100 g in mass is kept at 0°C. The amount of heat it requires to melt at 0°C is (take latent heat of melting of ice to be 333.6 J/g). **2016 I**
(a) 750.6 J (b) 83.4 J
(c) 33360 J (d) 3.336 J

22. After a hot sunny day, people sprinkle water on the roof-top because **2016 I**
(a) water helps air around the roof-top to absorb the heat instantly
(b) water has lower specific heat capacity
(c) water is easily available
(d) water has large latent heat of vaporisation

Atomic Structure

23. Neutrons are present in all atoms except
(a) He (b) C (c) H (d) Ne

24. Positive rays in a discharge tube with perforated cathode travel
(a) parallel to the anode
(b) parallel to the cathode
(c) from cathode to anode
(d) from anode to cathode

25. The increasing order (lowest first) for the value of e/m (charge/mass) is
(a) e, p, n, α
(b) n, p, e, α
(c) n, p, α, e
(d) n, α, p, e

26. Which of the following properties of the element can be a whole number?
(a) Atomic mass (b) Atomic number
(c) Atomic radius (d) Atomic volume

27. There are 15 protons and 16 neutrons in the nucleus of an element. The element is represented by the symbol
(a) $^{16}_{15}X$ (b) $^{30}_{15}X$ (c) $^{31}_{16}X$ (d) $^{31}_{15}X$

28. Isotopes of the same elements have
(a) same number of neutrons
(b) same number of protons
(c) same atomic mass
(d) different chemical properties

29. The atomic number of an element of mass number 27, which has 13 neutrons is
(a) 10 (b) 14 (c) 12 (d) 13

30. Which of the following statement(s) is/are true?
(a) Atomic number is always greater than mass number
(b) Mass number may be equal to atomic number
(c) Mass number is equal to the sum of number of protons and electrons
(d) Mass number never be equal to atomic number

31. How many electrons are present in the M shell of an atom of an element with atomic number 24?
(a) 5 (b) 6 (c) 12 (d) 14

32. The atom of an element contains 2 electrons in its M shell. The element is
(a) aluminium (b) sodium
(c) chlorine (d) magnesium

33. While performing cathode ray experiment, it was observed that there was no passage of electric current under normal conditions. Which of the following can account for this observation?
(a) Dust particles are present in air
(b) Carbon dioxide is present in air
(c) Air is a poor conductor of electricity under normal conditions
(d) All of the above

34. If K and L shells of an atom are full, then what would be the total number of electrons in the atom?
(a) 8 (b) 10 (c) 15 (d) 6

35. Taking into account three isotopes of hydrogen and three isotopes of oxygen occurring in nature, how many different kinds of water molecules can we expect on the earth?
(a) Six (b) Nine
(c) Twelve (d) Eighteen

36. Composition of two nuclei is shown below.

	X	Y
Atomic number	7	9
Mass number	15	17

X and Y are
(a) isobars (b) isotones
(c) isotopes (d) isomers

37. **Statement I** X-rays are not deflected by electric field.
Statement II X-rays travels with the velocity of light.
Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

38. Consider the following statements
1. An atom consists of a positively charged sphere and the electrons are embedded in it.
2. The negative and positive charges are equal in magnitude, therefore the atom as a whole is electrically neutral.

Select the correct statement(s).
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

39. Consider the following statements.
1. Cathode rays are originated from cathode and anode rays from anode.
2. Both travel in straight line.
3. Electrons are the fundamental particle of an atom.
4. e/m ratio of anode rays does not depend upon the nature of gas taken in discharge tube.

Select the correct statement(s).
(a) 1 and 4 (b) 2 and 3
(c) 1, 3 and 4 (d) 2, 3 and 4

40. Match the following Columns.

Column I	Column II
A. Proton	1. Chadwick
B. Neutron	2. Millikan
C. Charge of electron	3. Goldstein
D. Shelled nature of atom	4. Rutherford
	5. Madam Curie

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	4	3	(b)	3	1	5	4
(c)	5	2	1	4	(d)	3	1	2	4

Previous Years' Questions

41. Which one among the following statements about an atom is not correct? **2012 I**
(a) Atom always combine to form molecules
(b) Atoms are the basic units from which molecules and ions are formed
(c) Atoms are always neutral in nature
(d) Atoms aggregate in large numbers to form the matter that we can see, feel and touch

42. Which one among the following is not a correct statement? **2013 II**
(a) Cathode rays are negatively charged particles
(b) Cathode rays are produced from all the gases
(c) Electrons are basic constituents of all the atoms
(d) Hydrogen ions do not contain any proton

43. In tritium (T), the number of protons (p) and neutrons (n) respectively, are **2014 II**
(a) 1 p and 1 n (b) 1 p and 2 n
(c) 1 p and 3 n (d) 2 p and 1 n

44. The distribution of electrons into different orbits of an atom, as suggested by Bohr, is **2014 II**
(a) 2 electrons in the K-orbit, 6 electrons in the L-orbit, 18 electrons in the M-orbit
(b) 2 electrons in the K-orbit, 8 electrons in the L-orbit, 32 electrons in the M-orbit
(c) 2 electrons in the K-orbit, 8 electrons in the L-orbit, 18 electrons in the M-orbit
(d) 2 electrons in the K-orbit, 8 electrons in the L-orbit, 16 electrons in the M-orbit

45. Who among the following proposed that atom is indivisible? **2015 I**
(a) Dalton (b) Berzelius
(c) Rutherford (d) Avogadro

46. The atomic theory of matter was first proposed by **2015 II**
(a) John Dalton
(b) Rutherford
(c) J.J. Thomson
(d) Niels Bohr

47. Which one of the following statements is not correct? **2016 II**
(a) Atoms of different elements may have same mass numbers
(b) Atoms of an element may have different mass numbers
(c) All the atoms of an element have same number of protons
(d) All the atoms of an element will always have same number of neutrons

Radioactivity

48. In which categories did Marie Curie win her two different Nobel Prizes?
(a) Physics and Chemistry
(b) Chemistry and Medicine
(c) Physics and Medicine
(d) Chemistry and Peace

49. The α-particles are
(a) high energy electrons
(b) positively charged hydrogen ions
(c) high energy X-ray radiations
(d) double positively charged helium nuclei

50. Gamma rays have
(a) zero mass and no charge
(b) zero mass and positive charge
(c) unit mass and zero charge
(d) unit mass and negative charge

51. An α-particle consists of which of the following?
(a) 2 protons and 2 neutrons
(b) 1 proton and 1 electron
(c) 2 protons and 4 neutrons
(d) 1 proton and 1 neutron

52. Which of the following particles cannot be accelerated?
(a) Electrons (b) α-particles
(c) Neutrons (d) Protons

53. A nuclear reactor produce nuclear energy by
(a) nuclear fusion
(b) spontaneous fission
(c) uncontrolled chain reaction
(d) controlled chain reaction

54. In nuclear reactors, the speed of neutrons is slowed down by
(a) heavy water (b) ordinary water
(c) zinc rods
(d) molten caustic soda

55. The energy generation in stars is
(a) mainly due to the fusion of heavy nuclei
(b) mainly due to the fusion of lighter nuclei
(c) solely due to the fusion of heavy nuclei
(d) due to both fission and fusion of lighter nuclei

56. The difference between a nuclear reactor and atomic bomb is that
(a) no chain reaction takes place in nuclear reactor while in the atomic bomb, there is a chain reaction
(b) the chain reaction in nuclear reactor is controlled
(c) the chain reaction in nuclear reactor is uncontrolled
(d) no chain reaction takes place in atomic bomb while it takes place in nuclear reactor

57. 'Yellow cake' an item of smuggling across the border is
(a) a crude form of heroin
(b) a crude form of cocaine
(c) uranium oxide
(d) unrefined gold

58. Hydrogen bomb is based on the principle of
(a) nuclear fission
(b) nuclear fusion
(c) natural radioactivity
(d) artificial radioactivity

59. In treatment of cancer, which of the following is used?
(a) $^{131}I_{53}$
(b) $^{32}P_{15}$
(c) $^{60}Co_{27}$
(d) $^{2}H_{1}$

60. **Statement I** Hydrogen bomb is more powerful than atomic bomb.
Statement II In hydrogen bomb, reaction is initiated by fission followed by fusion.
Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

61. Consider the following statements
1. Atomic number decreases by 2 and mass number increases by 4 on α-particle emission.
2. Atomic number increases by 1 and mass number remains unchanged on β-particle emission.

Choose the correct statement(s)
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) None of the above

62. Which one of the following is not needed in a nuclear fission reactor?
1. Moderator 2. Accelerator
3. Coolant

Choose the correct option.
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) 2 and 3

63. Match the following Columns.

Column I (Nuclear reactor component)	Column II (Substance used)
A. Moderator	1. Uranium
B. Control rods	2. Graphite
C. Fuel rods	3. Boron
D. Coolant	4. Lead
	5. Sodium

Codes

	A	B	C	D
(a)	2	1	3	5
(b)	2	3	1	5
(c)	3	2	1	5
(d)	3	4	1	2

64. Consider the following statements In nuclear power reactors,
1. uranium-235 is used as the fission material.
2. graphite is used as a moderator.
3. rods of lead are used to control the rate of nuclear reaction.

Which of the above statements are correct?
(a) 1 and 3 (b) 1 and 2
(c) 2 and 3 (d) All of these

65. Consider the following statements
1. Gamma rays are not the constituents of nuclei but they are emitted when a nucleus in an excited state returns to its normal state.
2. The neutron to proton ratio for stable nuclei is always less than one.
3. Beta rays consist of ordinary electrons which are of nuclear origin but do not revolve in orbits.
4. X-rays and gamma rays are electromagnetic radiations.

Which of the above statements are correct?
(a) 2, 3 and 4
(b) 1, 3 and 4
(c) 1, 2 and 4
(d) 1, 2 and 3

66. Match the following Columns.

Column I (Radioisotopes)	Column II (Medicinal uses)
A. ^{60}Co	1. Leukemia
B. ^{131}I	2. Anaemia
C. ^{59}Fe	3. Cancerous tumours
D. ^{32}P	4. Disorders of thyroid gland

Codes

	A	B	C	D		A	B	C	D
(a)	3	4	1	2	(b)	1	2	3	4
(c)	3	4	2	1	(d)	4	3	2	1

67. Which one of the following is heavy water used in nuclear reactor?
(a) Water having molecular weight 18u
(b) Water having molecular weight 20u
(c) Water at 4°C but having molecular weight 19u
(d) Water below the ice in a frozen sea

68. The light emitted by firefly is due to
(a) the radioactive substance
(b) chemiluminiscence process
(c) the photoelectric process
(d) burning of phosphorus

Previous Years' Questions

69. Age of fossil may be found out by determining the ratio of two isotopes of carbon. The isotopes are **2012 I**
(a) C-12 and C-13 (b) C-13 and C-14
(c) C-12 and C-14
(d) C-12 and carbon black

70. Which one of the following reactions is the main cause of the energy radiation from the sun? **2012 I**
(a) Fusion reaction
(b) Fission reaction
(c) Chemical reaction
(d) Diffusion reaction

71. In an atomic explosion, release of large amount of energy is due to the conversion of **2012 II**
(a) chemical energy into nuclear energy
(b) nuclear energy into heat
(c) mass into energy
(d) chemical energy into heat

72. Carbon or graphite rods are used in atomic reactors as moderators for sustained nuclear chain reaction through nuclear fission process. In this process, **2014 II**
(a) the neutrons are made fast
(b) the protons are made fast
(c) the neutrons are made slow
(d) the protons are made slow

73. In an observation, α-particles, β-particles and γ-rays have same energies. Their penetrating power in a given medium in increasing order will be **2015 I**
(a) α, β, γ (b) β, γ, α
(c) α, γ, β (d) β, α, γ

Chemical Bonding and Redox Reactions

74. In case of a cation,
(a) number of electrons are more than the number of protons
(b) number of electrons are equal to the number of protons
(c) number of electrons are more than the number of neutrons
(d) number of electrons are less than the number of protons

75. Atomic numbers of *A* and *B* are 2 and 12, respectively. Valency (or number of electrons taking part in bonding) is two in case of
(a) Only *A* (b) Only *B*
(c) Both *A* and *B* (d) Neither *A* nor *B*

76. A bond formed by the transfer of electrons between atoms of the elements is called
(a) ionic bond (b) covalent bond
(c) coordinate bond (d) hydrogen bond

77. The compound which contains ionic bond is
(a) CH_4 (b) N_4
(c) $CaCl_2$ (d) CCl_4

78. Which of the following is a covalent compound?
(a) NaCl (b) $CaCl_2$
(c) $MgCl_2$ (d) C_2H_2

79. Strongest bond is
(a) $>C-C<$ (b) $>C=C<$
(c) $-C\equiv C-$ (d) All of these

80. Molten NaCl is a good conductor of electricity due to the presence of
(a) free electrons
(b) free ions
(c) free molecules
(d) None of the above

81. Which one has hydrogen bonding?
(a) HCl (b) HBr
(c) HF (d) HI

82. Hydrogen bonding is maximum in
(a) ethanol (b) diethyl ether
(c) ethyl chloride (d) triethyl amine

83. Salt is soluble in water because
(a) water forms hydrogen bond with it
(b) water is a polar solvent and interact with the ions of salt
(c) water is a non-polar solvent and exists as water of crystallisation
(d) water loses electrons and form ionic compound with salt

84. In the following reaction,
$Zn + [Ag(CN)_2]^- \longrightarrow [Zn(CN)_4]^{2-} + Ag$
(a) Zn is reduced and Ag^+ is oxidised
(b) Zn is oxidised and Ag^+ is reduced
(c) Zn is reduced and Ag is oxidised
(d) Neither Zn nor Ag is oxidised

85. Reducing agent can
(a) accept protons
(b) donate protons
(c) accept electrons
(d) donate electrons

86. Starch iodide paper is used to test for the presence of
(a) iodine (b) iodide ion
(c) oxidising agent (d) reducing agent

87. Oxidation number of N in N_3H is
(a) -3 (b) $+3$
(c) $+\frac{1}{3}$ (d) $-\frac{1}{3}$

88. A redox reaction is
(a) proton transfer reaction
(b) ion combination reaction
(c) a reaction in solution
(d) electron transfer reaction

89. From the following elements, which one has the same oxidation state in all of its compounds?
(a) Hydrogen (b) Fluorine
(c) Carbon (d) Oxygen

90. Consider the following statements about ionic compounds.
1. Ionic compounds are insoluble in alcohol.
2. Ionic compounds in the solid state are good conductor of electricity.

Which of these statement(s) is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

91. The fatty substances can be protected from being rancid by
1. storing them in refrigerator.
2. keeping them in an inert atmosphere.
3. adding some antioxidant.

The correct option is
(a) 1 and 2 (b) 2 and 3
(c) 3 and 1 (d) All of these

92. **Statement I** Nitric acid , hydrogen peroxide and potassium dichromate are reducing agents.
Statement II The substance whose oxidation number increases is oxidised and it is a reducing agent.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

93. Consider the following statements
1. The oxidation state of C in CO_2 is +4 which is its maximum oxidation state.
2. Nitrous acid (HNO_2) sulphurous acid (H_2SO_3) and sulphur dioxide (SO_2) act as as both reducing as well as oxidising agent.

Which of the statement(s) given above is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

94. Which of the following statement(s) is/are true?
1. The process of oxidation leads to a gain of electrons.
2. The process of oxidation leads to a loss of electrons.
3. The process of reduction leads to a gain of electrons.
4. The process of reduction leads to a loss of electrons.

Select the correct answer using the codes given below.
(a) Only 1 (b) Only 4
(c) 2 and 3 (d) 1 and 4

95. Match the following Columns

Column I	Column II
A. Ionic bond	1. NH_3
B. Covalent bond	2. Diamond
C. Molecular bond	3. NaCl
D. Hydrogen bond	4. Iodine

Codes

	A	B	C	D
(a)	1	2	4	3
(b)	3	2	4	1
(c)	2	3	1	4
(d)	4	1	3	2

Previous Years' Questions

96. $NaOH + HCl \longrightarrow NaCl + H_2O$
In the given chemical reaction, **2013 II**
(a) sodium is oxidised and oxygen is reduced
(b) sodium is oxidised and chlorine is reduced
(c) sodium and hydrogen are oxidised
(d) None of them are oxidised or reduced

97. Date of manufacture of food items fried in oil should be checked before buying because oils become rancid due to **2014 I**
(a) oxidation
(b) reduction
(c) hydrogenation
(d) decrease in viscosity

Gas Laws and Solutions

98. At constant temperature, the product of pressure and volume of a given amount of a gas is constant. This is
(a) Gay-Lussac law
(b) Charles' law
(c) Boyle's law
(d) Avogadro's law

99. Charles' law is represented by
(a) $V \propto \frac{1}{T}$ (b) $V = T$
(c) $V \propto T$ (d) $V \propto \frac{1}{T^3}$

100. At constant temperature and pressure, the volume of a gas is directly proportional to the number of molecules. This law had been given by
(a) Boyle
(b) Charles
(c) Graham
(d) None of the above

101. When a balloon is filled with air, its pressure and volume both increases, so it is a violation of
(a) Boyle's law (b) Charles' law
(c) Avogadro's law (d) All of these

102. An ideal gas is one which obeys
(a) Boyle's law (b) Charles' law
(c) Avogadro's law (d) All of these

103. In general gas equation $pV = nRT$, V is the volume of
(a) n moles of a gas
(b) any amount of a gas
(c) one mole of a gas
(d) one gram of a gas

104. Which one of the following is not the value of R ?
(a) 1.99 cal K^{-1} mol^{-1}
(b) 0.0821 L atm $K^{-1}mol^{-1}$
(c) 9.8 kcal K^{-1} mol^{-1}
(d) 8.3 JK^{-1} mol^{-1}

105. Cooking of chapatti (roti) is a practical implementation of
(a) Boyle's law (b) Charles' law
(c) Avogadro's law (d) All of these

106. Absolute zero is the temperature where all the gases are expected to have
(a) different volumes
(b) same volume (c) zero volume
(d) None of the above

107. The density of the gas is equal to
(a) np (b) Mp/RT (c) p/RT (d) M/V

108. Deviations from ideal behaviour will be more if the gas is subjected to
(a) low temperature and high pressure
(b) high temperature and low pressure
(c) low temperature
(d) high temperature

109. The rate of diffusion of hydrogen is about
(a) one half that of helium
(b) 1.4 times that of helium
(c) twice that of helium
(d) four times that of helium

110. Rate of diffusion of a gas is
(a) directly proportional to its density
(b) directly proportional to its molecular mass
(c) directly proportional to the square of its molecular mass
(d) inversely proportional to the square root of its molecular mass

111. A bottle of dry ammonia and one of dry hydrogen chloride are connected through a long tube. The stoppers at both ends of the tube are opened, simultaneously. The white ammonium chloride ring will first formed
(a) at the centre of the tube
(b) near the hydrogen chloride bottle
(c) near the ammonia bottle
(d) throughout the length of the tube

112. A solution which contains the maximum amount of the solute that is dissolved in a given amount of solvent at a particular temperature is called
(a) saturated solution
(b) unsaturated solution
(c) supersaturated solution
(d) None of the above

113. The number of moles of solute per kg of a solvent is called
(a) mole fraction (b) normality
(c) molality (d) molarity

114. 8 g of NaOH is dissolved in l L of solution. Its molarity is
(a) 0.8 M (b) 0.4 M
(c) 0.2 M (d) 0.1 M

115. The molarity of pure water at 298 K is
(a) 5.5 M (b) 5.55 M
(c) 5.50 M (d) 55.55 M

116. In which mode of expression, the concentration of the solution remains independent of temperature?
(a) Normality (b) Molality
(c) Molarity (d) Formality

117. At a given temperature, the solubility of a gas in liquid increases with
(a) increase in temperature
(b) reduction is gas pressure
(c) increase in gas pressure
(d) amount of liquid taken

118. Low concentration of oxygen in the blood and tissues of people living at high altitude is due to
(a) the low temperature
(b) the low atmospheric pressure
(c) the high atmospheric pressure
(d) both low temperature and high atmospheric pressure

119. Which one of the following is involved for the desalination of sea water?
(a) Reverse osmosis
(b) Simple osmosis
(c) Use of sodium aluminium silicate as zeolite
(d) Use of ion selective electrodes

120. Scuba divers are at risk due to high concentration of dissolved gases while breathing air at high pressure under water. The tanks used by scuba divers are filled with
(a) air diluted with helium
(b) O_2
(c) N_2
(d) a mixture of N_2 and helium

121. Which of the following statement(s) is/are correct?
1. During the pressure of osmosis, the solvent travels from the concentrated solution to the dilute solution.
2. In the reverse osmosis, external pressure is applied to the dilute solution.

Codes
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

122. **Statement I** A chemical reaction becomes faster at higher temperature.
Statement II At high temperature, molecular motion becomes more rapid.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

123. Match the following Columns

Column I (Concentration of solution)	Column II (Unit)
A. Molarity	1. mol kg^{-1}
B. Molality	2. mol L^{-1}
C. Normality	3. Gram equiv L^{-1}

Codes

	A	B	C		A	B	C
(a)	1	2	3	(b)	3	2	1
(c)	2	1	3	(d)	2	3	1

Previous Years' Question

124. By what mechanism does scent spread all over the room if the lid is opened? **2013 II**
(a) Pressure in the bottle
(b) Compression from the bottle
(c) Diffusion
(d) Osmosis

Acids, Bases and Salts

125. An acid is a substance which
(a) donates a proton
(b) accepts an electron pair
(c) gives H^+ in water
(d) All of the above

126. Formic acid is obtained from
(a) red ants (b) fats
(c) vinegar (d) orange

127. An aqueous solution of NH_4Cl is
(a) basic (b) neutral
(c) acidic (d) amphoteric

128. Burning of substance X produces a gas, solution of which turns red litmus paper blue. The substance X is
(a) base (b) acid
(c) salt (d) None of these

129. Which is not a Lewis base?
(a) H_2O (b) NH_3
(c) CO_2 (d) BF_3

130. Out of the three solutions, i.e. A, B and C, only one turns blue litmus red. This shows that the remaining two are
(a) acidic
(b) definitely neutral
(c) either acidic or basic
(d) definitely basic

131. The negative logarithmic value of hydrogen ion concentration is called
(a) pH (b) pOH (c) pK_a (d) pK_b

132. Water is neither acidic nor alkaline because
(a) it boils at a high temperature
(b) it cannot donate or accept electrons
(c) it can dissociate into equal number of hydrogen and hydroxyl ions
(d) it cannot accept or donate protons

133. The pH value of wine is
(a) 6.5 (b) 2.8 (c) 8.5 (d) 7.0

134. The pH value of sea water is
(a) 8.5 (b) 2.6 (c) 3.0 (d) 2.5

135. What is the pH value of pure water?
(a) 1 (b) 6 (c) 7 (d) 10

136. pH values of four solutions X, Y, Z, and P are 5, 7, 9 and 2, respectively. Which of these is the strongest acid?
(a) X (b) Z (c) P (d) Y

137. *Aqua-regia* used by chemists to separate silver and gold is a mixture of
(a) hydrochloric acid (concentrated) and nitric acid (concentrated)
(b) hydrochloric acid (concentrated) and sulphuric acid (concentrated)
(c) nitric acid (concentrated) and sulphuric acid (concentrated)
(d) hydrochloric acid (dilute) and sulphuric acid (dilute)

138. Consider the following statements about pH value
1. Human blood has pH value in the range of 7.36-7.42.
2. A change in pH of blood by 0.2 blood by 0.2 units results in death.

Which of the statement(s) given above is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

139. When lime juice is dropped on baking soda, brisk effervescence takes place because the gas evolved is
1. hydrogen 2. oxygen
3. carbon dioxide

Choose the correct option.
(a) Only 1 (b) Only 3
(c) 1 and 3 (d) All of these

140. **Statement I** Solution of blue vitriol ($CuSO_4$) turns red litmus blue.
Statement II It is a salt of strong acid and weak base.
Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

141. Match the following Columns.

Column I (Acid)	Column II (Use)
A. Boric acid	1. Explosives (TNT)
B. Oxalic acid	2. Preservatives
C. Benzoic acid	3. Rust stain remover
D. Nitric acid	4. Eye wash

Codes

	A	B	C	D
(a)	1	3	2	4
(b)	4	3	2	1
(c)	4	2	3	1
(d)	4	1	2	3

142. **Statement I** Addition of water to an aqueous solution of HCl decreases the pH.
Statement II Addition of water suppresses the ionisation of HCl.
Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

143. Match the following Columns.

Column I (Acid)	Column II (Source)
A. Lactic acid	1. Tamarind
B. Tartaric acid	2. Orange
C. Oxalic acid	3. Tomato
D. Citric acid	4. Sour curd

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	1	4	(b)	2	1	3	4
(c)	4	3	1	2	(d)	4	1	3	2

Previous Years' Questions

144. Which one among the following statements is correct? **2012 I**
(a) All bases are alkali
(b) None of the bases is alkali
(c) There are no more bases except the alkalis
(d) All alkalis are bases but all bases are not alkalis

145. The pH of fresh ground water slightly decreases upon exposure to air because **2012 I**
(a) carbon dioxide from air is dissolved in the water
(b) oxygen from air is dissolved in the water
(c) the dissolved carbon dioxide of the ground water escapes into air
(d) the dissolved oxygen of the ground water escapes into air

146. Antacids are commonly used to get rid of acidity in the stomach. A commonly used antacid is **2012 II**
(a) sodium hydrogen phthalate
(b) magnesium hydroxide
(c) calcium hydroxide
(d) manganese acetate

147. On the labels of the bottles, some soft drinks are claimed to be acidity regulators. They regulate acidity by using **2012 II**
(a) carbon dioxide
(b) bicarbonate salts
(c) Both (a) and (b)
(d) carbon dioxide and lime

148. Which of the following solutions will not change the colour of blue litmus paper to red?
1. Acidic solution
2. Basic solution
3. Common salt solution

Select the correct answer using the codes given below **2014 I**
(a) 1 and 3 (b) 2 and 3
(c) Only 1 (d) Only 2

149. **Statement I** During indigestion, milk of magnesia is taken to get rid of pain in the stomach.
Statement II Milk of magnesia is a base and it neutralises the excess acid in the stomach.
Codes **2015 I**
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

Chemical Thermodynamics and Surface Chemistry

150. A system which can exchange energy with the surroundings but no matter is called
(a) a heterogeneous system
(b) an open system
(c) a closed system
(d) an isolated system

151. When a gas is subjected to adiabatic expansion, it gets cooled due to
(a) no change in entropy
(b) loss in kinetic energy
(c) decrease in velocity
(d) energy spent in doing work

152. If a refrigerator's door is kept open, then
(a) room will be cooled
(b) room will be heated
(c) may get cooled or heated depending upon the weather
(d) no effect on room

153. When ammonium chloride is dissolved in water, the solution becomes cold. The change is
(a) endothermic (b) exothermic
(c) supercooling (d) None of these

154. Which one is true?
(a) 1 calorie > 1 erg > 1 joule
(b) 1 erg > 1 calorie > 1 joule
(c) 1 calorie > 1 joule > 1 erg
(d) 1 joule > 1 calorie > 1 erg

155. Out of the following given systems, in case of which exchange of energy takes place but not of matter?
(a) Jet engine
(b) Tea placed in steel kettle
(c) Pressure cooker
(d) Rocket engine during propulsion

156. When a strong beam of light is passed through a colloidal solution, the light is
(a) reflected (b) scattered
(c) Both (a) and (b) (d) None of these

157. Colloidal solution commonly used in the treatment of eye disease is
(a) colloidal silver
(b) colloidal gold
(c) colloidal antimony
(d) colloidal sulphur

158. Which of the following colloidal solution is commonly used as germ killer?
(a) Colloidal sulphur
(b) Colloidal gold
(c) Colloidal silver
(d) Colloidal antimony

159. Which one of the following is correctly matched?
(a) Aerosol-smoke (b) Foam-mist
(c) Emulsion-curd (d) All of these

160. In case of which of the following, one phase is not dispersed in another?
(a) Chlorophyll (b) Smoke
(c) Ruby glass (d) Milk

161. Which of the following makes path of the light visible when passing through it?
(a) Air (b) A gold ring
(c) Smoke (d) Salt water

162. Fog is an example of colloidal system of
(a) liquid in a gas (b) gas in a liquid
(c) gas in a solid (d) solid in a liquid

163. An emulsion consists of
(a) two liquids (b) two solids
(c) two gases (d) two salts

164. Artificial rain is caused by spraying
(a) neutral charged colloidal dust over a cloud
(b) similar charged colloidal dust over a cloud
(c) Both (a) and (b)
(d) opposite charged colloidal dust over a cloud

165. The bleeding of a wound is stopped by the application of ferric chloride because
(a) blood starts flowing in the opposite direction
(b) ferric chloride seals the blood vessels
(c) blood reacts and a solid is formed which seals the blood vessels
(d) blood is coagulated and the blood vessels are sealed

166. Dialysis can separate which of the following in addition to the glucose from human blood?
(a) Fructose (b) Starch
(c) Protein (d) Sucrose

167. Generally, the path of light becomes visible when we enters in a cinema hall. The similar effect is observed in case of all except
(a) gold sol (b) sugar solution
(c) emulsions (d) suspension

168. A catalyst increases the rate of a chemical reaction by
(a) increasing the activation energy
(b) decreasing the activation energy
(c) reacting with reactants
(d) reacting with products

169. TEL minimises the knocking effect when mixed with petrol. Here, it acts as a/an
(a) positive catalyst (b) negative catalyst
(c) autocatalyst (d) induced catalyst

170. The catalyst used in the hydrogenation of oils is
(a) Ni (b) Cu (c) Fe (d) Pb

171. The temperature at which the catalytic activity of the catalyst is maximum, is called
(a) critical temperature
(b) room temperature
(c) absolute temperature
(d) optimum temperature

172. In the Haber's process for the synthesis of NH_3,
(a) Mo acts as a catalyst and Fe as a promotor
(b) Fe acts as a catalyst and Mo as a promotor
(c) Fe acts as an inhibitor and Mo as a catalyst
(d) Fe acts as a promotor and Mo as autocatalyst

173. The enzymes are killed
(a) at a very high temperature
(b) during chemical reaction
(c) at low temperature
(d) under atmospheric pressure

174. Enzyme catalysis is an example of
(a) autocatalysis
(b) heterogeneous catalysis
(c) homogeneous catalysis
(d) induced catalysis

175. Which of the following statements is true?
(a) All the catalyst are enzymes
(b) All the enzymes are biocatalysts
(c) All other catalysts except enzymes are called biocatalysts
(d) Biocatalysts are non-proteinous in nature

176. Which of the following properties is/are shown by colloid?
1. Absorption 2. Tyndall effect
3. Paramagnetism

Choose the correct code
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) All of these

177. Gelatin is often used as an ingredient in the manufacture of ice cream. This is due to
1. to cause the mixture to solidify
2. to stabilise the colloid and prevent crystal growth

Choose the correct reason(s) and select the correct code.
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

178. **Statement I** In a chemical reaction, a catalyst increases the rate of reaction.
Statement II Catalyst possesses high activation energy.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

179. Consider the following statements
1. Emulsifiers stabilise the emulsion.
2. Soaps and detergents are emulsifiers.
3. Cleansing action of soap is due to the formation of emulsions.

Which of the above statements is/are correct?
(a) Only 2 (b) 1 and 2
(c) 2 and 3 (d) All of these

180. Which of the following statements are true regarding a catalyst?
1. It is a substance which increases the rate of reaction.
2. It is a substance which reduces the activation energy.
3. It is a substance which increases the activation energy.
4. It is a substance which is consumed in the reaction.

Select the correct answer using the codes given below
(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 4

181. Match the following Columns.

Column I	Column II
A. Smoke	1. Dispersion of gas in liquid
B. Gel	2. Dispersion of solid in solid
C. Emulsion	3. Dispersion of liquid in liquid
D. Foam	4. Dispersion of liquid in solid
	5. Dispersion of solid in gas

Codes

	A	B	C	D		A	B	C	D
(a)	4	2	3	1	(b)	5	4	3	1
(c)	2	4	1	5	(d)	5	2	1	3

Previous Years' Questions

182. What is the role of positive catalyst in a chemical reaction? **2013 I**
(a) It increases the rate of reaction
(b) It decreases the rate of reaction
(c) It increases the yield of the products
(d) It provides better purity of the products

183. Which type of mixture is smoke? **2013 II**
(a) Solid mixed with a gas
(b) Gas mixed with a gas
(c) Liquid mixed with a gas
(d) Gas mixed with a liquid and a solid

Electrochemistry

184. Electrolysis finds its applications in all the places except in
(a) production of hydrogen
(b) electrorefining of copper
(c) Leclanche cell
(d) galvanisation

185. During the electrolysis of fused NaCl which reaction will take place at anode?
(a) Na^+ ions are oxidised
(b) Cl^- ions are oxidised
(c) Na^+ ions are reduced
(d) Cl^- ions are reduced

186. Pure water does not conduct electricity because it is
(a) almost non-ionised
(b) low boiling
(c) neutral
(d) readily decomposed

187. Which of the following aqueous solutions will conduct an electric current quite well?
(a) Glycerol (b) HCl
(c) Sugar (d) Pure water

188. Which one of the following metals is less reactive than hydrogen?
(a) Barium
(b) Copper
(c) Lead
(d) Magnesium

189. The life span of a Daniell cell may be increased by
(a) large Zn electrode
(b) large Cu electrode
(c) lowering the temperature
(d) lowering the concentrations

190. The device which converts chemical energy of fuels such as CO, CH_4 directly into electrical energy is called
(a) concentration cell
(b) galvanic cell
(c) fuel cell
(d) Both (b) and (c)

191. The source of electrical energy on the Apollo moon flights was
(a) a generator set
(b) fuel cell
(c) Ni-Cd cell
(d) lead storage batteries

192. When a lead storage battery is charged, it acts as
(a) a primary cell
(b) a galvanic cell
(c) a concentration cell
(d) an electrolytic cell

193. Iron rusts in the presence of
(a) vacuum
(b) moisture
(c) oxygen
(d) Both (b) and (c)

194. The depolariser used in dry cell batteries is
(a) manganese dioxide
(b) NH_4Cl
(c) sodium triphosphide
(d) potassium hydroxide

195. The electrolytes which is generally used in one time batteries that are used in cameras, hearing aids etc., is
(a) ammonium chloride
(b) zinc oxide
(c) alkaline zinc oxide
(d) paste of potassium hydroxide and zinc oxide

196. Corrosion is basically a/an
(a) interaction
(b) union between two light metals and a heavy metal
(c) altered reaction in the presence of H_2O
(d) electrochemical phenomenon

197. Which one of the following pairs of materials serves as electrodes in rechargeable batteries commonly used in devices like torchlights, electric shavers, etc?
(a) Nickel and cadmium
(b) Zinc and carbon
(c) Lead peroxide and lead
(d) Iron and cadmium

198. To protect iron against corrosion, the durable metal plating on it, is
(a) tin plating
(b) copper plating
(c) zinc plating
(d) nickel plating

199. In order to prevent corrosion, iron pipes are often coated with a layer of zinc. This process is known as
(a) electroplating (b) annealing
(c) galvanisation (d) vulcanisation

200. When items or jewellery made of metals such as copper of nickel are placed in a solution having a salt of gold, a thin film of gold is deposited by
1. cooling to below 0°C
2. heating above 100°C
3. passing an electric current

Choose the correct code.
(a) 1 and 2 (b) Only 3
(c) 1 and 3 (d) Only 1

201. Sacrificial anode protects iron or ships, underground pipelines etc., from rusting, a process known as cathodic protection Which of the following metals cannot be used as a sacrifical anode?
1. Tin 2. Zinc
3. Magnesium 4. Aluminium

Choose the correct code.
(a) Only 1 (b) Only 4
(c) 2 and 3 (d) All of these

202. **Statement I** Zinc displaces copper and silver from solutions of their salts.
Statement II Zinc has a lower oxidation potential than copper and silver.

Codes
(a) Both the Statements are correct and Statement II is not correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

203. **Statement I** Zinc is used for galvanisation of iron.
Statement II Zinc is an amphoteric element.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

204. Which of the following statements about the commonly used automobile battery are true?
1. It is usually a lead-acid battery.
2. It has six cells with a potential of 2V each.
3. Its cells work as galvanic cells while discharging power.
4. Its cells work as electrolytic cells while recharging.

Select the correct answer using the codes given below.
(a) 3 and 4 (b) 1, 2 and 3
(c) 2 and 4 (d) All of these

205. Match the following

List I (Batteries)	List II (Cathode)
A. Lechlanche cell	1. Paste of HgO and carbon
B. Lead storage battery	2. Nickel dioxide
C. Nickel-Cadmium cell	3. Carbon (graphite) rod
D. Mercury cell	4. Lead packed with PbO_2

Codes

	A	B	C	D		A	B	C	D
(a)	1	3	2	4	(b)	4	3	2	1
(c)	3	4	2	1	(d)	1	2	4	3

Previous Years' Questions

206. Food cans are coated with tin but not with zinc because **2013 I**
(a) zinc is costlier than tin
(b) zinc has a higher melting point than tin
(c) zinc is more reactive than tin
(d) tin is more reactive than zinc

207. Iron sheet kept in moist air covered with rust. Rust is **2014 I**
(a) an element
(b) a compound
(c) a mixture of iron and dust
(d) a mixture of iron, oxygen and water

208. Which one among the following metals is prominently used in mobile phone batteries? **2014 II**
(a) Copper (b) Zinc
(c) Nickel (d) Lithium

209. Electricity is produced through dry cell from **2015 I**
(a) chemical energy
(b) thermal energy
(c) mechanical energy
(d) nuclear energy

Inorganic Chemistry

210. Magnalium is an alloy of magnesium with
(a) silicon (b) chlorine
(c) aluminium (d) calcium

211. Composition of bauxite is
(a) $Al_2O_3 \cdot 2H_2O$ (b) $Al_2O_3 \cdot H_2O$
(c) $Al_2O_3 \cdot 4H_2O$ (d) $Al_2O_3 \cdot 3H_2O$

212. Which of the following occurs in the native state?
(a) Silver (b) Copper
(c) Both Ag and Au (d) Carbon

213. Most abundant metal in earth's crust is
(a) carbon (b) steel
(c) iron (d) aluminium

214. The percentage of carbon is highest in
(a) tool steels (b) steel
(c) cast iron (d) solder

215. Carnallite is an ore of
(a) Cu (b) Co
(c) Mg (d) Al

216. The formula of Plaster of Paris is
(a) $CaSO_2 \cdot 2H_2O$
(b) $CaSO_4 \cdot \frac{1}{2}H_2O$
(c) $2CaSO_2 \cdot 8H_2O$
(d) $CaSO_4 \cdot 8H_2O$

217. Bone black is obtained by
(a) treating bones with H_2SO_4
(b) destructive distillation of bones
(c) heating bones
(d) heating wood

218. Chief ore of iron is
(a) bauxite (b) magnetite
(c) haematite (d) dolomite

219. Iron used for casting is
(a) steel (b) wild steel
(c) pig iron (d) wrought iron

220. Which is found in transistors?
(a) Ge (b) Ag (c) Pb (d) Kr

221. Noble gases are
(a) monoatomic (b) diatomic
(c) triatomic (d) pentaatomic

222. Nitrous oxide is called
(a) laughing gas (b) tear gas
(c) producer gas (d) methane gas

223. Stainless steel is a/an
(a) element (b) compound
(c) alloy (d) None of these

224. Which one is the mineral of aluminium?
(a) Malachite (b) Azurite
(c) Bauxite (d) Galena

225. What is the jeweller's rouge?
(a) Ferric oxide
(b) Ferrous oxide
(c) Ferrous carbonate
(d) Ferric carbonate

226. 'Misch metal' is widely used in the manufacture of which of the following?
(a) Material of car brake
(b) Smoke detectors
(c) Cigarette lighters
(d) Emergency lights

227. German silver is an alloy of
(a) gold and silver
(b) copper and silver
(c) copper, zinc and silver
(d) copper, zinc and nickel

228. Which one of the following is a mordant used in dyeing?
(a) Calcium hydroxide
(b) Aluminium sulphate
(c) Calcium carbonate
(d) Zinc phosphate

229. The following substances are used in low temperature applications
(a) liquid argon (b) liquid nitrogen
(c) dry ice (d) liquid air

230. Permanent magnets can be made from
(a) cobalt (b) aluminium
(c) zinc (d) lead

231. The most electronegative element among the following is
(a) fluorine (b) oxygen
(c) sodium (d) sulphur

232. The element which forms ions in dimeric state is
(a) Hg (b) Cu (c) Be (d) Ni

233. Diamond used in jewellery is
(a) a compound
(b) a metal
(c) a mixture of compounds
(d) an element

234. Dry ice is
(a) supercooled ice
(b) solid water with zero humidity
(c) solid carbon dioxide
(d) solidified ammonia

235. Which one of the following varieties of coal has the highest amount of carbon in it?
(a) Anthracite (b) Bituminous
(c) Lignite (d) Peat

236. Which of the following has largest electron affinity?
(a) F (b) Cl (c) Br (d) I

237. Lead painting and lead articles when exposed to atmosphere turn black due to reaction with
(a) carbon dioxide
(b) hydrogen sulphide
(c) oxygen
(d) sulphur dioxide

238. Which of the following is not a transition metal?
(a) Silver (b) Lead
(c) Tungsten (d) Manganese

239. Zero group elements are also called
(a) noble gases
(b) rare earth elements
(c) alkali metals
(d) Transition metals

240. The process of extraction of metal from ore is called
(a) calcination
(b) conductivity
(c) concentration
(d) metallurgy

241. Gypsum ($CaSO_4 \cdot 2H_2O$) is added to clinker during cement manufacturing to
(a) decrease the rate of setting of cement
(b) bind the particles of calcium silicate
(c) facilitate the formation of colloidal gel
(d) impart strength to cement

242. What is the main constituent of a Pearl?
(a) Calcium carbonate and magnesium carbonate
(b) Calcium sulphate only
(c) Calcium oxide and calcium sulphate
(d) Calcium carbonate only

243. Sodium thiosulphate ($Na_2S_2O_3$) solution is used in photography to
(a) remove reduced silver
(b) reduce silver bromide (AgBr) grain to silver
(c) remove undecomposed AgBr as a soluble silver thiosulphate complex
(d) convert the metallic silver to silver salt

244. Which one of the following is the softest?
(a) Sodium
(b) Aluminium
(c) Iron
(d) Copper

245. Which one of the following is the secondary source of light in a fluorescent lamp?
(a) Neon gas
(b) Argon gas
(c) Mercury vapour
(d) Fluorescent coating

246. Ordinary water, when compared to the heavy water used in nuclear reactors is
(a) several times lighter
(b) marginally lighter
(c) heavy
(d) as heavy because chemically both are the same

247. Carbon exists in a three-dimensional tetrahedral structure, which is the hardest among all the known solids. This solid is kept under the category of
(a) metallic crystal (b) molecular crystal
(c) ionic crystal (d) covalent crystal

248. As which one of the following, does carbon occur in its purest form in nature?
(a) Carbon black (b) Graphite
(c) Diamond (d) Coal

249. Which one of the following vitamins contains cobalt?
(a) Vitamin B_1 (b) Vitamin B_2
(c) Vitamin B_6 (d) Vitamin B_{12}

250. Which of the following metals are present in haemoglobin and chlorophyll, respectively?
(a) Fe and Mg
(b) Fe and Zn
(c) Mg and Zn
(d) Zn and Mg

251. The process of heating ore in absence of air is called
(a) smelting (b) decomposition
(c) roasting (d) calcination

252. What is the purpose of adding baking soda to dough?
(a) To generate moisture
(b) To give a good flavour
(c) To give good colour
(d) To generate carbon dioxide

253. Naturally occurring metal containing compound is called
(a) matrix (b) ore
(c) gangue (d) mineral

254. Which of the following properties generally decreases along a period?
(a) Atomic size
(b) Electron affinity
(c) Ionisation energy
(d) Electronegativity

255. Long form of periodic table is based on the properties of elements as a function of
(a) atomic size (b) atomic mass
(c) electronegativity (d) atomic number

256. The elements belonging to second period are called
(a) normal elements
(b) noble gases
(c) transition elements
(d) None of the above

257. On extracting a compound from a mixture, if it remain also associated with oxide of dihydrogen, then the best scientific method to test the presence of this liquid is
(a) smell
(b) taste
(c) use of litmus paper
(d) use of anhydrous copper sulphate

258. Which one of the following is not a periodic property, i.e. does not show any trend on moving from one side to the other in the periodic table?
(a) Atomic size (b) Valency
(c) Radioactivity (d) Electronegativity

259. Heavy water implies
1. water which is used in heavy industries such as thermal power plants
2. water which contains SO_4^{2-} and Cl^- of calcium and magnesium

Choose the correct code.
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

260. Water is a good coolant and is used to cool the engines of cars, buses, trucks etc. It is because water has a
1. high specific heat
2. high boiling point

Choose the correct code.
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

261. Consider the following statements about graphite.
1. It is a good conductor of electricity.
2. It is an allotropic modification of carbon.
3. It has a high refractive index.
4. It is present in the non-volatile part of crude petroleum.

Which of the above statements are correct?
(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1, 3 and 4

262. Consider the following elements.
1. Cobalt 2. Gold
3. Nickel 4. Silver

Which of these are magnetic substances?
(a) 1 and 2 (b) 1 and 3
(c) 2 and 4 (d) 3 and 4

263. Consider the following statements
1. Nitric acid is used in the production of fertilisers.
2. Sulphuric acid is used in the production of explosives.

Which of the statements given above is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

264. **Statement I** On mixing with water, plaster of Paris hardens.
Statement II By combining with water, plaster of Paris is converted into gypsum.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

265. **Statement** I During the setting of cement, the structure has to be cooled by spraying water.
Statement II The constituents of cement undergo hydration during setting of cement and it is an exothermic reaction.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

266. Arrange the following in the descending order of their carbon content.
1. Cast iron 2. Wrought iron
3. Steel

Codes
(a) 1, 2, 3 (b) 2, 1, 3
(c) 3, 2, 1 (d) 1, 3, 2

267. **Statement I** It is dangerous to pour water on hot oil while cooking.
Statement II Boiling point of water is higher than that of cooking oil.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

268. Match the following Columns.

Column I	Column II
A. Zeolite	1. Purgative
B. Sodium thiosulphate	2. Safety matches
C. Magnesium sulphate	3. Dry cell
D. Graphite	4. Purification of water
	5. Photography

Codes

	A	B	C	D
(a)	3	5	2	4
(b)	3	2	1	5
(c)	4	1	2	3
(d)	4	5	1	3

269. Match the following Columns.

Column I (Alloy)	Columns II (Composition)
A. Bronze	1. Lead, antimony, tin
B. Brass	2. Copper, zinc, nickel
C. German silver	3. Copper, zinc
D Type metal	4. Copper, tin
	5. Copper, zinc, nickel, aluminium

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	2	5	(b)	4	3	2	1
(c)	5	3	2	1	(d)	4	3	5	1

270. Match the following Columns.

Column I (Alloy)	Column II (Constituent)
A. Solder	1. Iron and carbon
B. Brass	2. Copper and zinc
C. Bronze	3. Copper and tin
D. Steel	4. Lead and tin

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	2	3	1
(c)	1	3	2	4	(d)	4	3	2	1

271. Which of the following when dissolved in H_2O gives hissing sound?
1. Limestone
2. Slaked lime
3. Quickime

Choose the correct code
(a) Only 2 (b) Only 3
(c) 1 and 3 (d) All of these

Previous Years' Questions

272. The elements of a group in the periodic table 2012 I
(a) have similar chemical properties
(b) have consecutive atomic numbers
(c) are isobars
(d) are isotopes

273. The gas which turns lime water milky is 2012 II
(a) carbon dioxide
(b) carbon monoxide
(c) ammonia
(d) nitrogen dioxide

274. Why hard water does not give lather with soap? 2013 I
(a) Hard water contains calcium and magnesium ions which form precipitate with soap
(b) Hard water contains sulphate and chloride ions which form precipitate
(c) pH of hard water is high
(d) pH of hard water is less

275. Which one among the following metals is used in fireworks to make a brilliant white light? 2013 I
(a) Sodium (b) Magnesium
(c) Aluminium (d) Silver

276. Which allotropy of carbon is in rigid three-dimensional structure? 2013 I
(a) Graphite (b) Fullerene
(c) Diamond (d) Carbon black

277. What are the elements which are liquids at room temperature and standard pressure? 2013 II
1. Helium 2. Mercury
3. Chlorine 4. Bromine

Select the correct answer using the codes given below.
(a) 2 and 3 (b) 2, 3 and 4
(c) 2 and 4 (d) 1 and 3

278. Which element forms the highest number of compounds in the periodic table? 2013 II
(a) Carbon (b) Oxygen
(c) Silicon (d) Sulphur

279. A compound that is a white solid which absorbs water vapour from the air is 2013 II
(a) sodium nitrate
(b) calcium chloride
(c) sodium carbonate
(d) calcium sulphate

280. Which of the following are the two main constituents of granite? 2014 I
(a) Iron and silica
(b) Iron and silver
(c) Silica and aluminium
(d) Iron oxide and potasium

281. Which one of the following elements is present in green pigment of leaf? 2014 I
(a) Magnesium (b) Phosphorus
(c) Iron (d) Calcium

282. Which method of water purification does not kill microorganism? 2014 I
(a) Boiling (b) Filtration
(c) Chlorination (d) UV-irradiation

283. What causes dough (a mixture of flour, water, etc.) to rise when yeast is added to it? 2014 I
(a) An increase in temperature
(b) An increase in the amount of the substance
(c) An increase in the number of yeast cells
(d) Release of carbon dioxide gas

284. Which one of the following gases is supporter of combustion? 2014 I
(a) Hydrogen
(b) Nitrogen
(c) Carbon dioxide
(d) Oxygen

285. **Statement I** Clay layers are poor aquifers.
Statement II The inter-particle space of clay minerals is the least.
Codes 2014 I
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

286. Tungsten is used for the construction of filament in electric bulb because of its 2014 II
(a) high specific resistance
(b) low specific resistance
(c) high light emitting power
(d) high melting point

287. Inactive nitrogen and argon gases are usually used in electric bulbs in order to 2014 II
(a) increase the intensity of light emitted
(b) increase the life of the filament
(c) make the emitted light coloured
(d) make the production of bulb economical

288. When carbon dioxide is passed through lime water, the solution turns milky, but, on prolonged passage the solution turns clear. This is because 2014 II
(a) the calcium carbonate formed initially is converted to soluble calcium bicarbonate on passage of more carbon dioxide
(b) the reaction is reversible and lime water is regenerated
(c) the calcium bicarbonate formed initially is converted to soluble calcium carbonate on passage of more carbon dioxide
(d) the initially formed insoluble compound is soluble in carbonic acid

289. Which of the following is a good lubricant? 2014 II
(a) Diamond powder
(b) Graphite powder
(c) Molten carbon
(d) Alloy of carbon and iron

290. The form of carbon known as graphite 2014 II
(a) is harder than diamond
(b) contains a higher percentage of carbon than diamond
(c) is a better electrical conductor than diamond
(d) has equal carbon-to-carbon distances in all directions

291. Which of the following is not correct about baking soda? 2014 II
(a) It is used in soda-acid fire extinguisher
(b) It is added for faster cooking
(c) It is a corrosive base
(d) It neutralises excess acid in the stomach

292. Sodium metal should be stored in 2015 I
(a) alcohol
(b) kerosene oil
(c) water
(d) hydrochloric acid

293. Which one among the following is used in making lead pencils? 2015 I

(a) Charcoal (b) Graphite
(c) Coke (d) Carbon black

294. When hard water is evaporated completely, the white solid remains in the container. It may be due to the presence of 2015 I

1. carbonates of Ca and Mg
2. sulphates of Ca and Mg
3. chlorides of Ca and Mg

Select the correct answer using the codes given below.

(a) 1 and 2 (b) 1 and 3
(c) Only 3 (d) All of these

295. How many elements are there in the 5th period of modern periodic table? 2015 I

(a) 2 (b) 8 (c) 18 (d) 36

296. Which one of the following is not true for diamond? 2015 II

(a) Each carbon atom is linked to four other carbon atoms
(b) Three-dimensional network structure of carbon atoms is formed
(c) It is used as an abrasive for sharpening hard tools
(d) It can be used as a lubricant

297. Which one of the following statements is not correct? 2015 II

(a) Water starts boiling when its vapour pressure becomes equal to atmospheric pressure
(b) Water is known as universal solvent
(c) Permanent hardness of water is due to presence of $MgCl_2$, $CaCl_2$, $MgSO_4$, $CaSO_4$
(d) Density of ice is greater than that of water

298. Red phosphorus is used in the manufacture of safety matches. This is due to the fact that 2015 II

(a) it shows phosphorescence
(b) at ordinary temperature , it is less reactive than other varieties of phosphorus
(c) it cannot be converted to white phosphorus on heating
(d) it does not react with halogen on heating

299. Which of the following statements with regard to Portland cement are correct? 2015 II

1. Silica imparts strength to cement.
2. Alumina makes the cement quick setting.
3. Excess of lime increases the strength of cement.
4. Calcium sulphate decreases the initial setting time of cement.

Select the correct answer using the codes given below.

(a) 2 and 4 (b) 1 and 3
(c) 1,2 and 4 (d) 1 and 2

300. Which one of the following statements is not correct?

(a) Hydrogen is an element 2016 I
(b) Hydrogen is the lightest element
(c) Hydrogen has no isotopes
(d) Hydrogen and oxygen form an explosive mixture

301. Vitamin-B_{12} deficiency causes pernicious anaemia Animals cannot synthesise vitamin-B_{12}. Humans must obtain all their vitamin-B_{12} from their diet. The complexing metal ion in vitamin-B_{12} is 2016 I

(a) Mg^{2+} (magnesium ion)
(b) Fe^{2+} (iron ion)
(c) Co^{3+} (cobalt ion)
(d) Zn^{2+} (zinc ion)

302. German silver is used to make decorative articles coinage metal, ornaments etc. The name is given because 2016 I

(a) it is an alloy of copper and contains silver as one of its components
(b) Germans were the first to use silver
(c) its appearance is like silver
(d) it is an alloy of silver

Organic Chemistry

303. The first organic compound synthesised in the laboratory from its element is

(a) urea (b) methane
(c) ethylene (d) acetic acid

304. The compound which cannot be obtained from plants or animals but can be obtained from its elements is

(a) ammonium cyanate
(b) marsh gas
(c) urea
(d) cane sugar

305. Which of the following is used as a hypnotic?

(a) Paraldehyde (b) Methaldehyde
(c) Acetaldehyde (d) Formaldehyde

306. The main point of difference between an organic and inorganic compound is that

(a) the former contains carbon but the later is not
(b) the later contains carbon but former is not
(c) the later may or may not contain carbon but the former always contain it
(d) the former may or may not contain carbon but the later always contain it

307. The isomerism observed in alkanes is

(a) metamerism
(b) chain isomerism
(c) position isomerism
(d) geometrical isomerism

308. General formula for the alkenes is

(a) C_nH_{2n} (b) C_nH_{2n+2}
(c) C_nH_{2n-2} (d) $C_{2n}H_{2n+1}$

309. The offending substance in the liquor tragedies leading to blindness etc., is

(a) ethyl alcohol (b) amyl alcohol
(c) benzyl alcohol (d) methyl alcohol

310. Which of the following is an active component of oil of clove?

(a) Menthol (b) Eugenol
(c) Methanol (d) Benzaldehyde

311. In cold countries, ethylene glycol is added in the water used in the radiators of cars during winter. This results in

(a) lowering in freezing point
(b) reducing the viscosity
(c) reducing the specific heat
(d) making water a better conductor of electricity

312. Denatured spirit is mainly used as a

(a) drug
(b) good fuel
(c) material in preparing
(d) solvent in preparing varnishes

313. Which of the following is generally present in tonics?

(a) Ethanol (b) Ether
(c) Ethanal (d) Chloral

314. The most important ingredient of dynamite is

(a) nitrobenzene (b) picric acid
(c) nitroglycerine (d) TNT

315. Dynamite is prepared by mixing nitroglycerine with

(a) saw dust and ammonium nitrate
(b) cellulose nitrate
(c) saw dust alone
(d) conc. sulphuric acid

316. Wine contains
(a) CH_3OH (b) C_2H_5OH
(c) C_6H_5OH (d) glucose

317. Chloroform is used as
(a) antiseptic (b) insecticides
(c) antipyretic (d) anesthetic

318. A sample of chloroform before using as an anaesthetic is tested by
(a) Fehling's solution
(b) ammoniacal cuprous chloride
(c) ammoniacal silver nitrate solution
(d) silver nitrate solution

319. The name fire damp is given to
(a) methane (b) ethane
(c) propane (d) butane

320. Denatured spirit is a mixture of ethyl alcohol, methyl alcohol and
(a) acetic acid (b) pyridine
(c) benzene (d) water

321. Vinegar contains acetic acid
(a) 10-20% (b) 100%
(c) 7-8% (d) 25%

322. Liquor poisoning is due to the presence of
(a) methyl alcohol
(b) ethyl alcohol
(c) carbonic acid
(d) bad compound in liquor

323. The chemical name of tear gas is
(a) acetophenone
(b) benzophenone
(c) bromoacetophenone
(d) chloroacetophenone

324. Petroleum is found
(a) on the surface of the earth
(b) in the atmosphere
(c) in Arctic ocean
(d) deep under the surface of the earth

325. Which one among the follwing fuels is used in gas welding?
(a) LPG (b) Ethylene
(c) Methane (d) Acetylene

326. CaC_2 react with water to give
(a) C_2H_4 (b) CH_4
(c) C_2H_2 (d) CO_2

327. Which one of the following is used in welding industry?
(a) Methane (b) Ethane
(c) Acetylene (d) Benzene

328. The fuel used in spirit lamp is
(a) ethanol (b) methanol
(c) propanol (d) butanol

329. Grain alcohol is common name of
(a) amyl alcohol (b) ethyl alcohol
(c) methanol (d) None of these

330. Wood spirit is known as
(a) ethanol (b) methanol
(c) acetone (d) benzene

331. Ethanol containing some methanol is called
(a) methylated spirit (b) rectified spirit
(c) absolute spirit (d) None of these

332. Lucas test is performed for
(a) amines (b) alkyl halides
(c) ethers (d) alcohols

333. Lemon is sour due to
(a) citric acid (b) tartaric acid
(c) oxalic acid (d) acetic acid

334. Which one of the following is likely to be a pollutant free alternative to petrol for automobiles?
(a) Ethanol (b) Acetylene
(c) Butane (d) Propane

335. The tracking of people by trained dogs is based on the recognisation of which of the following compounds, in the sweat from feet?
(a) Carboxylic acids (b) Uric acid
(c) Sugar (d) Salt

336. The law enforcement agencies use a chemical test to approximate a person's blood alcohol level. The mouthpiece of a bag containing sodium dichromate solution in acidic medium. A chemical reaction with ethanol changes the colour of the solution from
(a) orange to green
(b) orange to colourless
(c) yellow to orange
(d) colourless to orange

337. Consider the following statements
1. Saturated hydrocarbons undergo substitution reaction.
2. Carbon black is obtained when methane is heated in the absence of air

Which of the above statement(s) is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

338. Consider the following chemicals,
1. Benzene
2. Carbon tetrachloride
3. Sodium carbonate

Which of the above is/are used as dry cleaning chemicals?
(a) Only 1
(b) Only 2
(c) 1 and 2
(d) All of the above

339. **Statement I** Large cold storage plants use ammonia as refrigerant while domestic refrigerators use chlorofluorocarbons.
Statement II Ammonia can be liquefied at ambient temperature and low pressure.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

340. Match the following Columns.

Column I (Chemical name)	Column II (Molecular Formula)
A. Ammonia	1. C_6H_6
B. Benzene	2. NH_3
C. Ethyl alcohol	3. CH_3OH
D. Methyl alcohol	4. C_2H_5OH

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	2	1	4	3
(c)	4	3	2	1	(d)	1	4	3	2

341. Which of the following statements are correct about chloroform?
1. Liquid fuel
2. Anaesthetic in nature
3. Produces phosgene
4. Fire extinguisher

Choose the correct code
(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) 1 and 4

342. The sources of alkanes are
1. coal 2. petroleum
3. natural gas 4. minerals

The correct sources are
(a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 1, 2, and 4

343. Match the following Columns.

Column I (Haloalkane/arene)	Column II (Applications)
A. Iodoform	1. CF_4
B. BHC	2. Antiseptic
C. Freon-14	3. Moth repellent
D. Halothanes	4. Inhalative anaesthetic
E. *p*-dichlorobenzene	5. Termite pesticide

Codes

	A	B	C	D	E		A	B	C	D	E
(a)	2	4	5	3	1	(b)	2	5	1	4	2
(c)	3	4	2	1	5	(d)	1	3	5	2	4

Previous Years' Questions

344. Which one among the following non-toxic gases helps in formation of enzymes which ripen fruit? **2012 I**

(a) Acetylene (b) Ethene
(c) Methane (d) Carbon dioxide

345. Addition of ethylene dibromide to petrol **2014 I**

(a) increases the octane number of fuel
(b) helps elimination of lead oxide
(c) removes the sulphur compound in petrol
(d) serves as substitute of tetraethyl lead

346. Bagasse, a byproduct of sugar manufacturing industry, is used for the production of **2014 II**

(a) glass (b) paper
(c) rubber (d) cement

347. End of detergent have

(a) ester group
(b) aldehyde
(c) amine group
(d) sodium sulphate

Man Made Materials

348. Soaps can be classified as

(a) carbohydrates
(b) ethers
(c) salt of fatty acids
(d) None of the above

349. Soft soaps are

(a) sodium salt
(b) calcium salt
(c) magnesium salt
(d) potassium salt

350. Made synthetic fibre is

(a) wool (b) rayon
(c) nylon (d) cotton

351. Natural fibre is

(a) polyester (b) wool
(c) nylon (d) cashmillon

352. Polyethene is a polymer of

(a) ethane
(b) ethene
(c) ethyne
(d) methane

353. Which one of the following pairs is not correctly matched?

Plastic	Type
(a) Nylon fibres	Fibres
(b) PVC	Thermoplastic
(c) Phenol formaldehyde	Thermoplastic
(d) Polypropylene	Thermoplastic

354. Which one of the following fuels causes minimum environmental pollution?

(a) Diesel (b) Hydrocarbon
(c) Hydrogen (d) Kerosene

355. Which one of the following has the highest fuel value?

(a) Hydrogen (b) Charcoal
(c) Natural gas (d) Gasoline

356. The gas supplied in cylinders for cooking is

(a) marsh gas
(b) LPG
(c) mixture of CH_4 and C_2H_6
(d) mixture of ethane and propane

357. Which of the following is biodegradable?

(a) Polythene (b) BHC
(c) Paper (d) Copper

358. Oil gas is a mixture of

(a) $H_2 + CH_4 + CO$
(b) $H_2 + CH_4 + C_2H_4 + CO$
(c) $CO + N_2$
(d) All of the above

359. An ideal fuel should have

(a) high calorific value
(b) low ignition temperature
(c) regulated and controlled
(d) All of the above

360. The quality of gasoline sample is determined by its

(a) iodine value (b) cetane number
(c) octane number (d) mass density

361. The process by which vegetable ghee is manufactured is known as

(a) saponification (b) hydrogenation
(c) esterificaiton (d) hydrolysis

362. The major portion of combustable part of gobar gas is

(a) methane (b) ethane
(c) ethylene (d) acetylene

363. Octane number of fuel can be increased by

(a) isomerisation (b) alkylation
(c) reforming (d) All of these

364. Candles contains a mixture of

(a) bees wax and paraffin wax
(b) bees wax and stearic acid
(c) paraffin wax and stearic acid
(d) higher fatty acid

365. Ammonium sulphate and lime should be applied to the soil at the same time because

(a) nitrogen would be lost as ammonia
(b) it support fungal growth
(c) soil structure would be adversely affected
(d) harmful bacterial population would get activated

366. Commercial vulcanisation of rubber involves

(a) sulphur (b) carbon
(c) phosphorus (d) selenium

367. The order of appearance of the following with increasing temperature during the refining of crude oil is

(a) kerosene, gasoline, diesel
(b) diesel, gasoline, kerosene
(c) gasoline, kerosene, diesel
(d) gasoline, diesel, kerosene

368. Which one among the following is a strong smelling agent added to LPG cylinder to help in the detection of gas leakage?

(a) Ethanol (b) Thioethanol
(c) Methane (d) Chloroform

369. The cleaning of dirty clothes by soaps and detergents is due to a type of molecules, called surfactants, which are present in soaps and detergents.The surfactant molecules remove the dirt by

(a) making the cloth slippery
(b) producing some gases between the dirt and the cloth
(c) dissolving the dirt
(d) forming some aggregates of themselves and take away the dirt in the core of the aggregates

370. Consider the following statements

1. Superphosphate of lime can be assimilated by plants which is soluble in water.
2. Soaps do not form lather with water containing salts of calcium and magnesium.

Choose the correct statement(s)

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) None of the above

371. Which one of the following is used in the preparation of antiseptic solution?

1. Potassium nitrate
2. Iodine

Choose the correct code

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) None of the above

372. **Statement I** Phenyl is used as a household germicide.
Statement II Phenyl is phenol derivative and phenol is an effective germicide.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

373. Some statements about the benefits of organic farming are given below.
1. It reduces CO_2 emission.
2. It does not lead to toxic effect.
3. It improves the water-retention capacity of the soil.

Which of the above statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 3 (d) 2 and 3

374. Arrange the following fertilisers according to the decreasing order of their nitrogen content.
1. Ammonium sulphate
2. Ammonium nitrate
3. Potassium nitrate
4. Urea

Codes
(a) 2, 4, 3, 1 (b) 4, 3, 2, 1
(c) 4, 2, 1, 3 (d) 4, 2, 3, 1

375. Match the following Columns.

Column I	Column II
A. Sodium palmitate	1. Ore
B. Galena	2. Fertiliser
C. NPK	3. Soap
D. Cellulose	4. Natural polymer
	5. Artificial fibre

Codes

	A	B	C	D
(a)	5	4	1	2
(b)	5	2	1	4
(c)	3	1	2	4
(d)	3	1	4	2

376. Consider the following statements
1. Liquefied Natural Gas (LNG) is liquefied under extremely cold temperatures and high pressure to facilitate storage or transportation in specially designed vessels.
2. First LNG terminal in India was built in Hassan.
3. Natural Gas Liquids (NGL) are separated from LPG and these include ethane, propane, butane and natural gasoline.

Which of the statement(s) given above is/are correct?
(a) Only 1 (b) 1 and 3
(c) 2 and 3 (d) All of these

377. Consider the following statements regarding the properties and uses of glass wool.
1. Glass wool has tensile strength greater than steel.
2. Glass wool is fire proof.
3. Glass wool has high electrical conductivity and absorbs moisture.
4. Glass wool is used to prepare fibre glass.

Which of the statements given above are correct?
(a) 1 and 2 (b) 1, 2 and 4
(c) 2 and 4 (d) 1, 3 and 4

378. **Statement I** The main constituent of the liquified petroleum gas is methane.
Statement II Methane can be used directly for burning in homes and factories where it can be supplied through pipelines.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

379. Match the following Columns.

Column I (Fuel gases)	Column II (Major constituents)
A. CNG	1. Carbon monoxide, hydrogen
B. Coal gas	2. Butane, propane
C. LPG	3. Butane, ethane
D. Water gas	4. Hydrogen, methane, CO

Codes

	A	B	C	D
(a)	2	1	3	4
(b)	3	4	2	1
(c)	2	4	3	1
(d)	3	1	2	4

380. Which one of the following polymeric materials is used for making bullet proof jacket?
(a) Nylon-66
(b) Rayon
(c) Kevlar
(d) Dacron

381. **Statement I** Superphosphate of lime can be assimilated by plants.
Statement II Superphosphate of lime is soluble in water.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

Previous Years' Questions

382. Vermicompost is an/a **2012 I**
(a) inorganic fertiliser
(b) toxic substance
(c) organic bio-fertiliser
(d) synthetic fertiliser

383. The macro nutrients provided by inorganic fertiliser are **2012 I**
(a) carbon, iron and boron
(b) magnesium, manganese and sulphur
(c) magnesium, zinc and iron
(d) nitrogen, phosphorus and potassium

384. Which one among the following polymers is used for making bulletproof material? **2012 I**
(a) Polyvinyl chloride (b) Polystyrene
(c) Polyethylene (d) Polyamide

385. Soaps cannot be used in acidic condition because they lose their cleansing effect due to formation of insoluble **2012 II**
(a) esters
(b) alcohols
(c) hydrocarbons
(d) long chain fatty acids

386. Chromium oxide in paints makes the colour of paint **2014 II**
(a) green (b) white (c) red (d) blue

387. The main constituent of Gobar gas or bio gas is
(a) ethane (b) methane
(c) propane (d) acetylene

388. Which one among the following fuels is used in gas welding? **2015 I**
(a) LPG (b) Ethylene
(c) Methane (d) Acetylene

389. Which one of the following is a non-renewable resource? **2015 II**
(a) Solar energy (b) Coal
(c) Water (d) Fisheries

390. The synthetic rubber has replaced natural rubber for domestic and industrial purposes. Which one of the following is the main reason behind that? 2016 I
(a) Natural rubber is unable to meet the growing demand of different industries.
(b) Natural rubber is grown in tropical countries only
(c) Raw materials for synthetic rubber are easily available
(d) Natural rubber is not durable

391. The handle of pressure cookers is made of plastic because it should be made non-conductor of heat The plastic used there is the first man-made plastic, which is 2016 I
(a) polythene (b) terylene
(c) nylon (d) bakelite

Environment and its Pollution

392. How much amount of sun rays reaching the earth is used by plants?
(a) 90% (b) 1% (c) 15% (d) 48%

393. The pollutants which come directly in the air from sources are called primary pollutants. Primary pollutants are sometimes converted into secondary pollutants. Which of the following belongs to secondary air pollutants?
(a) CO
(b) Hydrocarbon
(c) Peroxyacetyl nitrate (PAN)
(d) NO

394. Ozone is formed in the upper atmosphere from oxygen by the action of
(a) thermal radiation from sunlight
(b) cosmic rays
(c) ultraviolet rays
(d) infrared rays

395. Which of the following statements is wrong?
(a) Ozone is not responsible for greenhouse effect
(b) Ozone can oxidise sulphur dioxide present in the atmosphere to sulphur trioxide
(c) Ozone hole is thinning of ozone layer present in stratosphere
(d) Ozone is produced in upper stratosphere by the action of UV rays on oxygen

396. Which one of the following is the major cause of depletion of ozone strata of the atmosphere?
(a) Carbon dioxide
(b) Chlorofluoro carbon
(c) Sulphur dioxide
(d) Nitrogen oxide

397. Which of the following statements is correct?
(a) Ozone hole is a hole formed in stratosphere from which ozone ooze out
(b) Ozone hole is a hole formed in the troposphere from which ozone ooze out
(c) Ozone hole is thinning of ozone layer of stratosphere at some places
(d) Ozone hole means vanishing of ozone layer around the earth completely

398. Oxidation of sulphur dioxide into sulphur trioxide in the absence of a catalyst is a slow process but this oxidation occurs easily in the atmosphere. Which substance catalyse the reaction?
(a) Oxygen (b) Particulate
(c) UV rays (d) IR rays

399. Dinitrogen and dioxygen are main constituents of air but these do not react with each other to form oxides of nitrogen because
(a) the reaction is endothermic and requires very high temperature
(b) the reaction can be initiated only in the presence of a catalyst
(c) oxides of nitrogen are unstable
(d) N_2 and O_2 are unreactive

400. In acid rain, the acid present in highest concentration is
(a) nitric acid (b) hydrochloric acid
(c) sulphuric acid (d) carbonic acid

401. Which of the following statements is not true?
(a) London smog is a mixture of smoke and fog
(b) London smog is oxidising in nature
(c) Photochemical smog causes irritation in eyes
(d) Photochemical smog results in the formation of PAN

402. Which of the following statements is not true about classical smog?
(a) Its main components are produced by the action of sunlight on emissions of automobiles and factories
(b) Produced in cold and humid climate
(c) It contains compounds of reducing nature
(d) It contains smoke, fog and sulphur dioxide

403. Which of the following compounds caused tragedy of Bhopal in 1984?
(a) Phosphene
(b) Methyl isocyanate
(c) Carbon monoxide
(d) Methyl cyanate

404. SO_2 and NO_2 causes pollution by increasing
(a) alkalinity
(b) neutrality
(c) acidity
(d) buffer action

405. Freon used as refrigerants is chemically known as
(a) chlorinated hydrocarbon
(b) fluorinated hydrocarbon
(c) chlorofluoro hydrocarbon
(d) fiuorinated aromatic compound

406. Which one of the following is considered as point source for water pollution?
(a) Factories
(b) Construction site
(c) Acid rain
(d) Field lawns

407. Sewage containing organic waste should not be disposed in water bodies because it causes major water pollution. Fishes in such a contaminated water die because of
(a) large number of mosquitoes
(b) increase in the amount of dissolved oxygen
(c) decrease in the amount of dissolved oxygen
(d) clogging of gills by mud

408. Biochemical Oxygen Demand (BOD) is a measure of organic materials present in water. BOD value less than 5 ppm indicates a water sample to be
(a) rich in dissolved oxygen
(b) poor in dissolved oxygen
(c) highly polluted
(d) not suitable for aquatic life

409. For dry cleaning, in the place of tetrachloroethane, liquefied carbon dioxide with suitable detergent is an alternative solvent. What type of harm to the environment will be prevented by stopping use of tetrachloroethane?
(a) It results in tropospheric pollution
(b) It causes depletion of ozone layer
(c) It causes particulate pollution
(d) Both (a) and (b)

410. From which of the following water sources, the water is likely to be contaminated with fluoride?

1. Ground water
2. River water

Choose the correct code
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

411. Consider the following statements.

1. Ultraviolet rays coming from sun causes skin cancer.
2. Taj Mahal is threatened by pollution from SO_2

Choose the correct statement and select the correct code.
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

412. **Statement I** Oxides of sulphur and nitrogen present in high concentration in air are dissolved in rain drop.
Statement II Oxy acids of sulphur and nitrogen makes rain water acidic.

Codes
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

413. Which of the following occupational health hazards commonly faced by the workers of ceramics, pottery and glass industry.

1. Stone formation in gall bladder
2. melanoma
3. silicosis

Choose the correct code
(a) Only 2
(b) Only 3
(c) 1 and 2
(d) All of the above

414. Ordinary dry air consists of the following.

1. Nitrogen 2. Argon
3. Oxygen 4. Carbon dioxide

What is the decreasing sequence of these in percentages?
(a) 1, 3, 2 and 4 (b) 1, 2, 4 and 3
(c) 2, 1, 3 and 4 (d) 2, 1, 4 and 3

415. Match the following Columns.

Column I (Pollutant)	Column II (Source)
A. Microorganisms	1. Chemical fertilisers
B. Plant nutrients	2. Abandoned coal mines
C. Sediments	3. Domestic sewage
D. Mineral acids	4. Erosion of soil by strip mining
	5. Detergents

Codes

	A	B	C	D		A	B	C	D
(a)	2	5	3	1	(b)	3	1	4	2
(c)	4	2	5	1	(d)	1	3	2	4

416. Which one of the following is associated with the formation of brown air in traffic congested cities?
(a) Sulphur dioxide
(b) Nitrogen oxide
(c) Carbon dioxide
(d) Carbon monoxide

417. Which one of the following chemicals is commonly used by farmers to destroy weeds?
(a) DDT (b) Malathion
(c) Methyl bromide (d) 2,4-D

418. From which one among the following water sources, the water is likely to be contaminated with fluoride?
(a) Ground water (b) River water
(c) Pond water (d) Rain water

419. Which of the occupational health hazards commonly faced by the workers of ceramics, pottery and glass industry is
(a) stone formation in gall bladder
(b) melanoma
(c) silicosis
(d) stone formation in kidney

Previous Years' Questions

420. Which one among the following is responsible for the expansion of water in the ocean? **2013 I**
(a) Carbon dioxide
(b) Nitrogen dioxide
(c) Carbon monoxide
(d) Sulphur dioxide

421. Which one among the following gases readily combines with the haemoglobin of the blood? **2013 I**
(a) Methane
(b) Nitrogen dioxide
(c) Carbon monoxide
(d) Sulphur dioxide

422. Which of the following gases in the atmosphere is/are responsible for acid rain? **2013 I**

1. Oxides of sulphur
2. Oxides of nitrogen
3. Oxides of carbon

Select the correct answer using the codes given below.
(a) 1 and 2
(b) 1 and 3
(c) Only 2
(d) All of the above

423. Match the following Columns. **2013 I**

Column I (Agent)	Column II (Disease)
A. Arsenic	1. Fluorosis
B. Fluoride	2. Melanosis
C. Dust	3. Presbycusis
D. Noise	4. Silicosis

Codes

	A	B	C	D
(a)	3	1	4	2
(b)	3	4	1	2
(c)	2	1	4	3
(d)	2	4	1	3

424. **Statement I** Chlorine radicals $Cl^{\cdot}$ initiate the chain reaction for ozone depletion.
Statement II Gaseous hypochlorous acid and chlorine are photolysed by sunlight.

Codes **2013 I**
(a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

425. Nitric oxide pollution can lead to all of the following, except **2014 I**
(a) leaf spotting in plants
(b) bronchitis related respiratory problems in human
(c) production of corrosive gases through photochemical reaction
(d) silicosis in human

426. The technique of inducing rain from cloud is called? **2014 II**
(a) Cloud computing
(b) Cloud control
(c) Cloud engineering
(d) Cloud seeding

427. Match the following Columns. 2015 I

Column I (Air pollutants)	Column II (Effect)
A. Chlorofluorocarbon	1. Acid rain
B. Sulphur dioxide	2. Depletion in ozone layer in the atmosphere
C. Lead compound	3. Harmful for human nervous system
D. Carbon dioxide	4. Topmost contribution to greenhouse effect.

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	1	2	(b)	4	1	3	2
(c)	2	1	3	4	(d)	2	3	1	4

428. Which one of the following is the most appropriate and correct practice from the point of view of a healthy environment? 2015 I

(a) Burning of plastic wastes to keep the environment clean
(b) Burning of dry and fallen leaves in a garden or field
(c) Treatment of domestic sewage before its release
(d) Use of chemical fertilisers in agricultural fields

429. Which of the following statement(s) is/are correct?

1. Acid rain reacts with buildings made from limestone.
2. Burning of sulphur containing coal can contribute to acid rain.
3. Eutrophication is an effective measure to control pollution.

Select the correct answer using the codes given below. 2016 I

(a) 1 and 2
(b) 2 and 3
(c) Only 1
(d) All of the above

430. Methyl isocyanate gas, which was involved in the disaster in Bhopal in December, 1984, was used in the Union Carbide factory for production of 2016 I

(a) dyes
(b) detergents
(c) explosives
(d) pesticides

ANSWERS

1	a	2	c	3	b	4	b	5	a	6	c	7	b	8	a	9	b	10	a
11	b	12	c	13	c	14	b	15	a	16	c	17	d	18	a	19	a	20	a
21	c	22	d	23	c	24	d	25	d	26	d	27	d	28	b	29	b	30	b
31	d	32	d	33	c	34	b	35	d	36	b	37	b	38	c	39	a	40	d
41	a	42	d	43	b	44	c	45	a	46	a	47	d	48	a	49	d	50	a
51	a	52	c	53	d	54	a	55	b	56	b	57	c	58	b	59	c	60	b
61	b	62	b	63	b	64	b	65	b	66	c	67	b	68	b	69	c	70	a
71	c	72	c	73	a	74	d	75	b	76	a	77	c	78	d	79	c	80	b
81	c	82	a	83	b	84	b	85	d	86	c	87	d	88	d	89	b	90	a
91	b	92	d	93	c	94	c	95	b	96	d	97	a	98	c	99	c	100	d
101	a	102	d	103	a	104	c	105	b	106	c	107	b	108	a	109	b	110	d
111	b	112	a	113	c	114	c	115	d	116	b	117	c	118	b	119	a	120	a
121	d	122	a	123	c	124	c	125	d	126	a	127	c	128	a	129	d	130	c
131	a	132	c	133	b	134	a	135	c	136	c	137	a	138	c	139	b	140	d
141	c	142	c	143	d	144	d	145	a	146	b	147	b	148	b	149	a	150	c
151	d	152	b	153	a	154	c	155	b	156	b	157	c	158	a	159	a	160	a
161	c	162	a	163	a	164	d	165	d	166	c	167	b	168	b	169	d	170	a
171	d	172	b	173	a	174	d	175	b	176	c	177	b	178	c	179	d	180	a
181	b	182	a	183	a	184	c	185	b	186	a	187	b	188	b	189	b	190	c
191	b	192	d	193	d	194	a	195	d	196	d	197	a	198	c	199	c	200	b
201	a	202	c	203	c	204	d	205	c	206	c	207	b	208	d	209	a	210	c
211	a	212	c	213	d	214	c	215	c	216	b	217	b	218	c	219	c	220	a
221	a	222	a	223	c	224	c	225	a	226	c	227	d	228	b	229	c	230	a
231	a	232	a	233	d	234	c	235	a	236	b	237	b	238	b	239	a	240	d
241	a	242	d	243	c	244	a	245	d	246	b	247	d	248	a	249	d	250	a
251	d	252	d	253	d	254	a	255	d	256	d	257	d	258	c	259	d	260	a
261	a	262	b	263	c	264	a	265	a	266	c	267	c	268	d	269	b	270	b

271	*b*	272	*a*	273	*a*	274	*a*	275	*b*	276	*c*	277	*c*	278	*a*	279	*b*	280	*c*
281	*a*	282	*b*	283	*d*	284	*d*	285	*a*	286	*c*	287	*b*	288	*a*	289	*b*	290	*c*
291	*c*	292	*b*	293	*b*	294	*d*	295	*c*	296	*d*	297	*d*	298	*b*	299	*d*	300	*c*
301	*c*	302	*c*	303	*a*	304	*a*	305	*a*	306	*c*	307	*b*	308	*a*	309	*d*	310	*b*
311	*a*	312	*d*	313	*a*	314	*c*	315	*a*	316	*b*	317	*d*	318	*d*	319	*a*	320	*b*
321	*c*	322	*a*	323	*d*	324	*d*	325	*d*	326	*c*	327	*c*	328	*a*	329	*b*	330	*b*
331	*a*	332	*d*	333	*a*	334	*a*	335	*a*	336	*a*	337	*c*	338	*c*	339	*a*	340	*b*
341	*c*	342	*b*	343	*b*	344	*b*	345	*d*	346	*b*	347	*d*	348	*c*	349	*d*	350	*c*
351	*b*	352	*b*	353	*c*	354	*c*	355	*a*	356	*b*	357	*c*	358	*b*	359	*d*	360	*c*
361	*b*	362	*a*	363	*d*	364	*c*	365	*a*	366	*a*	367	*c*	368	*b*	369	*d*	370	*c*
371	*b*	372	*a*	373	*d*	374	*c*	375	*c*	376	*b*	377	*d*	378	*d*	379	*b*	380	*c*
381	*a*	382	*c*	383	*b*	384	*c*	385	*d*	386	*a*	387	*b*	388	*d*	389	*b*	390	*c*
391	*d*	392	*b*	393	*c*	394	*c*	395	*a*	396	*b*	397	*c*	398	*b*	399	*a*	400	*c*
401	*b*	402	*a*	403	*b*	404	*c*	405	*c*	406	*a*	407	*c*	408	*a*	409	*b*	410	*b*
411	*c*	412	*a*	413	*b*	414	*a*	415	*b*	416	*b*	417	*d*	418	*b*	419	*c*	420	*a*
421	*c*	422	*a*	423	*c*	424	*b*	425	*b*	426	*d*	427	*c*	428	*c*	429	*a*	430	*c*

Solutions

5. At STP, 44 g (1 mole) carbon dioxide

$$= 6.02 \times 10^{23} \text{ molecules}$$

7. Density of liquid is generally less than that of its solid form. But solid ice is lighter than water as it floats on water,

i.e. the density of solid form of water (ice) is less than that of the liquid form of water.

14. Pure water reaches at its maximum density, when cooled at approximately 4°C (39°F). As it is cooled further, at 0°C, water expands to become less dense.

This unusual negative thermal expansion is attributed to strong, orientation dependent, intermolecular interactions (hydrogen bonds).

16. Chemical changes are generally irreversible. So, the frying of an egg is a chemical change.

17. Molecular weight of NaCl $= 23 + 35.5 = 58.5$

Equivalent weight

$$= \frac{\text{Molecular weight}}{\text{Valence factor}} = \frac{58.5}{1} = 58.5$$

. **21.** Latent heat is the amount of heat required per mass during the change of phase of a substance.

Sl unit Jkg^{-1}

$$\text{Latent heat (L)} = \frac{\text{Quantity of heat } (Q)}{\text{Mass } (m)}$$

Here, $m = 1100$ g of ice

$L = 333.6$ J/g

So, amount of heat (Q) $= m \times L = 100 \times 333.6$

$Q = 33360$ J

109. $\dfrac{r_H}{r_{He}} = \sqrt{\dfrac{M_{He}}{M_H}} = \sqrt{2}$

$r_H = 1.4\, r_{He}$

114. $$\text{Molarity} = \frac{\text{Mass of solute}}{\text{Molar mass of solute} \times \text{volume of solution (L)}}$$

$$= \frac{8}{40 \times 1} \qquad (\because \text{NaOH} = 23 + 16 + 1 = 40 \text{ g mol}^{-1})$$

$$= \frac{1}{5} = 0.2\text{M}$$

115. Molarity of pure water $= \dfrac{1000}{18} = 55.55$ M

120. The tanks used by scuba divers are filled with air diluted with helium (i.e. He—O_2 mixture) because unlike nitrogen, helium is not soluble in blood even under high pressure.

147. Soft drinks regulate acidity with bicarbonate salts. Bicarbonate of soda is effective in regulating the pH levels of other substances, it ensures that the substance is neither too alkaline nor too acidic.

148. The blue litmus paper turns red in acidic solution and the red litmus paper turns blue in basic solution. The neutral solution does not affect the litmus paper. Thus, only acidic solution can change the colour of blue litmus paper to red.

241. Gypsum ($CaSO_4 \cdot 2H_2O$) is added to clinker during cement manufacturing to decrease the rate of setting of cement so that it gets sufficiently hardens.

242. Pearl consists of approximately 85% calcium carbonate hence, the main constituent of a pearl is calcium carbonate.

243. Sodium thiosulphate ($Na_2S_2O_3$) solution is used in photography as a fixer since, it removes undecomposed AgBr as a soluble silver thiosulphate complex. $Na_2S_2O_3$

$$AgBr + 2Na_2S_2O_3 \longrightarrow \underset{\text{Sodium argento thiosulphate}}{Na_3[Ag(S_2O_3)_2]} + NaBr$$

245. Fluorescent coating on the glass is the secondary source of light in a fluorescent lamp.

258. Radioactivity is not a periodic property. It is a nuclear 'property. Atomic size, valency and electronegativity are periodic properties because they show a trend on moving from one side to the other in the periodic table.

259. Heavy water or deuterium oxide (D_2O) is an oxide of deuterium which is an isotope of hydrogen. It is a colourless, odourless and tasteless liquid. It contains 0.014% normal water (H_2O). It is used as moderator in nuclear reactor to slow down the neutrons.

264. On mixing with water, Plaster of Paris hardens because it takes up the water of crystallisation again and thus converts back into the dihydrate, i.e. gypsum and sets to a hard mass with slight expansion.

$$\underset{\text{Plaster of Paris}}{CaSO_4 \cdot \frac{1}{2}H_2O} + 1\frac{1}{2}H_2O \longrightarrow \underset{\text{Gypsum}}{CaSO_4 \cdot 2H_2O}$$

270.

Alloy	Constituent
Solder	Lead and tin
Brass	Copper and zinc
Bronze	Copper and tin
Steel	Iron and carbon

274. The hardness of water is mainly due to the presence of Ca or Mg ions. When hard water containing these ions is treated with soap solution, it reacts to form white crudy precipitate, known as scum.

275. Magnesium burns with a very bright white light, so it is used to add white sparks or improve the overall brilliance of a firework.

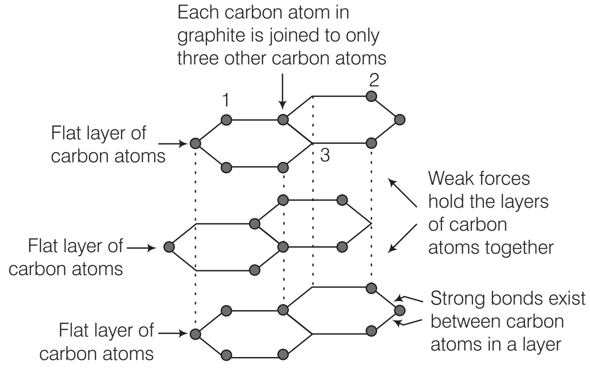
276. Carbon atoms in diamond form a rigid, three dimensional structure with each carbon atom bonded to four other carbon atoms.

282. Boiling kills most of the microorganisms. Filtration removes most of the bacteria but not the viruses less than 0.3μm diameter. These viruses are removed/killed by chlorination or UV-treatment.

286. Tungsten is uniquely suitable for using as a filament in electric bulbs because of its high melting point (3695 K), high resistance, lowest vapour pressure and tensile strength.

290. The two common allotropes of carbon are diamond and graphite. Graphite is a good conductor of electricity whereas diamond is a non-conductor of electricity. In a graphite crystal, each carbon atom is joined to only three other carbon atoms by covalent bonds.

The fourth valence electron of each carbon atom is 'free' to move, which is responsible for the conduction of electricity. In a diamond crystal, each carbon atom is linked to four other carbon atoms by strong covalent bonds.



Structure of Graphite

03

BIOLOGY

After analysing the previous year question papers, we have seen that 40-45 questions are asked from science section, out of these 10-15 questions were asked from Biology. From biology section, around 2-3 questions are asked from classification, genetics and evolution, 3-4 questions from human physiology, and 1 to 2 questions each from topics like cell, human health and diseases and applied biology.

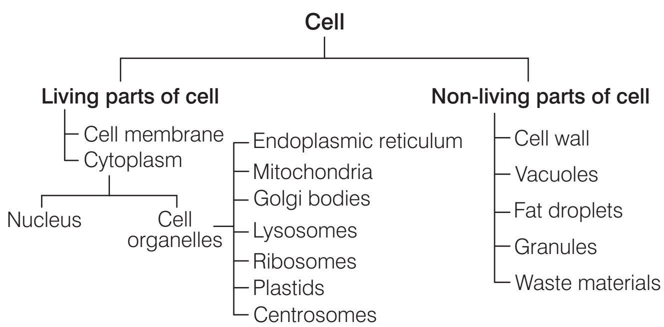
CELL : THE UNIT OF LIFE

Biology is the branch of science concerned with the study of living organisms. The term 'biology' is coined by **Lamarck** and **Treviranus**. The study of plants is called **Botany,** while study of animals is called **Zoology**.

THE CELL

Cell is the basic structural and functional unit of all the living organisms. The word 'cell' comes from the Latin word *cellula* which means a 'small room'.

- Living organisms may consist of one or more cells, according to this there are two types of organisms, i.e. **unicellular** (composed of single cell) and **multicellular** (composed of many cells).



- **Robert Hooke** in 1665 discovered the cell. He is the Father of Cytology.
- Cell theory was given by **Schleiden** and **Schwann**.
- **Matthias Jacob Schleiden** (1838) a German botanist said that, all plants are composed of different kinds of cells, which in turn form plant tissues.
- **Theodor Schwann** (1839) said that, all animals and plants are made up of cells. Cell theory is not applicable for viruses.
- According to **Virchow** 'new cells arise from pre-existing cells', i.e. *Omnis cellula-e-cellula*.
- Smallest cell is mycoplasma or Pleuropneumonia Like Organism (PPLO). Largest cell is ostrich's egg and the longest cell is nerve cell.
- The smallest human cell is the red blood cell. Bacteria was discovered by **AV Leeuwenhoek**.
- First compound microscope was invented by **Jansen** and **Jansen**.
- Electron microscope was invented by **Knoll** and **Ruska**.
- Vital stains are used for staining living cells or living parts of cell, e.g. Janus green for mitochondria.
- Protoplasm is the living component of cell that includes cytoplasm, cell organelles and nucleoplasm.
- Cytoplasm term was given by **Strasburger**.

- Mitochondria, chloroplast, nucleus are double-membraned structures.
- Lysosome, Golgi body, Endoplasmic Reticulum (ER) are single-membraned structures.
- Centrosome, flagella, ribosome are non-membranous in structure.
- Glycogen is referred to as animal starch.

Some Discoveries about the Cell

Scientist	*Discovery*
Robert Brown (1831)	Nucleus
Purkinje (1837)	Named protoplasm to be a content of cell
W Flemming (1882)	Observed mitotic cell division, splitting of chromosome and named the term chromatin
F Miescher	Discovered the nucleic acid
C Benda (1897)	Mitochondria
Beneden and Boveri (1887)	The chromosome number is constant in a species
Camillo Golgi (1891)	Golgi body
WS Sutton (1902)	Significance of reductional division
F Meves (1904)	Mitochondria in plant cell
JB Farmer and Moore (1905)	Named meiosis
FA Johanssen (1909)	Chiasma formation
TH Morgan (1933)	Role of chromosomes in heredity

Types of Cells

On the basis of the organisation, complexity and variety, all cells can be grouped into two types:

(*i*) **Prokaryotic cell** It is a simpler cell, lacking a true nucleus, i.e. nuclear membrane is absent in these cells, e.g. bacteria, mycoplasma, etc.

Bacteria are the microscopic prokaryotic organisms whose single cell have neither a membrane enclosed nucleus nor other membrane enclosed organelles like mitochondria and chloroplasts.

- The bacterial cell wall is strong and rigid.
- Murein or peptidoglycan or mucopeptide is present in the cell wall.
  Bacteria are mainly of two types:
  (i) **Archaebacteria** They are obligate anaerobes.
  (ii) **Eubacteria** They are true bacteria.

Some examples are as follow:

- *Staphylococcus aureus* causes boils, pneumonia, food poisoning and other diseases.
- *Staphylococcus thermophilus* gives yoghurt its creamy flavour.
- *S. lactis* often added to milk for the production of cheese.

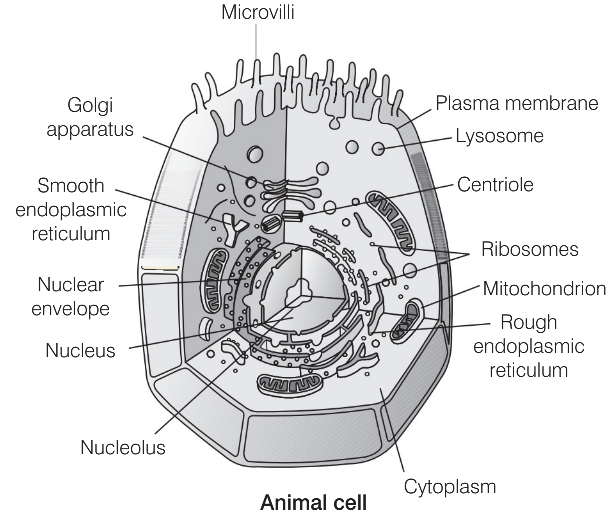
(*ii*) **Eukaryotic cell** It is a complex cell with double membrane bound nucleus, e.g. plant and animal cells. Eukaryotic organisation is seen in all the protists, plants, fungi and animals.

Differences between Prokaryotic and Eukaryotic Cell

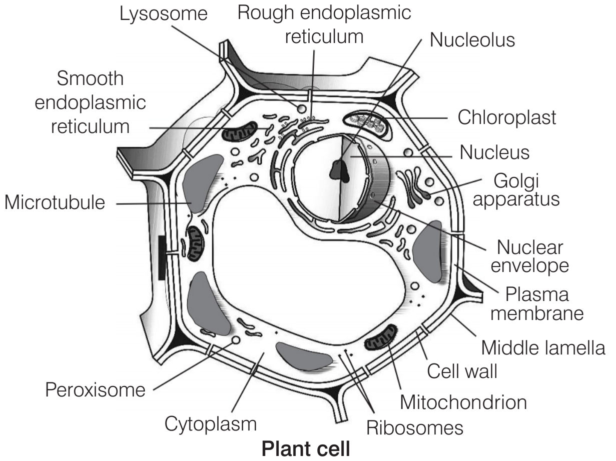
Prokaryotic cell	*Eukaryotic cell*
Simplest and primitive in nature.	Advanced and comparatively complex in nature.
Nucleus is absent.	Nucleus is present.
Membrane bound cell organelles are absent.	Membrane bound cell organelles are present.
Single naked chromosome is present.	Many chromosomes are present.
Cell division is direct.	Cell division by mitosis or meiosis.

Structure of Animal and Plant Cell

Animal and plant cell structures are described below



Animal cell



Plant cell

Differences between Plant and Animal Cell

Plant cell	*Animal cell*
Cell wall is present in plant cell.	Cell wall is usually absent.
Plastids are found.	Plastids are usually absent.
Centrioles are found in some algae and fungi.	Centrioles are found in all cells.
Big vacuoles are present but they are less in number.	Vacuole is very small in size but present in more number.
Cell shape is usually round or irregular.	Cell shape is rectangular and fixed.

Composition of Cell

Representation of the chemical composition of living tissue is as follows:

Component	*The total cellular mass in %*	*Component*	*The total cellular mass in %*
Water	70-90%	Lipids	2%
Proteins	10-15%	Nucleic acids	5-7%
Carbohydrates	3%	Ions	1%

Cell Organelles

There are some organelles as follow:

Cell Wall

It is a rigid and solid covering that gives shape and strong structural support to the cell. It is a non-living structure, which forms the outer covering of the plasma membrane in plants and fungi, but is absent in animal cells. Cell wall consists of following parts:

- **Middle lamella** It is composed of calcium pectate (chief component) and magnesium pectate.
- **Primary cell wall** It is composed of cellulose, hemicellulose and lignin. Rigidity of cell wall is due to lignin.
- **Secondary cell wall** It is composed of cellulose and hemicellulose.
- **Tertiary cell wall** It is present in tracheids of gymnosperms.
- Bacterial cell wall is composed of **peptidoglycan** or **murein**. Bacterial cell wall contains D-amino acids.
- Fungal cell wall is composed of chitin.

Functions of Cell Wall

- It helps in preventing cell from bursting or collapsing and allows the material to pass in and out of the cell.
- It wards off the attack of pathogens like viruses, bacteria, fungi and protozoans.

Cell Membrane or Plasmalemma

It is the innermost layer of cell envelope that is semipermeable in nature and is responsible for the interaction of cell with the outside environment.

- It consists of protein (52%), lipid (40%) and carbohydrate (5-10%). It is elastic in nature because of the presence of lipids.
- The most popular theory 'fluid mosaic model' of membrane structure was given by **Singer** and **Nicolson**.
- It states that integral and peripheral proteins are found in cell membrane.
- **Pinocytosis** (cell drinking) In this process, cells ingest extracellular fluid.
- **Phagocytosis** (cell eating) Engulfing and digestion of extracellular solid food by cell.

Nucleus

The term 'Nucleus' is given by **Waldeyer** (1888). Nucleus is a specialised largest principal cell organelle of the cell, which contains all genetic information for controlling all essential processes related to metabolism and transmission. It was first discovered by **Robert Brown** in 1831.

- Most of the eukaryotic cells are uninucleate except *Paramecium*, which is binucleate.
- Nucleus is made up of nuclear membrane, nucleoplasm, nucleolus, chromatin. Chromatin is composed of nucleic acid and protein (histone and non-histone chromosome).
- Chromosome of eukaryotic cell is composed of DNA + histone protein + some amount of RNA. Normally each cell contains 23 pair of chromosomes in humans.
- Nucleic acid, the genetic material was discovered by **Friedrich Miescher**. The term 'Nucleic acid' was given by **Altman**.

Chromosome

During the cell division, chromatin shrinks (compressed) and gets divided into various smaller, thick and consolidated form, known as chromosomes.

Chromosome Number in Some Organisms

Organisms	*Chromosome number in each body cell*	*Organisms*	*Chromosome number in each body cell*
Roundworm	2	Mouse	40
Mosquito	6	Rat	42
Drosophila	8	Human beings	46
Garden pea	14	Potato	48
Onion	16	Dog	64
Maize	20	Pigeon	80
Frog	24	Gold fish	100
Rice	26	*Amoeba*	250
Sunflower	34		

Study of Cell Organelles

Cell organelle	*Nature/Composition*	*Function*	*Present (P)/Absent (A)*	
			Animal cell	*Plant cell*
Cell membrane	Double layer of lipid molecules with proteins molecules, differentially permeable.	Gives shape to the cell, maintains distinct composition of the cell.	P	P
Cell wall	Freely permeable, made up of fibrous polysaccharide (cellulose). Bacteria and unicellular eukaryotes have peptidoglycan compound in their cell wall.	Gives strength and rigidity to the cell.	A	P
Cytoplasm	Consists of aqueous ground substance, having starch, glycogen, etc.	Provides medium for suspension of materials and cell organelles.	P	P
Vacuoles	Fluid-filled spaces without any definite shape and size. Larger in plant cell.	Store water, minerals, salts, food pigments, wastes, etc.	P	P
Nucleus	Oval-shaped, bounded by double membrane having pores possess genetic material.	Controls the metabolic activities, regulates cell division, synthesises and stores proteins, also called as the **controller of the cells.**	P	P
Mitochondria	Bounded by a double membrane, found in cytoplasm. They contain DNA also.	Cellular respiration or oxidation of food, ATP is stored in it, so called as **powerhouse of the cell.**	P	P
Endoplasmic Reticulum (ER)	Membranous network of fluid-filled with lumen with or without ribosomes, absent in RBCs.	Forms supporting skeletal framework of the cell also called **skeleton of the cell**.	P	P
Ribosomes	Dense and spherical particles, found either attached to endoplasmic reticulum or freely in cytoplasm. It is the smallest non-membranous organelles.	Play a vital role in protein synthesis also called **protein factory of the cell**.	P	P
Lysosomes	Tiny spherical bag-like structures found in cytoplasm with hydrolytic enzymes.	Helps in intracellular digestion, digest worn out organelles, also called **suicidal bags** of the cell.	P	P
Chloroplasts	Sac-like double walled with protein matrix generally called **plastids,** contain photosynthetic pigments called **chlorophyll** and **carotenoids.** These also contains DNA.	Sites of photosynthesis, give colour to organ, also called as **kitchen of the cell**.	A	P
Golgi bodies	Consist of tubules and vesicles.	Secrete hormones, enzymes, involved in synthesis of cell wall, plasma membrane, etc.	P	P
Centriole	Hollow and cylindrical structures made up of microtubules.	Helps in cell division in animal cell.	P	A
Microtubules	Tubulin protein	Spindle formation	P	P

CELL CYCLE

- Cell cycle is an orderly sequence of events or stages by which a cell duplicates its genome, synthesise the other constituents of the cell and eventually divides into two daughter cells. It is a cyclic process, which involves cell growth and cell division.
- **Cell division** is an essential process in all living organisms. The mode of cell division is fundamentally similar in all organisms. During cell division, the processes like DNA replication and cell growth must take place in a sequential coordinated manner to ensure the correct division and formation of progeny cells with intact genome. Cell cycle involves two phases, i.e. **interphase** followed by **M-phase**.

1. Interphase

It is the period between the end of one cell division to the beginning of the next cell division (i.e. between true successive M-phase). Interphase is further divided into following three substages on the basis of various synthetic activity.

(*i*) **G_1-phase** (Gap 1) It corresponds to the duration between the mitosis (M-phase) and initiation of replication of DNA. Cell grows, becomes metabolically active, but does not undergo DNA replication.
Some time after G_1-phase cell may enter S-phase and continue the cell cycle or it may enter into G_0-phase. In G_0-phase the cells leave cell cycle.

(*ii*) **S-phase** (Synthesis) It is known to be the phase, in which actual synthesis or replication of DNA takes place. In case of animal cell, during S-phase DNA replication begins inside the nucleus, while the duplication of centrioles takes place in the cytoplasm.

(*iii*) **G_2-phase** (Gap 2) This phase is also called **post-synthetic** or **pre-mitotic phase.** During this stage, the synthesis of DNA stops and proteins required for M-phase are being synthesised, while the growth of the cells continues. It prepares the cell to undergo divisions.

2. M-phase

M-phase is the actual phase of cell division. It may occur in two way depending upon the type of cell, it is taking place in, i.e. mitosis (vegetative cell) or meiosis (generative cell).

(A) Mitosis

A type of cell division, in which chromosomes replicate themselves and get equally distributed into daughter nuclei, i.e. the chromosome number in the parental and the progeny becomes same.

The four phases of mitosis are:

(*i*) **Prophase** Chromosome deviation in chromatids starts and nuclear membrane and nucleolus disappear.

(*ii*) **Metaphase** Chromosomes line up in the middle and spindle fibres from centriole connect to each chromatid (half of chromosome).

(*iii*) **Anaphase** Chromatids are pulled apart to separate poles and the membrane begins to pinch off in the middle.

(*iv*) **Telophase** Complete division of cytoplasm (cytokinesis) and two cells are formed. Two daughter cells, which are formed by mitosis have same/equal number of chromosomes. It results in formation of somatic cells.

(B) Meiosis

A type of cell division that results in the formation of four daughter cells each with half the chromosome number of the parent cell.

In the process of meiosis, there are two main stages:

Meiosis-I

It is further divided into four phases:

(*i*) **Prophase-I** The longest phase of meiosis-I.

It is composed of five substages:

(a) **Leptotene** Chromatins condense to form chromosome.

(b) **Zygotene** Pairing of homologous chromosomes called **synapsis**, in this phase, synaptonemal complex is formed.

(c) **Pachytene** Crossing over between non-sister chromatids.

(d) **Diplotene** Longest duration phase, in which chiasmata formation takes place.

(e) **Diakinesis** It involves slipping of chiasmata called **terminalisation**.

(*ii*) **Metaphase-I** Lining of chromosomes in the middle. Formation of spindle apparatus takes place and each centromere is joined by chromosomal fibres.

(*iii*) **Anaphase-I** Separation of homologous chromosomes (disjunction) to form two cells with haploid number of the chromosomes.

(*iv*) **Telophase-I** This is the last stage of meiosis-I, two cells with *n* number of chromosomes are formed.

Meiosis-II

It is very similar to mitosis.

(*i*) **Prophase-II** The shortening and thickening of the chromatids.

(*ii*) **Metaphase-II** Lining of chromosomes in the middle.

(*iii*) **Anaphase-II** Separation of chromosomes into chromatids.

(*iv*) **Telophase-II** Uncoiling and lengthening of the chromosomes, four daughter cells with haploid set of chromosomes are formed.

Differences between Mitosis and Meiosis

Mitosis	*Meiosis*
Found in somatic cells.	Found in diploid reproductive cells.
Forms two identical diploid cells.	Forms four haploid cells.
Every chromosome behaves independently.	Homologous chromosomes show pairing.
Chromosome number remains constant.	Chromosome number becomes half.
Crossing over does not occur.	Crossing over occurs.

IMPORTANT POINTS

- Chromoplasts are yellow, orange and red coloured.
- Leucoplasts are colourless plastids.
- **Golgi bodies** are absent in bacteria, blue-green algae, mature sperm and RBCs.
- Dinomitosis is failure of meiosis-II to occur after meiosis-I. It solely occurs in the class–Dinophyceae, a class of Alga with chlorophyll-*c*.

CLASSIFICATION OF PLANTS AND ANIMALS

TAXONOMY

The classification results in an organised system, which is essential for the naming and cataloging of future specimens and reflects inter-relationships between living organisms. Biologists use method of classification to understand the evolutionary history and it provides an organised way of studying characteristics of living organisms.

- **Classification** means the ordering of organisms into groups. The branch of science that deals with the principles and procedures of biological classification is called **taxonomy**.
- **Carolus Linnaeus** is the Father of Taxonomy. '*Systema Naturae*' and '*Species Plantarum*' books are written by Carolus Linnaeus.

- **Nomenclature** means providing distinct and proper names to organisms, so that they can be easily recognised and differentiated from others. The two codes for nomenclature are
  - **ICBN** International Code of Botanical Nomenclature
  - **ICZN** International Code of Zoological Nomenclature
- **Binomial nomenclature** was given by Linnaeus. It is the system of nomenclature using two terms, the first indicating the **genus** and the second one indicating the **species**, e.g. *Mangifera indica* (mango).
  - The categories used in classification are kingdom, phylum, class, order, family, genus and species.
  - Genus and species terms were proposed by **John Ray**.
  - The smallest unit of classification is species.
- **Living organisms** have been classified by different scientists. Some of the main classification systems are:
  (*i*) Two system classification was given by **Linnaeus** in 1758. It was the first classification. He divided all the living organisms into two kingdoms. These are Plantae and Animalia.
  (*ii*) **Copeland** designed four kingdom classification system in 1956.
  (*iii*) **Whittaker** gave five kingdom classification system in 1969.

CLASSIFICATION OF PLANTS

Many systems for classifying the plant kingdom have been formulated by various researchers and scientists, from time to time. Scientist **Eichler** has classified the plant kingdom into two main groups.

- **Plant Kingdom**
  - Cryptogamae
    - Thallophyta (Algae)
    - Bryophyta
    - Pteridophyta
  - Phanerogamae
    - Gymnosperms
    - Angiosperms
      - Monocot
      - Dicot

1. Cryptogamae

These plants never bear flowers and seeds. These are the most primitive. This division includes three main groups:

(i) Thallophyta

These are the simplest and primitive plants. Plant body is an undifferentiated mass of cells called as **thallus**. They lack vascular tissues.

This group includes algae and lichens (a symbiotic combination of an alga with a fungus).

(a) **Algae** These are plant-like organisms that are usually photosynthetic and aquatic in nature.
These are chlorophyll containing thallophytes, in which vascular tissue (xylem and phloem) are absent.
Some examples are as follow:
- *Chlorella* used as component of sewage oxidation tanks.
- Antibiotic chlorellin is obtained from *Chlorella.*
- Iodine is obtained from *Laminaria.*
- Agar is obtained from *Gelidium* and *Gracilaria*.
- *Spirogyra* is also called as **pond silk**.

(b) **Fungi** These are non-vascular, heterotrophic, spore forming eukaryotic organisms.
They have stored food material as glycogen. Cell wall is made up of chitin or fungal cellulose.
Some examples are as follow:
- *Albugo candida* – White rust of crucifers
- *Puccinia graminis tritici* – Black rust of wheat
- *Rhizopus stolonifer* – Fumatic acid
- *Aspergillus* – Produces aflatoxin (natural mycotoxin)
- *Agaricus* – Mushroom, these have filamentous structure called **hyphae**.
- In Whittaker's five kingdom classification, fungi has been placed in a separate kingdom – Fungi. It is mainly because fungi do not have chlorophyll.

(c) **Lichens** Symbiotic relationship between algae and fungi. These are the indicators of water pollution.

(ii) Bryophyta

These are called as the amphibians of plant kingdom.

(a) They do not possess vascular tissue.
(b) Roots are generally absent in them. They are present in damp and shady places.
(c) Some examples are as follow:
- *Riccia, Marchantia*, *Funaria*.
- *Zoopsis* is the smallest bryophyte.
- *Dawsonia* is the largest moss.
- *Sphagnum* is employed for dressing of wound.

(iii) Pteridophyta

The term 'Pteridophyta' comes from Greek word *Pteris* means fern and *phyton* means plant.

(a) These are seedless vascular plants that have plant body differentiated into true roots, stems and leaves. Secondary growth is absent in them.
(b) Some examples are as follow:
- *Selaginella, Equisetum.*
- *Dryopteris* (male shield fern).
- *Azolla, Marsilea, Salvinia* are aquatic ferns.
- *Adiantum* (maiden hair fern).
- *Cyathium* (tree fern).

2. Phanerogamae

These are also known as **spermatophytes**. They include those plants, which produce seeds. These are divided into two major types:

(i) Gymnosperms

The term 'Gymnosperm' comes from Greek word *Gymno* means naked and *sperma* means seed. These are vascular plants. Vascular plants means xylem and phloem are present in them for transportation of water and food, respectively.

Gymnosperms do not have outer covering around their seeds, i.e. their seeds are naked. Pollination type is anemophily (pollination by wind).

Some examples are as follow:

- *Cycas* (sagopalm)-*Nostoc* /*Anabaena* show symbiotic association with *Cycas* (most primitive gymnosperm).
- *Pinus* – Resin and terpentine obtained from *Pinus chilgoza* is obtained from *Pinus*.
- *Ginkgo* (maiden hair tree) – Living fossil.
- *Ephedra* – Drug ephedrine is obtained from the stem of *Ephedra*.
- *Sequoia gigantia* (red-wood tree), tallest (largest tree in term of total volume).
- *Rhynia* (most primitive gymnosperms).

(ii) Angiosperms

These are seed bearing, flowering vascular plants. Their seeds are enclosed within fruits. Double fertilisation takes place in angiosperms. Flowers are generally bisexual, e.g. mustard plant (*Brassica campestris*).

They are further divided into two types:

(a) **Monocots** These are angiospermic or flowering plants, which are characterised by the presence of single cotyledon in their seeds, e.g. maize, grass, etc.

(b) **Dicots** These are angiospermic flowering plants characterised by the presence of two cotyledons in their seeds, e.g. pea, rose, cotton, sunflower, etc.

CLASSIFICATION OF ANIMALS

Animals are classified in a variety of ways. This helps scientists to study the relationships in animal groups and to see the whole animal family as it has developed through time. Phylum is the grouping of different classes of animals, which have some common characteristic features like mode of nutrition, reproduction, locomotion, etc.

Storer and **Usinger** classified animals into following phylums:

Phylum–Protozoa

- These are unicellular eukaryotic organisms.
- In these, all the metabolic activities like digestion, respiration, excretion and reproduction take place in unicellular body.
- Respiration and excretion take place through diffusion, e.g. *Amoeba*, *Paramecium*, *Plasmodium* (malarial parasite).
- *Euglena* have characters of both animals and plants.
- Phylum–Protozoa has now been placed in kingdom–Protista.

Phylum–Porifera

- These are multicellular animals.
- These all are found in marine water and have porous body. The pores are called **ostia**.
- Their skeleton is made up of minute calcareous or silicon spicules, e.g. *Sycon*, sponge, etc.

Phylum–Coelenterata

- These are aquatic animals that possess thread-like structures called **tentacles** around the mouth, which help in holding the food.
- They have specialised cnidoblast cells to help in catching the food.
- *Hydra* has a tendency of regeneration of body organs, ***e.g.*** *Hydra*, jellyfish, sea anemone, etc.

Phylum– Platyhelminthes

- These animals are also called **flatworms**.
- Excretion takes place by flame cells.
- There is no skeleton, respiratory organ, circulatory system, etc., but alimentary canal is present.
- These are hermaphrodite animals.
  e.g. *Planaria*, liver fluke, tapeworm, etc.
- These organisms are found as internal parasites. For example, tapeworm is a common parasite found in intestine of human beings.

Phylum–Aschelminthes

- These are long cylindrical, unsegmented and unisexual worms, also called **Nematoda** or round worms.
- Their alimentary canal is complete, in which mouth and anus both are present.
- There is no circulatory and respiratory systems, but nervous system is well-developed.
- Excretion takes place through protonephridia.
- Most forms are parasitic, but some are free -living in soil and water, e.g. *Ascaris*, threadworm, etc.
- **Threadworm** is found mainly in the anus of child. Children feel itching and often vomit. Some children urinate on the bed at night.

- *Wuchereria bancrofti* (filaria) is human parasitic roundworm, which causes elephantiasis (filariasis) that spread through larvae. The **mosquito** (*Culex*) is the intermediate host.

Phylum–Annelida

- These are segmented worms.
- Their body is long, thin and soft.
- Alimentary canal is well-developed, nervous system is normal and the colour of the blood is red.
- They respire through skin, excretion occurs by nephridia.
- They move through setae made up of chitin, e.g. earthworm, *Neries*, leech, etc.
- Earthworm has four pair of hearts and it possess both male and female sex organ, i.e. it is hermaphrodite.
- Earthworm's body is sensitive for acidic, basic temperature and humidity.

Phylum–Arthropoda

- This is the largest phylum of Animalia.
- Jointed leg is their main feature. Their body is divided into three parts, i.e. head, thorax and abdomen. It is covered by chitinous exoskeleton.
- Circulatory system is open type.
- Cockroach's heart has **13 chambers** and has spiracles.
- Trachea, book lungs, body surface are respiratory organs.
- Insects generally have six feet and four wings, e.g. cockroach, prawn, crab, bug, housefly, mosquito, bees, etc.
- Arachnids possess 8 legs, e.g. spider.
- Ant is a social animal, which reflects labour division.
- Termite is also a social animal, which lives in colony.
- Insect bite releases formic acid, which can cause itching, swelling, redness, etc.

Phylum–Mollusca

- Their body is soft unsegmented and divided into head, muscular foot and visceral hump.
- Body is covered by a hard calcareous shell.
- Their alimentary canal is well-developed.
- Respiration takes place through gills or ctenidia.
- Blood is blue due to the presence of haemocyanin.
- Excretion takes place through kidneys. e.g. *Pila, Aplysia* (sea rabbit), *Doris* (sea lemon), *Octopus* (devilfish), *Sepia* (cuttlefish).

Phylum–Echinodermata

- All the animals in this group are marine.
- They have water vascular system.
- Brain is not developed in nervous system.
- They have a special capacity of regeneration, e.g. starfish, sea urchin, sea cucumber, etc.

Phylum –Hemichordata

Hemichordata was earlier placed as a sub-phylum under the phylum-Chordata. But now, it is considered as a separate phylum (under non-chordata). These are also called

Phylum–Chordata

- They have notochord, a dorsal hollow tubular nerve cord and paired pharyngeal gill slits.
- This phylum is subdivided into two **subphylum**, i.e. Protochordata and Vertebrata.

Some main groups of phylum– Chordata are:

Group–Pisces

- These are aquatic animals (cold-blooded animals).
- Their heart pumps only impure blood and have two chambers.
- Class–Cyclostomata includes jawless fishes, e.g. lampreys.
- Cartilaginous fishes come under class–Chondrichthyes, e.g. Indian shark.
- Class–Osteichthyes includes bony fishes, e.g. sea-horse (*Hippocampus*)
- Respiration takes place through gills, gill slits, etc. e.g. *Scoliodon* (dogfish), *Torpedo,* etc.

Group–Tetrapoda

Group Tetrapoda classified into four clasess

(i) Amphibia

- These can live both on land and water. All these animals are cold-blooded or ectothermic.
- Ectothermic organisms can not regulate their body temperature.
- Respiration takes place through gill, skin and lungs. e.g. frog, toad, *Necturus*, *Icthyophis*, Salamander.
- In water frog respires through the moist skin, while on land it respires through lungs.

(ii) Reptilia

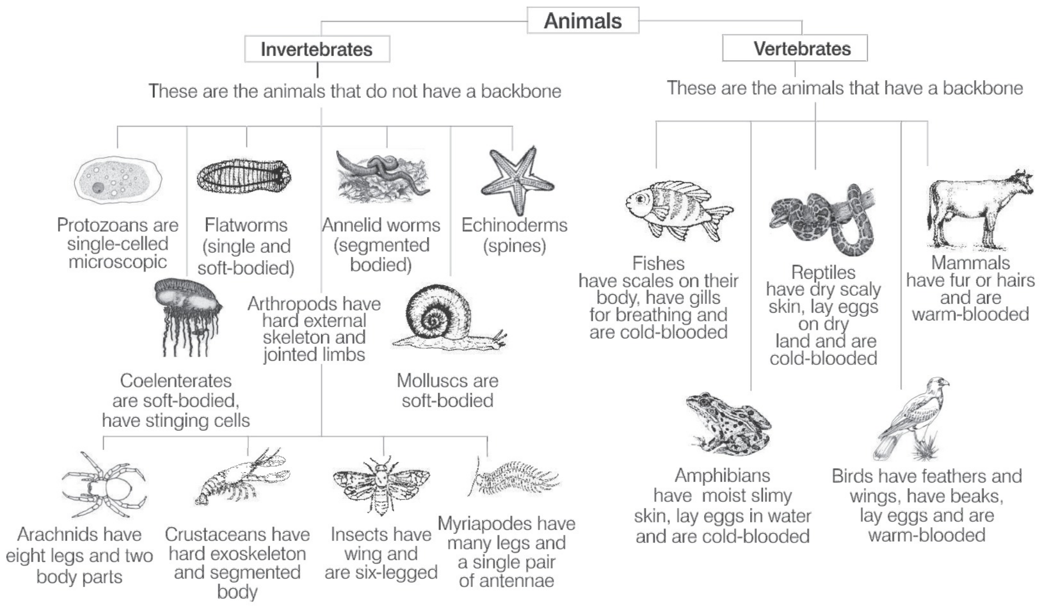
- These are crawling animals, which are cold-blooded and contain two pair of limbs.
- Their skeleton is completly flexible.
- Respiration takes place through lungs.
- Their eggs are covered with shell made up of calcium carbonate.
  e.g. lizard, snake, tortoise, crocodile, turtle, *Sphendon*, etc.
- Cobra is the only snake, which makes nests.
- Sea snake is also called *Hydrophis*. It is world's most poisonous snake.
- *Heloderma* is the only poisonous lizard.

(iii) Aves

- The animals of this group are warm-blooded (endothermic), tetrapod vertebrates with flight adaptation and show the property of Homeothermy.
- Homeothermy is the thermoregulation that maintains internal body temperature regardless of external influence.
- Their forefeet are modified into wings for flying.
- They respire through lungs.
- Birds have no teeth, beak helps in feeding, e.g. crow, peacock, parrot, etc.
- Flightless birds are kiwi and emu. Largest alive bird is Ostrich.
- Smallest bird is humming bird.
- Largest zoo in India is Alipur (Kolkata) and the largest zoo of the world is 'Cruiser National Park' in South Africa.

(iv) Mammalia

- They are heterotrophs, mostly omnivorous organisms.
- These are warm-blooded or endothermic animals.
- Tooth comes twice in these animals (diphyodont).
- There is no nucleus in its red blood cells (except in camel and llama). Skin of mammals have hairs. External ear is present, e.g. human, whale, platypus, kangaroo, etc.
- Mammals are divided into three subclasses:
  - **Prototheria** It lays eggs, e.g. *Echidna.*
  - **Metatheria** It bears the immature child, e.g. kangaroo.
  - **Eutheria** It bears the well-developed child, e.g. humans, cow, etc.
- Mammals give birth to young ones but *Echidna* and duck-billed platypus are the only egg laying mammals.
- **Bats** have poor vision, but can fly in dark due to the production of ultrasonic waves by them.
- **Cheetah** is the fastest mammal.
- **Blue whale** is the largest mammal.
- **Spiny anteater** is the most primitive mammal.
- **Gorilla** is largest ape.
- *Panthera tigris* is the largest animal of cat family.
- **Dolphin** is the national aquatic animal of India.
- *Homo sapiens* is the commonest mammal.



Classification of animals

GENETICS AND MOLECULAR BIOLOGY AND EVOLUTION OF LIFE

GENETICS

Gregor Johann Mendel is known as the **Father of Genetics**. The term 'Genetics' was first used by **William Bateson** in 1905.

- Genetics is the process of transfer of character (traits) from one generation to next generation through genes.
- A **gene** is known as the molecular unit of heredity of a living organism.
- **Dr. Hargovind Khorana** (1968) discovered the genetic code and its function in protein systhesis.

Mendel's Experiments

GJ Mendel for the first time conducted experiments to understand the pattern of inheritance of variation in living organisms.

- He conducted hybridisation (crossing) experiments on garden pea (*Pisum sativum*) for seven years and proposed the laws of inheritance in living organisms.
- Mendel conducted artificial pollination/cross-pollination experiments using several true-breeding lines of pea.
- A **true-breeding** line refers to one that have undergone continuous self-pollination and expressed stable traits for several generations.
- Mendel selected 14 true-breeding pea plant varieties, which were similar except for one character with contrasting traits.

These characters are listed in table given below:

A List of Contrasting Traits Studied by Mendel in Pea Plant

Characters	*Contrasting traits*	
	Dominant	*Recessive*
Stem height	Tall	Dwarf
Flower colour	Violet	White
Flower position	Axial	Terminal
Pod shape	Inflated	Constricted
Pod colour	Green	Yellow
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green

In 1900, Mendel's work was 'rediscovered' by three European scientists, **Hugo de Vries**, **Carl Correns** and **Erich von Tschermak**.

Mendel's Laws

Mendel's laws of inheritance are based on his observations on monohybrid and dihybrid crosses. He summarised his findings into following three laws:

Law of Dominance

(a) Law of dominance states that, when two alternative forms of a trait or character (genes) are present in an organism, only one factor expresses itself in F_1 progeny and is called **dominant**, while the other whose expression remain masked is called **recessive**.

(b) This shows that crossing between pure organisms for two contrasting characters produce only one character in F_1-generation and both in F_2-generation.

Law of Segregation

(a) This law states that, though, F_1 hybrid, alleles for dominant and recessive characters remain together, but these alleles do not show any blending and both the characters are recovered as such in the F_2-generation.

(b) These two co-existing alleles of an individual for contrasting trait segregate (separate) during gamete formation, so that each gamete receive only one of the two alleles.

(c) Alleles again unite during fertilisation of gametes.

Law of Independent Assortment

(a) It is also known as inheritance law and it states that alleles for one character assort independently of alleles for another character during gamete formation.

(b) When two pairs of traits are combined in a hybrid, segregation of one pair of character is independent of the other pair of characters during gamete formation.

➡ **Note** Back cross is performed between F_1 hybrid and one of its parent whether dominant or recessive, while test cross is a cross between F_1 hybrid and recessive parent only.

Exceptions to Principle of Dominance and Principle of Paired Factors

Incomplete dominance is the phenomenon, in which phenotype of the F_1 hybrid offspring does not resemble any of the parent, but it is an intermediate between the expression of two alleles in their homozygous state.

Red and white flowering true breeding lines of *Mirabilis jalapa* are when crossed, they shows F_1 hybrid having pink flowers, i.e. incomplete dominance (exception to dominance law).

Codominance is the phenomenon, in which two alleles of a hybrid express themselves independently. In this, offsprings show resemblance to both the parents. ABO blood group system in humans is the well-known example of codominance.

A, B, O BLOOD GROUP

The ABO blood group system was discovered by the Austrian scientist **Karl Landsteiner**. ABO blood grouping is controlled by the gene I. The plasma membrane of the Red Blood Cells (RBCs) has sugar polymers that protude from its surface. These gene controls the kind of sugar protruding from RBC. The gene I has three alleles, I^A, I^B and i. The alleles I^A *and* I^B produce a slightly different form of sugar, while allele i does not produce any sugar.

On the basis of the presence or the absence of the antigen A or B, four types of blood groups are present, which are A, B, AB and O.

The Genetic Basis of Blood Groups in Human Population

Allele from parent 1	*Allele from parent 2*	*Genotype of offsprings*	*Blood types of offsprings*
I^A	I^A	$I^A I^A$	A
I^A	I^B	$I^A I^B$	AB
I^A	i	I^A i	A
I^B	I^A	$I^A I^B$	AB
I^B	I^B	$I^B I^B$	B
I^B	i	I^B i	B
i	i	i i	O

- For transfusion purpose the Rhesus factor and ABO grouping are the two most important compatibility factors to consider.
- An individual may be Rh^+ or Rh^--factor. In simpler term, if an individual's blood type is A and positive for the Rhesus factor, then he or she is deemed to be 'A^+'.

SEX DETERMINATION IN HUMANS

In humans, each cell contains 46 chromosomes (44 autosomes + 2 sex chromosomes). Male sex chromosome is XY, whereas female sex chromosome is XX. During gamete formation, in female, all gametes contain only one type of chromosome, i.e. X and in male, half of the sperm contain X-chromosome, while half contain Y-chromosome.

Thus, when a male gamete, i.e. sperm carrying X-chromosome fertilises an ova, the zygote develops into female and when a sperm carrying Y-chromosome fertilises an ova, the zygote develops into a male. Any increase or decrease in the number of sex chromosomes or autosomes causes genetic disorder.

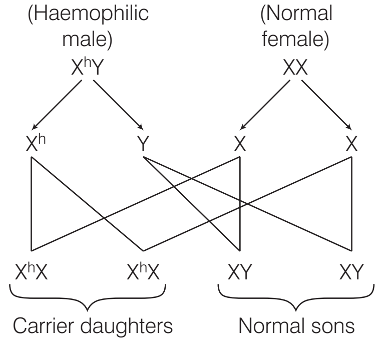
SOME HEREDITARY DISEASES

These are anatomical or physiological abnormalities present from birth, also called **congenital diseases**. These diseases can be caused by genetic mutations or environmental factors. **Mutation** is nothing but the sudden change in genetic make up due to addition, deletion or replacement of any gene.

Diseases by genetic mutation can be transmitted to next generation, whereas by environmental factors are non-transmissible.

Haemophilia-A

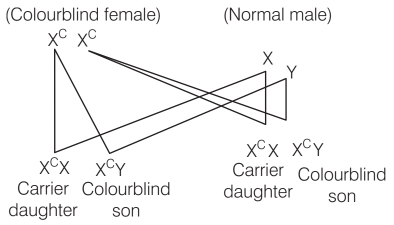
- It is the most common X-linked recessive genetic disorder associated with serious bleeding.
- It is caused by a reduction in the amount and activity of coagulation factor VIII.



Haemophilia-A

Colour Blindness

- It is an X-linked recessive genetic disorder, which results in the inability to perceive differences between some of the colours that others can distinguish.
- It does not mean not seeing any colour at all infact it leads to the failure in discrimination between red and green colour.



Colour blindness

Cri-du-Chat Syndrome

- It is a rare genetic disorder due to a missing part of chromosome 5.
- It is characterised by having a **cat-like cry** of affected children.

Sickle-Cell Anaemia

- It is an autosome linked recessive trait that can be transmitted from parents to the offsprings, when both the partners are carriers for the gene.
- It is a genetic blood disorder characterised by an abnormal, sickle-shape red blood cells.
- It is caused by the substitution of glutamic (Glu) acid by valine (Val) at the sixth position of the β-globin chain of the haemoglobin molecules.

Down's Syndrome/ Trisomy-21

- It is a chromosomal disorder caused by the presence of all or a part of an extra 21st chromosome.
- It is associated with impairment of cognitive ability and physical growth and a particular set of facial characteristics.
- In this syndrome, the person exhibit **Mongolism**.

Klinefelter's Syndrome

- It is a condition, in which human males have an extra X-sex chromosome (44 + XXY).
- Effects are development of small testicles and reduced fertility.
- Such persons are sterile males or feminised males with undeveloped testis and some feminine characteristics like enlarged breasts, etc. They may be mentally retarded aslo.

Duchenne Muscular Dystrophy

- It is an X-linked disease associated with muscular dystrophy (mutations in the dystrophin gene).
- It is characterised by rapid progression of muscle degeneration, eventually leading to loss in ambulation, respiratory failure and death.

Turner's Syndrome

- It is a condition, in which human female has only one sex chromosome (XO).
- Effects are rudimentary ovaries and lack of secondary sexual character in women.

Patau Syndrome/Trisomy-13

- A syndrome, in which a patient has an additional copy of autosomal chromsome 13 due to a non-disjunction of chromosomes during meiosis.
- Its effects are mental retardation, cut mark in the lip.

Alkaptonuria

It is a rare inherited genetic disorder of phenylalanine and tyrosine metabolism.

Phenylketonuria

It is an autosomal recessive metabolic genetic disorder characterised by mutation in the gene for hepatic enzyme phenylalanine hydroxylase.

Progeria

Caused due to the point mutation in position LMNA gene replacing cytosine for thymine. Creating a truncated from progenin of prelamin a protein. Early aging, small face and jaw, pinched nose, loss of eye sight, hair loss are some of major symptoms.

Blue Baby Syndrome

It is used to describe newborns with cyanotic heart lesions. It is caused basically due to high nitrate contamination in ground water resulting in decreased O_2 carrying capacity of haemoglobin in babies leading to death. It is also called as **methaemoglobinemia**.

MOLECULAR BASIS OF INHERITANCE

Nucleic acid is the genetic material found in living beings. It was discovered by **Friedrich Miescher**. The term 'Nucleic acid' was given by **Altman**. Nucleic acids are polymers as nucleotides. Basically two types of nucleic acids are found in living systems, i.e. Ribonucleic Acid (RNA) and Deoxyribonucleic Acid (DNA).

Ribonucleic Acid (RNA)

RNA was the first genetic material to be discovered even before DNA evolved as a genetic material.

- The *2′–OH* group of ribonucleotides is a reactive group that makes RNA, a catalyst. It is evident that the essential life processes such as metabolism, translation and splicing, etc., have evolved, around RNA being a catalyst was reactive and hence, unstable.
- Therefore, DNA has evolved from RNA with chemical modification that makes it more stable. RNAs are of three main types, i.e. messenger RNA (*m*RNA), transfer RNA (*t*RNA) and ribosomal RNA (*r*RNA), which participate in the process of transcription and translation also.
- RNA helps in the formation of protein, i.e. **protein synthesis** or **translation**.

Deoxyribonucleic Acid (DNA)

DNA acts as a genetic material in most organisms.

- It is a long polymer of deoxyribonucleotides.
- The number of nucleotides or base pair (bp) present in the DNA determines its length.
- DNA is made up of two polynucleotide chains and its back bone is constituted by sugar and phosphate. From this backbone the nitrogenous bases project inwards.
- The two strands of DNA are paired through hydrogen bonds (H-bonds) between bases.

Differences between DNA and RNA

DNA	RNA
It usually occurs inside nucleus and in some cell organelles like mitochondria and chloroplast.	Very little RNA occurs inside nucleus. Most of it is found in the cytoplasm.
DNA is the genetic material.	RNA is not the genetic material except in certain viruses, e.g. HIV, retrovirus.
It is double-stranded with the exception of some viruses like polyomavirus.	RNA is single-stranded with the exception of some viruses, e.g. double- stranded in T_2, T_4, T_6 bacteriophage, etc.
DNA shows regular helical coiling.	There is no regular coiling except in some parts of RNA.
It contains deoxyribose sugar.	It contains ribose sugar.
Nitrogen base thymine occurs in DNA along with three other adenine, cytosine and guanine.	Thymine is replaced by uracil in RNA. The other three are adenine, cytosine and guanine.
It replicates to form new DNA molecules.	It cannot replicate itself except in RNA-RNA viruses.
DNA controls heredity, evolution, metabolism, structure and differentiation.	RNA controls only protein synthesis.

Structure of a Polynucleotide Chain (DNA/RNA)

RNA and DNA both are comprised of polynucleotide chain. A nucleotide is composed of a nitrogenous base, pentose sugar and a phosphate group.

Nitrogenous Base

It is a nitrogen containing organic molecule having physical properties similar to a base.

- There are two types of nitrogenous bases:
    1. **Purines** (Adenine and Guanine)
    2. **Pyrimidines** (Cytosine, Thymine and Uracil)
- Adenine and thymine are bonded by double bond (A═T).
- Cytosine and guanine are bonded by triple hydrogen bond (C≡G).
- Out of pyrimidines, cytosine is common for both DNA and RNA, while thymine is present in DNA only.
- Uracil is present in RNA in place of thymine.

A Pentose Sugar

- Nitrogenous base is attached to the pentose sugar by N-glycosidic linkage to form nucleoside.
- Two types of sugars are present in RNA and DNA respectively, i.e. ribose in case of RNA and deoxyribose in case of DNA.

A Phosphate Group

When a phosphate group is attached to 5′–OH of a nucleoside through phosphodiester linkage, a nucleotide is formed.

Discoveries Related to Structure of DNA and Composition of DNA

- **Friedrich Miescher** (in 1869), identified DNA as an acidic substance present in nucleus and termed it nuclein.
- **James Watson** and **Francis Crick** (in 1953) proposed the double helix model of the DNA structure.
- **Erwin Chargaff** observed that for a double-stranded DNA, the ratios between adenine, thymine and guanine, cytosine are constant and equal to one.

$$\frac{A+T}{G+C} = 1$$

Genetic Code

The relationship between the sequence of nucleotide on *m*RNA (i.e. messenger RNA) and sequence of amino acids in the polypeptide is called **genetic code**.

- In this nitrogenous bases in *m*RNA encode the information required for protein synthesis.
- The sequence of nitrogenous bases (nucleotides) in *m*RNA for coding a single amino acid is called as **codon**.
- A codon is triplet, i.e. it is formed of three nucleotide bases, specific, continuous, i.e. without comma or punctuation and universal in nature.

ORIGIN OF LIFE

The origin of life is considered a unique event in the history of universe. The study of history of life forms on the earth is called **evolutionary biology**. The important theories of origin of life are as follows:

Theory of Spontaneous Generation

- It states that life originated from decaying and rotting matter like straw, mud, etc., spontaneously, i.e. organism is formed automatically from non-living matter.
- **Van Helmont** (1652) stated that young mice could arise from wheat grains, when these are put in the dark along with a moist shirt.

Theory of Biogenesis

- Theory of spontaneous generation (life) was rejected by **Francesco Redi** (1668), **Lazzaro Spallanzani** (1765) and **Louis Pasteur** (1861).
- Spontaneous generation of flies from rotting meat was disproved by F Redi.
- Spallanzani stated that air carried microorganisms.
- Pasteur is famous for 'Germ Theory of Diseases'.
- Theory of biogenesis states that the living things can only arise from previously existing living things.

Modern Theory (Oparin's Hypothesis)

- This theory is also known as modern theory or abiogenic origin or naturalistic theory or physiochemical evolution.
- According to this theory, life originated on earth after a long period of 'abiogenic molecular evolution'.
- It was hypothesised by **AI Oparin**. Oparin wrote a book '*The Origin of Life*' in **1936**.
- The first experimental support to Oparin-Haldane's theory of origin of life came from **Urey** and **Miller's** simulation experiment in 1853.
- They demonstrated clearly that the UV radiations or electrical discharge or combination of these can produce complex organic compounds from a mixture of CH_4, NH_3, H_2O *and* H_2 with ratio of methane, ammonia and hydrogen was 2 : 1 : 2 in the experiment.
- The most important condition for the origin of life is the presence of water.
- Hydrogen atoms were most numerous and most active in primitive atmosphere.

EVOLUTION

The term 'Evolution' has been derived from the Latin word *evolvere* (*e*–out; *volvere*–to roll) means to unfold or to unroll to reveal hidden potentialities. Evolution is a process by which different kinds of living organisms develope from earlier froms.

Geological Time Scale and Events in the Evolution

By studying fossils occurring in different strata of rocks, geologists are able to reconstruct the time and course of evolutionary change.

Time Scale of Earth

Era	*Period Age in million years from present*	*Epoch Age in million years from present*	***Fauna** (Animals)*	***Flora Plants***
Cenozoic (Era of modern life)	Quarternary 0-2	Recent (Holocene) (0.01)	Modern man dominant; Modern mammals, birds, fishes, insects.	Rise of herbaceous plants, decline of woody plants.
Age of mammals and angiosperms		Pleistocene 2	Extinction of great mammals; Humans appeared evolution of human society and culture.	
	Tertiary 2-65	Pliocene 6-7 (Age of mammals)	Formation and adaptive radiation of modern mammals.	Adaptive radiation of flowering plants.
		Miocene 26	Mammals at peak, first man-like apes formed.	
		Oligocene 38	Rise of first monkeys and apes.	
		Eocene 54	Diversification of placental mammals, origin of horse.	Angiosperm dominance.
		Palaeocene 65	Rise of first primates.	
Mesozoic (Era of medieval life)	Createaceous 135		Extinction of dinosaurs and toothed birds; Rise of and placental mammals.	Dominance of angiospermic plants.
Age of reptiles and gymnosperms	Jurassic 145 (Age of reptiles)		Origin of toothed birds (first birds) dinosaurs dominant.	Origin of angiosperm. Dominance of gymnosperms.
	Triassic 225		Origin of dinosaurs and mammals.	Abundance of gymnosperms.
Palaeozic (Era of ancient life)	Permian 280		Extinction of trilobites. Origin of mammals-like reptile (therapsids) and most modern orders of insects.	Origin of conifers.
	Carboniferous 350 (Age of amphibians)		Origin of reptiles and winged insects. Amphibians dominant.	Abundance of tree ferns, forming coal forests.
	Devonian 400 (Age of fishes)		Origin of amphibians; fishes abundant spiders appeared.	Earliest mosses and ferns. First speed plants appeared.
	Silurian 440		Origin of jawed fishes and wingless insects; earliest coral reefs.	Earliest spore-bearing plants.
	Ordovician 500 (Age of invertebrates)		Origin of vertebrates (Jawless fishes); invertebrates abundant, also called age of giant molluscs.	
	Cambrian 570		All invertebrate phyla established; origin of trilobites.	
Proterozoic (Era of early life)	1000, 2000, 3000		Ancient metazoans (sponges, cnidarians), primitive eukaryotes; scanty fossils (prokaryotes).	
Archaezoic (Era of invisible life)	3500		Origin of life; no recognisable fossils.	
Azoic (Era of no life)	4600		Origin of solar system; no life.	

Evidences of Organic Evolution

- **Homologous organs** are similar in basic structure and origin, but dissimilar in functions, e.g. wings of bat, cat's paw, front foot of horse, human hands and wings of birds are homologous organs.
- **Analogous organs** are the organs similar in shape and function, but their origin, basic development plan are dissimilar, e.g. wings of insects, birds and bats are analogous organs. Analogous organs are also called as **homoplastic organs**.
- **Vestigial organs** are degenerated, non-functional organs, which were functional earlier. Human body has been discribed to possess about 90 vestigial organs.
- Some of these are muscles of ear pinna, canine teeth and third molar teeth, body hairs, vermiform appendix (degenerated terminal part of caecum) nictitating membrane of eye, caudal vertebrae (coccyx or tail bone), etc.
- **Atavism or reversion** is the sudden reappearance of some ancestral features. Appearance of thick body hair, large canines, monstral face, short temporary tails, additional pairs of nipples, etc., are examples of atavism.
- Connecting links are organisms that has features of two existing group of organisms. These also provide an evidence for evolution.

Organism	*Connecting link between*
Virus	Living and non-living
Euglena (Protozoa)	Plants and animals
Proterospongia (Protozoa)	Protozoa and Porifera
Peripatus (Arthropoda)	Annelida and Arthropoda
Neopilina (Mollusca)	Annelida and Mollusca
Balanoglossus (Chordata)	Non-chordata and Chordata
Dipnoi (Lung fish)	Pisces and Amphibia
Archaeopteryx (Aves)	Reptiles and Aves
Prototheria (Mammalia)	Reptiles and Mammalia

Theories of Evolution

Evolution is generally progressive and variation is most important requirement for it. The ultimate source of organic variation is mutation (sudden change in genes).

These are some important theories of evolution:

Lamarckism

The French biologist **Jean Baptiste de Lamarck** (1744-1829) suggested a complete theory of evolution. Lamarckian theory is also known as 'Theory of Inheritance of Acquired Characters' or 'Theory of Use and Disuse of Organs'.

(*i*) Lamarck's theory was published in his book '*Philosophie Zoologique*' in 1809.

(*ii*) Lamarck arranged his theory in four postulates.
- Internal forces tend to increase size of the body.
- Formation of new organs is the result of the need.
- Development of organs is based on the continuous use and disuse.
- All changes acquired by the organism are transmitted to offspring by the process of inheritance.

(*iii*) Some examples of uses and disuses of organs are:
- Webbed feet of swimming birds.
- Rudimentary eyes of cave dwellers.
- Elongated limbless body of snake.
- Long thin neck in giraffe.
- Vestigial organs of living animals.

Darwinism

Charles Darwin (1809-1882) and **Alfred Russel Wallace** (1823-1913) proposed, the 'Theory of Natural Selection'.

Darwin's theory is based on five principles:

(*i*) Over production is shown by every organism.
(*ii*) Organism show struggle for existence.
(*iii*) Struggle for existence leads to variations and their inheritance.
(*iv*) The variations in the organisms lead to the survival of the fittest.
(*v*) Natural selection and species formation is due to variations.

One major criticism against Darwin's theory was his failure to give an explanation for variations.

Mutation Theory

Hugo de Vries (1845-1935) gave much importance to the discontinuous variation or saltatory variations and proposed that new species arise not by the accumulation of minor and continuous variation through the natural selection, but by saltatory variations that appear suddenly for, which he coined the term 'mutation'.

(*i*) After that he proposed the 'theory of mutation'.
(*ii*) Mutations are discontinuous variation.
(*iii*) These are due to changes in chromosomes, genes and DNA. These changes may or may not be inherited.

Synthetic Theory

The modern theory of origin of species or evolution is known as **modern synthetic theory** of evolution.

(*i*) Initial basis of synthetic theory was given by **Dobzohansky** (1937). Modern synthetic theory of evolution was designated by **Huxley** in 1942.

(*ii*) According to it the five basic factors are:
- Gene mutation
- Changes in chromosome structure and number
- Genetic recombinations
- Natural selection
- Reproductive isolation

First three factors are responsible for genetic variability.

Hardy-Weinberg Equilibrium

Hardy-Weinberg law is the fundamental law of population genetics, which provides the basis for studying the Mendelian population. According to this law,

(*i*) Mutations introduce new genes into a species resulting into a change in gene frequencies.

(*ii*) In certain conditions gene frequencies remain constant. The conditions necessary for gene frequency to remain constant are:
- Mating must be completely random.
- Mutation must not occur.
- The population must be very large.
- All genes must have an equal chance of being passed to the next generations.

(*iii*) Constant gene frequencies over several generations indicate that natural selection and evolution are not taking place.

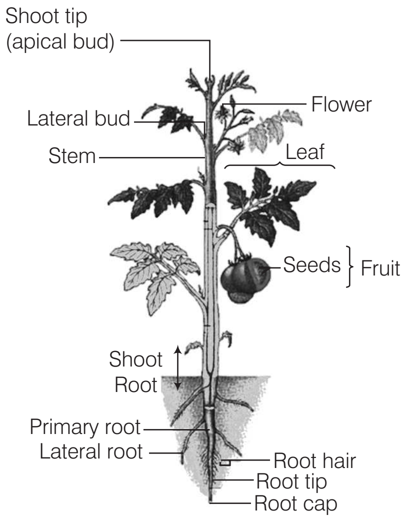
(*iv*) Changing gene frequencies would indicate that evolution is in progress.

PLANT MORPHOLOGY AND PHYSIOLOGY

Angiosperms are the seed bearing, flowering plants in which seeds are formed inside fruits; and male and female reproductive parts are organised into flowers.

MORPHOLOGY OF ANGIOSPERMS

Morphology (*Morphe*–form; *logos*–study) is the branch of biology, which deals with the study of external forms and features of different plant organs like roots, stems, leaves, flowers, seeds, fruits, etc. The body of typical angiospermic plant is differentiated into an underground root system and an aerial shoot system.



Morphology of plant

Types of Plants

On the basis of nutrition, plants are of following types:

(*i*) **Autotrophic plants** These are green and capable of making their own food by the process of photosynthesis, e.g. all green plants.

(*ii*) **Parasitic plants** Depend on other plants for food and water. They have special roots for absorption of food and water. These roots are called **sucking roots** or **haustoria,** e.g. *Cuscuta, Duranta*.

Type of parasitic plants are as follow:

(*i*) **Obligate parasite**, e.g. *Nuytsia floribunda*.

(*ii*) **Facultative parasite**, e.g *Rhinanthus*.

(*iii*) **Root parasite**, e.g. *Rafflesia,* sandal wood.

(*iv*) **Stem parasite**, e.g. *Cuscuta, Loranthus.*

(*v*) **Saprophytic humus plants** Grow on dead organic matter, e.g. fungi.

(*vi*) **Symbiotic plants** Symbiosis or mutually beneficial partnership of two organisms, e.g. mycorrhiza associated with roots of higher plants and legumes associated with *Rhizobium* (N_2-fixing bacteria).

Different organ systems present in angiosperms are as follow:

Root System

Roots develop from radicle of seed. They are non-green, underground, geotropic, phototropic and hydrotropic in nature and they do not bear buds, nodes and internodes. They have unicellular root hairs. Lateral roots arise endogenously from pericycle.

Types of Roots

Roots are of two types:

(*i*) **Tap root** It develops from radicle. The primary root grows and gives rise to secondary and tertiary roots forming tap root system, e.g. dicot.

(*ii*) **Adventitious root** These roots develop from any part of the plant body other than the radicle, e.g. monocot.

Functions of Roots

- Roots help in fixation and absorption of water and minerals.
- Storage of food is one of the major function of roots, conduction of water takes place by roots.
- It helps in photosynthesis and respiration by providing supply of water to all parts of plant.

Shoot System

Shoot system consists of following parts:

Stem

Stem is the ascending part, of plant, bearing branches, leaves, flowers and fruits, formed by the prolongation of the plumule of embryo. It bears nodes and internodes and has buds. It may also bear multicellular hairs on external surface.

Modifications of Stem

Stem is modified in the following forms:

- **Rhizome** It grows parallel or horizontal to soil surface. It bears nodes, internodes, buds and leaves, e.g. ginger.
- **Tuber** It is the terminal portion of underground stem branch, which is swollen due to accumulation of food, e.g. potato.
- **Corm** It grows vertically to soil surface and is covered by thin sheathing leaf bases of dead leaves called **scales**, e.g. *Colocasia, Gladiolus*.
- **Phylloclade** It is green, flattened or rounded succulent stem with leaves either feebly developed or modified into spines, e.g. *Opuntia, Euphorbia.*
- **Cladode** Phylloclade with one or two internodes are called cladode, e.g. *Asparagus*.
- **Thorn** It is the modification of axillary bud. Thorns not only reduce transpiration, but also check browsing by animals, e.g. *Citrus, Duranta*.

Leaves

A leaf is a flat, lateral outgrowth of the stem or the branch, arising from a node and usually having a bud in its axil. It originates from shoot apical meristem. These are the most important organ for photosynthesis. The chief functions of the leaf include photosynthesis and transpiration.

Venation

The arrangement of veins and the veinlets in the lamina of leaf is called venation. These are of two main types:

- **Reticulate venation** The branches or veins form a network, e.g. dicots.
- **Parallel venation** The veins and veinlets remain parallel to each other, e.g. monocots.

Phyllotaxy

It is the arrangement of leaves on the stem or branch and is adopted, so that each leaf is properly exposed to sunlight.

Types of Phyllotaxy

Phyllotaxy is usually of three types:

- **Alternate** When a single leaf arises at each node it is called as alternate, e.g. *Hibiscus* sp.
- **Opposite** When two leaves arise at each node and occurs opposite to each other it is called as opposite, e.g. *Syzygium, Calotropis*.
- **Whorled** If more than two leaves arise at a node and form a whorl. Heterophylly is occurrence of more than one type of leaves on the same plant, e.g. *Ranunculus* sp.

Flower

A flower is the reproductive unit in angiosperms, which is meant for sexual reproduction. A flower arises in the axil of a leaf-like structure called **bract**, e.g. *Bougainvillea.*

Structure of a Flower

Flower consists of four distinct parts, calyx, corolla, androecium and gynoecium.

- The calyx and corolla are accessory parts while the androecium and gynoecium are essential parts.
- When calyx, corolla, androecium and gynoecium are all present on same flower it is called as **complete flower**.
- A flower can be **unisexual** or **bisexual**.
- The symmetry of a flower depends upon the size, shape and arrangement of floral parts.
- On the basis of symmetry flower can be:
  (*i*) **Actinomorphic** When a flower can be divided into two exactly equal halves by any verticle section passing through the centre of a flower, e.g. mustard.
  (*ii*) **Zygomorphic** When a flower can be divided into two identical halves through only one particular plane, e.g. *Cassia.*
- An extra whorl of sepal-like green coloured structures called **bracteoles** occurs on the outside of calyx in members of Malvaceae. It is called as **epicalyx** fused sepals and petals form **perianth**.
- Stamens and pistils are the male and female reproductive organs of a flower. Stamen consists of anthers and filaments while pistil consists of stigma, style and ovary.
- Anther consisting of four microsporangia (tetrasporangiate) is called **dithecous**. Anthers are reniform or kidney-shaped and consisting of two microsporangia (bisporangiate) is called as **monothecous**.
- In Cruciferae, stamens are six (2 outer shorts and 4 inner longs). This condition is called **tetradynamous**.
- **Monoadelphous stamens** are found in members of Malvaceae.
- **Axile placentation** is found in members of Malvaceae, Liliaceae and Solanaceae.
- **Basal placentation** is present in members of Asteraceae.

Types of Flower

On the basis of position of ovary in flowers, these can be of different types such as:

(*i*) When the ovary is at the top of thalamus, flowers are **hypogynous** and ovary is superior, e.g. *Brassica, Citrus.*

(*ii*) When calyx and corolla arise from the upper side of ovary and surrounded by thalamus, ovary is inferior and flower is **epigynous**, e.g. *Cucurbita,* Sunflower.

(*iii*) When ovary is half superior and half inferior then it is called as **perigynous**, e.g. rose, plum, etc.

The flowers, in which both male and female reproductive organs are present and called as bisexual flowers, whereas the flowers having either male or female reproductive organ are called as unisexual flowers.

On the basis of the presence of male and female flowers, plants are of following types:

(*i*) The plants, in which both male and female flowers are present are called as **monoecious plants**, e.g. *Cocos, Ricinus, Zea, Colocasia, Acalypha*.

(*ii*) The plants, in which either male or female flowers are found, are called as **dioecious plants**, e.g. mulberry, papaya.

(*iii*) When unisexual (male or female), bisexual flowers are present on the same plant the plant is called **polygamous plants**, e.g. *Polygonum,* mango.

Largest flower is *Rafflesia* and smallest is *Wolffia*.

Inflorescence

The arrangement of flowers and mode of distribution of flowers on the shoot system of a plant is called **inflorescence**.

Types of Inflorescence

(*i*) **Racemose** (Indefinite) Main axis of inflorescence does not end in a flower, but continues to grow. The development of flowers is acropetal. The opening of flowers is centripetal.

(*ii*) **Cymose** (Definite) Main axis ends in a flower. The development of flowers is basipetal and opening of flowers is centrifugal.

(*iii*) **Special type capitulum** It is characteristic type of inflorescence in members of Compositae (Asteraceae).

Fruits

A fruit is a part of a flowering plant that is derived from specific tissues of the flower. The ovary of the flower develops into the fruit. It is the characteristic feature of flowering plant. The term 'fruit' refers to the mature or ripened ovary that contains seeds.

Parthenocarpy is the formation of fruits without fertilisation, such fruits are **seedless**.

Types of Fruits

Fruits can be broadly classified into following types:

(*i*) **Simple fruits** These are the fruits develop from the single simple or compound ovary of a flower. These can be dry (pericarp dry) or fleshy fruits.

(*ii*) **Aggregate fruits** They are the group of fruitlets, which develops from a flower having polycarpellary free gynoecium, also called **etaerio**.

(*iii*) **Multiple** (Composite) **fruits** These are composite fruits, which develop from an entire inflorescence.

Types and Edible Parts of some Common Fruits

Common name	*Botanical name*	*Type*	*Edible parts*
Simple Fruits			
Mango	*Mangifera indica*	Drupe	Fleshy mesocarp
Coconut	*Cocos nucifera*	Drupe	Endosperm
Almond	*Prunus amygdalus*	Drupe	Seeds
Walnut	*Juglans regia*	Drupe	Cotyledons
Apple	*Pyrus malus*	Pome	Fleshy thalamus
Pear	*Pyrus communis*	Pome	Fleshy thalamus
Betel nut	*Areca catechu*	Berry	Seeds
Pea	*Pisum sativum*	Legume	Seeds
Cashew nut	*Anacardium occidentale*	Nut	Cotyledons and fleshy thalamus
Litchi	*Litchi chinensis*	Nut	Aril
Water chestnut	*Trapa bispinosa*	Nut	Seeds
Pomegranate	*Punica granatrum*	Balausta	Succulent testa
Bengal quince (bael)	*Aegle marmelos*	Amphisarca	Inner fleshy layer of pericarp and placenta
Grape	*Vitis vinifera*	Berry	Pericarp and placenta
Papaya	*Carica papaya*	Berry	Mesocarp
Tomato	*Solanum lycopersicum*	Berry	Mesocarp
Banana	*Musa paradisicica sp.*	Berry	Pericarp and placenta
Watermelon	*Citrullus lanatus*	Pepo	Mesocarp and endocarp
Lemon	*Citrus limon*	Hesperidium	Juicy placenta and juicy hairs developed from endocarp
Wheat	*Triticum aestivum*	Caryopsis	Starchy endosperm
Aggrate Fruits			
Lotus	*Nelumbo nucifera*	Etaerio of achenes	Fleshy thalamus and seeds
Strawberry	*Fragaria vesca*	Etaerio of achenes	Fleshy thalamus and seeds
Custard apple	*Anona squamosa*	Etaerio of berries	Inner layer of pericarp and thalamus
Multiple or Composite Fruits			
Mulberry	*Morus alba and M. nigra*	Sorosis	Succulent perianth and fleshy axis
Pineapple	*Ananas comosus*	Sorosis	Fleshy axis, bracts, fused peria and pericarp
Jack fruit	*Artocarpus heterophyllus*	Sorosis	Fleshy axis, bracts, perianth and seeds
Fig	*Ficus carica*	Syconus	Fleshy receptacle or thalamus

Seeds

A seed is an embryonic plant enclosed in a protective outer covering called **seed coat**, usually with some stored food. The ovary of the flower contains the ovules. The fertilised ovules develop into seeds. The study of seed is called **Spermology**. The seeds are of mainly two types:

(*i*) Non-endospermic or exalbuminous.

(*ii*) Endospermic or albuminous.

- **Vivipary** When the seeds of a plant germinate before they detach from the parent, e.g. mangroves.
- **Parthenogenesis** When an unfertilised egg develops into embryo (false embryo), the process is called **parthenogenesis**.
- **Parthenogamy** When sexual reproduction fails, the gametes directly behave as spores, these are called **parthenospores** or **zygospores**. The **phenomenon** is called parthenogamy.
- **Polyembryony** The occurrence of more than one embryo in the seed is called **polyembryony**.

Plant Anatomy

The branch of botany dealing with the internal organisation of plants is called **anatomy**. **N Grew** (1682), 'Father of Plant Anatomy', gave the terms tissue and parenchyma. **Nageli** (1850) coined the terms meristem, xylem and phloem. A group of similar or dissimilar cells that perform a common function and have a common origin is called **tissue**. The tissues have been classified into two groups:

(*i*) **Meristematic tissues** constitute an undifferentiated mass of cells, that are in a continuous state of division or retain their power of division.
  - These tissues divide to form new cells, which differentiate to give rise to permanent tissues.
  - Procambium is the meristematic tissue, which forms primary xylem, phloem and vascular cambium.

(*ii*) **Permanent tissues** do not have the power of division. They include protective tissue (e.g. epidermis), supporting tissue (e.g. parenchyma, collenchyma, sclerenchyma) and conducting tissue (e.g. xylem, phloem).

- Phloem, consists of three types of cells are called **sieve element**, companion cells and phloem parenchyma.
  - Xylem consists of tracheids, vessels, xylem parenchyma and xylem fibres.

➡ **Note** *Marchantia* belongs to Bryophyta and conducting tissues, i.e. xylem and phloem are absent in them for water absorption.

- In monocot leaf, mesophyll is undifferentiated and made up of only spongy parenchyma.
- In bicollateral vascular bundles, xylem is sandwiched between external and internal phloem.
- Parenchyma is a simple permanent tissue. When cells of this tissue have large intercellular spaces they are called as **aerenchyma**, e.g. hydrophytes.

PLANT PHYSIOLOGY

Plant physiology is concerned with the functioning of plants. It is the science, which is connected to the material and energy exchange, growth and development as well as movement of plant.

Plant Nutrients

The minerals required for proper growth and development plants are called as **nutrients**. There are two types of nutrients:

(*i*) **Macronutrients** Minerals needed in large amount, e.g. C, H, O Ca, P, Mg, S, K, N.

(*ii*) **Micronutrients** Minerals needed by plants in traces, e.g. Fe, Zn, Cl, Cu, Mo, B, Mn, Ni.

- C, O, H and N are non-mineral elements that are abundent in plants.

Imbibition

It is the phenomenon of absorption of water or liquid by the solid particles of a substances without forming a solution.

- The solid particles, which absorb water are called **imbibants**. These are hydrophillic colloids.
- The liquid, which is imbibed is called **imbibate**. The pressure that an imbibant develops after being imbibed is called **imbibition pressure/matric potential** (ψ_m).
- Increase in the pressure will increase the imbibition.

Diffusion

It is the net movement of particles from a region of its higher concentration to a region of its lower concentration due to their own kinetic energy.

- Rate of diffusion $\propto$ temperature $\propto \dfrac{1}{\textit{Size or mass of particle}}$
- **Diffusion Pressure Deficit** (DPD) The difference between diffusion pressure of the solution and its solvent at a particular temperature and atmospheric conditions, is called DPD.

Permeability

Permeability is the ability of a membrane to permits the movement of molecules across it.

Types of membrane on the basis of permeability are:

1. **Permeable membrane** Allows the diffusion of both the solvent and solute molecules across it.
2. **Impermeable membrane** Neither the solute nor solvent molecules can diffuse across membrane.

3. **Semipermeable membrane** Allows movement of solvent molecules across it, but not the solute molecules.
4. **Selectively permeable membrane** Membrane allowing diffusion of only selected molecules across it.

Osmosis

It is the movement of solvent (water) molecules from a region of lower concentration to a region of higher concentration across a semipermeable membrane. It was discovered by **Nollet** in 1748.

Types of Osmosis

Osmosis are of following types:

(*i*) When a cell is placed in hypotonic (less concentrated) solution water enters into the cell. This is called as **endosmosis**.

(*ii*) When a cell is placed in hypertonic (high concentrated) solution water exits the cell. It is called as **exosmosis**.

- Isotonic solutions are the solutions of same concentration. No net movement of molecules occurs between isotonic solution.
- The actual pressure that develops in a solution, when it is separated from pure water by means of a semipermeable membrane is called **osmotic pressure**.
- Osmotic pressure of a plant cell can vary from 4-5 atm.

Turgor Pressure

As a result of osmosis, water enters into the cell sap. Due to this a pressure is developed by cell sap against cell well making the cell turgid. This pressure is called **turgor pressure/hydrostatic pressure/pressure gradiant** (ψ_p).

$$\psi_p = \psi_s$$

Where, ψ_s = Concentration gradiant.

Plasmolysis and Deplasmolysis

The phenomenon of shrinkage of protoplasm away from the cell wall under the influence of hypertonic solution is called **plasmolysis**. **Deplasmolysis** occurs enters into the cell sap, due to this the cell becomes turgid resulting into protoplasm that assumes its normal shape.

TRANSPIRATION

The loss of water vapour from the living tissues of aerial parts of plant into the atmosphere is referred to as **transpiration**.

- The transpiration occurs from the leaves all the time and water released from leaves absorbs heat from the surrounding atmosphere to vapourise (similar to evaporation) due to which, area becomes cool nearby.
- It is also the ultimate cause of water movement in a plant stem against the gravity.

Types of Transpiration

Transpiration are of following types:

(*i*) **Stomatal transpiration** Stomata is a tiny pore that is used for gas exchange. Air containing CO_2 and O_2 enters the plant through these openings and utilised by the plant. It accounts for 80-90% of total water loss from the plant.

(*ii*) **Cuticular transpiration** Cuticle is the relatively impermeable covering of plant. If it is thin and green then upto 10-20% transpiration take place through it.

(*iii*) **Lenticular transpiration** Transpiration through lenticels, i.e. an airy aggregation of cells within the structural surfaces of stem, roots and other parts of vascular plants. Very little transpiration occurs through lenticels.

Factors Affecting Rate of Transpiration

Two factors affect rate of transpiration, which are as follow:

External Factors

- **Temperature** Transpiration increases with increase in atmospheric temperature.
- **Light** Rate of transpiration, i.e. rate of transpiration increases in light and decreases in dark.
- **Wind velocity** Upto 20-30 km/hr the rate of transpiration increases, while wind velocity of 40-50 km/hr decreases transpiration.
- **Humidity** The rate of transpiration is inversely proportional to the relative humidity.

  $$\text{Transpiration} \propto \frac{1}{\text{Humidity}}$$

Internal Factors

- **Area of leaf** Rate of transpiration is directly proportional to the area of leaf. Transpiration $\propto$ Area of leaf
- **Leaf structure** Some modifications of leaf structure such as thick cuticle on surface of plant parts, wax, resin and suberin coating on plants, sunken stomata in leaves contributes to reduction in transpiration.

PHOTOSYNTHESIS

Photosynthesis (Gk. *photon*–light; *synthesis*–putting together) is the anabolic process by which green plants synthesise complex carbohydrates from simple substances like carbon dioxide and water with the help of light energy.

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2$$

Site of Photosynthesis

Photosynthesis takes place in chloroplasts. These are the green plastids, which occur in green parts of plants.

- Maximum number of chloroplasts are present in leaves.
- It is a double membranous organelle enclosing a liquid matrix called **stroma**.
- In the stroma, tiny, membranous sac-like lamellae present. These are called as **thylakoids**.
- Thylakoids are stacked over one another to form **grana**.
- Stroma is the site of dark reaction and thylakoids for light reaction.

Photosynthetic Pigments

Pigments are the organic molecules that absorb the light of specific wavelength in the visible region. Different types of pigments are:

Chlorophylls (Gk. *Chlor*–green, *phyll*–leaf)

These are the green photosynthetic pigments present in all photosynthetic autotrophic organisms. There are about ten types of chlorophylls-*a*, *b*, *c*, *d* and *e*, bacteria chlorophyll-*a*, *b*, *c* and *d* and bacteriovirdin. Chl-*a* is found in all the oxygen evolving organisms. It is the **primary photosynthetic pigment**.

Structure of Chlorophyll

The structure of chlorophyll consists of porphyrin head and phytol tail.

- The porphyrin head is made up of four pyrole rings linked together by methane groups, forming a ring system.
- In the centre of tetrapyrole ring a divalent ion Mg^{2+} is present, which is complexed with the nitrogen atoms of four pyrole rings.
- The phytol tail is made up of 20 carbon alcohol and bound by ester linkage to the 4th pyrole ring.

Carotenoids

These are accessory pigments that help in photosynthesis in plants. Two major classes of carotenoids are carotenes (contain no oxygen) and xanthophylls (contain oxygen atom in their terminal ring).

Phycobilins

These pigments are found in red algae and blue-green algae, but not in higher plants. The phycoerythrin and phycocyanin are the two main phycobilins found in red algae and blue-green algae, respectively.

Mechanism of Photosynthesis

Photosynthesis is a two step process:

- The **first step** is dependent on light and responsible for accumulation of assimilatory power. It is called as **light reaction**.
- The **second step** is called **dark reaction** as light is not required for the purpose. It is responsible for CO_2-fixation into carbohydrate.

> - If vaseline is applied on both the surfaces of the leaf of a plant photosynthesis, transpiration and respiration in plant will get affected as all these processes require the direct contact between the leaf and atmosphere.
> - Mushroom (white fungi) plant is not capable of manufacturing its own food.

RESPIRATION

It is the process of exchange of respiratory gases (O_2 and CO_2) between an organism and its environment. Respiration is the most important catabolic process, which involves the liberation of energy by oxidation and breakdown of food inside living cell.

On the basis of O_2, respiration is of two types:

(*i*) **Aerobic respiration** The process that leads to a complete oxidation of organic food in the presence of O_2 releasing CO_2, H_2O and a large amount of energy.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + 686 \text{ kcal}$$

(*ii*) **Anaerobic respiration** Organic food is broken down incompletely to release energy in the absence of oxygen and the products are CO_2, C_2H_5OH and $C_3H_6O_3$ (lactic acid).

Glycolysis

The breakdown of a glucose molecule into two molecules of pyruvic acid is called **glycolysis**. Net yield of glycolysis is 6 ATP molecules.

TCA or Kreb's Cycle

TCA cycle starts with the condensation of acetyl group with Oxaloacetic Acid (OAA) and water to yield citric acid.

$$\text{Pyruvic acid} + 4NAD^+ + FAD^+ + 2H_2O + ADP + Pi \xleftrightarrow{\text{Mitochondrial matrix}} 3CO_2 + 4NADH + 4H^+ + FADH_2 + ATP$$

$$\textbf{Respiration Quotient } (RQ) = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ absorbed}}$$

PLANT GROWTH

All living organisms show various changes in their weight, shape, size and volume during their entire life cycle (birth to death). The process of increase in the size, weight and shape is known as **growth**. The growth of plants is regulated by certain chemical substances, which are synthesised by them and these are called **growth hormones** or **growth regulators**. Plant growth regulators are also called as **phytohormones**.

Plant growth regulators are as follow:

Auxins

- Auxin was initially isolated from the urine of human, but later on its presence was also found in plants.
- They promote cell elongation, cell division and cell enlargement.
- IAA is natural auxin, while IBA, NAA and 2, 4-D are synthetic auxins. Auxins can inhibit lateral bud formation.

Gibberellins

- Gibberellins are weakly acidic growth hormone.
- It causes cell elongation and increases length and is produced in embryos, roots and young leaves near the shoot tip.
- It is helpful in flowering, enzyme synthesis and fruit growth.

Cytokinins

- Natural cytokinins are known to be synthesised in the regions, where rapid cell division takes place, e.g. root apex, young fruits, etc.
- They promote cytokinesis (cell division).
- Kinetin was first isolated from degraded sample of DNA.
- Zeatin was isolated from maize endosperm.
- It is also responsible for cell division, cell enlargement, prevention of senescence and enzyme synthesis.

Ethylene

- It is a gaseous hormone, which is produced from the ripening fruits and mainly acts as growth inhibitor.
- It fastens ripening of fruits and promotes ageing of plant organs.
- It promotes senescence and break dormancy.

Abscisic Acid (ABA)

- It is a growth inhibitor as it counteract other hormones.
- It is responsible for dormancy in buds and seeds, ageing in leaves, it inhibits mitosis, abscission of leaves, flowers and fruits.
- It causes partial closure of stomata and therefore, called **anti transpirant**.

REPRODUCTION

It is a biological process, in which organisms produce young ones (offspring) similar to themselves.
Reproduction is of two types:

(*i*) **Asexual reproduction** When offsprings are produced by a single parent without the involvement of gametic fusion, e.g. unicellular (monerans and protists), many plants and simple animals.
Types of asexual reproduction are fission, budding, fragmentation, regeneration, vegatative propagation (potato), etc.

(*ii*) **Sexual reproduction** It is the process of development of a new individual and the formation and fusion of two gametes, e.g. all types of plants and animals.

- Flowers are the sit of sexual reproduction in angiosperms.
- Androecium consisting of stamens represents the male reproductive parts. Gynoecium consisting of pistils represents the female reproductive parts.
- Pollen grains (male reproductive organ) develop inside the microsporangia (anther).
- The pistil has three parts–**stigma**, **style** and **ovary**. Ovules are present in the ovary.
- Pollination is the mechanism to transfer pollen grains from the anther to the stigma.
- A cell of the archesporium, the megaspore mother cell divides meiotically and one of the megaspore forms the embryo sac (the female gametophyte).
- Angiosperms exhibit double fertilisation.
- The mature embryo sac is 7-celled and 8-nucleated.
- The developing embryo through different stages develops into seed.

IMPORTANT POINTS

- The study of structure and development of embryo is called Embryology. The great embryologist of India is **P Maheshwari**.
- **Tissue culture** It is the technique to exploit the property of totipotency of the plant cells.
- Rose, *Bryophyllum* and marigold can be propagated by cutting.

ECONOMIC BOTANY

- Natural rubber is para rubber, obtained from *Hevea brasiliensis*.
- Seeds of groundnut have 23-30% proteins.
- Starch is a polymer of glucose used as a thickening agent or emulsifier in pharmaceutical or detergents.
- Coir is a natural fibre obtained from fibrous mesocarp of coconut. It is used in products such as floormats, doormat’s, etc.
- Sunflower is cultivated for oil and ornamental flowers.
- Red and black seeds of *Abrus* (Ratti) are used as jeweller’s weight.
- *Eucalyptus* grows very fast and its stem is used in paper and pulp industry.
- Chief source of sugar in India is shoot of sugarcane (*Saccharum officinarum*).
- Coffee is obtained from seeds of *Coffea arabica* (family–Rubiaceae).

- Sunnhemp is the plant used for green manuring in India.
- Botanical name of tea is *Thea sinensis* (family–Theaceae).
- Most important cereals are rice, wheat, maize, etc.
- Banana, mango and *Citrus* are indigenous to India.
- Pungent smell in garlic is due to allicin compound.
- 'Black gold of India' is pepper.
- Cutting and peeling of onions bring tears to eyes because onion acids combine with sulphur to form amino acid sulphoxides.
- Quinine is a natural white crystalline alkaloid used in the treatment of malaria. It has antipyretic (fever reducing), antimalarial, analgesic and anti-inflammatory properties and found naturally in the bark of *Cinchona* tree.
- Bagasse is the fibrous matter remains after sugarcane stacks are crushed to extract their juice. It is used as a biofuel and as a substitute for wood in many subtropical anatropical countries for the production of pulp, paper, board, etc.
- Cereals are rich in carbohydrates and belong to family–Gramineae.

Common Medicinal Plants

Common name	*Botanical name*	*Part used*
Aconite	*Aconitum napellus*	Tuberous roots
Belladona	*Atropa belladona*	Leaves
Sarpgandha	*Rauwolfia serpentina*	Roots
Quinine	*Cinchona officinale*	Bark
Opium	*Papaver somniferum*	Fruits
Ashwagandha	*Withania somnifera*	Roots
Isubgol	*Plantago ovata*	Fruits and seeds
Ephedrine	*Ephedra*	Bark

ANIMAL PHYSIOLOGY

Animal physiology is the biological study of the functions of animals and their parts, including all physical and chemical processes. Physiology basically focuses principally at the level of organs and organ systems. Human physiology studies how our cells, muscles, organs work together and how they interact.

It involves growth and development, absorption and processing of nutrients, synthesis and distribution of nutritive substances and functioning of different tissues, organs and other anatomical structures.

NUTRITION

It is the sum of all those activities, which are concerned with ingestion, digestion and absorption of digested food into blood or lymph followed by egestion.

There are mainly two types of nutrition:

1. **Autotrophic** or **Holophytic nutrition** Preparation of organic food from inorganic material by organism itself, e.g. all green plants, some unicellular organisms, e.g. like *Euglena*, *Chlamydomonas* and *Volvox* are autotrophs.
2. **Heterotrophic nutrition** Taking the readymade organic food material synthesised by autotrophs is termed as heterotrophic nutrition, e.g. animals, parasites and saprophytes are heterotrophs.

   Food or **Diet** is the nutritive substances taken by an organism for growth, work, repair and maintaining life processes. It is a mixture of various components, in which the quality and quantity of components may vary.

The major seven components of food are as follow:

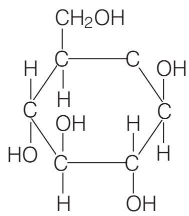
(*i*) Carbohydrates (*ii*) Lipids
(*iii*) Proteins (*iv*) Minerals
(*v*) Vitamins (*vi*) Water
(*vii*) Roughage

Carbohydrates

Carbohydrates are polyhydroxy aldehyde or ketones, usually composed of C, H and O in the ratio of 1 : 2 : 1 with some exceptions. General formula of carbohydrates is $C_n(H_2O)_n$ (n = number of carbon atoms).

1g of carbohydrate = 17 kJ of energy.

Carbohydrates release energy more quickly as compared to fats.


Structure of Carbohydrate (glucose)

Sources of Carbohydrates

Main sources of carbohydrates include potatoes, fruits, cereals (rice, wheat, maize), sugar, honey, bread, milk, etc.

Categories of Carbohydrates

On the basis of number of sugar units present, carbohydrates are of mainly following three types:

Monosaccharides

Definition	*Examples*	*Functions*
These are the simple carbohydrates, which are made up of only one sugar molecule.	Glucose, fructose, ribose, etc.	Glucose is the blood sugar, ribose, (5-carbon monosaccharide), forms the backbone of genetic molecule RNA.

Disaccharides

Definition	*Examples*	*Functions*
These are formed of two or more unit of monosaccharides inter-linked by glycosidic bonds.	Sucrose (sugarcane), maltose, lactose, etc.	Sucrose is the table sugar and used in preparation of desserts and can also act as a food preservative.

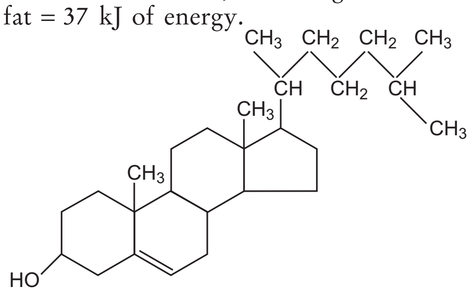
Polysaccharides

Definition	*Examples*	*Functions*
These are formed of large number (9<) of monosaccharides inter-linked by glycosidic bond.	Starch, glycogen, chitin, cellulose, hyaluronic acid, etc.	Polysaccharides serve for the storage of energy (e.g. starch and glycogen) and as a structural component (e.g. cellulose in plants and chitin in animals).

Lipids

Lipids are esters of long chain fatty acids and an alcohol called **glycerol**. They are insoluble in water, but soluble in non-polar solvents, e.g. chloroform, benzene or ether.

- Hydrolysis of fats occurs by a pancreatic enzyme called **amylase**.
- Animal fats are semi-solid, while vegetable fats are oils.
- 2 g of fat = 37 kJ of energy.



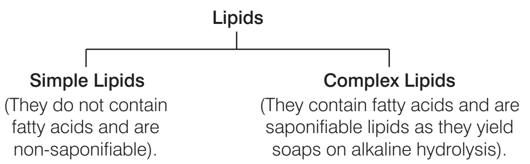
Structure of lipid

Sources of Lipids

Fats are supplied to our body by many food sources such as butter, ghee, cheese, milk, egg-yolk, meat, nuts, etc.

Categories of Lipids

On the basis of the presence or absence of fatty acids, lipids can be divided into two types :



Excessive intake of fat can lead to obesity and high cholesterol level gives rise to **hypercholesterolemia** resulting in cardiac damage. Fat deficiency results in drying of skin and deficiency of fat soluble vitamins. Fats are also required in cell membrane formation.

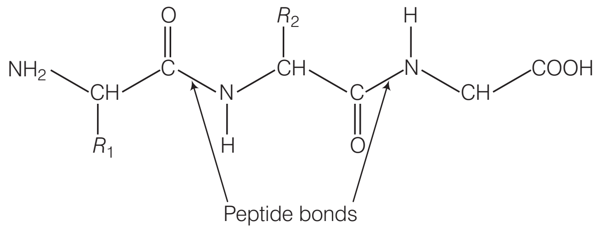
Sterols are also organic compounds. The most common sterol in animals is cholesterol.

The sterols found in plants are called as phytosterols, e.g. stigmasterol, sitosterol, etc.

Proteins

Proteins are the biochemical compounds consisting of one or more polypeptides folded in a typical way. Polypeptide is a single linear polymer chain of amino acids bonded together by peptide bonds.

- Proteins are composed of carbon, hydrogen, oxygen and nitrogen.
- Some proteins also contain sulphur, phosphorus and iron.
- There are 20 essential amino acids.
- The term proteins was given by J Berzelius (1938).



Structure showing peptide bonds between amino acids

Sources of Proteins

Main sources of proteins are cereals, pulses, fishes, eggs, milk, leafy vegetables, fruits, soyabean, peas, beans, cheese, curd, etc.

- Deficiency of proteins causes two diseases in children, i.e. **Marasmus** and **Kwashiorkor**.
- Silk is a protein fibre secreted by pupa of silk insect.

Some Essential Proteins

Body proteins	*Functions*
Enzymes	These are the biocatalysts, which help in biochemical reactions. All enzymes are proteinaceous in nature.
Hormones	These are non-virulent chemicals, which regulate body functions.
Transport proteins	Haemoglobin transports oxygen throughout the body.
Structural proteins	These help in the formation of cells and tissues.
Protective proteins	Antibodies help to fight against infection.
Contractile proteins	These proteins are responsible for contraction of muscles and movements, e.g. myosin, actin, etc.

Proteins are also found in the body of living organisms as enzymes. Thus, all enzymes are proteins only, but all proteins are not enzymes.

Minerals (Inorganic Salts)

The metals, non-metals and their salts are called **minerals**. They form about 4% of our body weight. Minerals are generally of two types:

(a) **Macronutrients** (needed by body in large amount), e.g. calcium (Ca), phosphorus (P), potassium (K), sulphur (S), sodium (Na), chlorine (Cl) and magnesium (Mg).

(b) **Micronutrients** (needed by body in small amounts), e.g. iodine (I), iron (Fe), zinc (Zn), manganese (Mn), copper (Cu), cobalt (Co), fluorine (F), molybdenum (Mo) and selenium (Se).

Sources of Minerals

These are obtained from the food and drink consumed by animals. These are also present in milk, eggs, meat, fruits, vegetables and tables salts.

Sources and Functions of Different Minerals and their Associated Deficiency Diseases

Minerals	*Daily requirement*	*Sources*	*Functions*	*Deficiency diseases*
Sodium	Adequate amount in ordinary diet	Table salt, vegetables	Essential for nerve impulse conduction	Nervous depression, improper muscle contraction
Potassium	Adequate amount in ordinary diet	Vegetables	Required for muscle contraction and nerve impulse conduction	Nervous disorder, poor muscles leading to paralysis
Iron	18 mg	Leafy vegetables	Helps in oxygen transport as part of haemoglobin and myoglobin	Anaemia
Iodine	150 mg	Sea foods, water, onions	Essential for metabolic control as part of thyroid hormone thyroxine	Goitre
Calcium	1200 mg	Cereals, meat	Bone and teeth formation	Nervous disorder
Phosphorus	1200 mg	Meat, vegetables	Bone and teeth formation	Tetany and rickets
Magnesium	400 mg	Soyabean, green leafy vegetables	Activator of enzymes	Heart disease in infants
Zinc	15 mg	Beet, cheese	Cofactor of enzymes	Retardation of growth and loss of appetite
Manganese	5 mg	Peanuts	Cofactor of enzymes	Impairs glucose metabolism
Copper	Adequate amount in ordinary diet	Peanuts, beet	Cofactor of enzymes	Haematological and neurological disorder
Cobalt	Traces	Meat, milk	Part of vitamin-B_{12}	Leads to deficiency of vitamin- B_{12}
Fluorine	Traces	Water	Prevention of dental carries	Dental carries
Molybdenum	Traces	Water	Part of enzyme system	Neurological damage

Vitamins

Vitamins are the compounds essential for the metabolic activities of our body, but in very small quantities. These do not provide energy, but control energy yielding reactions of our body. The term 'vitamin' was coined by **Casimir Funk**.

On the basis of solubility, vitamins are of two types:

(a) **Fat soluble vitamin**, i.e. Vitamin-A, D, E and K.

(b) **Water soluble vitamin**, i.e. Vitamin-B complex and C.

Our body can synthesise vitamin-D and K.

Fat Soluble Vitamins

Fat in diet is essential otherwise these vitamins will not be absorbed.

Vitamin-A (Retinol)

- **Steenbock** (1919) discovered the vitamin-A and **Karrear** (1931) determined the structure of vitamin-A.
- It is also called as **antiinfective vitamin**.
- It is necessary for healthy eye sight (normal vision).
- It is destroyed by strong light.
- The main sources are yellow or green leafy vegetables (spinach), carrot, papaya, ripe mango, maize, milk, ghee, cod liver oil, etc.
- Deficiency causes night blindness (patient cannot see the object in dim light) and xerophthalmia or keratomalacia (dryness and wrinkleness of outer layer of eyeball).

Vitamin-D (Calciferol)

- It is called poorman's vitamin and is a sterol derivative.
- Its formation takes place under the skin in the presence of sunlight thats why also called as **sun-shine vitamin** or **antiricket vitamin**.
- It is needed for strong bones and teeth, helps in DNA synthesis, absorption of calcium and phosphorus.
- Some main sources are egg, milk, fish liver oil, etc.
- It affects the bones and causes rickets and osteomalacia in children and adults, respectively.

Vitamin-E (Tocopherol)

- It is also known as **beauty** or antisterility vitamin.
- It acts as an oxidant, helpful in making RBCs and is necessary for normal functioning of reproductive system in male and female both.
- The most important sources are green vegetable, wheat, vegetable oil and animal food.

- Its deficiency destroys the muscles and causes abnormal functioning of the reproductive system in males as well as in females.

Vitamin-K (Phylloquinone)

- It was discovered by **Henrik Dam** (1935).
- It is also called as **naphthoquinone** and is synthesised in body by some bacteria.
- It is a coagulation vitamin, that is why it helps in clotting of blood.
- The main sources are cauliflower, spinach, tomatoes, soyabean, etc.
- Its deficiency delays clotting of blood and causes haemorrhage that is why also called **antihaemorrhagic vitamin**.

Water Soluble Vitamins

These are the vitamins, which are soluble in water that is why excess of these vitamins are normally excreted through the urine.

Vitamin-B_1 (Thiamine)

- The name of vitamin-B_1 was given by **Funk** and discovered by **Eijkman**.
- It is destroyed on cooking and get dissolved in cooking water.
- It is helpful in normal functioning of nerve cells and metabolism.
- It also helps in maintaining normal appetite and digestion.
- Sources are yeast, rice, wheat, grains, beans, soyabean, liver oil, milk, etc.
- Its deficiency causes beri-beri (loss of appetite, paralysis of legs and head, etc).

Vitamin-B_2 (Riboflavin)

- It was first isolated by **Warsung** and **Christian** from animal tissue.
- These are destroyed on cooking and in strong sunlight. It helps in protein and fat metabolism.
- It is necessary for normal growth of body.
- The main sources of vitamins are milk, eggs, liver, green leafy vegetables, pulses, cheese, etc.
- Its deficiency causes inflammation of tongue, cornea, cracks appear at the corner of the mouth.

Vitamin-B_3
(Niacin or Nicotinic Acid)

- It is also known as antipellagra factor.
- It helps in oxidation of carbohydrates, proteins and fats.
- The main sources are cereals, liver, maize, fruits, milk, egg, meat, yeast, kidney, etc.
- Its deficiency causes pellagra, dermatitis, diarrhoea.

Vitamin-B_5 (Pantothenic Acid)

- It is a part of coenzyme-A required for cell respiration, normal nerves and skin.
- Its sources include yeast, peas, liver, eggs, kidney, etc.
- Its deficiency can leads to anaemia, dermatitis and nerve degeneration.

Vitamin-B_6 (Pyridoxine or Pyridoxal)

- It is an essential part of coenzymes for amino acid synthesis.
- Extreme excess of this vitamin leads to poor control and coordination.
- The main sources include meat, milk, liver, eggs, etc.
- Its deficiency can lead to disturbance of central nervous system and anaemia.

Vitamin-B_7 (Biotin)

- It is a sulphur containing substance also called **vitamin-H**.
- It serves as coenzyme for fatty acid synthesis.
- Sources include fruits, vegetables, liver, milk, eggs, etc.
- Its deficiency can lead to scaly and itchy skin, muscle pain and weakness.

Vitamin-B_9 (Folic Acid)

- It is also called **vitamin-M.**
- It was first isolated from the spinach leaves.
- It is essential for growth and maturation of red blood cells.
- This vitamin plays a vital role in various metabolic disorders. Some main sources are green leafy vegetables, yeast, banana, pulses, cauliflower, meat, liver, etc.
- Its deficiency causes macrocytic anaemia in man.

Vitamin-B_{12} (Cyanocobalamine)

- It is a dark coloured cobalt based pigment, also known as **cobamide cyanide**.
- It helps in RBC formation and proper functioning of nerve tissues.
- The main sources are liver, cheese, milk, meat, fish, eggs, kidney, etc.
- Its deficiency can lead to **pernicious anaemia**.
- It is not found in plants, however it has been considered that alga and *Spirulina* contain this vitamin.

Vitamin-C (Ascorbic Acid)

It is an earliest known vitamin, which is essential for keeping teeth, gums and body joints healthy.

- It prevents the formation of stones in gall-bladder.
- It is quickly and easily destroyed by heat and increases resistance of our body against infections, i.e. viral infections and common cold.
- The main sources are amla, *Citrus* fruits (lemon and orange), guava, tomatoes, peppers, etc.
- Its deficiency can lead to scurvy (bleeding and loosing of gums).

Water

It is an inorganic compound made up of H and O in the ratio of 2 : 1.

- Human body contains about 65% water.
- It regulates body temperature by sweating and evaporation.
- It helps in digestion, transportation and excreting body wastes.
- The main sources are metabolic water, liquid food and drinking water.
- Abnormal depletion of body fluid leads to dehydration (loss of water).

Roughage

It is fibrous material present in plants.

- It does not take part in growth as we cannot digest it.
- It adds bulk in food item, to prevent constipation and helps in retaining water in body and aids in proper working of digestive system.
- The main sources are salads, outer layers of grains cereals vegetables and porridge (dalia).

Balanced Diet

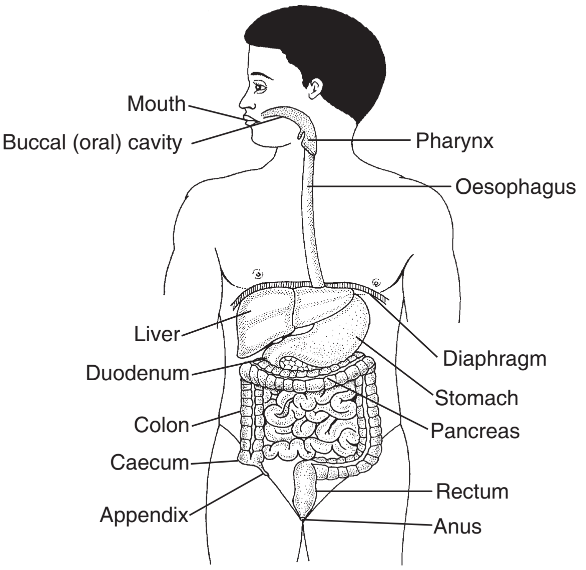
A diet Which contains all the essential nutrients in required quantity.

- It is related to state of one's age, health and occupation.
- Body requires carbohydrates, proteins and fats in the approximate ratio of 4 : 1 : 1 for good health.
- Growing child needs more protein than a grown up man. Person doing physical work (athlete, carpenter, rickshaw puller, etc.) need more carbohydrates and fats.
- A pregnant woman needs more proteins, calcium, potassium in her diet.
- Person recovering from illness need more proteins, minerals and vitamins.
- The eating of a particular type of diet due to the food habit leads to deficiency diseases.

DIGESTIVE SYSTEM

Parts of body concerned with digestion, absorption and assimilation of food followed by elimination of undigestible remains is called **digestive system**.

- Digestion involves splitting of food molecules by hydrolysis into smaller molecules that can be absorbed through the epithelium of the gastrointestinal tract.
- Man and other animals have holozoic nutrition, i.e. solid form of food.



Structure of digestive system

Dentition

The oral or buccal cavity of vertebrates contain masticatory organs called **teeth** or **dentes**. Mammal contains two sets of teeth. The first set is called **milk teeth** and second set is called **permanent teeth**.

- There are 32 permanent teeth in man.
- These are of following four type:
  (*i*) Incisors (for cutting) four in numbers.
  (*ii*) Canines (for tearing) two in numbers.
  (*iii*) Premolars (for grinding) four in numbers.
  (*iv*) Molars or wisdom teeth (for grinding) six in numbers.
- In elephants, the tusks are the incisors of upper jaw.
- Maximum number of teeth are present in **horse** and **pig**.
- Hardest part in the body is tooth enamel.
- Main bulk of tooth is formed of dentine.

Dental Formula of some Common Mammals

Mammal	*Dental formula*	*Total number of teeth*
Man (child)	2102/2102	20
Man (adult)	2123/2123	32
Horse	3143/3143	44
Dog	3142/3143	42
Cow and sheep	0033/3133	32
Cat	3121/3121	30
Rabbit	2033/1023	28
Mouse	1003/1003	16

Digestion

The food we intake is in the form of biomacromolecules and thus, cannot be utilised by our body. The complex food substances are converted into simpler absorbable form in the process called **digestion**.

Digestion takes place in following five steps:
(*i*) Ingestion of food (*ii*) Digestion of food
(*iii*) Absorption of digested food
(*iv*) Assimilation and egestion of unwanted food

Ingestion of Food

Ingestion takes place in buccal cavity. Food is taken through mouth cavity. Salivary glands lubricate the food and bind the food particles together to form bolus. Salivary glands have starch splitting enzyme called **ptyalin**. Food is masticated by teeth and swallowed.

Digestion of Food

Process of converting complex, insoluble food particles into simple soluble and absorbable form is called digestion. It takes place in various parts of digestive system.

Digestion in Mouth

Salivary glands secrete mucus to moist the food and to digest starch.
- In mouth, salivary amylase acts on starch.

$$\underset{\text{(Complex form)}}{\textit{Starch}} \xrightarrow{\textit{Ptyalin}} \underset{\text{(Simple form)}}{\textit{Maltose}}$$

- pH of saliva = 6-7.4

Digestion in Stomach

The food passes down through the oesophagus into stomach. Now, food is mixed with gastric juice and hydrochloric acid, which disinfect the food and create acidic medium.
- The pH of gastric juice is 4.5.
- Pepsin digests proteins and converts them into peptones and renin converts milk to curd. Digested food now is called **chyme**.
- The opening of stomach into small intestine is called **pylorus**. Alcohol is easily diffusible across the plasma membrane.

Digestion in Small Intestine

Now, chyme moves to duodenum. In small intestine, food receives three alkaline secretions–bile from liver, pancreatic juice from pancreas and intestinal juice from intestinal glands.
(*i*) **Bile** Food is mixed with bile (liver) to breakdown fats into smaller globules.
(*ii*) **Pancreatic Juice** It contains following three enzymes:
(a) **Trypsin** Which acts upon proteins and breaks them into peptides.
(b) **Amylase** Which converts starch into simple sugar.
(c) **Lipase** Which converts fats into fatty acids and glycerol.
(*iii*) **Intestinal Juice** Food passes into ileum and mixes with intestinal juice. It contains following three types of enzymes:
(a) **Maltase** Which converts maltose into glucose.
(b) **Lactase** Which converts lactose into glucose.
(c) **Sucrase** Which converts sucrose into glucose.
- Food is now called **chyle**.

Absorption and Assimilation of Digested Food

Ileum's internal surface has finger-like folds called **villi**. There is a dense network of blood capillaries and lymph capillaries in each villus. It helps in absorption of food.
- Absorption occurs by two types of processes, i.e. **passive** and **active absorption**.
- Passive absorption occurs down the concentration gradient and **active absorption** is independent of concentration gradient.
- Assimilation is followed thereafter, i.e. the absorbed food materials are incorporated into tissue cells.

Egestion of Unwanted Food

- After complete assimilation digested food passes into large intestine. Large intestine cannot absorb food, but absorbs much of the water. The remaining semi–solid waste is called **faeces** and is passed into rectum. It is expelled out through anus.
- Sometimes, the acids in stomach increases causing indigestion to get rid of the stomach aches milk of magnesia, a base is taken to neutralise the excess acid in stomach.

RESPIRATION

Respiration is an oxidising and energy liberating process, during which complex organic compounds are broken down into simpler substances. It is brought about by respiratory organs.

Respiratory organs are different in different animals:
(*i*) **Lungs** In man, frog, birds, lizards, camel and cattles, etc.
(*ii*) **Skin** In frog, earthworm.
(*iii*) **Gills** In fishes, prawn.
(*iv*) **Trachea** In insects.

General body surface In *Amoeba, Euglena, Spirogyra and Chlamydomonas*.

Types of Respiration

Respiration is of two types based on the presence or absence of O_2.

1. Aerobic Respiration

It is the process, in which respiratory substrates are oxidised into CO_2, water and energy, in the presence of oxygen. This is common in higher organisms.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + 2870 \text{ kJ}$$

One molecule of glucose is converted into two molecules of pyruvic acid, the process is known as **glycolysis**. It takes place in cytoplasm of cells.

- Pyruvic acid releases energy + CO_2 + H_2O (Krebs' cycle).
- 1 molecule of glucose = 36 ATPs in eukaryotic cell.
- 1 molecule of glucose = 38 ATPs in prokaryotic cell.

2. Anaerobic Respiration

It is the process, in which the respiratory substrates are incompletely oxidised into CO_2 and alcohol.

It occurs in the absence of oxygen. It is also called as **fermentation**.

$$C_6H_{12}O_6 \xrightarrow{\text{Yeast}} 2CO_2 + 2C_2H_5OH + 56 \text{ kcal}$$

It occurs in lower organisms such as bacteria and fungi and in higher organisms under the condition when O_2 is limiting.

Respiratory Quotient (RQ)

The ratio of volume of CO_2 evolved to volume of O_2 consumed in respiration is called RQ.

$$RQ = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}}$$

Respiratory substrate	*RQ*
Carbohydrates	1
Fats	0.7
Organic acids	4

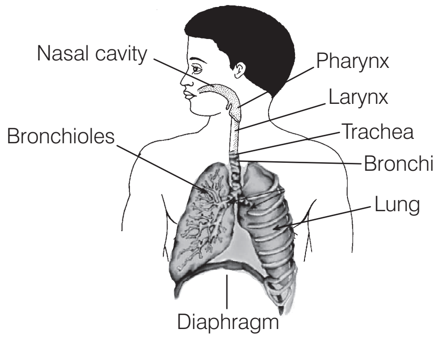
Respiration in Human

Respiration in humans takes place in two phases:

External Respiration (Breathing)

It comprises of inspiration and expiration of air.

(*i*) **Inspiration** This is the phenomenon, in which intake of air takes place. Muscles of the diaphragm contract and diaphragm flattens. The lower ribs are raised upward and outwards. The chest cavity enlarges, the air pressure in the lungs is decreased, air rushes into the lungs.



Human respiratory system

(*ii*) **Expiration** This is the phenomenon, in which air is exhaled out.

- Relaxation of muscles of the ribs and diaphragm takes place. Diaphragm again becomes dome-shaped.
- Chest cavity is reduced and air is forced outward through nose and trachea.
- Normal breathing rate in man is about 18-20 per min.

Internal Respiration

It is a complex process, in which food is broken down to release energy.

- Biochemical phase takes place inside the cell.
- Overall passage of air in human is as follows:
  - Nostrils → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli → Cells → Blood.

CIRCULATORY SYSTEM

Circulation of body fluids is also referred to as **circulatory system** or **internal transport**.

In higher and multicellular organisms, there is no direct supply of the useful materials and the removal of wastes from the body cells, so they need a transport system called **circulatory system**. Circulation of nutrients and hormones takes place in different directions.

- It transports nutrients, i.e. glucose, fatty acids, vitamins, etc., from the site of absorption to the different parts of the body.
- It transports nitrogenous wastes, oxygen, hormones, water, chemical substances, etc., from different parts of the body to organ tiussue/cells of direction.

Types of Circulatory System

Blood circulatory system is of two types:

1. Open Circulatory System

Blood may be present in blood vessels for sometime, but finally comes out from the blood vessels in open spaces called **sinuses**. It is found in leeches, among the annelids, cockroach, prawns, insects, spiders, starfish, etc.

- Blood flows with very slow velocity and at low pressure.
- In cockroach, blood circulation is completed in 5-6 minutes.
- Blood is in direct contact with the tissues.
- Respiratory pigment if present is dissolved in plasma.

2. Closed Circulatory System

Blood flow in closed vessels. It is found in earthworm, *Nereis*, molluscs and all vertebrates. Blood flows with high speed and at high pressure.

- Exchange of materials occurs through the tissue fluids.
- Blood does not come in direct contact with tissue cells. Respiratory pigment is present in RBCs.

Blood Vascular System

Blood vascular system consists of three components:

1. Blood Vessels

These are elastic muscular tubes that carry blood.

These are:

(*i*) **Arteries** These are the thick-walled blood vessels, which carry the blood away from the heart to various body parts.
They are deep seated in body and have no valves in them and carry oxygenated blood in them, except the pulmonary artery, which carries deoxygentated blood to the lungs. Blood flows at high pressure and high speed.

(*ii*) **Veins** These are thin-walled blood vessels, which carry blood from various body parts to the heart. Superficial in position and have valves in them to prevent back flow of blood in them, blood flows at low pressure and lower speed. They carry deoxygenated blood in them, except the pulmonary vein, which carries oxygenated blood to the heart.

(*iii*) **Capillaries** These are the thinnest blood vessels, which connect arteries to the veins. Each capillary is lined by a single layer of flat cells. These help in exchange of materials like the nutrients, gases, waste products, etc., between blood and cells.

2. Blood

Blood is a red vascular connective tissue. An average adult person has about 4-6L of blood. It forms 6-10% of body weight and 30-35% of extracellular fluid. Blood have specific gravity of 1.06. It is a viscous fluid due to the presence of formed elements.

It has a saltish taste and is mildly alkaline with a pH = 7.4. The alkaline nature of blood contributes to the oxygen carrying capacity of haemoglobin as in acidic pH this is decreased. It is bright red when oxygenated and purple when deoxygenated. The study of blood is called **haematology**. People at higher altitudes have more blood than those at lower altitudes. This extra blood supplies the additional oxygen to body cells. It is sticky and opaque fluid with a viscosity of 4.7.

The blood is formed of two parts:

(i) Plasma (ii) Blood corpuscles

(i) Plasma

Plasma is faint yellow-coloured non-living fluid present in blood. It is slightly alkaline and viscous. It constitutes about 55-60% of the blood volume. It is composed of following components in the table below:

Components of Plasma

Water	90-92%
Inorganic salt	1-2%
Plasma protein in colloidal state	6-7%
Other organic compounds	1-2%

Functions of Plasma

- It transports simple food components (glucose, amino acids, etc.) from intestine and liver to other body parts.
- It transports metabolic wastes like urea, uric acid, etc., from body tissues to the kidneys or their excretion.
- It transports hormones from endocrine glands to the target organs.
- It helps in keeping a constant pH of blood.
- Blood proteins and fibrinogen, present in plasma, help in blood clotting at the injury. It forms the tissue fluid, which keeps the tissue moist and helps in exchange of material in between the blood and cells.

(ii) Blood Corpuscles or Blood Cells

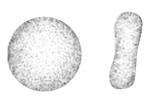
Blood contains certain floating bodies called **formed elements**. These floating elements include blood corpuscles or blood cells. These form 40-45% of blood.

- The percentage of blood cells is called **haematocrit value** of packed cell volume.
- They are of three types:
  (*i*) Red Blood Corpuscles (RBCs) or **Erythrocytes**
  (*ii*) White Blood Corpuscles (WBCs) or **Leucocytes**
  (*iii*) Blood platelets

Red Blood Corpuscles (RBCs)

Shape in all vertebrates except mammals, these are oval, biconvex, nucleated and non-nucleated.

- In mammals except camel and llama, the mature RBCs are circular, biconcave and non-nucleated.
- **Number** The RBCs are much more in number than WBCs. Normal RBC count is slightly lower in a woman (i.e. 4500000/cu mm) than a man (i.e. 5000000/ cu mm).
- The instrument used to determine RBC count is **haemocytometer**.
- Thrombocytes, also called **spindle cell**. These are spindle-shaped and found in blood of vertebrates other than mammals. These have oval nucleus and help in blood clotting in vertebrates.
- Physiological state and high altitudes increase the RBC count. RBC count decreases due to haemorrhage, haemolysis, etc., and is called **anaemia**. If RBC increases much more than the normal level, it is called **polycythemia**.
- **Lifespan** Average lifespan of human RBC is 120 days, while it is 100 days in frog and 50-70 days in rabbit.
- **Haemoglobin** (*Haem*–iron containing pigment, *globin* – protein). One RBC has about 280 haemoglobin molecules.
- **Structure** A red blood cell is bounded by elastic semipermeable membrane. It does not possess cell organelles. This feature enables it to hold more haemoglobin due to availability of space.



Structure of erythrocytes

- **Colour** RBC appears yellow when seen singly, but these appear red when in bulk due to the haemoglobin.
- **Formation** In the developing foetus, the formation of RBC takes place in liver and spleen, while after the birth, RBCs are mainly produced in the bone marrow.
- Maturation of RBC is controlled by folic acid and vitamin-B_{12}. The excess of RBCs are stored in the spleen, which acts as a blood bank of the body.

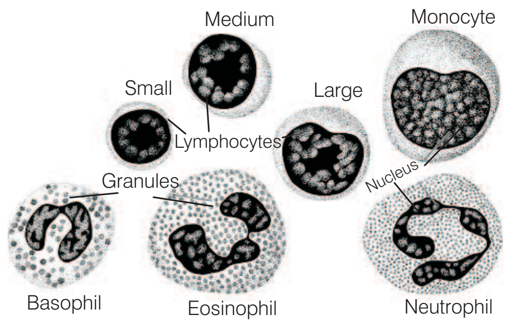
Functions of RBCs

- Haemoglobin helps in maintaining a constant pH.
- Haemoglobin also transports carbon dioxide from the tissues to the lungs and oxygen from the lungs to body tissues.

White Blood Corpuscles (WBCs)

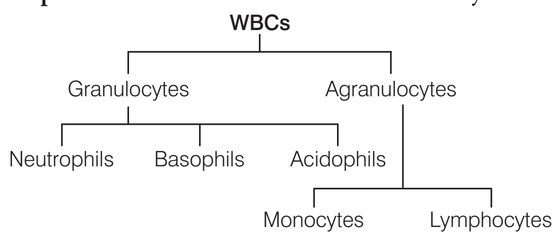
WBCs are also called **leucocytes** or **leukocytes**. These are the cells, which are important part of immune system and involved in protecting the body against both infectious disease and foreign invaders.

- **Shape** These are rounded or amoeboid (irregular), nucleated, non-pigmented cells. WBCs are larger in size than RBCs.
- **Number** WBCs are much less in number than RBCs (1:6).



Structure of WBC

- **Leukemia** (blood cancer) is a pathological increase in number of WBCs. Fall in WBC count is termed as **leucopenia**.
- **Colour** WBCs are colourless.
- **Formation** WBCs are formed in red bone marrow.
- **Lifespan** WBCs survive for 8-96 hours only.



Classification of WBCs

Granulocytes These are about 65% of total leucocytes. These are subdivided on the basis of shapes of their nuclei and the staining reactions of their granules.

(*i*) **Neutrophils**
- These are about 62% of total number of WBCs.
- They originate from bone marrow.
- The cytoplasm of these contains fine granules.
- Their nucleus is 3-5 lobed and lifespan is about 10-12 hours.
- These also help in sex differentation and act as soldiers of body.

(*ii*) **Basophils**
- These are also called **cyanophils**. Their nucleus is 2 or 3-lobed or S-shaped and lifespan is about 8-12 hours.
- These secrete heparin and histamine, thus help in local anticoagulation.

(*iii*) **Acidophils**
- These are also called **eosinophils**.
- Their lifespan is about 14 hours.
- These increase in number in allergic diseases.
- These help in healing of wounds.

Agranulocytes Formation of agranulocytes is also called **agranulopoiesis**. These form about 35% of the total WBCs. These are divided in two subtypes:

(*i*) **Monocytes**
- These are the largest sized leucocytes and form about 5.3% of all the leucocytes. The nucleus is oval, kidney or horse shoe-shaped and is excentric.
- These are formed in the lymph nodes and the spleen.
- These are highly motile and their lifespan is of 3-4 days.

(*ii*) **Lymphocytes**
- These form about 30% of leucocytes.
- Their nucleus is large and rounded, so that cytoplasm forms a thin peripheral layer. They are formed in bone marrow.
- These are formed in the thymus, lymph nodes, spleen, tonsils, etc.
- Their lifespan is of 3-4 days.
- These produce antibodies and opsonin, so play an important role in immune system.

Blood Platelets

- **Shape** These are oval-shaped and possess discoidal cytoplasmic fragments.
- **Number** Their number varies from 250000-5000000. A decrease in the number of platelets in the blood is called **thrombocytopenia**. Increase in number of blood platelets is called **thrombocytosis**.
- **Structure** Platelets are flat, non-nucleated large cells in the bone marrow. They are bounded by a membrane and contain few organelles in cytoplasm. These play an important role in blood clotting.
- **Colour** They are colourless, formed in red bone marrow.
- **Lifespan** about one week.

Blood Groups

Landsteiner recognised three kinds of blood groups 'A', 'B' and 'O'. Fourth and most rare 'AB' blood group was discovered by **Von Decastello** and **Sturle** (1902).

Landsteiner (1900) discovered two kinds of antigens called a and b. Antigens a and b are not proteins, but are mucopolysaccharides.

Characteristics of Human Blood Groups

Blood group	*Antigens in RBCs*	*Antibodies in plasma*	*Can donate to groups*	*Can receive from groups*
A	a	Anti-b	A, AB	O, A
B	b	Anti-a	B, AB	O, B
AB	a and b	None	AB	O, A, B, AB
O	None	Anti-a Anti-b	O, A, B, AB	O

- Persons with 'O' blood group are called as **universal donors** and AB are **universal recepient**.

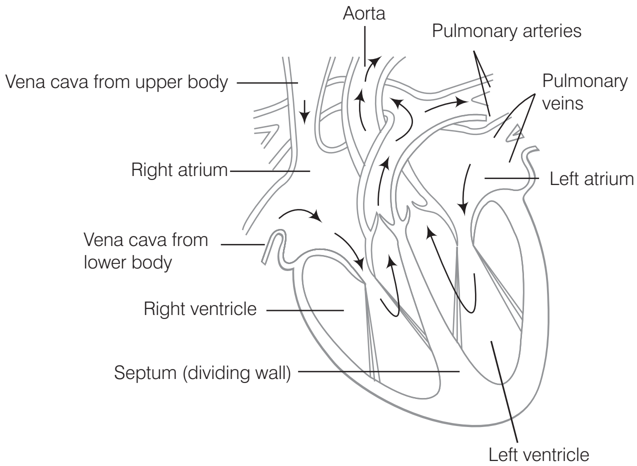
Rh-Factor

Rh-factor was first of all reported in RBCs of a monkey by **Landsteiner** and **Wiener** in 1940.

- Rh-factor is actually the Rh antigen present on the surface of red blood cells.
- Person having Rh-factor is called $Rh^+$and without Rh factor is Rh^-.
- For an example, if person of blood group A has Rh antigen on surface of RBC, then he is A^+/A positive.
- About 85% of world's people are Rh^+.
- Percentage of Rh^+ people in India is 97% of the population.
- Rh-factor is found in man and rhesus monkey, it has not been reported from other animals.

3. Heart

It is a muscular organ in both humans and other animals, which help in pumping blood *via* blood vessels of the circulatory system.



Internal Structure of Heart

- In amphibians, heart is three-chambered.
- Reptilian heart is structurally three-chambered, but is functionally four-chambered (i.e. incompletely four-chambered) except in crocodile.
- In crocodile, birds and mammals the heart is divided into four chambers (two auricles and two ventricles).
- Every person's heart is of about the same size as that of person's first. The human heart is four-chambered, two auricles and two ventricles.
- A thin, muscular wall called the interatrial septum separates the left and right atria, whereas thick-walled interventricular septum separates the right and left ventricles.
- The atrium and ventricle of same side are separated by atrioventricular septum.
- Opening between right atrium and right ventricle is tricuspid valve, formed of 3 cusps and opening between left atrium and left ventricle is called **bicuspid** or **mitral** valve.
- Pulmonary artery and aorta are provided with semilunar valves to prevent backflow of blood.
- Sino Atrial Node (SAN) is present in right upper corner of right atrium. Atrio Ventricular Node (AVN) is present in lower left corner of right atrium.

Cardiac Cycle

The sequence of events, which occur from the beginning of one heartbeat to the beginning of next is called **cardiac cycle**. Systole refers to contraction and diastole refers to relaxation. Single cardiac cycle involves four steps:

1. **Atrial Systole**

- Right atrium receives deoxygenated blood from superior and inferior vena cava.
- Left atrium receives oxygenated blood through pulmonary veins.
- The atria contract due to wave of contraction stimulated by SA node.

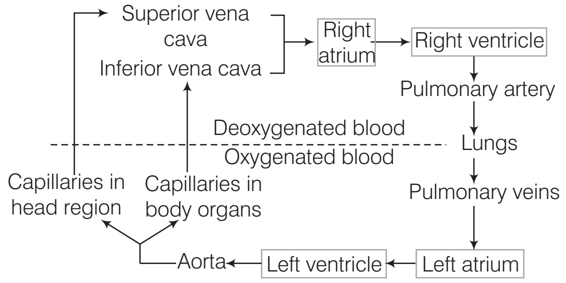
2. **Ventricular Systole**

- Action potential passes from SA node to AV node. The blood is then forced into ventricles leading to contraction of ventricles.
- At this time first heart sound is produced due to closure of valves.
- When the contraction is complete blood flows to pulmonary arteries carrying deoxygenated blood to lungs.

3. **Ventricular Diastole**

- Semilunar valves close to produce second heart sound.
- The ventricular relax. AV valve opens to allow blood to pass from atria to ventricles again.

Left ventricles supply oxygenated blood to aorta, which in turn supplies blood to various body parts.



Circulation of blood through the whole body

Blood Pressure (BP)

The pressure exerted by the blood on the walls of the blood vessels due to the repeated pumping of heart is called blood pressure. The rate of pulsation increases during excitement.

- Blood pressure is recorded as systolic/diastolic.
- Blood pressure in a normal adult person = 120/80 mm Hg.
- If a person has persistent high blood pressure then it is called **hypertension**. To avoid hypertension good cholesterol, i.e. high density lipoprotein should be consumed.

> **IMPORTANT POINTS TO REMEMBER**
> - Haemoglobin can readily combine with carbon monoxide.
> - Blood does not coagulate inside the body due anti-coagulant heparin.
> - The crew and passanger of aircraft usually suffer from pulmonary problems because of the presence nitrogen oxide in the atmosphere.
> - Coronary arteries supply blood to the right atrium.
> - Hepatic portal system and renal portal system supply and take blood from liver and kidney respectively.

- Hypotension is condition of low blood pressure.
- **ECG** or Electrocardiogram is a graphic record of the electric current produced by the excitation of cardiac muscles.
- The normal rate of heartbeat at rest is about 72-80 beats per minute.
- In a newly born baby, heartbeat rate is about 140 beats per minute.
- During heavy exercises, it may be high as 170-200 beats per minute.
- **Sphygmomanometer** measures BP.

EXCRETORY SYSTEM

The process of removal of wastes from the body is called **excretion**. The organs of excretion are called **excretory organs**.

Excretory Organs in Different Animals

- **Plasmalemma** Protozoans like *Amoeba*.
- **General body surface** Porifera (sponges), coelenterates (*Hydra*).
- **Flame cells** Platyhelminthes (*Taenia* and *Fasciola*).
- **Nephridia** Annelida.
- **Malpighian tubules** Arthropods (cockroach).
- **Coxal gland** Spiders.
- **Kidney** Main excretory organ in all vertebrates.
- **Antennal gland** or **green gland** Crustaceans crayfish.

Types of Excretion

Depending upon the form of nitrogenous waste excreted, there are three modes of excretion:

1. Ammonotelism

- It is the elimination of nitrogenous end product (waste) mainly as ammonia.
- Ammonotelic excretion is found in aquatic animals like protozoans, e.g. *Amoeba*, *Paramecium*, sponges (e.g. *Sycon*), coelenterates (e.g. *Hydra*), aquatic arthropods (e.g. *Prawn*), most aquatic molluscs (e.g. *Pila*), bony fishes (e.g. *Labeo*) and frog's tadpole.
- Those organisms, which secrete ammonia are called **ammonotelic animals**.

2. Ureotelism

- It is the process, in which main nitrogenous waste is urea.
- It is a common method of excretion in man, whales, seals, camels, kangaroo, toads, frogs, sharks, etc.

3. Uricotelism

- Main nitrogenous waste is crystals of uric acid.
- It is common in birds, land reptiles (snakes and lizards), insects, snails, etc.

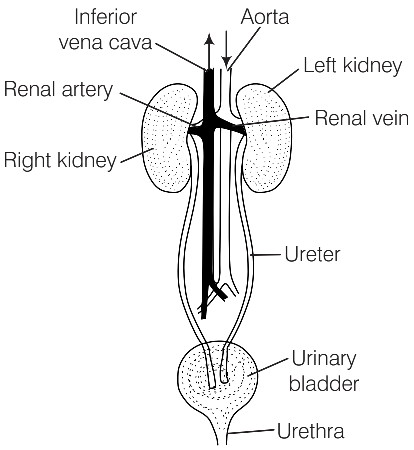
Excretory System of Human

Excretory system of human consists of the following parts:

(a) Kidneys (two)
(b) Ureter (two)
(c) Urinary bladder (one)
(d) Urethra (one)

1. Kidneys

- It is a bean-shaped, chocolate brown structure lying in the abdomen, one on each side of the vertebral column just below the diaphragm.
- These form the urine and control osmotic pressure within the organism with respect to its external environment.
- The left kidney is placed a little higher than the right kidney (but reverse in rabbit).
- Concavity of kidney called **hilum** or **hilus**, is always inward directed.



Structure of human excretory system

2. Ureters

- These are a pair of narrow tube arising from the hilum and is called as **ureter**.
- It is about 30 cm in length.
- They bring the urine downwards and open into urinary bladder.

3. Urinary Bladder

- It is a medium pear-shaped sac situated in the lower or pelvic region of the abdominal cavity within the body wall.
- Each ureter opens in urinary bladder.
- It temporarily stores urine.
- It is absent in birds. In both reptiles and birds, ureters and rectum open into a common sac called **cloaca**.

4. Urethra

- It is a muscular and tubular structure, which extends from neck of bladder to outside.
- In females, this tube is small and serves as a passage of urine only, while in males, it is long and functions as a common passage for urine and spermatic fluids.

Nephron

- It is the structural and functional unit of kidney.
- These are also called as **renal tubules** or **uriniferous tubules**.
- It is differentiated into four regions having different anatomical features and physiological role.
  (*i*) Bowman's capsule.
  (*ii*) Proximal Convoluted Tubule (PCT).
  (*iii*) Henle's loop U-shaped.
  (*iv*) Distal Convoluted Tubule (DCT).

Bowman's Capsule

- It is a double-walled cup and is lined by thin flat cells called **podocytes**.
- It contains group of capillaries called **glomerulus**.
- Glomerulus in the kidney acts as a dialysing bag.

Proximal Convoluted Tubule (PCT)

- It is a highly coiled (convoluted) tubular structure.
- It is about 12-24 mm in length.
- Almost whole of vitamins, glucose, amino acids, sodium and potassium, etc., is reabsorbed here, mainly by active transport.

Henle's Loop

- It is a U-shaped segment of nephron located in renal medulla.
- Loop of Henle is long in mammals and birds, which secrete hypertonic urine, but short in other vertebrates like reptiles, etc.

Distal Convoluted Tubule (DCT)

- It is similar to PCT region, i.e. greatly curved twisted.
- It is connected to the collecting duct.
- Active reabsorption of some Na^+ takes place here.
- It is impermeable to H_2O.
- Renal tubules along with large intestine accounts for most of the absorption of water in body.

Urine

Urine is formed after glomerular filtration, reabsorption and secretion. It is pale-yellow coloured transparent fluid due to the presence of urochrome pigment. It is acidic in nature and is slightly heavier than water.

- It has a faint aromatic odour due to urinod.
- Daily urine output in normal adult is 1.5-1.8 L.
- Chemical composition of urine, i.e. water is 95-96%, urea is 2% and some other substances like uric acid, creatinine, etc., are 2-3%.

Skeleton System

The human skeleton consists of both fused and individual bones supported and supplemented by ligaments, tendons, muscles and cartilage, ligaments and tendons are connective tissue made up of collagen fibres.

It is divided into two parts:

(i) Axial Skeleton (80 Bones)

- It includes skull, vertebral column and bones of chest.
- Vertebral column is responsible for the upright position of the human body.
- Most of the body weight is located in back of the spinal column.
- It provides flexibility to the neck and protection to spinal cord.
- The total number of bones of head is 29.
- The total number of bones in vertebral column is initially
  33 and after development is 26.
- The total number of bones of ribs is 24.

(ii) Appendicular Skeleton (126 Bones)

- It consists of pelvic and pectoral girdles.
- Girdles are the structures, which connect the limb bones with the vertebral column.
- Their functions are to make locomotion possible and to protect the major organs of locomotion, digestion, excretion and reproduction.
- Calcium and phosphorous are two minerals that helps in giving strength to bones. Deficiency of these may cause we opening of bones.

NERVOUS SYSTEM

The nervous system provides the fastest means of communication within the body, so that suitable response to stimuli can be made at once.

Nervous system is found only in animals and absent in plants. It is made up of nervous tissue. Nervous system controls the body by using a series of tissues throughout the body formed by a network of electrically conducting cells called **neurons** or **nerve cells**.

Neuron

It is the structural and functional unit of nervous system.

Neurons are of three types:

(*i*) **Motor** Conducts messages from central nervous system to effector organs.

(*ii*) **Sensory** Conducts information from sensory organ to central nervous system.

(*iii*) **Mixed** Works both as sensory and motor neuron.

Structure of Neuron

A neuron is a microscopic structure composed of three major parts, i.e. cell body, dendrites and axon.

- **Dendrites** Cytoplasmic processes from cell body. They carry impulses towards cell body.
- **Cell body** Also called **cyton** or **soma**. It contains cytoplasm and certain granular bodies called **Nissl's granules**.
- **Axon** Single long extension of cell body that protected by membrane neurolemma.
- **Synapse** It is the junction between the dendrites of one neuron with axon terminal knobs of other neuron.

Classification of Neuron/Nerve Cell

Neurons are of two types:

1. **Myelinated** Axon is composed of Schwann cell coated by myelin sheath. The gap between adjacent myelin sheath is called **nodes of Ranvier**.

On the basis of number of axon and dendrites
- Unipolar (with one axon)
- Bipolar (with one axon and one dendrite)
- Multipolar (with one axon and two or more dendrites)

2. **Non-myelinated** Axon is composed of **Schwann** cells not enclosed by myelin sheath.

Types of Nervous System

Main parts of nervous system are as follow:

Main parts of vertebrate nervous system
- Central nervous system
  - Brain
  - Spinal cord
- Peripheral nervous system
  - Somatic nervous system
  - Autonomic nervous system
    - Parasympathetic nervous system
    - Sympathetic nervous system

Central Nervous System (CNS)

The CNS is the site of information processing and control. It consists of brain and spinal cord.

- Grey matter is made up of cell bodies of neuron and white matter has only nerve fibres.
- Brain lies in the cranial cavity of skull. Brain and spinal cord are covered by two meninges in frog and three meninges in mammals.
- Cerebrospinal fluid is present in and around brain and spinal cord.

Human Brain

Brain is soft, whitish in colour and somewhat a flattened organ.

- It is the only thing in the universe, which is attempting to understand itself.
- It is mainly divided into three main parts:

Brain
- Forebrain
  - Cerebrum
  - Olfactory lobes
  - Thalamus (Hypothalamus)
- Midbrain
  - Corpora quadrigemina
  - Corpora diencephalon
- Hindbrain
  - Cerebellum
  - Medulla oblongata
  - Pons

Forebrain

It forms the greatest part of brain. It consists of following types:

- **Cerebrum** It is the largest part of the brain covered by cerebral hemispheres. The roof of cerebrum is cerebral cortex. Cerebrum leads to consciousness, storage of memory of information.
- **Olfactory lobes** These are pair of small sized structure, completely covered by cerebrum. These are the centre of skull.
- **Thalamus** It is for integrity of sensory impulses sent from sense organs like eyes, ears and skin. It deals with pain, pressure and temperature.

- **Broca's area** It is present in brain and is related with speech, while Wernick's area of brain is related with understanding speech.
- **Hypothalamus** It deals with water balance in body, behavioural patterns of sex, sleep, stress, emotions, etc. It also regulates pituitary hormones and metabolism of fat, carbohydrate and water.

Midbrain

It is located between hypothalamus and pons of the hindbrain.

- It consists of two pairs of rounded swelling lobes called **corpora quadrigemina** (soild structures).
- It deals with visual analysis, auditory, etc.
- Midbrain of rabbit consists of four optic lobes.

Hindbrain

It consists of cerebellum, pons varolii and medulla oblongata.

- **Cerebellum** It is in the bottom part of the head and back of it. It controls the coordination of accurate movements and balancing. It is considered as small brain.
- **Medulla oblongata** It is the lowermost part of brain, which is connected to spinal cord. It deals with control of heartbeats, blood vessels, breathing, salivary secretion and most of reflex and involuntary (uncontrolled) movements.
- **Pons Varolii** It is an oral mass that lies above medulla oblongata. It interconnects two cerebellar hemispheres and also joins the medulla with higher brain centres.

Spinal Cord

It is a long cord-like structure present at the back, centrally located and well-protected by bony vertebral column.

- It gives out 31 pairs of spinal nerves in man.
- It conducts impulses to and from the brain.
- It is the centre for reflex actions and provides means of communication between spinal nerves and the brain.

Reflex Action

It is unconditional, inborn and unlearned reflex to a stimulus.

Examples Blinking of the eye when an object comes near to our eyes suddenly, rapid withdrawl of hand, while burned, sneezing, coughing, yawning, knee-jerk reflex, etc.

Acquired Reflex Action

It is also called as **conditioned reflex** and dependent on past experience, training and learning. These are under cerebral control during learning. First demonstrated by **Ivan Petrovitch Pavlov** in hungry dog.

e.g. Learning of dancing, cycling, swimming, singing and driving, etc.

Peripheral Nervous System

It is composed of 12 pairs of cranial nerves and 31 pairs of spinal nerves.

- Cranial nerves emerge from brain, while spinal nerves arise from spinal cord. Different number of cranial nerves are present in different organism.
  e.g. 10 pairs of cranial nerves are present in fishes and amphibians.
  - Number of cranial nerves found in rabbit is 12 (same as man) but 37 spinal nerves are present in rabbit.
  - The first 10 pairs are common for frog and rabbit.

List of Cranial Nerves

Cranial	*Number*	*Function*	*Distribution*
Olfactory	I	Sensory	Nose
Optic	II	Sensory	Retina of eye
Occulomotor	III	Motor	4 muscles of eyeballs
Trochlear (smallest)	IV	Motor	Superior oblique muscle
Trigeminal (largest)	V	Mixed	
Opthalmic	...		Snout
Maxillary	...		Upper jaw
Mandibular	...		Lower jaw
Abducens	VI	Motor	External rectus
Facial Vestibulocochlear	VII	Mixed	Neck, pinna, tongue, lower jaw
(auditory)	VIII	Sensory	Internal ear
Glossopharyngeal	IX	Mixed	Tongue, pharynx
Vagus	X	Mixed	Larynx, heart, lungs, stomach, etc.
Spinal accessory	XI	Motor	Neck
Hypoglossal	XII	Motor	Neck and tongue

Autonomous Nervous System

It was discovered by **Langley**. It is entirely motor and operates without conscious control.

Autonomic nervous system consists of two divisions:

(*i*) **Sympathetic nervous system** This increases defence system of body against adverse conditions and active in stress conditions, e.g. pain, fear and anger.

(*ii*) **Parasympathetic nervous system** It provides relaxation, comfort, pleasure at the time of rest and helps in the restoration and conservation of energy.

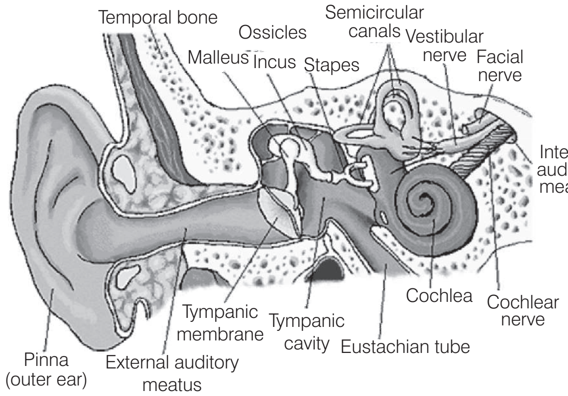
SENSE ORGANS

The sensory organs (receptors) enable us to detect all types of changes in the environment and send appropriate signals to the CNS, where all the inputs are processed and analysed signals are then sent to different centres of brain.

The different types of sense organs are as follows:

Human Ear

Ears are a pair of stato-acoustic organs meant for both sensory functions, i.e hearing and maintainance of body balance. The study of structure, function and disease of ear is called **otology**.



Structure of human ear

- These are located on sides of head.
- Hearing is controlled by auditory area of temporal lobe of cerebral cortex.
- The measuring unit of sound frequency is decibel. Human ear can listen the sound of 60-80 decibel.
- Human ear is sensitive to sound frequency 50-20000 cycles/sec and most sensitive to a frequency of 1000-3000 cycles/sec.
- Human ear is divided into three parts as:
- **External ear** (pinna + external auditory meatus), **middle ear** (tympanic cavity), **internal ear** (membranous labyrinth).

(i) External Ear

- The external organ consists of pinna and auditory canal.
- **Pinna** is designed to collect sound waves.
- **Auditory canal** contains few hairs and specialised sebaceous glands called **ceruminous glands**. These glands secrete cerumen (ear wax), it prevents foreign particles from entering the ear.

(ii) Middle Ear

- Tympanic membrane (ear drum) separates middle ear from external ear.
- The middle ear consist of three auditory ossicles. These are
  (a) **Malleus** Outer and hammer-shaped.
  (b) **Incus** Middle and oval-shaped.
  (c) **Stapes** Inner and stirrup-shaped attached on the outer surface of fenestra ovalis membrane.
- The smallest bone in human body is stapes.
- The passage connecting the middle ear with pharynx is Eustachian tube.
- **Eustachian tube** equalises pressure in both sides (external and middle ear) or tympanic membrane.

(iii) Internal Ear

The inner ear consists of a labyrinth of fluid-filled chambers within the temporal lobe of the skull. Labyrinth consists of two main divisions:

(a) **Bony labyrinth** is filled with a fluid called **perilymph**, while membranous labyrinth contains endolymph.
(b) **Membranous labyrinth** is concerned with both balancing and hearing.

- Internal ear is formed of three parts:
  (a) Body proper (utricules and sacculus)
  (b) Semicircular canals
  (c) Cochlea
- Cochlea is small snail-shaped structure.
- It makes $2\frac{1}{2}$ turns in rabbit and $2\frac{2}{4}$ turns in man.
- Three chambers in cochlea are **scala vestibuli**, **scala media** and **scala tympani**.
- Scala media contains the organ of hearing named organ of Corti (discovered by **Alfanso Corti**).
- Defects of ear are otalgia earache (pain in ear) acute infection of middle ear, labyrinthine disease (malfunction of ear).

Human Eye

The organ of sight is a pair of eyes in human. Eyes are situated in deep bony cavities called **orbits** or **sockets** of the **skull**. These are the sensitive detectors of light. It is most sensitive to yellow colour.

- The study of structure, functions and diseases of eye is called **ophthalmology**.
- Eyeball is made basically of three concentric layers:
- **Outermost fibrous layer** composed of sclera and cornea.
- **Cornea** is thin transparent front part of sclera, which locks blood vessels, but is rich in nerve endings.
- **Sclera** outermost, tough and opaque part. It maintains the shape of eye.
- **Middle layer** consists of choroid, ciliary body and iris.
- **Choroid** chocolate coloured, pigmented (with melanin) present along innerside of sclera.
- **Iris** consists of circular sphincter and radial dilators. It adjusts the size of pupil.
- **Pupil** is the black hole in the centre of the iris, through it light enters the eyeball. It also controls the light entering into the eye.
- **Conjunctiva** Cornea is covered by a thin transparent layer called **conjunctiva**. It refracts the incident light rays to focus on the retina.
- **Lens** is a biconvex, transparent, circular, solid structure and is proteinaceous in nature.
- **Aqueous humor** is filled between cornea and lens.
- **Vitreous humor** is filled in space between the lens and the retina.

- **Retina** is innermost and incomplete layer. It consists of a nervous tissue layer and a pigmented layer. Image of object is formed on retina.
- Retina is composed of two types of cells, which are rod cells and cone cells. The image formed on it is real and inverted.
- **Rods** are longer, slender and filamentous, while cones are shorter and thicker. These are highly sensitive of dim light, contain a reddish-purple pigment called **rhodopsin**.
- **Cones** are sensitive to bright light, hence differentiate the colours (i.e. red, green, blue).
- **Yellow spot** is in the exact centre of the retina also called **macula lutea**. The fovea centralis is the area of sharpest vision due to high concentration of cones.
- **The blind spot** (optic disc) has no rods and cones cells, hence no image forms in this region.
- **Tapetum lucidum** increases the sensibility of vision. It is made up of zinc, cysteine and guanine.
- The eyes of carnivores like, e.g. cat, dog, lion, seal, etc., glow in night due to the tapetum lucidum.
- **Atropine** is a chemical used by doctors to dilate the pupil.
- **Night blindness** is caused by deficiency of rhodopsin in rod cells.
- Colour blindness is caused due to deficiency of cones.

The visual pigments for colour are:

(a) Erythropsin (sensitive to red)
(b) Chloropsin (sensitive to green)
(c) Cyanopsin (sensitive to blue)

Diseases of Eyes

Some common diseases of eye are as follows:

- **Myopia** (short-sightedness) In this, the image of distant object being formed in front of retina. It is corrected by using concave lens.
- **Hypermetropia** (long or far-sightedness) Image is formed behind the retina, corrected by using convex lens. The image of nearer object becomes blurred.
- **Astigmatism** Due to irregular cornea or lens, corrected by using cylindrical lens.
- **Cataract** This is an eye disease generally occurs in older people. Lens becomes opaque and can be corrected by operation and using biconvex glasses.
- **Conjunctivitis** Inflammation of conjunctiva by bacteria.

Nose (Olfactory Organ)

Nose is a sense organ for smell or olfaction. The nose contains mucus coated olfactory (smell) receptors.

The nasal cavity is lined by olfactory epithelium lines.

Olfactory epithelium have three principal kinds of cells:

(*i*) Olfactory bipolar neurons
(*ii*) Columnar epithelial cells
(*iii*) Bowman's gland

Tongue (Gustatory Organ)

Organs for gustation are taste buds, which are present mainly in the mucosa of taste papilae present on upper surface of tongue.

Taste buds are numerous on the tongue, but also found on the soft palate, pharynx and epiglottis. A papila may contain few to many hundreds of taste buds. Taste buds are more numerous in children than in adults. Location of taste buds for basic tastes:

- **Sweet** On the tip of tongue.
- **Salty** On upper surface of anterior half.
- **Sour** Lateral side of tongue.
- **Bitter** Near the base of tongue.
- Bitter taste is evolved by many organic substances, i.e. quinine, morphine, caffeine and urea.

GLANDS SYSTEM

In the vertebrate body glands may be classified on the basis of the presence or absence of ducts. Glands are of two types:

(i) Endocrine Glands

The word endocrine derives from Greek words, '*endo*' means within and '*crines*' means to secrete. The endocrine system is the collection of glands that produce hormones that regulate metabolism, growth and development, tissue function, sexual function, reproduction, etc.

- The science dealing with the study of endocrine glands and their secretion is called **endocrinology**.
- These glands do not have ducts (tubes) to carry their secretion to the target organs.
- These are also called as **ductless glands**.
- Their secretion is called **hormone** or **internal secretion**.
- These glands secrete their secretion into the blood for their transportation to the sites of action, e.g. thyroid, pituitary, pancreas, hypothalamus, adrenal, etc.

(ii) Exocrine Glands

- These glands have ducts.
- These glands secrete their secretions in ducts to carry them to sites of action, e.g. sweat and oil glands of skin, salivary gland, liver, etc.

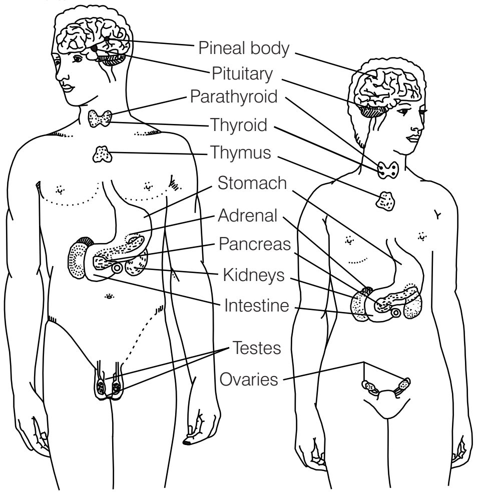
Hormones

- These are the chemical messengers that can regulate biological processes. They are steroids, proteins, peptides of amino acid derivatives.
- They can act as coenzymes.
- These are the organic compounds secreted in small amount by endocrine glands of body.
- Hormone was first discovered by **Bayless** and **Ernst H Starling** in 1903.
- Secretin was the first hormone to be discovered.

Endocrine Glands of Human

There are total nine endocrine glands in human beings. Most of them are common to male and female.

(*i*) Pituitary (master gland) (*ii*) Thyroid
(*iii*) Parathyroid (*iv*) Thymus
(*v*) Adrenals (supra-adrenals) (*vi*) Pancreas
(*vii*) Pineal gland (*viii*) Gonads
(*ix*) Hypothalamus



Human endocrine glands

Pituitary Gland

- It is the smallest gland, which is about the size of a pea, but plays a major role in endocrine system.
- It is also called as the **master gland**.
- It is slightly larger in woman than in man.
- It is located in region of forebrain.

Thyroid Gland

- It is the largest endocrine gland.
- It is found on ventral and lateral sides of upper part of the trachea in the neck.
- It is brownish-red, shield-shaped, bilobed gland.

Hormones secreted by it are given in the table below:

Parts of pituitary	Hormones	Target organs	Actions/Effects
Anterior pituitary	Growth Hormone (GH)	All tissues	Normal body growth.
	Adrenocorticotropin Hormone (ACTH)	Adrenal cortex	Glucocorticoid secretion.
	Thyrotropin Stimulating Hormone (TSH)	Thyroid gland	Thyroxine secretion.
	Prolactin	Alveolar cells of mammary glands	Milk secretion.
	Follicle Stimulating Hormone (FSH)	Ovarian follicles	Growth of ovarian follicle and oestrogen secretion in females and spermatogenesis in male.
	Leuteinising Hormone (LH)	Testes or ovaries	Development of corpus luteum and progesterone secretion in females, testosterone secretion in males.
Intermediate pituitary	Melanocyte Stimulating Hormone (MSH)	Skin	Synthesis of melanin in the skin.
Posterior pituitary	Oxytocin (protein)	Uterus, mammary glands	Milk ejection, uterine contraction, contraction of smooth muscles.
	Vasopressin (ADH)	Kidneys and arteries	Hypertonic urine, absorption of water, arterioler constriction.

Hormones secreted by it are given in the table below:

Hormones	Target organs	Actions/Effects
Calcitonin (CT)	Bones, kidney	Fall in blood calcium, excretion of Ca^{2+}.
Thyroxine	Heart, liver, kidney, skeletal muscles, mast cells.	Tissue metabolism, growth, IQ, differentation of gonads, metamorphosis in amphibians.

Parathyroid Glands

- These are four in numbers and are embedded in thyroid glands.
- These are small glands, which are oval-shaped and yellow coloured and present in the human neck.
- Parathyroid gland produces parathyroid hormone.

Hormones	Target organs	Actions/Effects
Parathormone (PTH)	Bones, kidney	Rise in blood calcium level, phosphorus check of calcium secretion.

Pancreas

- It is a mixed gland with both exocrine and endocrine function and located within the curve of duodenum.
- Three kinds of cells are found in islets of Langerhans namely alpha, beta and delta cells.
  (*i*) **α-cells** Larger and peripheral cells, produce glucagon hormone.
  (*ii*) **β-cells** Central and smaller cells, produce insulin that a peptide hormone.
  (*iii*) **δ-cells** Middle cells produce somatostatin hormone.

Hormones secreted by the pancreas are given in the table below:

Hormones	Target organs	Actions/Effects
Insulin	All cells	Decreases blood sugar level, stimulates glycogenesis, increases stored protein in tissue. Hypersecretion results into diabetes mellitus.
Glucagon	Liver	Rise in blood sugar.

Adrenal Glands

- Our body has paired endocrine glands, located superior to kidneys. It is structurally and functionally divided into adrenal cortex and adrenal medulla.
- It is also called as **4S gland** because of its four functions including salt retention, sugar metabolism, as a source of energy and as a sex organ.

Hormones secreted by the adrenal gland are given in table below:

Hormones	*Target organs*	*Actions/Effects*
Glucocorticoids	Tissues, mast cells	Carbohydrate, fat and protein metabolism.
Mineral corticoids	Kidney	Sodium and potassium metabolism, water absorption.
Sex corticoids	Body cells	External sex characters.
Adrenaline and noradrenaline	Mast cells	Heartbeat, blood pressure rise, contraction and relaxation of muscles.

Pineal Gland

- It is a small gland in the brain derived from the embryonic ectoderm. It secretes hormones in duct response to nervous activity.
- In man, it starts to degenerate at about age of 7 years.
- In adult, it is largely a fibrous tissue.

Hormone secreted by pineal gland is given below:

Hormone	*Target organ*	*Action/Effect*
Melatonin	Melanophores	Dispersion of melanin.

Thymus Gland

- It is a lymphoid gland that plays an important role in the development of immune system.
- It is situated in front of the heart.
- Thymus is active in young ones, but gradually becomes inconspicuous after sexual maturity.
- It arises from the endoderm of the embryo and consists of peripheral cortex and central medulla.

Hormone secreted by thymus gland is given below:

Hormone	*Target organ*	*Action/Effect*
Thymosin	—	Immune system of body produces lymphocytes.

Hypothalamus

The complete endocrine system works more or less under the influence of hypothalamus.

- It is located in the floor of forebrain.
- Neurohormones were first discovered by **Guillemin** and **Schally**.

Hormones secreted by hypothalamus are given below:

Hormones	*Target organs*	*Actions/Effects*
Thyrotropin releasing hormone	Thyroid, adrenal gland	Thyrotropin secretion, Corticotropin secretion
Corticotropin releasing hormone	Pituitary	Secretion of pituitary gonadotropin
Somatostatin	Pituitary	Initiation of growth hormone secretion

Gonads

- The main function of gonads is to produce gametes.
- They also secrete sex hormones.
- Sex hormones are mostly steroids.
- The testes in male and ovaries in females are gonads.

Hormone secreted by gonads are given below:

Endocrine gland	*Hormones*	*Target organs*	*Actions/Effects*
Testes	Testosterones	Male reproductive organ	Male secondary sex organs and external male characters.
Ovaries	Oestrogen	Female reproductive organ	Female secondary sex organ and external female characters.
Corpus luteum (ovaries)	Progesterone	Uterus, mammary glands	Pregnancy changes in female sex organs.

Apart from these, hormones are also secreted from heart kidney and gastro-intestinal tract.

Organ	*Hormone*	*Role*
Heart	Atrial natriuretic factor	Dilation of blood vessels
Kidney	Erythropoietin	Erythropoiesis
Gastro-intestinal tract	Gastrin, Secretin Cholecystokinin	Secretion of HCl Action pancreas for the secretion of water acts on gall bladder and pancreas for bile juice secretion.

REPRODUCTION IN ANIMALS

The process by which new individuals are produced from their parents is called **reproduction**. Reproduction is of two kinds:

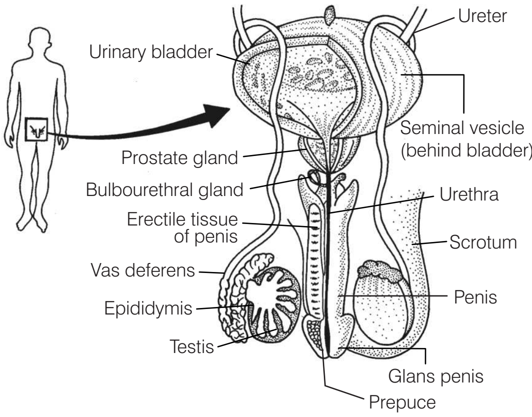
Reproduction

- **Asexual reproduction** Production of offspring by a single parent without the formation and fusion of gametes.
  - **Binary fission** It is the division of parent into two equal sized daughter cells,
  - **Multiple fission** It is the division of parent cells into a number of daughter cells, e.g. *Amoeba*.
  - **Budding** Formation of daughter cell from a small projection called **bud**, e.g. *Hydra,* sponges.
  - **Fragmentation** Breaking up of animal's body into two or more pieces, each of which grows into a new indvidual, e.g. coelenterates.
- **Sexual reproduction** Production of offspring by two parents, male and female.
  - **Syngamy** It involves fusion of two entire gametes to form a zygote.
    - **Endogamy** (self-fertilisation) e.g. tapeworm.
    - **Exogamy** (cross-fertilisation) e.g. frog and man.
  - **Conjugation** Temporary pairing of two points e.g. *Paramecium*.

Human Male Reproductive System

Male reproductive system consists of following main parts:

- **Testes** These are the primary sex organ of males, which perform dual role, i.e. function as endocrine gland and also act as male sex organs. These are suspended in thin skin bag called **scrotal sacs**.
- **Epididymis** It is highly coiled, thin tube formed by union of seminiferous tubules.
- **Penis** It is muscular organ, richly supplied with the blood. Externally, the penis has a bulge called **glans penis**, which is covered by prepuce (a skin fold).
- Male can produce spermatozoa (sperm) throughout their life starting from the age of 13-14 years.
- The growth of hairs on body is due male sex hormones, i.e. androgen (testosterone).



Male reproductive system

Reproductive Organs, Function and their Numbers in Man

Reproductive organs of male	*Number*	*Function performed*
Testes	2	To produce sperms and testosterone.
Sperm duct	2	To conduct the sperms from the testes to urethra.
Seminal vesicles	2	To secrete seminal plasma.
Epididymis	2	To temporarily store sperms and provide mobility.
Urethra	1	To conduct urine, sperms.
Prostate gland	2	To secrete an alkaline fluid to neutralise the acidity of urethra.
Cowper's gland	2	To secrete an alkaline white fluid.
Penis	1	To pass urine and deposite sperms in female genital tract.

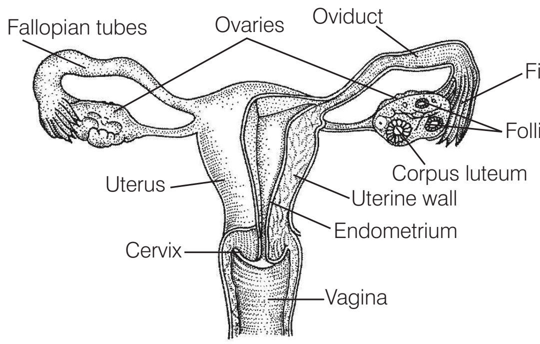
Human Female Reproductive System

The female reproductive system consists of the following parts:

- **Ovary** These are primary sex organs that produces females gametes (ovum) and several steroid hormone.
- After maturity, the ovary releases an ovum (egg cell) after every 28 days.

Reproductive Organs, Function and their Number in Female

Reproductive organ of female	*Number*	*Function performed*
Ovaries	2	To produce ova and hormones.
Oviducts	2	To move the ovum towards uterus.
Uterus	1	To provide space for developing child.
Vagina	1	To receive the sperms.



Female reproductive system

- **Lactation** Child is delivered after its development and mother produces milk to nourish the child.
- **Placenta** It is the connection between developing embryo and mother. It supplies blood, nutrition, etc., to the foetus.
- **Gestation period** The embryo develops for nine months in uterus. This period is called as **gestation period**.
- **Amniocentesis** A technique to detect chromosomal abnormalities, if any in the developing foetus by analysing the cells in the amniotic fluid. This technique is often misused to know the sex of the foetus. Ultrasound can also be used to ditect sex of foetus.

Menstrual Cycle

- **Reproductive period** of a human female extends from puberty (9-13 years) to menopause (45-50 years).
- **Menopause** is stopping of ovulation and menses. It normally occurs between the ages of 45 and 50. In this stage, women lose the ability to reproduce. Menopause is characterised by hot flushes.
- The periodic vaginal bleeding during menstrual cycle is called **menstruation**.
- Menstrual cycle is controlled by FSH, LH, oestrogen and progesterone.
- On an average, menstrual cycle is completed in 28 days.
- It is absent during pregnancy, may be suppressed during lactation and permanently stops at menopause.
- About 13 mature eggs are released from two ovaries of female in an year.

Contraception Methods

Contraception is the prevention of union of sperm and ovum.

Some common contraceptive methods are as follow:

(*i*) **Natural contraception** Avoid copulation during ovulation.
(*ii*) **Mechanical contraception** Use devices to avoid union of sperm and ovum, e.g. condoms, diaphragm, Intra Uterine Devices (IUDs).
(*iii*) **Chemical contraception** Fertilisation is avoided by the use of drugs, spermicidal creams, tablets, foam, etc.
(*iv*) **Vasectomy** It is the surgical removal of sperm ducts (vas deferens) in males.
(*v*) **Tubectomy** Cut and tied of Fallopian tubes in females.

HUMAN GROWTH

Human population is growing fast and its present rate is about 2% per annum. **Malthus** (1798) gave the concept of **human population**. Study of human population is called **demography**.

(*i*) The physiological capacity of organisms is called **biotic potential**.
(*ii*) The maximum number of individuals, which an environment can support of sustain is called **carrying capacity**.
(*iii*) Population growth is determined by the number of organisms added to the population (through birth) minus the number of organisms lost (through death).
(*iv*) When the number of individuals lost is called **zero population**.

Growth curves are of following two types:

- **J-shaped curve** When there is no environmental resistance, a population grows exponentially and a J-shaped curve is obtained.
- **S-shaped curve** When environment resistance does not allow population growth to soar towards infinitely.

HUMAN HEALTH AND DISEASES

HEALTH

It is a state of complete physical, mental and social well-being. Any deviation from this state results in a disease.

- According to World Health Organisation (WHO) (1948), 'health is a state of complete physical, mental and social well-being and not merely an absence of disease'.
- There are some basic factors, which contribute to our good health. These are balanced diet, personal hygiene and regular exercise.
- Some other factors having major impact on our health are awareness about diseases and their effects on different functions of body, vaccination against infectious diseases, proper disposal of waste, maintenance of hygienic food and water resources.

HUMAN DISEASES

A disease is an abnormal condition affecting the body of an organism or in other words, we can also say that 'any change from normal state causing discomfort is called disease'. It is a specific condition associated with specific symptoms and signs.

- The factors, which affect human health can be categorised into two groups:

(*i*) **Intrinsic factors** The disease causing factors, which are within the human body are called **intrinsic factors**, e.g. malfunctioning of organs (like heart, kidney, etc.) genetic diseases, hormonal imbalance, etc.
(*ii*) **Extrinsic factors** The disease causing factors, which come from outside the human body are called **extrinsic factors**, e.g. disease caused by microorganism, tobacco, alcohol, narcotics and inadequate diet.

- The disease causing organisms are called as **pathogens**.
- Various microorganisms responsible for infectious diseases are bacteria, viruses, fungi, protozoans, worms, etc.
- The pathogens enter our body by various ways, multiply and disturb the normal metabolic activities, thus shattering the major organ systems.

Classification of Diseases

Diseases may be broadly classified into two types:

- Diseases
  - Acquired diseases
    - Communicable (spread from one person to another person)
      - Contagious (by direct contact)
      - Non-contagious (by indirect contact)
    - Non-communicable (do not spread from one person to another person)
  - Congenital diseases

Acquired Diseases

These diseases develope after birth and are not transferred from parents to offsprings.

They can be classified into two main types:

(*i*) Communicable (*ii*) Non-communicable

Communicable Diseases or Infectious Diseases

These diseases are caused by pathogens and can spread from an affected person to a normal person. Depending upon the pathogen communicable diseases are of following types:

(i) Bacterial Diseases

Some of the common bacterial diseases are:

Pneumonia (*Streptococcus*), cholera (*Vibrio cholerae*), tuberculosis (*Mycobacterium tuberculosis*), tetanus (*Clostridium tetani*), leprosy (*Mycobacterium lepreae*), food poisoning (*Salmonella*), plague (*Yersinia pestis*), etc.

- Bacterium *Staphylococcus* present in the nasal passage causes food poisoning by producing a heat resistant toxin in the food.
- Bacterium *Salmonella typhi* is a pathogenic bacterium that causes typhoid fever in human beings. It affects liver, pancreas, nervous system, gastrointestinal tract, kidney, heat, etc.
- Bacterium *Clostridium botulinum* causes food poisoning in canned food.
- Bacteria like *Streptococcus pneumoniae* and *Haemophillus influenzae* are responsible for pneumonia in human beings and infect alveoli of lungs.
- **Anthrax** It is an infectious disease caused by a bacteria called *Bacillus anthracis*. Anthrax cannot directly spread from one individual to other, it spread by its spores.
- **Diphtheria** It is an upper respiratory tract illness caused by *Cornebacterium diphtheriae*, which is characterised by sore throat, low fever and an adherent membrane on the tonsils, pharynx and nasal cavity.
- **Pertusis** (Whooping cough) It is a serious and common infection of children caused by **Gram negative bacteria**, *Bordetella pertusis*. It is an air borne infection with incubation period of 1-3 weeks.
- DPT is a vaccine used against diphtheria, whooping cough and tetanus.

(ii) Viral Diseases

Some of the common viral diseases are:

- Polio myelitis (polio) mumps (paramyxo virus), chickenpox (variola virus), measles (paramyxo virus), AIDS (HIV).
- The virus causing dengue fever or break bone fever is an RNA virus of the family– Flaviviridae. Symptoms of dengue include headache, muscle and joint pain.
- Dengue causes platelet count in blood to decrease. It occurs mainly because this virus infects endo thelial cells of bone marrow that give rise to platelets resulting in bone marrow suppression.
- Conjunctivitis (also called pink eye or madras eye) is inflammation of the conjunctiva (the outermost layer of eye and inner surface of eyelids) caused by adenoviruses. It can also be caused by bacterial infection.
- Rabies is a viral disease that causes acute encephalitis (inflammation of brain). Roughly 97% of human rabies cases come from dog bites, but rabid bats are also capable of causing this disease.
- Hepatitis is a medical condition defined by inflammation of liver. It is caused by hepatitis viruses.
- Influenza commonly called as **flu**, is an infectious disease caused by RNA viruses of family– Orthomyxoviridae.
- Measles also known as rubeola or morbilli is an infection of the respiratory system caused by a virus, specifically a paramyxovirus of the genus–*Morbillivirus.* Symptoms include fever, cough, runny nose, red eyes, etc.
- Mumps is a viral disease of human species caused by mumps virus. Painful swelling of salivary glands (parotid gland) is the most typical symptom.
- **AIDS** (Acquired Immuno Deficiency Syndrome) It weakens the immunity of body. It spreads through sexual contact, transfusion of blood infected by AIDS virus, through infected needles, child gets diseases from mother's milk. Best way to avoid it is to adopt safe sex and use of sterilised syringes.

Antibiotics

Antibiotics are the chemical substances produced by certain microorganisms that kill or inhibit the growth of other microorganisms. Penicillin was first antibiotic discovered by **Alexander Fleming**. It inhibits cell wall formation in bacteria.

- Antibiotic inhibit the metabolic activities of other microorganisms. Such as synthesis of cell wall protein synthesis, nuclear membrane synthesis, etc.
- The microorganisms, which produce antibiotics (anti-microbial substance) is called **antibiont**.
- The antibiotics can be of bacterial, fungal or Actinomycetes origin.

Some of the antibiotics obtained from different sources are shown below in the given table:

Microbial Antibiotics

Antibiotics	*Source*
Penicillin	*Penicillium notatum*, *P. chrysogenum*
Streptomycin	*Streptomyces griseus*
Neomycin	*Streptomyces fradiae*
Tetracycline	*Streptomyces erythraeus*

(iii) Diseases Caused by Protozoans and Worms

Malaria is caused by *Plasmodium* sp. It is transferred to the host by means of vectors. Insects that can transmit diseases to humans are referred to as vectors, e.g. *Anopheles* serves as a vector for malaria *Culex* for filaria, *Aedes* for dengue and sandfly for kala-azar.

- **Kala-azar** or dum-dum fever is caused by protozoan, *Leishmania donovani*, hence it is also called as **leishmaniasis**.
- The infection to man is caused by the bite of infected sandflies, *Phlebotomus papatasi* and *P. argentipus*.
- **Amoebiasis** It is caused by *Entamoeba histolytica* in man. This protozoan parasitises in large intestine of man, occurring world wide whenever sanitation and hygiene are poor.
- **Sleeping sickness** It is caused by a flagellate protozoan parasite belonging to the class–Mastigophora and genus–*Trypanosoma* and hence, it is also called **trypanosomiasis**. It is transmitted by blood sucking **tse-tse fly**.
- **Ascariasis** It is a common helminthic infection (worm) casued by *Ascaris lumbricoides*. The diseases is characterised by internal bleeding, muscular pain, fever, anaemia and blockage of intestinal passage.
- **Filariasis** It is caused by filarial nematodes, *Wuchereria bancrofti.*

(iv) Diseases Caused by Fungi

- Ringworm (skin problems)
- Food poisoning (*Aspergillus*)
- Athlete's foot
- Communicable diseases are of two sub-types on the basis of mode of infection

(a) **Contagious Diseases** These diseases are transferred through simple contacts, e.g. smallpox, ringworm, chickenpox, influenza, measles, leprosy, etc.

(b) **Non-contagious Diseases** These diseases are transferred through some physical carriers such as air, water, food, etc., e.g. cholera, taeniasis, ascariasis, typhoid, etc. Some of these diseases are transmitted through some **disease carrier vectors**, e.g. plague, AIDS, malaria, hepatitis, etc.

Non-communicable Diseases or Non-Infectious Diseases

These diseases remain confined only to the person who develops them. They can be of four types:

(*i*) **Organic** (degenerative) **diseases** caused due to organ malfunctioning.

(*ii*) **Deficiency diseases** caused due to deficiency of mineral, hormones, vitamins and nutrients.

(*iii*) **Allergic diseases** caused due to the hypersensitivity of body to foreign particle.

(*iv*) **Cancer** caused due to uncontrolled division of certain cells of body.

Diseases of Malfunctioning

Diseases caused by malfunctioning of organs are:

- Disease caused by improper functioning of heart–Cardiac failure.
- Disease caused by improper functioning of kidney–**Kidney failure**.
- Disease caused by improper functioning of bones–**Osteoporosis**.
- Disease caused by improper functioning of eyes–**Myopia**, **hypermatropia** and **cataract**.

Deficiency Diseases

The diseases caused due to defect in the production of hormones are deficiency diseases. These are as follows:

- **Cretinism** In children, occurs due to deficiency of thyroxine hormone, which makes a child mentally retarded.
- **Diabetes** Due to deficiency of insulin.
- **Dwarfism** Due to the sluggish activity of the pituitary gland, the person has diminished growth of the bones.
- **Gigantism** Due to the excessive activity of the pituitary gland, the person has excessive growth of the bones.
- **Goitre** It is caused by deficiency of iodine in diet. It is characterised by swelling of thyroid gland.
- **Anaemia** It is a decrease in number of RBCs or less than the normal quantity of haemoglobin in the blood. Anaemia leads to hypoxia (lack of oxygen) in organs.

Allergic Diseases

The unfavourable response of the body to certain things like dust, serum, drugs, fabrics and pollen, etc., is called **allergy**. Sneezing, irritation of throat, itching and skin rashes, etc., are the symptoms of allergic diseases, e.g. asthma is caused by allergy.

Cancer

It is an uncontrolled growth of normal cells, which have lost control on cell division as a result these cells spread in the body. Sometimes cells keep dividing and produce more cells even when they are not required. As a result, mass of tissue is formed, which is called as **tumour**. Tumours can be benign or malignant.

Types of Cancers

(a) Carcinomas, e.g. lung cancer, breast cancer, cancer of pancreas and stomach.

(b) Sarcomas, e.g. bone cancer.

(c) Leukemia, e.g. blood cancer.

Breast cancer It is common among women while lungs cancer is common among smokers.

- Radiations cause cancer, leukemia and skin injuries damage genes or chromosomes, induce unmarked cell division, destroy tissues, cells and blood cells.
- Cancer is common in older people bacause over the time their genomes tends to accumulate more mutations.

Diseases Caused by Addictive Substances

Alcohol Reduces alertness of mind and causes liver damage.

Narcotic drugs Narcotic drugs are morphine, cocaine, heroin, hashish, ganja, opium, marijuana, barbiturates and LAD (Lysergic Acid Diethylamide).

Diseases Caused due to Pollution

Chemical pollution Pneumoconiosis is caused by inhaling coal dust. Silicosis is caused by inhaling stone dust (silica). Asbestosis is caused by inhaling asbestos dust. Lung diseases and bronchitis are caused by smoke.

> **IMPORTANT POINTS TO REMEMBER**
>
> DOTS or directly observed treatment, short-course was launched by WHO as a strategy to control TB (Tuberculosis). It ensures a regular, uninterrupted supply of high quality anti-TB drug.

Congenital Diseases

These are anatomical or physiological abnormalities that come from birth. These diseases can be caused by genetic mutations or environmental factors. Diseases by genetic mutation can be transmitted to next generation, whereas diseases by environmental factors are non-transmissible.

Some common congenital diseases are phenylketonuria, alkaptonuria, haemophillia, sickle-cell anaemia, etc.

Chromosomal Aberrations

These are any variation from the 46 chromosome number of humans. These includes Turner's syndrom, klinfelter syndrom, etc.

VACCINATION

Louis Pasteur coined the term 'Vaccine' and vaccination for the first time.

- The word 'vaccination' has been derived from a Latin word '*vacca*' which means **cow**.
- A vaccine is either a cell suspension or a byproduct excreted by the cell.
- These vaccines, when introduced into the body, stimulate the production of antibodies.
- Serum is used for obtaining antibodies.
- This introduction of harmless antigenic material into healthy person for providing acquired immunity against the disease is known as **immunisation** and **vaccination**.

Diseases of Animals

Diseases	*Animal affected*	*Caused by*	*Organ affected*
Pox	Cattle, sheep, goat	Virus	Skin
Tuberculosis	Cattle, birds	Bacteria	Lungs, intestine
Rinderpest	Cattle, sheep, goat	Bacteria	Spleen
Anthrax	Cattle, sheep, goat	Bacteria	Skin, lungs, intestine
Foot and mouth	Cattle, pigs	Virus	Pharynx
Dermatitis	Goat, sheep, cattle	Virus	Skin
Mastitis	Cattle	Bacteria	Swollen udders
Ringworm	Cattle	Fungus	Body
Trypanosomiasis	Cattle	Protozoan	Body
Helminthic	Cattle	Worms	Body
Aspergillosis	Poultry birds	Fungus	Body
Cholera	Poultry birds	Bacteria	Body
Diarrhoea	Poultry birds	Bacteria	Body
Ranikhet	Poultry birds	Virus	Body

APPLIED BIOLOGY

FERTILISERS

Fertiliser is any material found naturally or formed synthetically and is applied to soil or plant tissues, to supply one or more plant nutrients essential to the growth of plants.

They are of mainly two types:

1. Chemical Fertilisers

These are salt or an organic compound containing the necessary plant nutrients. They may be organic or inorganic, e.g. ammonium sulphate is an inorganic fertiliser, while urea is an organic fertiliser. If excess fertilisers are applied to a plant without water, crop will die due to the plasmolysis.

Chemical fertilisers can be classified into following types:

(*i*) **Nitrogenous fertilisers** These are nitrogen containing fertilisers, e.g.
- Ammonium nitrate [NH_4NO_3]
- Ammonium sulphate [$(NH_4)_2SO_2$]
- Sodium nitrate [$NaNO_3$]
- Urea [NH_2CONH_2] (organic fertiliser)

(*ii*) **Phosphatic fertilisers** These are phosphorus containing fertilisers, e.g.
- Ammonium phosphate [$(NH_4)_3PO_4$]
- Ammonium hydrogen phosphate [$(NH_4)H_2PO_4$]
- Superphosphate [$Ca_2(H_2PO_4)_2$]

(*iii*) **Potassium fertilisers** These are potassium containing fertilisers, e.g.
- Potassium chloride [KCl]
- Potassium nitrate [KNO_3]

- Potassium sulphate [K_2SO_4]
- Calcium Ammonium Nitrate (CAN) also called as **Kisan Khad**.
- N, P and K are the macronutrients provided by inorganic fertilisers. Such fertilisers are also called as **complete fertiliser**.

2. Biofertilisers

The fertilisers, which are biologically originated are referred to as biofertilisers. In other words, biofertilisers are those organisms, which can bring about nutrient availability to crop plants. One of the important type of biofertiliser is manures.

Manures

- They are the semi-decayed substances, which are added to the soil to enhance soil's fertility.
- It is decomposed product of organic wastes.
- Manures improve the following properties of soil:
  - Fertility
  - Aeration
  - Water holding capacity
  - Crumb structure
- Farmers in our country are traditionally using cow dung as manure.
- They are also of different types based on their composition.

The three types of manures are as follows:

(a) **Farm Yard Manure** (FYM) Consist of mixture of cattle dung, crop residues, fallen leaves, twigs, etc.

(b) **Composite Manure** (compost) Consist of rotten vegetables, garbage, sewage sludge, animal refuse, etc.

(c) **Green Manure** Quick growing crops cultivated and ploughed into the soil. These can increase next crop yield by 30-50%.

Green Manuring

Green manure is a type of cover crop grown primarily to add nutrients and organic matter to soil.

- The technique of utilising green manure is called **green manuring**, e.g. leguminous plants such as *Crotalaria juncea*, *Glycine max*, *Indigofera linifolia*, etc., are used as green manures.
- *Anabaena azollae* (cyanobacteria) is used as biofertiliser in rice field.
- Sunnhemp plants are used for green manuring in India.

Advantages of Manuring

- It improves the fertility of soil and enhances the soil structure.
- It improves the soil aeration capacity.
- It helps in control of pests, insects, mites and growth of weeds.

Sources of Biofertilisers

- Natural processes fix about 190×10^{12} g per year of nitrogen through **biological nitrogen-fixation**.
- Biological nitrogen-fixation provides about 1750 million tonnes of nitrogen, free of cost naturally in the form of biofertilisers.
- Biofertilisers contain some organisms, which can bring about nutrient availability to the crop plants.

The main sources of biofertilisers are:

1. Free-living Nitrogen-fixing Bacteria

Non-symbiotic (free-living) nitrogen-fixing bacteria are found abundantly in rhizosphere (i.e. the region where soil and roots make contact). They may be aerobic or anaerobic.

(*i*) **Aerobic nitrogen-fixing bacteria** Aerobic nitrogen-fixing bacteria such as *Azotobacter* are thought to maintain reduced oxygen conditions (microaerobic conditions) through their high level of respiration. Others such as *Gloeothece* evolve O_2 photosynthetically during the day and fix nitrogen during the night.

(*ii*) **Anaerobic nitrogen-fixing bacteria** In anaerobic nitrogen-fixing bacteria, oxygen does not pose a problem, because it is absent in their habitat. They can be either photosynthetic (e.g. *Rhodospirillum*) or non-photosynthetic (e.g. *Clostridium*).

2. Symbiotic Nitrogen-fixing Bacteria

Many plants form a symbiotic relationship with nitrogen-fixing bacteria that contain an enzyme complex nitrogenase. Nitrogenase enzyme can reduce nitrogen into ammonia.

- The best known example of symbiotic nitrogen-fixation is that of the *Rhizobium* group of bacteria, which form nodules on the roots of leguminous plants.
- Nodules contain an oxygen binding haemoprotein called **leghaemoglobin**.
- Leghaemoglobin is present in the cytoplasm of infected nodules cells at high concentration and gives the nodules pink colour.
- Non-leguminous trees and shrubs such as *Alder*, *Ceanothus* and *Purshia* have the nitrogen-fixing Actinomycetes, i.e. *Frankia* as a symbiont.

3. Free-living Nitrogen- fixing Cyanobacteria

Anabaena, *Nostoc*, *Aulosira*, *Cylindrospermum*, *Trichodesmium*, etc., are some free-living nitrogen-fixing blue-green algae or cyanobacteria.

- They add 20-30 kg of nitrogen per hectare of soil and water.
- *Aulosira* is the most active nitrogen fixer in rice field, whereas *Cylindrospermum* is active in sugarcane and maize fields.

4. Symbiotic Nitrogen- fixing Cyanobacteria

Anabaena and *Nostoc* sp. of cyanobacteria are common symbionts in lichens, *Anthoceros*, *Azolla* and *Cycads* roots.

- The water fern *Azolla* has *Anabaena azollae* as the symbiont, which fixes nitrogen within fronds.
- They grow thickly in rice paddy water and often inoculated to rice fields for nitrogen-fixation.

5. Mycorrhiza

Mycorrhizae (sing. mycorrhiza) are the mutualistic association of a fungus and the roots of higher plants. Mycorrhizae are found in most groups of vascular plants. There are two types of mycorrhizae listed as follow:

(*i*) **Ectomycorrhiza** The mycorrhizal fungi is present on the root surface only. The cortical cells are surrounded by a network of fungal hyphae called **Hartig net**. It is found in roots of oak, *Pinus*, peach, *Eucalyptus*, etc.

(*ii*) **Endomycorrhiza** The mycorrhizal fungus enters the roots and spreads in the root cortex. It may be intercellular or intracellular.

The type of endomycorrhiza, which produce vesicles and arbuscules inside the cortical cells of roots are called **VAM** (Vesicular Arbuscular Mycorrhiza), e.g. orchids, common crop plants and grasses.

Weed

Wild plants, which grow along with a cultivated crop are called **weeds**. Common weeds found in wheat and rice field are *Phalaris minar* (mandusi), *Amaranthus* (chaulai), *Chenopodium* (bathua). Exotic weeds are *Parthenium hysterophorus*, *Eichhornia*, etc. Most troublesome terrestrial weed is congress grass (*Parthenium hysterophorus*).

Weedicides

These are the chemical substances that are used to kill weeds. Some weedicides are butachloro 2,4-D (2,4-dichlorophenoxy acetic acid), **MCPA** (2-Methyl 4-Chloro, 1-Phenoxy Acetic acid).

Pesticides

Pesticides are used to destroy pests (pathogens, insects, rodents, etc.) by spraying the crop.

- Some microbes such as bacteria, fungi, viruses, protozoans, etc., and their products used for pest control are known as **biopesticides**.
- These are inexpensive, cause no pollution, pick no risk to human health and have a long term capability to control pests.
- Some common pesticides are warfarin sulphur, malathion, **DDT** (Dichloro Diphenyl Trichloroethane), BHC (Benzene Hexachloride), etc.
- Some examples of pesticides are:
  - DDT and 2, 4-D were mainly used during the Second World War.
  - Endosulfan is an organochlorine pesticide and acaricide used on plants and insects.
  - **Bordeaux mixture** a fungicide, composed of $CuSO_4$, $Ca(OH)_2$ was discovered by **Millardet**.
  - **Methyl isocyanate gas**, which is responsible for Bhopal gas tragedy (3rd Dec, 1984), is used for synthesising carbaryl (selvin).
  - Rotenone is obtained from roots of *Derris eliptica* and *Lonechocarpus*.
  - Fumigation is another method of pest control used mainly to kill pests attacking stored grains. The substances used for fumigation are known as fumigants, e.g. methyl bromide, iodoform.

ANIMAL HUSBANDRY

Animal husbandry is the science of rearing, caring, feeding, breeding, improvement and utilisation of domesticated animals. It is an agricultural practice of breeding and raising **livestock**.

- The animals used for transport, milk, meat and agriculture are collectively called **livestock**.
- On the basis of utility, the livestock can be categorised into the following categories:

1. **Milk yielding animals** Cows, buffaloes and goats provide us milk, which is used to obtain animal protein and serves as a perfect natural diet.
2. **Meat and egg yielding animals** Sheeps, goats, pigs, chicken, ducks and fowls provide us meat and eggs.
3. **Animals utilised as motive power** Buffaloes, horses, donkeys, bullocks, camels and elephants are used in transport and ploughing the fields.
4. **Wool giving animals** Sheeps are reared for obtaining wool.
5. **Miscellaneous uses** The hides of cattles are used for making a variety of leather goods.

Animal Breeding

- It is an important aspect of animal husbandry, which aims to increase the yield of animals and to improve the desirable qualities of produce.
- **Breed** is a group of animals related by descent and similar in most characters like general appearance features, size, configuration, etc.
- Animal breeding can be classified as:

(*i*) **Inbreeding** The crossing of closely related animals within the same breed for 4-6 generation is called **inbreeding**.

(*ii*) **Outbreeding** Breeding between unrelated animals either of the same breed but not having common ancestors for 4-6 generations (called **outcrossing**) or of different breeds (**cross-breeding**) or even different species (**interspecific hybridisation**) is called **outbreeding**.

> **ARTIFICIAL INSEMINATION**
>
> It is the process in which the stored semen of a superior male animal is introduced into the genital tract of a selected female animal by using suitable instruments, for carrying out reproduction, to get a better breed of the animal or to solve the problem of infertility.

Pisciculture

The breeding, rearing and transplantation of fish by artificial means is called pisciculture. It involves raising fish commercially in tanks or enclosures usually for food.

Fishes are of two types:

(*i*) **Freshwater fishes** Rohu, *Catla*, carp, tirica, mali, singhara and calbasu.

(*ii*) **Marine fishes** Bombay duck, conger eel, promphert, Salmon, Sardine, hilsa.

Fishes are the rich source of protein and vitamin (A and D).

Poultry

The term 'Poultry' means rearing of fowls, ducks, geese turkeys and some varieties of pigeons, but more often is used for fowl rearing.

Fowls are reared for food or for their eggs.

(*i*) Poultry birds, reared for meat are called **broilers**.

(*ii*) **Layers** are the female fowls raised for egg production.

(*iii*) The fowls may be categorised into two breeds.

- **Indigenous breeds** The term 'Desi' includes all indigenous fowls exhibiting great variations in their shape, size and colour, e.g. Assel, Ahagus, etc.
- **Exotic breeds** The term 'Improved' is applied generally to the exotic and comparatively modern breeds of chickens, e.g. white leghorn, Rhode island red, etc.

SOME HIGH YIELDING BREEDS OF ANIMALS

- **Cows** Karan Swiss, Sunandini, Jersey–Sindhi, Brown Swiss– Sahiwal, Friesion-Sahiwal, etc.
- **Buffalo** Murrah, Surti, Jaffarabadi, Nili, Nagpuri, etc.
- **Goat** Kashmiri, Gaddi, Bengal, etc.
- **Sheep** Gaddi, Lohi, Bikaneri, Gurez, Nillore, etc.
- **Pig** Large white Yorkshire, Middle white Yorkshire, etc.
- Halstein is the cattle breed that produces highest amount of milk.
- Pashmina (fine wool) obtained from mountain (Kashmiri) goat.
- Indigenous Hens Aseel, Basara, Chittagong, Ghagus, Brahma and Cochin.
- Exotic high egg yielding breeds of hens White leghorn, Rhode island red, Black minorica, Plymoth rock and Light sussex.
- High yielding breeds of hens developed in India by cross breeding of indigenous varieties with the foreign breeds are ILS-82, HH-260 and B-77.

Beekeeping (Apiculture)

- It includes the maintenance of hives of honeybees for the production of honey and beeswax.
- **Honey** It can be used as a food of high nutritive value. A number of ayurvedic medicines contain honey.
- **Beeswax** Produced by honeybees is used in industry for the manufacture of cosmetics and polishes.
- Several species of honeybees are reported in different parts of India, but the most common species reared by beekeepers is *Apis indica*.

GENETIC ENGINEERING

Biochemist **Paul Berg** is the father of genetic engineering.

Genetic engineering or genetic manipulation involves the introduction of foreign DNA or synthetic genes into the organism of interest, so as to obtain desired product, e.g. Golden rice is a variety of *Oryza sativa* produced through genetic engineering to biosynthesise β-carotene, a precursor of provitamin-A in the edible parts of rice. Two β-carotene biosynthesis gene were transformed in golden rice.

(*i*) ψ *(psy)* (phytoene synthase) from daffodil.

(*ii*) *crt* I from soil bacterium *Erwinia uredovora*.

Later on, other modifications were also developed and golden rice was expected to reach market by the year 2013.

- **Polymerase Chain Reaction** (PCR) Discovered by **Kary Mullis** (1985). It is a technique of amplification of DNA segment with the help of *taq* polymerase (isolated from *Thermus aquaticus*).
- **Restriction enzymes** These are used to cut DNA at specifies sites, also called **molecular scissors**.
- **Bioasphalt** It is an asphalt alternative made from non-petroleum based renewable resources like sugar, natural tree and gum resins, etc.
- **Gene therapy** It is the insertion of genes into an individual's cell or tissues to treat disease(s).
- **Terminator technology** It is a technology that is used to alter seeds, which fail to germinate after the first generation.
- **ELISA** It is a test for AIDS, which is based on Ag-Ab reaction.

Some Achievements of Genetic Engineering

- **Sweet bug** A bacterium that turns coconut water into delicious confectionary. The bacterium is *Acetobacter oxylinum*.
- **Prof Anand Mohan Chakraborty** has developed a new strain of oil eating bacteria called 'Superbug' by using species of *Pseudomonas*.
- **Organic foods** Foods produced by using methods free of inputs such as synthetic pesticides, chemical fertilisers, chemical additives, irradiation, etc.
- **Biodegradable plastic films** (PSV) made by blending the polymers with an additive to provide an oxidation or starch. First human insulin **Humulin** was produced by genetic engineering.

DNA Fingerprinting

It is a technique employed by forensic scientists to assist in the identification of individuals by their respective DNA profiles. This technique is based upon the fact that the DNA constitution of an individual carries some specific sequences of nucleotides, which do not carry any information for protein synthesis. These are called VNTR regions, i.e. Variable Number Tandem Repeats or mini satellites.

DNA fingerprinting was invented by **Dr Alec Jeffreys**.

Process

- Collection of sample (most preferably buccal swab).
- **RFLP** (Restriction Fragments Length Polymorphism) analysis.
- **Southern blotting** DNA fingerprinting is a quite laborious method that requires large amount of undegraded sample so the method was replaced by what is used today based on STR (Short Tandem Repeats) and PCR.

Cloning

Cloning in biology is the process of producing similar organism of genetically identical individuals that occur in nature when organisms such as bacteria, insects or plants reproduce asexually.

There are two types of cloning:

(*i*) **Reproductive cloning** It is used for the production of offspring. In it a new individual is created from a single cell by replacing the nucleus in an egg cell with the nucleus (containing the genetic material) from another cell of the body. The cloned egg cell grows and develops into an embryo. The embryo is implanted inside a surrogate mother's womb to mature and produce a viable foetus.

(*ii*) **Therapeutic cloning** It is used for the development of embryonic stem cells. The inner cell mass of the blastocyst is then removed and used for the creation of an embryonic stem cell line that has the genetic make up of the donated nucleus.

SUCCESSFULLY DEVELOPED CLONED

- Dolly a sheep, the first mammal clone synthesised by Dr Wilmut, UK.
- **Noori** The world's first cloned Pashmina goat named as Noori, an Arabic word referring to light. Funded by World Bank, the clone project was jointly worked by SKAUST and Karnal based National Dairy Research Institute (NDRI).
- The world's first buffalo calf through the Handguided Cloning Technique was born on 6 February, 2009 at NDRI, Karnal.
- **Garima-II**, another cloned buffalo calf through the new and advanced 'Handguided Cloning Technique' was born at NDRI, Karnal.

Stem Cell Therapy

The most promising use of stem cells is due to their ability to be modified into different functional adult cell types and serve as a potential source of replacement cells to treat numerous diseases. Stem cell research involves the use of several different types of cells besides embryonic stem cell, such as adult stem cells from humans or animals or stem cells from foetuses, umbilical cord or amniotic fluid.

Transgenic or GM Crops

A transgenic crop is a crop, which contains and expresses a transgene (a gene that is transferred into an organism by genetic engineering). It involves introduction of new DNA into a crop to confer a new superior trait. A popular term for the transgenic crop is genetically modified crops or GM crops.

***Bt* Crops** *Bt* crops are produced by inserting the toxic genes that code for a toxin directly into the genetic code of the crop in order to protect the crop from insects along with this, section of gene known as a **promoter**, which would encourage the crop to produce the toxin and a **marker**, which could be used to track and identify modified crop are also introduced, e.g. *Bt* cotton, *Bt* brinjal, Golden rice, etc.

***Bt* Cotton** It is created by using a bacterium, *Bacillus thuringiensis* (*Bt* is short form). It produces a protein that kills certain insects, e.g. beetles, flies, budworm, etc.

Golden Rice It is a transgenic rice biosynthesising β-carotene, vitamin-A, essential for eye sight precursor in the edible parts of rice.

Advantages of GM Crops

- Provides high nutrient containing varieties, i.e. enriched in specific nutrients.
- GM crops are more tolerant to cold, insect, drought, salinity, etc.
- Resistance to pest, herbicides and diseases.

Hybridisation

It is a technique, which involves cross breeding of different varieties of crop to get a new crop with desired characteristics. Such crops/varieties are called **High Yielding Varieties** (HYV) or hybrid.

- High yielding varieties of **wheat** are Lerma, Sharbati Sonara, Kalyan Sona, Roja 64, Pusa Lerma, Hera-Moti, Sonara-64, Sonalika and Arjun.
- High yielding varieties of **paddy** (rice) are Jaya, Padma, Pusa 205, IR-6, T-141.
- High yielding varieties of **maize** are Ganga-101, Ranjit, Daccan hybrid.
- **Germ plasm** It is the living genetic resource such as seed or tissue maintained for the purpose of animal and plant breeding preservation and other research aspects.

Terms Related with Biotechnology

- **Genetic screening** is an experimental technique used to identify genes that may produce damaging changes in the individuals.
- It involves collecting sequencing and comparison of DNA genomes.
- **Biochips** DNA chips single-stranded DNA chains or repetitive DNA segments fierily struck to silica glass chip.
- **Bioinformatics** An interdisciplinary field, which addresses biological problems using computational techniques and makes the rapid organisation and analysis of biological data possible.
- **Blue biotechnology** An application of biotechnology in the marine and aquatic, but rarely used.
- **Green biotechnology** It is an application of biotechnology in the agricultural processes.
- **Red biotechnology** It is an application of biotechnology in the medical processes.
- **White biotechnology** It is a branch of biotechnology applied to industrial processes.
- **Biomarkers** (Biosignature) are specific proteins used to diagnose disease.

Green Revolution (1940-70)

Green revolution refers to a series of research, development and technology transfer initiatives that increased agricultural production around the world.

- Great increase in the production of food grain crops due to use of hybrid varieties, fertilisers and modern agricultural techniques took place.
- Father of green revolution in India is **MS Swaminathan** and in world is **NE Borlaug**.

Operation Flood (White Revolution)

Great increasement in the production of milk due to use of high milk yielding breads of cow and buffaloes.

- It resulted in making India the largest producer of milk and milk products and hence called **white revolution**.
- Operation flood was rural programme started by India's National Dairy Development Board (NDDB) in 1970.

Silver Revolution

Great increasement in the production of eggs due to use of exotic breeds.

Fermentation

Fermentation is the conversion of carbohydrates to alcohol and carbon dioxide or organic acids using yeasts, bacteria or a combination of there, under anaerobic conditions, e.g. *Saccharomyces cerevisiae* used in baking and alcohol industry.

- Cheese is formed by the process of coagulation of casein by enzyme rennin.
- Yoghurt is produced by fermentation using *Lactobacillus* and *Streptococcus*.

BIOGAS

Biogas is also called **Landfill Gas** (LFG) or **digester gas**.

It is the gas produced by biological breakdown of organic matter in the absence of O_2.

- It is methane rich and produced by anaerobic digestion of biomass with the help of methanogenic bacteria.
- It is mainly produced from animal wastes.
- **Non-biodegradable waste** cannot be broken down by microorganism, so cannot used for biogas production.
- **Organic wastes** such as dead plants or animal material, animal dung, and kitchen waste are also called **biodegradable waste** can be easily converted into gaseous fuel, i.e. biogas. These wastes include cow dung paddy-husk, vegetables/fruits, wastes, etc.
- It mainly comprises methane CH_4-(50-70%) and carbon dioxide CO_2 (25-50%) and hydrogen sulphide H_2S (0-3%), moisture and siloxanes.

Advantages of Biogas

- It is cheaper than other fuels and ideal for small scale application.
- It helps to clean environment because waste materials can be used as substrate.
- It is a renewable source of energy.

BIOREMEDIATION

It is the treatment of environmental problems by the use of biological methods. It may be defined as treatment, which include microorganisms or their enzymes to return the environment that has been to its original condition. It may be employed in order to attack specific contaminants, such as chlorinated pesticides that are degraded by bacteria.

PHYTOREMEDIATION

This is the treatment of environmental problems through the use of plants.

- The term phytoremediation is a combination of two words, i.e. **phyto** means plant and **remediation** means to remedy.
- It helps in depolluting contaminated soils, water and air with plants, able to degrade or eliminate metals, pesticides, solvents, explosives, crude oil and its derivatives.

PRACTICE EXERCISE

Cell : The Unit of Life

1. Which of the following organism does not obey the 'cell theory'?
(a) Virus (b) Bacteria
(c) Fungi (d) Plants

2. The nucleic acid in a cell was discovered by
(a) E Strasburger (b) A Kossel
(c) F Miescher (d) Huxley

3. The 'cell theory' for organisms was proposed by
(a) Purkinje and Von Mohl
(b) Schleiden and Schwann
(c) Carolus Linnaeus
(d) Felix Dujardin

4. The fact that new cells arise from pre-existing cells was proposed by
(a) JE Purkinje (b) F Dujardin
(c) R Virchow (d) E Abbe

5. The scientists who first recognised and named the nucleus was
(a) Robert Hooke
(b) Robert Brown
(c) Theodore Schwann
(d) Thomas Morgan

6. Which of the following is a physical basis of life?
(a) Mitochondria (b) Ribosome
(c) Protoplasm (d) Nucleus

7. The bacteria were discovered by which of the following person?
(a) H Urey
(b) E Haeckel
(c) AV Leeuwenhoek
(d) Ivanowski

8. The term 'Meiosis' is coined by
(a) Fleming
(b) Blackmann
(c) Liebig
(d) Farmer and Moore

9. The physical basis of life protoplasm was named by
(a) Purkinje (b) J Huxley
(c) Robert brown (d) Sherington

10. Which of the following is not present in prokaryotes?
(a) Ribosomes
(b) Nuclear membrane
(c) Cell wall
(d) Plasma membrane

11. Which one among the following is known as 'animal starch'?
(a) Cellulose (b) Glycogen
(c) Pectin (d) Chitin

12. Consider the given cell organelles.
1. Mitochondria 2. Plastids
3. Ribosome

Which of the cell organelle(s) given above is/are common in both plant and animal cells?
(a) Only 2
(b) 1 and 2
(c) 1 and 3
(d) All of the above

13. Rigidity of cell wall is due to
(a) chitin (b) lignin
(c) suberin (d) pectin

14. The plasma membrane of animal cells is highly elastic. It is mainly due to the presence of
(a) carbohydrates
(b) proteins
(c) lipids
(d) nucleic acids

15. The secretory product pectin of cell wall is chemically a
(a) protein
(b) carbohydrate
(c) lipid
(d) nucleic acid

16. Chromosomes are
(a) present only in the nucleus of a cell
(b) the largest in number in human cells
(c) made up of DNA as a main component
(d) visible in all cells at every time

17. Consider the following statements
1. Endoplasmic Reticulum (ER) is called as 'skeleton of cell'.
2. ER is divided into three types according to the presence or the absence of ribosome.
3. Smooth ER is rich in actively protein secreting cells.

Which of the statement(s) given above is/are correct?
(a) Only 1
(b) 1 and 2
(c) 2 and 3
(d) Only 3

18. Match the following Columns.

Column I (Cell organelle)	Column II (Common names)
A. Ribosome	1. Suicidal bags of cell
B. Lysosome	2. Protein factory of cells
C. Mitochondria	3. Controller of cells
D. Nucleus	4. Power house of cell

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	2	1	4	3
(c)	4	2	1	3	(d)	1	4	2	3

19. Which of the following cell organelles functions as the powerhouse of a living cell?
(a) Chloroplast (b) Mitochondria
(c) Ribosomes (d) Golgi apparatus

20. Which one of the following organelles is the smallest membrane bound organelle?
(a) Ribosome (b) Golgi bodies
(c) Lysosome (d) Nucleolus

21. When our cells are filled with dead or damaged organelles like mitochondria, ER, etc. The degradation of these organelles is mainly performed by
(a) ribosomes (b) lysosomes
(c) peroxisomes (d) glyoxysomes

22. Consider the following statements
1. Cortical granules originate from Golgi bodies.
2. Cortical granules form the fertilisation membrane.
3. Cortical granules are found only in ovum.

Which of the statement(s) given above is/are correct?
(a) Only 2 (b) 1 and 3
(c) 2 and 3 (d) All of these

23. Consider the following statements
1. Lysosomes have hydrolytic enzymes.
2. Lysosomes are autophagic.
3. Lysosomes can dissolve proteins, lipids and carbohydrates.
4. All lysosomes are same structurally and functionally.

Which of the statement(s) given above is/are correct?
(a) Only 1 (b) 1 and 2
(c) 1, 2 and 3 (d) All of these

24. Consider the following statements
1. Chloroplast is known as 'cell within a cell'.
2. Chloroplast is the largest cell organelle found in plant cells.
3. Chloroplasts have single-stranded extra chromosomal DNA.

Which of the statement(s) given above is/are correct?
(a) Only 2 (b) 1 and 2
(c) 1 and 3 (d) All of these

25. Which of the following pairs of cell organelles mainly concerned with fatty acid metabolism in both plants and animals?
(a) Ribosomes and nucleus
(b) Lysosomes and mitochondria
(c) Peroxisomes and mitochondria
(d) Glyoxysomes and peroxisomes

26. DNA is mainly found in the following organelles.
1. Mitochondria
2. Chloroplasts
3. Peroxisomes
4. Nucleus

Which of the statements given above are correct?
(a) 2 and 4 (b) 1, 2 and 4
(c) 2, 3 and 4 (d) All of these

27. Which of the following cell organelles is found in a cell that is very active in protein synthesis?
(a) Mitochondria
(b) Chloroplast
(c) Lysosome
(d) Ribosome

28. Consider the following organelles.
1. Chloroplast
2. Mitochondria
3. Peroxisome

Which of the organelle(s) given above is/are concerned with photorespiration?
(a) Only 3 (b) 1 and 2
(c) 2 and 3 (d) All of these

29. Number of chromosomes in *Drosophila* is
(a) eight (b) ten
(c) twelve (d) fourteen

30. How many linkage groups are present in man?
(a) 46 (b) 23
(c) 48 (d) 10

31. Which of the following phases is the longest phase in the meiotic cell division?
(a) Anaphase-II
(b) Telophase-I
(c) Prophase-I
(d) Metaphase-I

32. Which of the following is the correct sequence of the different phases of cell division?
1. Anaphase 2. Telophase
3. Prophase 4. Metaphase

Codes
(a) 3, 1, 4, 2 (b) 1, 3, 2, 4
(c) 3, 4, 1, 2 (d) 1, 2, 3, 4

33. Chromosomes are best seen in
(a) telophase (b) anaphase
(c) metaphase (d) prophase

34. Which type of division occurs in cleavage?
(a) Amitotic (b) Mitotic
(c) Meiotic (d) Both (b) and (c)

35. In which one of the following types of cell divisions crossing over of chromosomes takes place?
(a) Mitosis (b) Meiosis
(c) Amitosis (d) Cytokinesis

36. Which is the correct order of phases in prophase-I?
(a) Leptotene, Diakinesis, Pachytene, Diplotene, Zygotene
(b) Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis
(c) Diakinesis, Pachytene, Diplotene, Zygotene, Leptotene
(d) Diakinesis, Pachytene, Leptotene, Diplotene, Zygotene

37. Which of the following stages is associated with the formation of chiasmata during meiotic cell division?
(a) Leptotene (b) Zygotene
(c) Pachytene (d) Diplotene

38. Which of the following organisms have peptidoglycan compound as an important constituent in cell wall?
(a) Bacteria and cyanobacteria
(b) Bacteria and unicellular eukaryote
(c) Archaebacteria and eukaryote
(d) All members of Monera and Protista

39. Which of the following term is associated with the failure of meiosis-II to occur after meiosis-I?
(a) Brachymeiosis
(b) Dinomitosis
(c) Dimunition
(d) Karyokinesis

Previous Years' Questions

40. Which organelle in the cell, other than nucleus, contain DNA? 2013 (I)
(a) Endoplasmic reticulum
(b) Golgi apparatus
(c) Lysosome
(d) Mitochondria

41. Which of the following parts are found in both plant and animal cells? 2014 (II)
(a) Cell membrane, chloroplast, vacuole
(b) Cell wall, nucleus, vacuole
(c) Cell membrane, cytoplasm, nucleus
(d) Cell wall, chloroplast, cytoplasm

42. Which one among the following statement is correct? 2015 (I)
(a) Prokaryotic cells possess nucleus
(b) Cell membrane is present in both plant and animal cells
(c) Mitochondria and chloroplasts are not found in eukaryotic cells
(d) Ribosomes are present in eukaryotic cells only

43. The living content of cell is called protoplasm. It is composed of 2016 (I)
(a) only cytoplasm
(b) cytoplasm and nucleoplasm
(c) only nucleoplasm
(d) cytoplasm, nucleoplasm and other organelles

Classification of Plants and Animals

44. Study of principle and procedures of classification is called
(a) taxonomy (b) systematics
(c) nomenclature (d) identification

45. Which of the following does not contain chlorophyll?
(a) Fungi (b) Algae
(c) Bryophyta (d) Pteridophyta

46. Bryophytes are photosynthetic but do not have vascular tissue and true roots. This feature enables them to resemble with which of the following?
(a) Fungi (b) Algae
(c) Pteridophytes (d) Angiosperms

47. Vascular tissue is found in
(a) Thallophyta (b) Bryophyta
(c) Pteridophyta (d) Lichens

48. The gymnosperms are
(a) seed bearing plants
(b) non-flowering
(c) thallus bearing
(d) Both (a) and (b)

49. When any plane passing through the central axis of the body divides the organism into two halves that are approximately mirror images, it is called
(a) radial symmetry
(b) bilateral symmetry
(c) Both (a) and (b)
(d) None of the above

50. If we sprinkle common salt on an earthworm, it dies due to
(a) osmotic shock
(b) respiratory failure
(c) toxic effect of salt
(d) closure of pores of skin

51. Itching due to insect bite is caused by
(a) formic acid
(b) acetic acid
(c) lactic acid
(d) maleic acid

52. Which of the following is the group with closely related animal types?
(a) Sea star, sea lilly, sea hare
(b) Dogfish, silverfish, crayfish
(c) Leech, louse, snail
(d) Jellyfish, sea fan, sea anemone

53. Which one of the following animals breathe through the skin?
(a) Fish (b) Pigeon
(c) Frog (d) Cockroach

54. Match the following Columns.

Column I	Column II
A. Himalayan salamander	1. Amphibia
B. Indian shark	2. Chondrichthyes
C. Sea-horse	3. Osteichthyes
D. Tortoise	4. Reptilia

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	1	2	4	3
(c)	2	1	4	3	(d)	2	1	3	4

55. Consider the following statements
1. Heart is three-chambered in fishes.
2. Heart is four-chambered in birds.
3. Most animals of class—Amphibia are characterised by two pair of limbs.
4. In all reptiles respiration is by lungs only.

Which of the above statements are correct?
(a) 1, 2, 3, 4 (b) 1 and 3
(c) 2 and 4 (d) 2, 3, 4

56. Match the following Columns.

Column I	Column II
A. Gill slits	1. Birds
B. Scales	2. Osteichthyes
C. Operculum	3. Chondrichthyes
D. Hairs	4. Reptilia
E. Lungs	5. Mammalia

Codes

	A	B	C	D	E		A	B	C	D	E
(a)	3	4	2	5	1	(b)	1	2	4	3	5
(c)	5	4	2	1	3	(d)	1	2	5	4	3

57. Consider the following statements
1. Warm-blooded animals can remain active in cold environment in which cold-blooded animals can hardly move.
2. Cold-blooded animals require much less energy to survive than warm-blooded animals.

Which of the statement(s) given above is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

58. Bats can fly in dark because they
(a) have strong wings
(b) have sharp eyes
(c) produce ultrasonic waves
(d) are nocturnal

59. Which one of the following characteristics is common among parrot, *Platypus* and kangaroo?
(a) Oviparity
(b) Toothless jaws
(c) Homeothermy
(d) Functional postanal tail

Previous Years' Questions

60. Which one among the following groups of animals maintains constant body temperature in changing environmental conditions? **2013 (I)**
(a) Birds (b) Amphibians
(c) Fishes (d) Reptiles

61. Among the following animals, choose the one having three pairs of legs. **2014 (I)**
(a) Spider (b) Scorpion
(c) Bug (d) Mite

62. Which one among the following is the generic name of the causal organism of elephantiasis? **2014 (II)**
(a) Filaria
(b) Microfilaria
(c) *Wuchereria bancrofti*
(d) *Culex pipiens*

63. Which one among the following pairs is not correctly matched? **2014 (II)**
(a) Sandal wood plant — Partial root parasite
(b) *Cuscuta* — Parasite
(c) *Nepenthes* — Carnivorous
(d) Mushrooms — Autotroph

64. Match the following Columns. **2014 (II)**

Column-I	Column-II
A. *Acsaris*	1. Mammalia
B. Malarial parasite	2. Arthropoda
C. Housefly	3. Nematoda
D. Cow	4. Protozoa

Codes

	A	B	C	D
(a)	3	4	2	1
(b)	3	2	4	1
(c)	1	2	4	3
(d)	1	4	2	3

Genetics, Molecular Biology and Evolution of Life

65. The scientist who is considered as the father of genetics?
(a) Lamarck
(b) Charles Darwin
(c) Hugo de Vries
(d) Gregor Johann Mendel

66. Dr. Hargobind Khorana is credited for the discovery of
(a) synthesis of proteins
(b) synthesis of genes
(c) synthesis of nitrogenous bases
(d) None of the above

67. A pair of contrasting characters is called
(a) phenotype (b) genotype
(c) allele (d) None of these

68. Which one of the following ratio represents a test cross?
(a) 3 : 1 (b) 9 : 7
(c) 1 : 1 : 1 (d) 1 : 1 : 1 : 1

69. Addition of one chromosome to a diploid set is
(a) trisomy (b) monosomy
(c) nullisomy (d) tetrasomy

70. A couple with blood types A and B may have children with blood group
(a) A and B
(b) A, B and AB
(c) A, B, AB and O
(d) Only AB

71. Which one among the following is not correct about Down's syndrome?
(a) It is a genetic disorder
(b) Affected individual has early ageing
(c) Affected person has mental retardation
(d) Affected person has furrowed tongue with open mouth

72. What is the most conspicuous salient feature of people with 'Progeria'?
(a) More hairs on body
(b) Less immunity to opportunistic infections
(c) Faster rate of ageing
(d) Suffer from infertility

73. The nucleic acid in cell was discovered by
(a) E Strasburger (b) A Kossel
(c) F Miescher (d) Huxley

74. The sugar in RNA is
(a) ribose (b) deoxyribose
(c) sucrose (d) lactose

75. Which base replaces thymine in RNA?
(a) Adenine (c) Guanine
(c) Uracil (d) Cytosine

76. Transfer of genetic information from one generation to the other is accomplished by
(a) DNA (b) messenger RNA
(c) transfer RNA (d) Both (b) and (c)

77. Chromosomes are
(a) present only in the nucleus of a cell
(b) the largest in number in human cells
(c) made up of DNA as a main component
(d) visible in all cells at every time

78. The heredity unit gene is made up of
(a) RNA
(b) DNA
(c) protein
(d) endoplasmic reticulum

79. The correct model of DNA structure is proposed by
(a) Jacob and Monod
(b) Watson and Crick
(c) Khorana
(d) Baltimore and Temin

80. Enzyme responsible for DNA replication is
(a) polymerase (b) proteolytic
(c) dehydrogenase (d) carboxylase

81. DNA structure was discovered by
(a) Weismann
(b) Watson and Crick
(c) Sutton (d) Jacob

82. Which of the following is represented by X in figure?

DNA ⟶ C G A T C G C T
mRNA ⟶ G C X A G C G A

(a) Adenine (b) Guanine
(c) Uracil (d) Thymine

83. Genetic code is
(a) singlet (b) doublet
(c) triplet (d) None of these

84. Which of the following were first formed?
(a) Genes (b) Cells
(c) Eobionts (d) Coacervates

85. Life originated in
(a) air (b) land
(c) water (d) All of these

86. Genetic variation arises by
(a) recombination (b) mutation
(c) chromosomal aberrations
(d) All of the above

87. The evolution of human species took place mainly in
(a) Asia (b) Africa (c) Europe (d) China

88. The organs which are morphologically different but perform the same function are called
(a) homologous organs
(b) analogous organs
(c) vestigial organs
(d) None of the above

89. Vermiform appendix is vestigial in man due to
(a) cellulose digestion
(b) omnivorous diet
(c) heterodont condition
(d) cooking habit

90. Which of the following have not undergone much of a change during the process of evolution over millions of years?
1. Crocodile 2. Cockroach
3. Horse

Codes
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) All of these

91. Which of the following is the correct sequence in which the given animal groups appeared on the earth during the course of evolution?
(a) Porifera — Annelida — Coelenterata — Protozoa
(b) Protozoa — Coelenterata — Porifera — Annelida
(c) Annelida — Porifera — Protozoa — Coelenterata
(d) Protozoa — Porifera — Coelenterata — Annelida

92. A haemophilic man marries with a normal homozygous woman. What is the probability that their son will be haemophilic?
(a) 100% (b) 75%
(c) 50% (d) 0%

93. The nearest relative of man are
(a) lemurs
(b) apes
(c) new world monkeys
(d) old world monkeys

94. Which is the correct sequence in order of chronology?
1. Rediscovery of Mendel's law of inheritance.
2. Darwin's theory of evolution.
3. Synthetic theory of evolution
4. Hugo de Vries theory of mutation.

Codes
(a) 1, 2, 3, 4 (b) 2, 4, 1, 3
(c) 3, 2, 1, 4 (d) 4, 2, 3, 1

95. Which one of the following is an example of vestigial organ in man?
(a) Jaw apparatus (b) Ear muscles
(c) Canine teeth (d) Humerus

96. Arrange the following in the order of their evolution.
1. Amphibians 2. Fish
3. Reptiles 4. Birds

Codes
(a) 1, 2, 4, 3 (b) 2, 1, 4, 3
(c) 2, 1, 3, 4 (d) 4, 1, 3, 2

97. Which one of the following theories was not proposed by Charles Darwin?
(a) Natural selection
(b) Survival of the fittest
(c) Struggle for existence
(d) Inheritance of acquired characters

98. The theory of 'survival of the fittest' was propounded by
(a) Lamarck
(b) Mendel
(c) Charles Darwin
(d) Karl Landsteiner

99. According to Darwin, variations are
(a) heritable (b) non-heritable
(c) mutational (d) environmental

100. Who proposed the mutation theory of evolution?
(a) Huxley (b) Darwin
(c) Lamarck (d) Hugo de Vries

101. Synthetic theory was proposed by
(a) Morgan (b) Muller
(c) Darwin (d) Dobzohansky

102. Microevolution can be measured by comparing observed allelic frequencies. These can be predicted by
(a) chance
(b) the Hardy-Weinberg equilibrium
(c) differential mortality
(d) Mendelian ratio

Previous Years' Questions

103. The fossil of *Archaeopteryx* represents the evidence of origin of **2013 (I)**
(a) birds from reptiles
(b) mammals from reptiles
(c) reptiles from amphibians
(d) mammals from birds

104. Two strands of DNA are held together by **2014 (I)**
(a) hydrogen bonds
(b) covalent bonds
(c) electrostatic force
(d) van der Waals' forces

105. Match Column I with Column II and select the correct answer using the codes given below **2016 (I)**

Column I (Geological Time-Scale)	Column II (Life form)
A. Pleistocene	1. Mammals
B. Palaeocene	2. Human genus
C. Permian	3. Invertebrates
D. Cambrian	4. Frogs

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	4	3	(b)	2	4	1	3
(c)	3	4	1	2	(d)	3	1	4	2

106. Which of the following cause(s) variation in the genetic material of progeny? **2016 (I)**
1. Sexual reproduction
2. Asexual reproduction
3. Mutations
4. Epigenetic changes

Select the correct answer using the codes given below:
(a) Only 2 (b) 1, 2 and 3
(c) 1, 3 and 4 (d) 1 and 3

The Plant Morphology Physiology

107. Root hairs are the extension of
(a) epiblema cells
(b) cortex cells
(c) pericycle cells
(d) xylem tracheids

108. Match the following Columns.

Column I	Column II
A. Primary root	1. It is formed due to the repeated division of primary root.
B. Tap root system	2. It is primary root and its branches.
C. Fibrous root	3. Root arises any place other than the root system.
D. Adventitious root	4. It is direct prolongation of the radicle noticed nearly in all dicotyledonous plants.

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	3	4	2	1
(c)	2	1	4	3	(d)	4	2	1	3

109. Stem develops from
(a) plumule (b) radicle
(c) pericarp (d) procambium

110. Which one of the following statements regarding potato is correct?
(a) It is a root
(b) It is a normal stem
(c) It is a modified stem
(d) It is a modified root

111. A leaf without petiole is known as
(a) subpetiolate (b) sessile
(c) zygomorphic (d) heteromerous

112. The largest flower in the world is
(a) lotus (b) *Rafflesia*
(c) giant cactus (d) None of these

113. The fruit of coconut is
(a) drupe (b) hesperidium
(c) composite fruits (d) berry

114. Which of the following is correctly matched?
(a) Tomato—Pome (b) Banana—Berry
(c) Mango—Berry (d) Apple—Drupe

115. Match the following Columns.

Column I (Plants)	Column II (Seed dispersal mechanism)
A. Coconut	1. By animals
B. Drumstick	2. Explosive mechanism
C. Coklebur (*Xanthium*)	3. By water
D. Castor	4. By wind

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	4	3	(b)	2	4	1	3
(c)	3	4	1	2	(d)	3	1	4	2

116. In groundnut, the root is
(a) nodulated (b) napiform
(c) epiphytic (d) photosynthetic

117. Which one of the following is dicot endospermic seed?
(a) Maize (b) Bean
(c) Castor (d) Wheat

118. The apple fruit is called as false fruit because
(a) its pericarp is unconspicuous
(b) its actual fruit is located within edible fleshy thalamus
(c) its develops from the inferior ovary
(d) its endocarp is cartilaginous

119. Wood is the common name for
(a) vascular bundle
(b) cambium
(c) secondary xylem
(d) secondary phloem

120. The sharp and pointed outgrowths present on the stem of rose are called
(a) prickles (b) thorns
(c) spines (d) hooks

121. Cork cambium is
(a) primary meristem
(b) ground meristem
(c) lateral meristem
(d) intercalary meristem

122. The characteristic odour of garlic is due to which one of the following?
(a) Chlorine-containing compounds
(b) Fluorine-containing compounds
(c) Nitrogen-containing compounds
(d) Sulphur-containing compounds

123. Which one of the following nutrients is not a structural component of the plant?
(a) Nitrogen (b) Calcium
(c) Phosphorus (d) Potassium

124. Palak leaves are rich in
(a) vitamin-A (b) vitamin-D
(c) iron (d) carotene

125. Pulse crops can fix atmospheric nitrogen because of
(a) root nodules (b) deep roots
(c) aerial roots (d) root hair

126. The plant which is parasite in nature is
(a) *Cuscuta* (b) *Utricularia*
(c) *Rhizopus* (d) Green plants

127. Out of the following elements which is required in largest quantity?
(a) Phosphorus (b) Nitrogen
(c) Calcium (d) Sulphur

128. Interveinal necrosis in lemon leaf is caused by deficiency of
(a) boron (b) molybdenum
(c) copper (d) zinc

129. Absorption of water increased by which mineral?
(a) Magnesium (b) Zinc
(c) Calcium (d) Manganese

130. A limiting factor unique to a field planted with corn year after year is most likely
(a) temperature (b) water
(c) sunlight (d) soil nutrient

131. Which of the following are carnivorous plants?
1. Waterlilly 2. Pitcher plant
3. Sundew 4. *Begonia*

Codes
(a) 1 and 4 (b) 1 and 2
(c) 2 and 3 (d) 1, 2 and 3

132. Which one of the following plants is preferred for mixed cropping in order to enhance the bioavailability of nitrogen?
(a) Wheat (b) Gram
(c) Maize (d) Barley

133. Which factors affect the rate of diffusion?
(a) Permeability of membrane
(b) Temperature
(c) Pressure
(d) All of the above

134. Consider the following statements
1. Vapour pressure of a solution is always less than the vapour pressure of the pure solvent.
2. Osmotic pressure of a solution increases if the number of solute molecules is increased.
3. The temperature at which the liquid and solid states of a substance have the same vapour pressure is the freezing point.
4. Osmotic pressure of a solution is inversely proportional to the elevation in boiling point.

Which of the above statement(s) is/are correct?
(a) Only 1 (b) 2 and 4
(c) 3 and 4 (d) 1, 2 and 3

135. Osmosis takes place across
(a) semipermeable membrane
(b) impermeable membrane
(c) Both (a) and (b)
(d) None of the above

136. When soil is wet and atmosphere is humid plants lose water by
(a) photosynthesis (b) osmosis
(c) guttation (d) diffusion

137. Cohesion theory of ascent of sap was proposed by
(a) Dixon (b) Bore
(c) Pristley (d) Atkins

138. Cause of turgidity in plant cell is
(a) air (b) water
(c) hormones (d) All of these

139. If there is no movement of water into a cell from outside medium, the medium is known as
(a) hypertonic (b) hypotonic
(c) isotonic (d) transpiration

140. The diffusion through a semipermeable membrane is known as
(a) osmosis (b) imbibition
(c) guttation (d) transpiration

141. A membrane which permits selective movement of molecules through it, is called
(a) permeable membrane
(b) unit membrane
(c) semipermeable
(d) impermeable membrane

142. Roots can absorb minerals from the soil when they are in
(a) solid state (b) liquid state
(c) ionic state (d) gaseous state

143. Water absorption can be increased by keeping the plants
(a) in the shade
(b) in dim light
(c) under the fan
(d) covered with polythene bag

144. A cell neither shrinks nor swells when kept in a fluid, then the fluid in the cell in relation to ambient fluid is called
(a) hypertonic (b) hypotonic
(c) isotonic (d) hylaluronic

145. Transpiration is very low during storms due to
(a) presence of moisture in wind
(b) low temperature during storms
(c) high velocity of winds
(d) None of the above

146. Which of the following plants open their stomata during night and close during day time?
(a) Fleshy xerophytes
(b) Hydrophytes
(c) Night jasmine
(d) None of the above

147. Consider the statements and mark the correct one.
There is decrease in the rate of transpiration due to
(a) the presence of thick cuticle
(b) the presence of sunken stomata
(c) Both (a) and (b) are correct
(d) None of the above

148. Which of the following statements is correct?
(a) Maximum transpiration occurs through cuticle
(b) Transpiration takes place when outer atmosphere has per cent of moisture less than that of stomatal cavity
(c) There is increase in transpiration due to waxy coating on plants parts
(d) None of the above

149. Why do you feel cool under a tree but not so under a tin shed on a sunny day?
(a) The greenness of the tree gives the cool feeling
(b) Photosynthesis absorbs heat
(c) The leaves convert water vapours into water which is a heat absorbing process
(d) The leaves give out water which vapourises absorbing some heat as latent heat

150. In dry regions, the leaf size of a tree becomes smaller. It is so to
(a) reduce metabolism
(b) reduce transpiration
(c) maintain natural growth
(d) protect plant from animals

151. Tips of leaves in grasses and common garden plants show water drops in early morning hours. This water accumulation is obtained from
(a) atmosphere (b) stomata
(c) vascular bundles (d) hydathodes

152. Which of the following plants is not capable of manufacturing its own food?
(a) Algae (b) Mushroom
(c) Carrot (d) Cabbage

153. Conditions necessary for photosynthesis is/are
(a) chlorophyll and water
(b) CO_2
(c) light and suitable temperature
(d) All of the above

154. Absorption of light in photosynthesis is done by
(a) water (b) air
(c) chlorophyll (d) carbon dioxide

155. Which one of the following gases is released from rice fields in the most prominent quantities?
(a) Carbon dioxide
(b) Methane
(c) Carbon monoxide
(d) Sulphur dioxide

156. Which pigment plays a major role in seed germination?
(a) Chloroplast (b) Chlorophyll
(c) Xanthophyll (d) Phytochrome

157. Match the following Columns.

Column I (Cell organelles)	Column II (Physiological phenomena)
A. Mitochondria	1. Photosynthesis
B. Chloroplast	2. Transpiration
C. Stomata	3. Respiration
D. Cell membrane	4. Osmosis

Codes

	A	B	C	D		A	B	C	D
(a)	1	3	4	2	(b)	3	1	2	4
(c)	1	3	2	4	(d)	3	1	4	2

158. In which of the following cell organelles do photo and thermochemical reactions occur at different sites?
(a) Mitochondria (b) Chloroplasts
(c) Ribosomes (d) Lysosomes

159. Which one of the following statements about photosynthesis is correct?
(a) Carbohydrates are the source of electrons in photosynthesis
(b) CO_2 is the source of electrons in photosynthesis
(c) Water is the source of electrons in photosynthesis
(d) NADH is the source of electron in photosynthesis

160. CO_2 and O_2 molecules can move freely across a plasma membrane. What determines the direction of carbon dioxide and oxygen molecules movement?
(a) Orientation of cholesterol in the plasma membrane
(b) Concentration gradient across the membrane
(c) Configuration of phospholipid in the plasma membrane
(d) Location of receptors on the surface of the plasma membrane

161. Which one of the following is present in chlorophyll which gives a green colour to plant leaves?
(a) Calcium (b) Magnesium
(c) Iron (d) Manganese

162. Wavelengths of which of the following colour of the visible spectrum of light are maximally absorbed by green plants?
(a) Green and yellow (b) Red and blue
(c) Green and red (d) Blue and yellow

163. Which one of the following is not correctly matched?
(a) Adenine derivative – Kinetin
(b) Carotenoid derivative – ABA
(c) Terpenes – IAA
(d) Gas – Ethylene

164. The hormone which suppresses the growth of a plant?
(a) Auxins (b) Cytokinins
(c) Gibberellins (d) ABA

165. Consider the following statements with reference to human nutrition.
1. Banana is richer source of carbohydrates than apples.
2. Banana contains some amount of protein also.
3. Spinach has no protein at all.
4. Potatoes are richer source of protein than peas.

Which of the above statements are correct?
(a) 1 and 2 (b) 2, 3 and 4
(c) 3 and 4 (d) All of these

166. Eyes of potato are useful for
(a) nutrition
(b) respiration
(c) photosynthesis
(d) vegetative propagation

167. *Belladona* plant is the source of alkaloid
(a) auxin (b) atropine
(c) cocaine (d) nicotine

168. Stem cuttings are commonly used for regrowing
(a) cotton (b) banana
(c) mango (d) sugarcane

169. Vivipary is found in
(a) mango (b) *Brassica*
(c) mangrove plants (d) *Cycas*

170. Match the following Columns.

Column I (Mode of reproduction)	Column II (Plants)
A. Vegetative propagation by leaves	1. Rubber, mango and guava
B. Stem cuttings	2. *Bryophyllum* and *Begonia*
C. Grafting	3. Potato
D. Vegetative propagation by tubers	4. Sugarcane, rose and *Bougainvillea*

Codes

	A	B	C	D
(a)	2	4	1	3
(b)	1	4	3	2
(c)	3	2	4	1
(d)	4	2	1	3

171. Seedless berry is produced in which of the following plant?
(a) Date palm (b) Papaya
(c) Tomato (d) Banana

172. Which one among the following plants cannot be multiplied by cuttings?
(a) Rose (b) *Bryophyllum*
(c) Banana (d) Marigold

173. Which of the following is a living fossil?
(a) Blue-green algae (b) Fungus
(c) Green algae (d) *Ginkgo*

174. Which one among the following kinds of organisms resides in the roots of pulses to do nitrogen-fixation?
(a) Bacteria (b) Fungi
(c) Protozoa (d) Virus

175. Cloves, used as a spice, are derived from which of the following plant parts?
(a) Seeds (b) Fruits
(c) Flower buds (d) Young leaves

176. Which of the following is not a seed-borne disease?
(a) Brown leaf spot of rice
(b) Black arm of cotton
(c) Red rot of sugarcane
(d) Potato mosaic

177. Match the following Columns.

Column I (Plants)	Column II (Edible parts)
A. Pea	1. Mesocarp
B. Date palm	2. Cotyledons
C. Papaya	3. Bracts, perianth and receptacle
D. Pineapple	4. Fleshy pericarp

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	2	4	1	3
(c)	1	2	3	4	(d)	3	4	1	2

178. The leaves used as wrappers for bidis are obtained from which one of the following?
(a) Shikakai (b) Rudraksha
(c) Tendu (d) Lemon grass

179. Cutting and peeling of onions bring tears to the eyes because of the presence of
(a) sulphur in the cell
(b) carbon in the cell
(c) fat in the cell
(d) amino acid in the cell

180. The antimalarial drug quinine is made from a plant. The plant is
(a) neem (b) *Eucalyptus*
(c) cinnamon (d) cinchona

181. Which one among the following is a major source of sugar?
(a) Watermelon (b) Beetroot
(c) Sugarcane (d) Date

182. Golden fibre refers to
(a) hemp (b) cotton
(c) jute (d) nylon

183. Match the following Columns.

Column I (Mineral)	Column II (Major sources)
A. Iron	1. Banana, date
B. Potassium	2. Palak
C. Iodine	3. Iodised common salt
D. Calcium	4. Milk, egg

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	2	3	1	4
(c)	4	3	1	2	(d)	4	1	3	2

184. Quinine is a drug used in the treatment of malaria, from which part of the plant is it obtained?
(a) Roots (b) Stem (c) Bark (d) Leaves

185. The branches of this tree root themselves like new trees over a large area. The roots then give rise to more trunks and branches. Because of this characteristic and its longevity, this tree is considered immortal and is an integral part of the myths and legends of India. Which tree is this?
(a) Banyan
(b) Neem
(c) Tamarind (Imli)
(d) Peepal

186. Which one of the following is responsible for the stimulating effect of tea?
(a) Tannin
(b) Steroid
(c) Alkaloid
(d) Flavonoid

187. The genetically engineered 'golden rice' is rich in which of the following?
(a) Vitamin-A and nicotinic acid
(b) β-carotene and folic acid
(c) β-carotene and iron
(d) Vitamin-A and niacin

188. The plant dye henna imparts orange-red colour to skin and hairs due to its reaction with which of the following?
(a) Proteins and amino acids
(b) Lipids
(c) Carbohydrates
(d) Nucleic acids

Previous Years' Questions

189. The ultimate cause of water movement in a plant stem against gravity is **2013 (I)**
(a) osmosis (b) transpiration
(c) photosynthesis (d) diffusion

190. Vaseline was applied to both the surfaces of the leaves of a plant. Which of the following process/processes would be affected? **2013 (I)**
1. Photosynthesis
2. Respiration
3. Transpiration

Select the correct answer using the codes given below:
(a) 1 and 3 (b) Only 2
(c) 2 and 3 (d) All of these

191. Which one of the following elements is present in green pigment of leaf? **2014 (I)**
(a) Magnesium (b) Phosphorus
(c) Iron (d) Calcium

192. Which of the following structures of a plant is responsible for transpiration? **2014 (I)**
(a) Xylem (b) Root
(c) Stomata (d) Bark

193. Which of the following does not possess a specialised conducting tissue for transport of water and other substances in plants? **2014 (I)**
(a) *Marchantia* (b) *Marsilea*
(c) *Cycas* (d) Fern

194. Bagasse, a byproduct of sugar manufacturing industry, is used for the production of **2014 (II)**
(a) glass (b) paper
(c) rubber (d) cement

195. Which one among the following is a micronutrient present in soil for various crops? **2015 (I)**
(a) Calcium (b) Manganese
(c) Magnesium (d) Potassium

Animal Physiology

196. Vitamin-B complex represents a group of how many vitamins?
(a) 5 (b) 6 (c) 9 (d) 11

197. The richest source of vitamin-D is
(a) cod liver oil (b) spinach
(c) milk (d) cheese

198. Cow milk is a rich source of
(a) vitamin-A (b) vitamin-B
(c) vitamin-C (d) fatty acids

199. Which milk contains more fat?
(a) Cow (b) Buffalo
(c) Camel (d) Reindeer

200. Protein-calorie malnutrition causes
(a) malaria (b) hepatitis
(c) typhoid (d) kwashiorkor

201. Which of the following does not belong to vitamin-B group?
(a) Riboflavin (b) Tocopherol
(c) Cyanocobalamin (d) Nicotinic acid

202. Match the following Columns.

Column I (Vitamins)	Column II (Chemical names)
A. Vitamin-A	1. Riboflavin
B. Vitamin-B_2	2. Niacin
C. Vitamin-B_3	3. Retinol
D. Vitamin-C	4. Ascorbic acid

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	3	2	1
(c)	3	1	2	4	(d)	2	3	1	4

203. Consider the following statements regarding human nutrition.
1. Spinach is a good source of vitamin-A.
2. Excess vitamin-D in diet promotes hypercalcemia
3. One of the causes of pain in the joints is due to vitamin-C deficiency.
4. Diarrhoea is one of the symptoms of pellagra.

Which of the above statements are correct?
(a) 1, 2 and 4 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) All of these

204. Bleeding and rupturing of blood vessels is prevented by
(a) vitamin-E (b) vitamin-C
(c) vitamin-K (d) vitamin-B_2

205. Which enzyme is present in bile juice?
(a) Bilin (b) Bilirubin
(c) Biliverdin (d) None of these

206. Digestive gland of prawn is
(a) hepatopancreas
(b) liver
(c) thyroid
(d) pancreas

207. Omasum is absent in
(a) goat (b) cow
(c) camel (d) buffalo

208. Coagulation of milk is done by
(a) casein (b) rennin
(c) pepsin (d) trypsin

209. Consider the following substances
1. Ascorbic acid
2. Folic acid
3. Nicotinic acid
4. Pantothenic acid

Which of the above are vitamins?
(a) 1 and 3 (b) 1, 2, 3 and 4
(c) 2, 3 and 4 (d) 1, 2 and 4

210. Match the following Columns.

Column I	Column II
A. Vitamin-C	1. Night blindness
B. Vitamin-A	2. Scurvy
C. Vitamin-D	3. Ricket
D. Vitamin-B_{12}	4. Anaemia
	5. Destruction of muscles

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	5	4	(b)	4	3	2	1
(c)	2	1	3	4	(d)	2	1	4	5

211. Match the following Columns.

Column I	Column II
A. Enzyme	1. Retinol
B. Vitamin	2. Keratin
C. Protein	3. Maltase
D. Hormone	4. Adrenalins

Codes

	A	B	C	D		A	B	C	D
(a)	3	1	4	2	(b)	2	3	1	5
(c)	4	1	3	2	(d)	3	1	2	4

212. Honey has the largest percentage of
(a) water (b) starch
(c) glucose (d) sucrose

213. Which fruit can diabetic patients eat freely?
(a) Banana (b) Guava
(c) Orange (d) Mango

214. Which of the following vitamins and deficiency diseases is not correctly matched?
(a) Vitamin-A – Night blindness
(b) Vitamin-B_1 – Pellagra
(c) Vitamin-D – Deformities in bones
(d) Vitamin-K – Haemorrhage

215. The protein content of edible portion of egg is
(a) 15% (b) 13.3%
(c) 10.3% (d) 5%

216. Consider the following statements. Keratin is
1. a protein
2. a hormone
3. present in hair
4. present in nail

Which of the above statements(s) is/are correct?
(a) Only 1 (b) 1 and 3
(c) 2 and 3 (d) 1, 3 and 4

217. Match the following Columns.

Column I (Sources)	Column II (Rich in)
A. Fish liver oil	1. Vitamin-C
B. Egg	2. Vitamin-B complex
C. Lettuce	3. Vitamin-D
D. Cow milk	4. Vitamin-A

Codes

	A	B	C	D		A	B	C	D
(a)	2	4	3	1	(b)	2	4	1	3
(c)	3	2	1	4	(d)	4	2	3	1

218. Match the following Columns.

Column I (Sugar)	Column II (Source)
A. Cellulose	1. Honey
B. Fructose	2. Sugarcane
C. Maltose	3. Cotton wool
D. Sucrose	4. Starch

Codes

	A	B	C	D		A	B	C	D
(a)	3	1	4	2	(b)	3	4	1	2
(c)	2	1	4	3	(d)	2	4	1	3

219. The vitamin(s), which is/are generally excreted in urine, is/are
(a) vitamin-A (b) vitamin-E
(c) vitamin-C (d) vitamin-D and K

220. Match the following Columns.

Column I (Mineral)	Column II (Major Source)
A. Iron	1. Banana, date
B. Potassium	2. Palak
C. Iodine	3. Iodised common salt
D. Calcium	4. Milk, egg

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	2	3	1	4
(c)	4	3	1	2	(d)	4	1	3	2

221. Which among the following oils has the maximum protein content?
(a) Castor oil (b) Sunflower oil
(c) Soybean oil (d) Safflower oil

222. Anaemia is a common health problem especially in women. Which one of the following deficiencies is most frequently responsible for anaemia in India?
(a) Calcium (b) Iron
(c) Iodine (d) Zinc

223. Primary source of vitamin-D for human beings is
(a) citrus fruit (b) green vegetables
(c) yeast (d) sun

224. Match the following Columns.

Column I (Vitamin)	Column II (Chemical compound)
A. Vitamin-A	1. Thiamine
B. Vitamin-B_1	2. Retinol
C. Vitamin-C	3. Ascorbic acid
D. Vitamin-E	4. Tocopherol

Codes

	A	B	C	D		A	B	C	D
(a)	4	1	3	2	(b)	2	3	1	4
(c)	4	3	1	2	(d)	2	1	3	4

225. Which one of the following contains vocal cords?
(a) Larynx
(b) Pharynx
(c) Glottis
(d) Primary bronchus

226. Which one of the following is considered as the easily digestible source of protein?
(a) Egg albumin (b) Soybean
(c) Fish flesh (d) Red meat

227. Wisdom teeth are
(a) incisors (b) canines
(c) premolars (d) molars

228. Chemically silk fibres are predominantly
(a) protein
(b) carbohydrate
(c) complex lipid
(d) mixture of polysaccharide and fat

229. Which one of the following enzyme is present in saliva?
(a) Pepsin (b) Ptyalin
(c) Trypsin (d) Chymotrypsin

230. Which one of the following is true stomach?
(a) Ruman (b) Reticulum
(c) Omasum (d) Abomasum

231. Dental formula of human baby is
(a) 2102/2102 (b) 2103/2103
(c) 2100/2100 (d) 2123/2123

232. Acidity in stomach involves
(a) headache
(b) stomach aches
(c) asthma
(d) hormonal deficiency

233. The cells of pancreas that produce glucagon are
(a) α-cells
(b) γ-cells
(c) Both (a) and (b)
(d) β-cells

234. Metabolic rate in mammals is controlled by
(a) pancreas (b) liver
(c) pituitary (d) thyroid

235. Which one among the following will be absorbed fastest through the wall of digestive system?
(a) Black coffee as a hot beverage
(b) DDT taken as a poison
(c) Raw alcohol taken as a drink
(d) Ice-cream as a dessert

236. When we eat something we like, our mouth waters. This is actually not water, but fluid secreted from
(a) nasal glands (b) oval epithelium
(c) salivary glands (d) tongue

237. Which among the following is the correct increasing order of pH found in human body?
(a) Gastric juice, saliva, blood
(b) Blood, saliva, gastric juice
(c) Saliva, blood, gastric juice
(d) Gastric juice, blood, saliva

238. Consider the following statements
1. Iodine is necessary for the thyroid gland to make adrenaline.
2. Iodine deficiency leads to goitre in human beings.
3. Iodine is secreted by pancreas and helps in regulating cholesterol level.

Which of the statement(s) given above is/are correct?
(a) Only 2 (b) 1 and 2
(c) 1 and 3 (d) All of these

239. In human beings, the opening of the stomach into the small intestine is called
(a) caecum (b) ileum
(c) oesophagus (d) pylorus

240. Anaerobic respiration occurs in
(a) ants (b) earthworms
(c) tapeworms (d) echinoderms

241. Oxygen content in inspired air is
(a) 4% (b) 20% (c) 16% (d) 25%

242. Which is the main form in which CO_2 is transported by blood?
(a) Carbonic acid
(b) Oxyhaemoglobin
(c) Bicarbonates
(d) Carboxyhaemoglobin

243. The largest quantity of air that can be expired after a maximal inspiration is
(a) residual volume
(b) tidal volume
(c) lung volume
(d) vital capacity of lungs

244. Which one of the following helps in air conditioning?
(a) Nasal chambers (b) Nostrils
(c) Pharynx (d) Alveoli

245. Respiratory enzymes in bacteria are present in
(a) mitochondria
(b) Golgi complex
(c) plasma membrane
(d) endoplasmic reticulum

246. Which one of the following animals breathe through the skin?
(a) Fish (b) Pigeon
(c) Frog (d) Cockroach

247. Life of RBCs in human blood is of
(a) 30 days (b) 60 days
(c) 120 days (d) 15 hours

248. Element that is not found in blood is
(a) iron (b) magnesium
(c) copper (d) chromium

249. The blood pressure is the pressure of blood in
(a) arteries (b) veins
(c) auricles (d) ventricles

250. A reptile with a four-chambered heart is
(a) crocodile (b) turtle
(c) snake (d) lizard

251. Blood grouping was discovered by
(a) William Harvey (b) Landsteiner
(c) Robert Koch (d) Louis Pasteur

252. How many molecules of O_2 can associate with a molecule of haemoglobin in man?
(a) One (b) Two
(c) Three (d) Four

253. Between which one of the following sets of blood groups, is the blood transfusion possible?
(a) A and O (A donor)
(b) B and A (B donor)
(c) A and AB (A donor)
(d) AB and O (AB donor)

254. Which of the statements given below are correct?
1. A person having blood group-A can donate blood to persons having blood group-A and blood group-AB.
2. A person having blood group-AB can donate blood to persons having blood groups-A, B, AB or O.
3. A person with blood group-O can donate blood to persons having any blood group.
4. A person with blood group-O can receive blood from the person of any of the blood groups.

Select the correct answer using the code given below:
(a) 1 and 3 (b) 1 and 2
(c) 3 and 4 (d) All of these

255. Second heart sound is
(a) lubb at the end of systole
(b) lubb at the beginning of systole
(c) dupp at the end of diastole
(d) dupp at the beginning of diastole

256. Open type circulatory system is found in
(a) insects (b) leeches
(c) molluscs (d) All of these

257. The ECG is employed to detect
(a) cholesterol (b) blood pressure
(c) cardiac cycle (d) atherosclerosis

258. Membrane surrounding the heart is
(a) peritoneum
(b) pleura
(c) pericardium
(d) mucous membrane

259. The blood pressure is measured by
(a) haemoglobinometer
(b) stethoscope
(c) sphygmomanometer
(d) pulse rate

260. Heart with single circulation is found in
(a) mammals
(b) reptiles
(c) fishes and amphibians
(d) only fishes

261. Increase in the number of WBC is referred as
(a) anaemia (b) polythaemia
(c) leukopenia (d) leukemia

262. The lifespan of human WBC is about
(a) less than 10 days
(b) 24 hours
(c) 120 days
(d) 100 hours

263. Pacemaker in heart is
(a) AV node (b) SA node
(c) Purkinje fibres (d) Bundle of His

264. Vein carrying oxygenated blood is called
(a) pulmonary (b) all capillaries
(c) aorta (d) None of these

265. Which chamber of human heart pumps fully oxygenated blood to aorta and hence to the body?
(a) Right auricle (b) Left auricle
(c) Right ventricle (d) Left ventricle

266. The abnormal rise in RBC count is called
(a) lukemia (b) leukopenia
(c) thrombocytosis (d) polycythmia

267. Consider the following respiratory pigments.
1. Haemoglobin
2. Haemocyanin
3. Haemoerythrin
4. Haemocyanoglobin

Iron is contained in
(a) 1, 2, 3 and 4 (b) 1 and 3
(c) 1 and 2 (d) 1, 2 and 4

268. Match the following Columns.

Column I (Protein)	Column II (Type)
A. Haemoglobin	1. Structural proteins
B. Collagen	2. Contractile proteins
C. Albumin and glutamin	3. Transport proteins
D. Actin and myosin	4. Storage proteins

Codes

	A	B	C	D		A	B	C	D
(a)	1	3	4	2	(b)	2	1	3	4
(c)	3	1	4	2	(d)	1	2	4	3

269. Heart is innervated by
(a) trigeminal (b) vagus
(c) facial (d) glossopharyngeal

270. Which of the following parts of blood carry out the function of body defence?
(a) Red blood cells (b) White blood cells
(c) Platelets (d) Haemoglobins

271. What does sphygmomanometer measure?
(a) Blood pressure
(b) Velocity of fluids
(c) Temperature
(d) Curvature of spherical surfaces

272. White blood cells act
(a) as a defence against infection
(b) as source of energy
(c) for clotting blood
(d) as a medium for oxygen transport from lung to tissues

273. Which one among the following animal tissues transport hormones and heat and maintains water balance?
(a) Epithelial tissue (b) Muscular tissue
(c) Blood (d) Nervous tissue

274. Which one of the following is considered normal blood pressure in man?
(a) 120/80 mm water
(b) 120/80 mm blood
(c) 120/80 mm mercury
(d) 120/80 mm air

275. The yellow colour of urine is due to the presence of
(a) bile (b) lymph
(c) cholesterol (d) urochrome

276. Ammonia is the chief nitrogenous waste in
(a) mosquitoes
(b) tadpole of frog
(c) cartilaginous fishes
(d) desert mammals

277. Sweating during exercise indicates operation of which one of the following processes in the human body?
(a) Enthalpy (b) Homeostasis
(c) Phagocytosis (d) Osmoregulation

278. It has been observed that astronauts lose substantial quantity of calcium through urine during space flight. This is due to
(a) hypergravity
(b) microgravity
(c) intake of dehydrated food tablet
(d) low temperature in cosmos

279. The fibrous sheath that connects the bones is called
(a) tendon (b) aponeuroses
(c) periosteum (d) ligament

280. The white fibres are chemically formed of
(a) collagen (b) elastin
(c) myosin (d) reticulin

281. The longest bone in human body is
(a) femur (b) skull bone
(c) patella (d) fibula

282. The protein present in bone is
(a) chondrin (b) ossein
(c) actin (d) myosin

283. A band of white fibres which joins muscles to bone
(a) ligament (b) tendon
(c) elastin (d) actin

284. Muscle fatigue is due to the accumulation of
(a) pyruvic acid (b) lactic acid
(c) glycogen (d) succinic acid

285. Match the following Columns.

Column I	Column II
A. Shoulder and joints	1. Ball and socket joints
B. Ankle, knee and elbow	2. Hinge joints
C. Toe bones	3. Ellipsoid joints
D. Bones in palm	4. Gliding joints

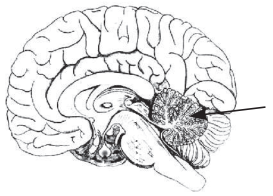
Codes

	A	B	C	D		A	B	C	D
(a)	2	4	3	1	(b)	1	3	4	2
(c)	1	2	3	4	(d)	2	1	4	3

286. Nissl's granules are found in
(a) cartilage cells (b) nerve cells
(c) muscle cells (d) bone cells

287. The junction of two neurons is called
(a) synapsis (b) synapticula
(c) synapse (d) synaptic cleft

288. There is different parts of brain in the figure. The darken portion of the figure represents



(a) cerebrum
(b) cerebellum
(c) cerebral hemisphere
(d) spinal cord

289. In human body, what is the number of cervical vertebrae?
(a) 5 (b) 7
(c) 8 (d) 12

290. Number of spinal nerves in man is
(a) 10 pairs (b) 12 pairs
(c) 21 pairs (d) 31 pairs

291. Outermost covering of brain of man is
(a) duramater (b) piamater
(c) arachnoid (d) choroid

292. Which of the following makes skin layer impervious to water?
(a) Collagen (b) Melanin
(c) Keratin (d) Chitin

293. Sweat glands occur in maximum number in the skin of the
(a) forehead (b) armpits
(c) back (d) palm of hands

294. Any chemical which causes loss of sensation is
(a) sedative (b) analgesic
(c) anaesthetic (d) stimulant

295. Human body's main organ of balance is located in
(a) inner part of ear
(b) middle part of ear
(c) front part of brain
(d) top part of vertebral column

296. Convex lenses are used for the correction of
(a) artigmatism
(b) short-sightedness
(c) cataract
(d) long-sightedness

297. The amount of light entering the eye is regulated by
(a) cornea (b) pupil
(c) iris (d) sclera

298. Dim-vision in the evening and night results from the deficiency of which one of the following?
(a) Vitamin-A (b) Vitamin-E
(c) Vitamin-B_{12} (d) Vitamin-C

299. Human ear ossicles are
(a) incus and stapes
(b) stapes
(c) incus, malleus and stapes
(d) incus and malleus

300. In neurons, action potential is produced due to
(a) Ca^{2+} (b) K^+ (c) Mn^+ (d) Na^+

301. Sensitive pigment layar of eye is
(a) sclerotic (b) retina
(c) cornea (d) None of these

302. The largest cell in the human body is
(a) nerve cell (b) muscle cell
(c) liver cell (d) kidney cell

303. Myopia can be corrected by
(a) convex lens
(b) concave lens
(c) cylindrical lens
(d) cornea replacement

304. Cones are with a pigment
(a) iodopsin (b) rhodopsin
(c) Both (a) and (b) (d) None of these

305. Rods and cones are present in
(a) iris (b) cornea
(c) sclerotic (d) retina

306. Match the following Columns.

Column I	Column II
A. Rods	1. Nervous coat of eye
B. Cones	2. Photoreceptor cells
C. Pupil	3. Regulate amount of light entering the eye
D. Retina	4. Iodopsin pigment

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	1	3	4	2
(c)	4	3	2	1	(d)	2	4	3	1

307. Due to contraction of eyeball, a long-sighted eye can see only
(a) farther objects which is corrected by using convex lens
(b) farther objects which is corrected by using concave lens
(c) nearer objects which is corrected by using convex lens
(d) nearer objects which is corrected by using concave lens

308. Consider the following statements.
1. A person with myopia can see distant objects distinctly but cannot see nearby objects clearly.
2. A person with hypermetropia cannot see distant objects clearly.
3. A person with presbyopia can see nearby objects without corrective glasses.

Which of the statement(s) given above is/are not correct?
(a) Only 3 (b) 1 and 2
(c) 1 and 3 (d) 1, 2 and 3

309. SA node of mammalian heart is known as
(a) autoregulator (b) pacemaker
(c) time controller (d) beat regulator

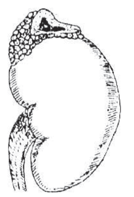
310. Match the following Columns.

Column I (Glands)	Column II (Body functions controlled)
A. Adrenal	1. Growth of bones
B. Pancreas	2. Level of blood calcium
C. Parathyroid	3. Salt and water balance in the body
D. Pituitary	4. Level of blood sugar

Codes

	A	B	C	D		A	B	C	D
(a)	1	4	3	2	(b)	3	1	2	4
(c)	3	4	2	1	(d)	4	2	3	1

311. Which of the following endocrine glands is shown in the below figure.



(a) Pituitary gland (b) Thymus gland
(c) Adrenal gland (d) Pineal body

312. Progesterone and relaxin are secreted by
(a) testes (b) pituitary
(c) thyroid (d) ovary

313. Spermatogenesis in mammalian testes is controlled by
(a) FSH (b) LH
(c) progesterone (d) ICSH

314. Which of the following statements correctly describe the properties of hormones?
1. They are steroids, proteins, peptides or amino acids derivatives.
2. They are not produced by body organs and are mostly taken as supplements.
3. They do not influence the working of those organs which have secreted them.
4. They act as coenzymes and help enzymes to perform their function.

Select the correct answer using the code given below:
(a) 1 and 4 (b) 1, 2 and 4
(c) 2 and 3 (d) All of these

315. A deficiency of which one of the following minerals is most likely to lead to an immunodeficiency?
(a) Calcium (b) Zinc
(c) Lead (d) Copper

316. The gestation period of cows is
(a) 150 day (b) 280 day
(c) 300 day (d) 365 day

317. Which of the following is an egg laying mammal?
(a) Bat (b) Penguin
(c) Whale (d) Spiny anteater

318. In the human body, Cowper's glands form a part of which one of the following?
(a) Digestive system
(b) Endocrine system
(c) Reproductive system
(d) Nervous system

319. Artificial insemination involves the use of
(a) natural semen and natural diluent
(b) natural semen and artificial diluent
(c) artificial semen and natural diluent
(d) artificial semen and artificial diluent

320. Mother's milk is preferred to cow's milk because it contains
(a) more fats and more lipids
(b) less fats and less lipids
(c) less fats and more lipids
(d) more fats and less lipids

321. Implantation of blastocyst occurs on
(a) 5th day (b) 4th day
(c) 6th day (d) 7th day

322. The release of seminal fluid in the vagina of female is called
(a) ejaculation (b) insemination
(c) coitious (d) implantation

323. The ovum released from the ovary is received by
(a) uterus (b) vagina
(c) urethra (d) ostium

324. Sperm entry in the ovum is assisted by
(a) hyaluronidase (b) hyaluronic acid
(c) fertilisin (d) antifertilisin

325. The cavity present in the Graafian follicle is
(a) amniotic cavity (b) archenteron
(c) antrum (d) ostium

326. Testosterone is secreted by
(a) Leydig's cells (b) Sertoli cells
(c) corpus luteum (d) oxyntic cells

327. Why are pregnant women recommended substantial intake of green leafy vegetables in their diet, especially in the 1st trimester?
(a) They are a rich source of chlorophyll
(b) They are a rich source of lecithin
(c) They are a rich source of folic acid which is required for DNA synthesis
(d) They are a rich source of essential fatty acids required for cell anabolism

Previous Years' Questions

328. Which one among the following statements is correct? **2013 (I)**
(a) All proteins are enzymes
(b) All enzymes are proteins
(c) None of the enzymes is protein
(d) None of the proteins is enzyme

329. Which one among the following statements about blood transfusion is correct? **2013 (I)**
Blood group B can give blood to
(a) blood group B and receive from group AB
(b) blood groups B and AB and receive from group B and O
(c) blood groups B and AB and receive from group A and O
(d) blood group O and receive from group B

330. Which one among the following vitamins is necessary for blood clotting? **2013 (I), 16 (I)**
(a) Vitamin-A (b) Vitamin-D
(c) Vitamin-K (d) Vitamin-C

331. The pH of human blood is normally around **2013(I), 16 (I)**
(a) 4.5-5.5 (b) 5.5-6.5
(c) 7.5-8.0 (d) 8.5-9.0

332. The crew and passengers of a flying aircraft suffer generally from chronic obstructive pulmonary disease due to the effect of **2013 (II)**
(a) solar radiation
(b) ozone concentration
(c) nitrogen oxide
(d) particulate pollutant

333. Which one among the following gases readily combines with the haemoglobin of the blood? **2013 (I)**
(a) Methane
(b) Nitrogen dioxide
(c) Carbon monoxide
(d) Sulphur dioxide

334. What would happen if human blood becomes acidic (low pH)? **2013 (II)**
(a) Oxygen carrying capacity of haemoglobin is increased
(b) Oxygen carrying capacity of haemoglobin is decreased
(c) RBC count increases
(d) RBC count decreases

335. The gastrointestinal hormones namely secretin and cholecystokinin secreted by duodenal epithelium activate respectively which organs to discharge their secretions? **2013 (II)**
(a) Pancreas and gall bladder
(b) Gall bladder and stomach
(c) Pancreas and stomach
(d) Stomach and small intestine

336. Human blood is a viscous fluid. This viscosity is due to **2013 (II)**
(a) proteins in blood
(b) platelets in plasma
(c) sodium in serum
(d) RBC and WBC in blood

337. In humans, which one among the following with reference to breathing is correct? **2013 (II)**
(a) During inhalation, diaphragm relaxes
(b) During exhalation, thorax cavity expands
(c) During inhalation, intra-pleural pressure becomes more negative
(d) Unlike inhalation, normal exhalation is an active process

338. Blood does not coagulate inside the body due to the presence of **2013 (II)**
(a) haemoglobin (b) heparin
(c) fibrin (d) plasma

339. A person feeds on rice and vegetable made up of potato only. He is likely to suffer from deficiency of **2013 (II)**
(a) carbohydrate and vitamins
(b) proteins
(c) carbohydrate and proteins
(d) proteins and fats

340. Match the following Columns. **2014 (I)**

Column I (Gland)	Column II (Hormone)
A. Pancreas	1. Cortisol
B. Pituitary	2. Vitamin-D
C. Adrenal	3. Thyroid stimulating hormone
D. Kidneys	4. Glucagon

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	1	2	(b)	4	1	3	2
(c)	2	1	3	4	(d)	2	3	1	4

341. Deficiency of which of the following elements is responsible for weakening of bones? **2014 (I)**
1. Calcium 2. Phosphorus
3. Nitrogen 4. Carbon

Select the correct answer using the codes given below:
(a) 1 and 2 (b) 1, 2 and 3
(c) Only 1 (d) Only 4

342. Which of the following statement(s) is/are correct?
1. Amnion contains fluid.
2. Ultrasound scan can detect the sex of an embryo. **2014 (I)**

Select the correct answer using the codes given below:
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

343. Which of the following statement(s) is/are correct regarding fats? **2014 (I)**
1. Fats are needed for the formation of cell membrane.
2. Fats help the body to absorb calcium from food.
3. Fats are required to repair damaged tissue.
4. Body cannot release energy in fats as quickly as the energy in carbohydrates.

Select the correct answer using the codes given below:
(a) 1 and 4 (b) Only 1
(c) 2 and 4 (d) 3 and 4

344. Which of the following statements are correct?
1. Coronary artery supplies blood to heart muscles.
2. Pulmonary vein supplies blood to lungs.
3. Hepatic artery supplies blood to kidneys.
4. Renal vein supplies blood to kidneys.

Select the correct answer using the codes given below: 2014 (II)
(a) 1, 2 and 3 (b) Only 1
(c) 2 and 4 (d) 1, 3 and 4

345. Which one among the following is the correct pathway for the elimination of urine? 2014 (II)
(a) Kidneys, ureters, bladder, urethra
(b) Kidneys, urethra, bladder, ureters
(c) Urethra, ureters, bladder, kidneys
(d) Bladder, ureters, kidneys, urethra

346. **Assertion** (A) During indigestion, milk of magnesia is taken to get rid of the stomach pain.
Reason (R) Milk of magnesia is a base and it neutralises the excess acid in the stomach. 2014 (II)
Codes
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

347. In human digestive system, the process of digestion starts in 2015 (I)
(a) oesophagus (b) buccal cavity
(c) duodenum (d) stomach

348. Which one among the following structures or cells is not present in connective tissues? 2015 (I)
(a) Chondrocytes (b) Axon
(c) Collagen fibre (d) Lymphocytes

349. Which one among the following statements is not true for mammals? 2015 (I)
(a) They possess hairs on the body
(b) Some of them lay eggs
(c) Their heart is three-chambered
(d) Some are aquatic

350. Neutrophils and lymphocytes originate from 2015 (II)
(a) kidney tubule
(b) spleen
(c) bone marrow
(d) lymph node

351. Absorption of water in the human body can be found in 2015 (II)
1. renal tubule in kidney.
2. hepatic cells in liver.
3. large intestine.
4. pancreatic duct.

Select the correct answer using the codes given below:
(a) 1, 2 and 3 (b) 1 and 3
(c) 2 and 4 (d) Only 3

352. Cobalt is associated with 2015 (II)
(a) growth hormone
(b) vitamin-B_{12}
(c) haemoglobin
(d) intestinal enzymes

353. Glucose is a source of energy. Which one of the following types of molecule is glucose? 2016 (I)
(a) Carbohydrate (b) Protein
(c) Fat (d) Nucleic acid

354. Which one of the following vitamins has a role in blood clotting? 2016 (I)
(a) Vitamin-A (b) Vitamin-B
(c) Vitamin-D (d) Vitamin-K

355. Which one of the following hormones contains peptide chain? 2016 (I)
(a) Oxytocin (b) Corticotrophic
(c) Insulin (d) Cortisone

356. In Artificial Insemination (AI) process, which of the following is/are introduced into the uterus of the female? 2016 (I)
(a) Only egg
(b) Fertilised egg
(c) Only sperms
(d) Egg and sperm

357. Plants contain a variety of sterols like stigmasterol, ergosterol, sitosterol, etc. which very closely resemble cholesterol. These plant sterols are referred as 2016 (I)
(a) phytosterols (b) calciferols
(c) ergocalciferols (d) lumisterols

358. Which of the following pairs of vitamin and disease is/are correctly matched? 2016 (I)
1. Vitamin-A – Rickets
2. Vitamin-B_1 – Beri-beri
3. Vitamin-C – Scurvy

Select the correct answer using the codes given below:
(a) Only 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

359. Which one of the following is considered as 'good cholesterol' with reference to individuals facing the risk of cardio-vascular diseases and hypertension? 2016 (I)
(a) High Density Lipoprotein (HDL)
(b) Low Density Lipoprotein (LDL)
(c) Triglyceride
(d) Fatty acids

Human Health and Diseases

360. DPT is a vaccine for
(a) diarrhoea, polio and typhoid
(b) diphtheria, whooping cough and tetanus
(c) diarrhoea, polio and tetanus
(d) diphtheria, whooping cough and typhoid

361. Which of the following cannot be controlled by vaccination?
(a) Smallpox (b) Diabetes
(c) Polio (d) Whooping cough

362. Which disease cannot be prevented by immunisation?
(a) Polio (b) Diphtheria
(c) Angina (d) Tuberculosis

363. Cholera *Bacillus* was discovered by
(a) Louis Pasteur (b) Ronald Ross
(c) Robert Koch (d) Joseph Lister

364. Match the following Columns.

Column I	Column II
A. Ascariasis	1. Bacteria
B. Giardiasis	2. Nematode
C. Hepatitis	3. Protozoan
D. Tetanus	4. Virus

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	4	1	(b)	4	3	2	1
(c)	2	1	4	3	(d)	4	1	2	3

365. A pregnant woman is advised to undergo abortion if she contracts a disease called
(a) measles (b) smallpox
(c) chickenpox (d) German measles

366. Which of the following diseases was prevalent among red Indians?
(a) Fibrosis (b) AIDS
(c) Syphilis (d) Gonorrhoea

367. Fever causing substance is called
(a) pathogen (b) pyrogen
(c) interferon (d) antigen

368. Match the following Columns.

Column I (Diseases)	Column II (Causative agents)
A. Syphilis	1. *Vibrio*
B. Sore throat	2. Bacilli
C. Cholera	3. Spirilla
D. TB (Tuberculosis)	4. Cocci

Codes

	A	B	C	D		A	B	C	D
(a)	3	4	2	1	(b)	2	4	3	1
(c)	3	4	1	2	(d)	4	2	1	3

369. Match the following Columns.

Column I (Scientist)	Column II (Work)
A. FG Banting	1. Vaccination for smallpox
B. J Lister	2. Germ theory
C. Louis Pasteur	3. Use of carbolic acid as an antiseptic
D. E Jenner	4. Discovery of insulin

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	2	1	(b)	4	2	1	3
(c)	3	4	2	1	(d)	1	4	3	2

370. Which of the following diseases are preventable by vaccine?
1. Tetanus 2. Polio
3. Leprosy 4. Pertussis

Select the correct answer using the codes given below:
(a) 1 and 3 (b) 2 and 4
(c) 1, 2 and 4 (d) All of these

371. Which one of the following causes the chikungunya disease?
(a) Bacteria (b) Helminthic worm
(c) Protozoan (d) Virus

372. Match the following Columns.

Column I (Vaccines)	Column II (Diseases)
A. BCG vaccine	1. Malaria
B. BPL vaccine	2. Sore throat
C. Chloroquine	3. Tuberculosis
D. Penicillin	4. Rabies

Codes

	A	B	C	D		A	B	C	D
(a)	3	4	1	2	(b)	3	4	2	1
(c)	4	3	2	1	(d)	4	3	1	2

373. Match the following Columns.

Column I (Diseases)	Column II (Causative agent)
A. Typhoid	1. Bacteria
B. Malaria	2. Virus
C. AIDS	3. Worms
D. Ringworm	4. Protozoa
	5. Fungi

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	5	1	(b)	1	4	2	5
(c)	3	1	2	5	(d)	1	2	3	4

374. To suspect HIV/AIDS in a young individual, which one among the following symptoms is mostly considered?
(a) Long standing jaundice and chronic liver disease
(b) Severe anaemia
(c) Chronic diarrhoea
(d) Severe persistent headache

375. Insects that can transmit diseases to human are referred to as
(a) carriers (b) reservoirs
(c) vectors (d) incubators

376. For which one among the following diseases no vaccine is yet available?
(a) Tetanus (b) Malaria
(c) Measles (d) Mumps

377. Match the following Columns.

Column I (Agent of transmission)	Column II (Disease transmitted)
A. *Anopheles* mosquito	1. Kala-azar
B. *Culex* mosquito	2. Dengue
C. *Aedes*	3. Malaria
D. Sandfly	4. Filaria

Codes

	A	B	C	D		A	B	C	D
(a)	3	2	4	1	(b)	1	4	2	3
(c)	1	2	4	3	(d)	3	4	2	1

378. Which of the following diseases is caused due to allergic reaction?
(a) Leprosy (b) Typhoid
(c) Asthma (d) Tetanus

379. Amnesia is related to
(a) sleeping sickness
(b) loss of sight
(c) loss of hearing
(d) loss of memory

380. Protein-calorie malnutrition causes
(a) malaria (b) hepatitis
(c) typhoid (d) kwashiorkor

381. Which of the following is not a mosquito borne disease?
(a) Dengue fever
(b) Malaria
(c) Sleeping sickness
(d) Filariasis

382. Which of the following is the science dealing with tumour?
(a) Carcinology (b) Serology
(c) Oncology (d) Chronology

383. The persons working in textile factories such as carpet weavers are exposed to which of the following occupational diseases?
(a) Asbestosis
(b) Asthma and tuberculosis
(c) Silicosis
(d) Siderosis

384. HIV mainly infects
(a) α-globulin
(b) cytotoxic T-lymphocytes
(c) helper lymphocytes
(d) killer lymphocytes

385. Minamata disease is caused by
(a) automobile exhausts containing lead
(b) air with sulphur
(c) industrial wastes having mercury compounds
(d) water from tanneries

386. Which of the following is a water borne disease?
(a) Tuberculosis (b) Cholera
(c) Influenza (d) Malaria

387. BCG is used against
(a) tuberculosis (b) typhoid
(c) hydrophobia (d) measles

388. A girl ate sweets contaminated with flies, due to this she suffered from a disease diagnosed as
(a) kwashiorkor (b) tuberculosis
(c) diphtheria (d) cholera

389. Which of the following statements is true with respect to leukemia?
(a) Number of RBCs increases in blood
(b) Number of WBCs increases in blood
(c) Number of both RBCs and WBCs decreases
(d) Deficiency of minerals

390. Myocardial infarction is a serious disease of
(a) heart (b) brain
(c) lungs (d) kidneys

391. Cancer located in connective tissues is called
(a) carcinoma (b) sarcoma
(c) leukaemia (d) metastasis

392. Which of the following diseases are transmitted from one person to another?
1. AIDS 2. Cirrhosis
3. Hepatitis-B 4. Syphilis

Select the correct answer using the codes given below:
(a) 1 and 2 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) All of these

393. Mosquito can be a vector for following diseases except
(a) yellow fever (b) dengue fever
(c) filaria (d) kala-azar

394. Which one among the following elements/ions is essential in small quantities for development of healthy teeth but causes mottling of the teeth if consumed in higher quantities?
(a) Iron (b) Chloride
(c) Fluoride (d) Potassium

Previous Years' Questions

395. Cancer is more common in older people because 2013
(a) their immune systems have degenerated
(b) the supply of certain hormones declines with age
(c) their bodies are unable to adjust to the changing environment
(d) they have accumulated more mutations

396. Consider the following statements regarding antibiotics. 2013 (I)
1. They are used to destroy disease-causing bacteria.
2. They can be applied to the skin, swallowed or injected to fight microorganisms inside the body.
3. They are effective against disease-causing viruses.
4. The first antibiotic to be discovered was tetracycline.

Which of the statement(s) given above is/are correct?
(a) 1 and 2 (b) 1, 2 and 4
(c) Only 1 (d) 2, 3 and 4

397. Which one of the following diseases in humans can spread through air? 2015 (I)
(a) Dengue (b) Tuberculosis
(c) HIV-AIDS (d) Goitre

398. The word 'vaccination' has been derived from a Latin word which relates to 2015 (I)
(a) pig (b) horse
(c) cow (d) dog

399. Measles is a disease caused by 2016
(a) bacteria (b) virus
(c) protozoan (d) worm

400. The mandate of the scheme entitled 'Directly Observed Treatment, Short-Course (DOTS)' launched by WHO is to ensure that
(a) doctors treat patients with medicine for a short duration
(b) doctors do not start treatment without a trial
(c) patients complete their course of drug
(d) patients voluntarily take vaccines

401. Penicillin inhibits synthesis of bacterial 2016 (I)
(a) cell wall (b) protein
(c) RNA (d) DNA

402. Most antibiotics target bacterial parasites interfering with various factors of growth or metabolism such as 2016 (I)
1. synthesis of cell wall.
2. bacterial protein synthesis.
3. synthesis of nuclear membrane.
4. mitochondria function.

Select the correct answer using the codes given below:
(a) 1, 2 and 3 (b) 1 and 4
(c) 2 and 3 (d) 1 and 2

403. Dengue virus is known to cause low platelet count in blood of patient by 2016 (I)
1. interfering in the process of platelet production in bone marrow.
2. infecting endothelial cells.
3. binding with platelets.
4. accumulating platelets in intestine.

Select the correct answer using the codes given below:
(a) 1 and 2 (b) 1 and 3
(c) 3 and 4 (d) 1, 2 and 3

Applied Biology

404. If excess fertiliser is applied to a plant without water, the plant will
(a) be stunted in growth
(b) develop modifications
(c) die due to plasmolysis
(d) remain unaffected

405. The free-living aerobic soil bacterium which fixes nitrogen
(a) *Azotobacter* (b) *Nostoc*
(c) *Clostridium* (d) *Frankia*

406. Match the following Columns.

Column I	Column II
A. Sugarcane	1. Quinine
B. *Cinchona*	2. Molasses
C. *Hevea*	3. Iodine
D. Sea kelp	4. Rubber

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	4	3	(b)	4	3	2	1
(c)	1	2	3	4	(d)	2	3	4	1

407. Consider the following microorganisms.
1. *Azotobacter*
2. *Clostridium*
3. *Bacillus polymyxa*
4. VAM (Vascular Arbuscular Mycorrhiza)

Which of the microorganism(s) given above is/are free-living nitrogen-fixing bacteria?
(a) Only 1 (b) 1, 2 and 3
(c) Only 3 (d) 2, 3 and 4

408. Consider the following plants.
1. Lentil 2. Cow pea
3. Sunnhemp 4. Wheat

Which of the plants given above is/are used as 'Green manure in India'?
(a) Only 3 (b) 1, 2 and 3
(c) 2 and 3 (d) All of these

409. What are the components of the fungicide 'Bordeaux mixture'?
(a) Magnesium sulphate and sodium hydroxide
(b) Copper sulphate and magnesium hydroxide
(c) Copper sulphate and calcium hydroxide
(d) Copper sulphate and sodium hydroxide

410. In the rice field, which of the following is used as biofertiliser?
(a) *Rhizobium* (b) *Pseudomonas*
(c) *Nitrobacter* (d) *Azolla*

411. From which plant listed below maximum amount of ethanol for biofuel is obtained?
(a) Wood (b) Tapioca
(c) Maize (d) Sugarcane

412. Plants die when over fertilised, because the fertiliser
(a) damage the wall of delicate root hairs
(b) blocks absorption of nitrogenous ions
(c) causes dehydration of plant by exosmosis
(d) increases the soil acidity

413. Which one among the following group of items contains only biodegradable items?
(a) Wood, grass, plastic
(b) Wood, grass, leather
(c) Fruit peels, lime juice, China clay cup
(d) Lime juice, grass, polystyrene cup

414. Endosulfan, which has been in news these days, is a/an
(a) pesticide (b) fertiliser
(c) sulpha drug (d) antibiotic

415. Which one among the following industries produces the most non-biodegradable wastes?
(a) Thermal power plants
(b) Food processing units
(c) Textile mills
(d) Paper mills

416. Which among the following statements about biofertilisers are correct?
1. *Azotobacter* is one of the nitrogen-fixing bacteria used as a biofertiliser.
2. They have to be applied to the leaves of the plant only.
3. They alter the chemical composition of the soil.
4. They can be used along with organic fertilisers.

Select the correct answer using the codes given below.
(a) 1 and 4 (b) 1, 2 and 4
(c) 2 and 3 (d) All of these

417. Which one of the following is not biodegradable?
(a) Woollen mat (b) Silver foil
(c) Leather bag (d) Jute basket

418. Which one of the following plants is used for green manuring in India?
(a) Wheat (b) Sunnhemp
(c) Cotton (d) Rice

419. Which among the following are the most important raw materials for manufacturing of soap?
(a) Fats and caustic alkali
(b) Fats and potash
(c) Fats and acid
(d) Vegetable oil and potash

420. Consider the following statements about Vechur cattle breed.
1. Vechur is the world's smallest cow.
2. It is indigenous breed found in Kerala.
3. Its milk protein has medicinal value.
4. Commonly used in farming.

Which of the statements given above are correct?
(a) 3 and 4 (b) 1, 2 and 3
(c) 2 and 3 (d) All of these

421. Cattle are capable of digesting cellulose present in the grass and/or fodder that they eat. This ability is attributed to the
(a) presence of cellulose degrading bacteria in the rumen
(b) production of cellulose by the cattle rumen
(c) acids present in the rumen
(d) prolonged retention of cellulose in the rumen

422. ELISA test is prescribed for
(a) cancer (b) typhoid
(c) polio (d) AIDS

423. Restriction enzymes are those enzymes that can
(a) cut RNA
(b) cut single-stranded DNA
(c) cut double-stranded DNA
(d) hydrolyse proteins

424. Which one among the following is the 'chemical knives (scissors)' used in genetic engineering?
(a) Polymerase (b) Ligases
(c) Endonucleases (d) Plasmid

425. Match the following Columns.

Column I (Techniques)	Column II (Compounds)
A. Northern blotting	1. RNAs
B. Southern blotting	2. Proteins
C. Western blotting	3. Lipids
D. Far-Eastern blotting	4. DNAs

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	2	3	1	4
(c)	1	2	4	3	(d)	1	4	2	3

426. Match the following Columns.

Column I (Scientists)	Column II (Discoveries)
A. Banting and Best	1. Hybridoma technology
B. Milstein Kohler	2. Isolation of insulin
C. Steward Miller	3. *In vitro* fertilisation
D. PC Steptoe	4. Coconut milk in tissue culture

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	2	1	4	3
(c)	4	2	1	3	(d)	4	1	2	3

427. By using which one of the following techniques, is DNA fingerprinting done?
(a) ELISA (b) RIA
(c) Northern blotting (d) Southern blotting

428. Which of the following techniques can be used to establish the paternity of a child?
(a) Protein analysis
(b) Chromosome counting
(c) Quantitative analysis of DNA
(d) DNA fingerprinting

429. Which among the following is the first hormone produced by genetic engineering?
(a) Insulin (b) Corticol
(c) Thyroxine (d) Testosterone

430. The genetically engineered 'Golden rice' is rich in which of the following?
(a) Vitamin-A and nicotinic acid
(b) β-carotene and folic acid
(c) β-carotene and iron
(d) Vitamin-A and niacin

431. Consider the following statements
1. Dolly, the cloned sheep.
2. George and Charlie, the cloned calves.
3. Headless frog.

Which of the statement(s) given above is/are incorrect?
(a) Only 1 (b) 1 and 2
(c) Only 3 (d) All of these

432. Match the following Columns.

Column I	Column II
A. Maize	1. Arjun
B. Paddy	2. Jaya
C. Wheat	3. Ranjit

Codes

	A	B	C		A	B	C
(a)	1	2	3	(b)	2	3	1
(c)	3	2	1	(d)	3	1	2

433. Which one of the following is responsible for converting milk into curd?
(a) Fungi (b) Bacteria
(c) Virus (d) None of these

434. Which of the following is used in the production of yoghurt?
(a) *Streptococcus thermophilous*
(b) *Lactobacillus bulgaris*
(c) *Streptococcus lactis*
(d) Both (a) and (b)

435. A milkman puts banana leaf in milk jar, because banana leaf
(a) gives a fresh flavour to milk
(b) makes the milk acidic and resistant to yeast
(c) makes the milk basic and resistant to yeast
(d) increases the whiteness of milk

436. Match the following Columns.

Column I	Column II
A. Wine	1. Barley
B. Beer	2. Sugarcane juice
C. Whisky	3. Grapes
D. Rum	4. Molasses

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	4	3	(b)	3	4	1	2
(c)	3	1	4	2	(d)	2	4	1	3

437. Which one of the following is not a constituent of biogas?
(a) Methane (b) Carbon dioxide
(c) Hydrogen (d) Nitrogen dioxide

438. Which of the following gases is released from rice fields in the most prominent quantities?
(a) Carbon dioxide
(b) Methane
(c) Carbon monoxide
(d) Sulphur dioxide

439. Consider the following statements about bioremediation.
1. It may be defined as any process that uses microorganisms or their enzymes to return the environment altered by contaminants to its original condition.
2. Bioremediation may be employed in order to attack specific contaminants, such as chlorinated pesticides that are degraded by bacteria.

Which of the statement(s) given above is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) Neither 1 nor 2

440. Match the following Columns.

Column I (Microorganisms)	Column II (Uses)
A. *Aeromonas hydrophila*	1. Cloning vector
B. Methanobacteria	2. Microbial fuel cells
C. *Agrobacterium tumefaciens*	3. Yogurt preparation
D. *Lactobacillus bulgaricus*	4. Biogas synthesis

Codes

	A	B	C	D		A	B	C	D
(a)	2	4	1	3	(b)	2	1	4	3
(c)	4	3	1	2	(d)	4	1	3	2

441. Which among the following are the major reasons behind preferring *Eucalyptus* tree in the planned forestation process?
1. Plantation grows very fast.
2. Plantation makes the soil more fertile.
3. Wood from *Eucalyptus* tree is easily converted into pulp for paper industry.

Select the correct answer using the codes given below:
(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) All of these

Previous Years' Questions

442. The macronutrients provided by inorganic fertilisers are **2012 (II)**
(a) carbon, iron and boron
(b) magnesium, manganese and sulphur
(c) magnesium, zinc and iron
(d) Nitrogen phosphorus and potassium

443. Which one among the following groups of items contains only biodegradable items? **2013 (II)**
(a) Paper, grass, glass
(b) Wood, flower, iron-scrap
(c) Sewage, plastic, leather
(d) Cowdung, paddy-husk, vegetable wastes

444. The main constituent of gobar gas or biogas is **2014 (I)**
(a) ethane (b) methane
(c) propane (d) acetylene

445. Which one of the following types of pesticides is convenient to control stored grain pests? **2014 (II)**
(a) Systemic pesticides
(b) Fumigants
(c) Contact poisons
(d) Stomach poisons

446. Which one among the following cattle breed produces highest amount of milk? **2014 (II)**
(a) Brown swiss (b) Holstein
(c) Dutch belted (d) Blaarkop

447. Who among the following is considered as the 'father of genetic engineering'? **2016 (I)**
(a) Philip Drinker
(b) Paul Berg
(c) Thomas Addison
(d) Alpheus S Packard Jr

448. Genetic screening is **2016 (I)**
(a) analysis of DNA to check the presence of a particular gene in a person
(b) analysis of gene in a population
(c) pedigree analysis
(d) screening of infertility in parents

449. Genetically Modified (GM) crops contain modified genetic material due to the **2016 (I)**
1. introduction of new DNA.
2. removal of existing DNA.
3. introduction of RNA.
4. introduction of new traits.

Select the correct answer using the codes given below:
(a) 1 and 2 (b) 1, 2 and 3
(c) 3 and 4 (d) 1, 2 and 4

450. Methyl isocyanate gas, which was involved in the disaster in Bhopal in December, 1984, was used in the Union Carbide factory for production of **2016 (I)**
(a) dyes (b) detergens
(c) explosives (d) pesticides

451. The germplasm is required for the propagation of plants and animals. Germplasm is the **2016 (I)**
1. genetic resources.
2. seeds or tissues for breeding.
3. egg and sperm repository.
4. a germ cell's determining zone.

Select the correct answer using the codes given below.
(a) Only 1 (b) 1, 2 and 3
(c) 2 and 3 (d) 2 and 4

ANSWERS

1	a	2	c	3	b	4	c	5	b	6	c	7	c	8	d	9	a	10	b
11	b	12	c	13	b	14	b	15	b	16	c	17	a	18	b	19	b	20	c
21	b	22	d	23	c	24	c	25	c	26	b	27	d	28	d	29	a	30	b
31	c	32	c	33	c	34	b	35	b	36	b	37	c	38	a	39	b	40	d
41	c	42	b	43	d	44	b	45	a	46	b	47	c	48	d	49	a	50	a
51	a	52	d	53	c	54	a	55	d	56	a	57	c	58	c	59	c	60	a
61	c	62	c	63	d	64	a	65	d	66	a	67	c	68	d	69	a	70	c
71	b	72	c	73	c	74	a	75	c	76	a	77	c	78	b	79	b	80	a
81	b	82	c	83	c	84	d	85	c	86	d	87	b	88	b	89	a	90	a
91	d	92	d	93	b	94	b	95	b	96	c	97	d	98	c	99	a	100	d
101	d	102	b	103	a	104	a	105	a	106	d	107	a	108	d	109	a	110	c
111	b	112	b	113	a	114	b	115	c	116	a	117	b	118	b	119	c	120	a
121	c	122	d	123	d	124	c	125	a	126	a	127	b	128	c	129	d	130	d
131	c	132	b	133	d	134	d	135	a	136	c	137	a	138	b	139	c	140	a
141	c	142	c	143	c	144	c	145	c	146	a	147	c	148	b	149	d	150	b
151	d	152	b	153	d	154	c	155	b	156	d	157	b	158	b	159	c	160	b
161	b	162	b	163	c	164	d	165	a	166	d	167	b	168	d	169	c	170	a
171	d	172	c	173	d	174	a	175	c	176	d	177	b	178	c	179	d	180	d
181	c	182	c	183	a	184	c	185	a	186	c	187	c	188	a	189	b	190	d
191	a	192	c	193	a	194	b	195	b	196	c	197	a	198	a	199	a	200	d
201	b	202	c	203	a	204	c	205	d	206	a	207	c	208	b	209	b	210	c
211	d	212	c	213	b	214	b	215	b	216	d	217	c	218	a	219	c	220	a
221	c	222	b	223	d	224	d	225	c	226	a	227	d	228	a	229	b	230	d
231	a	232	b	233	a	234	d	235	c	236	c	237	a	238	a	239	d	240	c
241	b	242	c	243	d	244	a	245	a	246	c	247	c	248	c	249	a	250	a
251	b	252	d	253	c	254	a	255	d	256	d	257	c	258	c	259	c	260	d
261	d	262	a	263	b	264	a	265	d	266	d	267	b	268	c	269	b	270	b
271	a	272	a	273	c	274	c	275	d	276	b	277	b	278	b	279	d	280	a
281	a	282	b	283	b	284	b	285	c	286	b	287	c	288	b	289	b	290	d
291	a	292	c	293	b	294	c	295	a	296	d	297	b	298	a	299	c	300	c
301	b	302	a	303	b	304	a	305	d	306	d	307	b	308	d	309	b	310	c
311	c	312	d	313	a	314	a	315	c	316	b	317	d	318	c	319	c	320	b
321	c	322	b	323	d	324	a	325	c	326	a	327	c	328	b	329	b	330	c
331	c	332	c	333	c	334	b	335	c	336	d	337	c	338	b	339	b	340	a
341	a	342	c	343	a	344	b	345	a	346	a	347	b	348	b	349	c	350	c
351	b	352	b	353	a	354	d	355	c	356	c	357	a	358	b	359	a	360	b
361	b	362	c	363	c	364	a	365	b	366	c	367	b	368	c	369	a	370	c
371	d	372	a	373	b	374	d	375	c	376	b	377	d	378	c	379	d	380	d
381	c	382	c	383	b	384	c	385	c	386	b	387	a	388	d	389	b	390	a
391	b	392	b	393	d	394	c	395	d	396	a	397	b	398	c	399	b	400	c
401	a	402	a	403	d	404	c	405	a	406	a	407	b	408	b	409	c	410	d
411	d	412	c	413	b	414	a	415	a	416	a	417	b	418	b	419	a	420	d
421	a	422	d	423	c	424	c	425	d	426	b	427	d	428	d	429	a	430	c
431	c	432	c	433	b	434	d	435	c	436	c	437	d	438	b	439	c	440	a
441	a	442	d	443	d	444	b	445	b	446	b	447	b	448	a	449	d	450	d
451	b																		